The Method

Cardioimmunology

The Method

Understanding Evidence and Building Knowledge in Cardioimmunology

Book 2

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Author and Scientific Editor

Editorial and Legal Notices

The Book

Cardioimmunology
Book 2. *The Method: Understanding Evidence and Building Knowledge in Cardioimmunology*

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Examples presented as fictional are entirely fictional: their numbers serve the exercise of reasoning and describe no study.

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References were checked in PubMed, and the most consequential statements were verified in the accessible articles; details that could not be verified were removed. Figures are reported with the design of the study that produced them; data obtained in animals or in culture name their model. Disagreements between sources are set out, not settled by authority.

This book contains no personal patient data; examples presented as fictional are entirely fictional.

Current guidelines remain the authority for treatment: this book organizes clinical reasoning; it does not replace it. Errors reported after publication will be listed and corrected in subsequent editions.

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General Introduction

I. Understanding What Warrants a Conclusion

I.1 A Question behind the Clarity of Exposition

KNOWING HOW ONE KNOWS

A medical treatise presents organized knowledge. It brings diseases together, explains their mechanisms and sets out the reasons for a diagnostic or therapeutic course of action. One may enter it through a practical question and look there for an answer. Yet another question remains, sometimes hidden behind the clarity of the exposition: how did anyone come to know what it says? This question concerns the experiments that made a hypothesis credible, the observations that shifted its limits, the trials that put an intervention to the test and the discussions that turned results into shareable knowledge. It also concerns the conclusions that were abandoned, those that are provisionally retained and those between which the sources do not yet allow a choice.

I.2 Passages between Scales and between Questions

CARDIOIMMUNOLOGY, SEVERAL SCALES

Cardioimmunology offers particularly rich ground for this examination. It requires following lines of reasoning that cross several scales: a signaling pathway, a cell, a tissue, a circulation, an organism, a population. A result obtained at one of these scales sometimes opens a question at another. It does not, by itself, supply the answer. These passages deserve particular attention, because it is there that the fruitfulness of research meets the fragility of certain interpretations. Understanding a publication means recovering the question it had the means to ask. An experiment can shed light on a mechanism without assessing the usefulness of a treatment. A study can show that a marker predicts an event without establishing that modifying this marker would prevent that event. A trial can test an intervention in a defined population without exhausting the biological question that motivated it. These distinctions determine what can be written and what must be left open.

THE QUESTION A STUDY CAN ASK

II. Four Readings for Understanding a Study and Its Scope

II.1 The Construction of Evidence and the Meaning of the Result

CRITICAL READING, BUILDING THE EVIDENCE

Critical reading examines how the evidence was constructed. It asks how participants or models were chosen, how comparisons were organized, how measurements were obtained and how results were analyzed. It is concerned with bias, but also with precision, with the relevance of the question and with the agreement between what was announced and what is finally concluded.

IMMUNOLOGICAL READING, WHAT A RESULT MEANS

Immunological reading seeks the meaning of the result. It places an observation within the relations among cells, mediators and tissues. It asks which explanations remain possible and which experiment would help to distinguish them. It resists the temptation to convert a description into a mechanism simply because the elements described bear familiar names.

II.2 The Scope for the Heart and for the Person

CARDIOLOGICAL READING, THE SCOPE

Cardiological reading examines the scope of this result. It asks what the result changes in the understanding of the heart and of cardiovascular diseases, and then what it contributes to diagnosis, prognosis or treatment. Knowledge can be important without immediately providing a clinical decision. Recognizing both this importance and this distance is one way of taking research seriously. Integrative clinical reading places this contribution within the overall situation of a person. It adopts the perspective of general practice: coexisting diseases, stated priorities, daily life, prevention and continuity of care. Depending on the article, it examines functional impact, treatment burden, coordination or access to information. It does not assume that the study measured all these dimensions; it distinguishes what the study illuminates from the additional questions that need examination.

INTEGRATIVE CLINICAL READING, THE PERSON

II.3 Readings That Correct One Another

FOUR READINGS THAT CORRECT EACH OTHER

These four readings correct one another. A biological interpretation must answer to the limits of the evidence; a clinical application must preserve the question actually studied. Integration consists in articulating this knowledge with the person's goals and constraints, while justifying the passages between them. It does not confer efficacy on an intervention that has not been evaluated. The fourth reading thus offers an original analysis of the overall scope of each text, without presenting itself as an additional result reported by the text's authors.

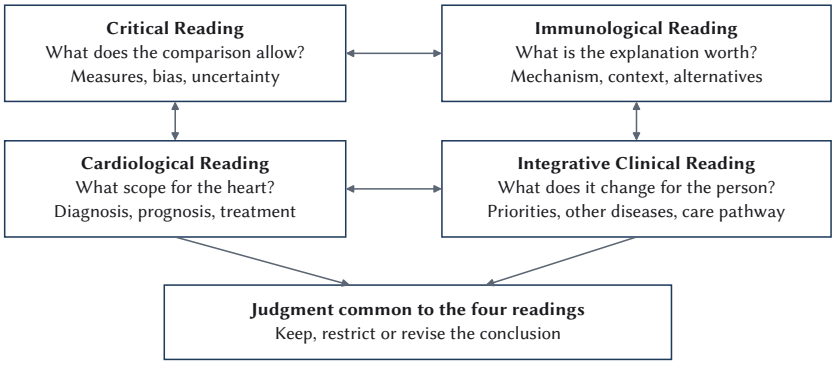


Figure 0.1. One proposition, four readings Original conceptual diagram. The four readings correct one another; this organization of reasoning represents neither a biological causality nor the demonstration of a clinical benefit.

III. Why Philosophy Belongs to This Reading

III.1 Concepts Already Present in Medical Reading

WHERE PHILOSOPHY BEGINS

Philosophy begins when the concepts and commitments that organize reasoning are examined. What is meant by a cause? What difference is drawn between explaining a disease and predicting its course? What kind of confidence is granted to a reproducible result? What becomes of a scientific judgment when it must support an action whose consequences affect a person? These questions are not added to medical reading from outside. They are already present when an author chooses an endpoint, describes a relationship as causal or presents a difference as clinically relevant. Leaving them implicit does not eliminate them; it only makes their discussion more difficult.

texts as interlocutors

Addressing them requires formulating arguments and examining the objections to them. A conception of knowledge will have to show what it makes understandable and acknowledge what it explains less well. A philosophical distinction will have to help in reading a specific difficulty. Philosophical texts will thus serve as interlocutors, with their own contexts, propositions and tensions.

III.2 Claude Bernard's Doubt and Its Precise Objects

Bernard's experimental doubt

Claude Bernard's experimental doubt provides a first interlocutor. In *An Introduction to the Study of Experimental Medicine*, the investigator must keep ideas open to the test of observations. Yet Bernard also maintains a commitment to the determinism of phenomena. Reading his thought requires holding together this critical openness and this explanatory horizon, rather than retaining only a general invitation to doubt.[1] Doubt must bear on a precise object: a measurement, a comparison, an explanation, a generalization or a decision. These questions have different objects and call for different answers. A useful method teaches how to distinguish them, so that disagreements become open to discussion.

IV. One Hundred Texts and Their Interlocutors

one hundred texts, no ranking, no sample

One hundred scientific texts are commented on for the questions they make it possible to work through, including a methodological analysis of the 2025 guidelines of the European Society of Cardiology (ESC). They represent neither a ranking by prestige nor a statistical sample of the whole literature. Their selection serves a purpose: to make intelligible the stages and difficulties of building knowledge in cardioimmunology. A text may be selected because it opens a field, introduces a way of measuring, clarifies a causal relationship, delimits a benefit or proves an expectation wrong. A contested work can be particularly instructive if its limits are established and its place explained. The selection must therefore give access to the objections that mattered as much as to the results that convinced. Each text receives a main location. It may return in another chapter when the question changes, but its analysis is not repeated. Complementary works serve to clarify it, contradict it or update it. The one hundred texts structure the path; they do not close the bibliography. Several of these texts are among the sources of the Treatise on Cardioimmunology. Their commentary examines the reasoning that links their results to a medical synthesis; it can be read independently of the Treatise. This documentary kinship does not amount to verifying every proposition in the Treatise.

the kinship with the Treatise

V. The Place of the Reader

following before mastering

It should be possible to follow the reasoning before all its instruments are mastered. Technical notions will be introduced at the point where they become necessary. Each calculation will be accompanied by its question, its data and its assumptions. A figure should make a relationship easier to understand; its simplicity should not hide an important uncertainty. Each commentary will lead to distinguishing what is established, what remains plausible, what cannot be concluded and what could change the judgment. This distinction does not claim to sort knowledge into fixed boxes. It makes the state of a discussion

established, plausible, not concluded

explicit so that it can be revisited with new elements. The exercises will ask for a conclusion to be formulated, a comparison to be identified or an objection to be discussed. Their answers will set out the reasons for the judgment. When several answers are defensible, it will be necessary to understand which commitments they rest on and what information would make it possible to choose between them. The path leads from the conditions of knowledge to mechanisms, from mechanisms to treatments and then from measuring instruments to medical judgment. Its coherence rests on a single requirement: to preserve, for as long as possible, the links between the question asked, the evidence obtained and the conclusion transmitted.

VI. Choosing a Path and Testing One's Understanding

A SHORTENED PATH

For a first reading, Chapters 1, 3 and 4 provide the landmarks of reasoning and of the examination of sources. Chapters 5 and 6 introduce the inflammatory and cellular problems; Chapter 9 then makes it possible to follow their clinical testing. Chapters 18 and 19 return to the interpretation of numbers and to the passage from the population to the person. This shortened path prepares for the complete reading; it does not replace the developments devoted to the other diseases. Someone already familiar with these fields can enter through a clinical controversy in Part III or Part IV. Such a person will benefit from writing down their own conclusion before reading the commentary, then returning to Chapters 2, 3 or 20 when the disagreement concerns the definition of disease, causality or the conditions of objectivity. To work on a text in a seminar, the reconstruction of the question can first be assigned to one reader and the most important objection to another. Each must point to the passage or result that supports their position. The discussion ends with a shared conclusion, or with the exact identification of the disagreement that remains. The exercise succeeds if the participants can explain what information would change their judgment. Reciting a list of possible biases is not enough to pass this test.

WORKING ON A TEXT IN A SEMINAR

References for the Introduction

1. Bernard C. Introduction à l'étude de la médecine expérimentale. 1865. Part One, chapter II, especially the sections on doubt and fixed ideas, and § VIII, "De la preuve et de la contre-épreuve."

PART I

Building Knowledge of Living Systems

A medical conclusion commits to more than the accuracy of its references. A sentence can cite its sources correctly and still go beyond what they establish. The word “normal” can designate a frequency, a preserved function or a state that is satisfactory for the person. The word “cause” adds to the observed succession a comparison that must be named. A bibliography, finally, becomes a body of evidence only when related to a precise question. These difficulties share a single form. Each time, a claim must be related to what supports it: proposition to reasons, definition to the task it serves, explanation to the comparison that could refute it, source to its version and content. Keeping these links visible does not make judgment more timid. It makes it possible to say what is established, what remains plausible and what a new observation could change.

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Knowing in Medicine

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THE VERB COMMITS TO THE FACTS

A medical sentence can be clear and documented and yet go beyond what its sources establish. The difficulty does not always come from a wrong figure. It may lie in the verb chosen, in the subject to which the result is applied or in a condition lost during transmission. “Is associated with,” “contributes to,” “causes” and “makes it possible to prevent” do not express the same commitment to the facts. Learning to know in medicine begins with this attention. Before asking whether a claim is true, it is necessary to understand exactly what it claims. It is also necessary to determine what type of observation could support it and what would count against it. A proposition whose meaning changes every time a result troubles it becomes difficult to test, even if it remains easy to defend.

1.1 What a Proposition Asserts and What Supports It

1.1.1 The Proposition before the Evidence

A PROPOSITION TO BE SPECIFIED

Consider a deliberately general proposition: an inflammatory pathway participates in a heart disease. To discuss it, the meaning of “participates” must be specified. Is the pathway more active in people with the disease? Does it precede the lesion? Does its disruption modify a feature of the disease? Does its

FOUR DISTINCT QUESTIONS

inhibition improve an outcome that matters to patients? These questions form a possible research program. None of them coincides with all the others. That a pathway is measurable in patients can justify studying it without demonstrating that it explains their disease. That an intervention modifies this pathway can strengthen a hypothesis without establishing that all the effects of the intervention pass through it. Precision in the proposition protects research against two excessive demands. The first asks a mechanistic study to provide immediate therapeutic proof. The second considers that a biological explanation suffices to establish that proof. In both cases, the work is judged by a

A SENTENCE TRACED TO THE WORK

question it did not necessarily ask. A simple discipline follows: write the conclusion as a sentence in which each term can be traced to an element of the work. The population, intervention, comparator, outcome and time must keep their place in it when omitting them would change the meaning. The sentence may be less brilliant; it becomes more exact and more useful.

1.1.2 The Source and the Reason to Believe

1.1.2.1 An exact reference is not yet a reason

REFERENCE, PASSAGE, REASON

A reference identifies a document. For it to support a claim, a passage, analysis or result in that document must also be relevant to the claim in question. The quality of the reasons it provides must then be examined. These operations are linked, but they are not interchangeable. A publication can be located exactly and made to say what it does not say. Its result can be cited correctly while a limitation that changes its scope is neglected. Its interpretation can, finally, be faithfully reported without any examination of whether that interpretation is justified. Documentary fidelity and scientific soundness must therefore be

TRACING BACK THE LINEAGES

assessed separately. The situation becomes more delicate when several texts repeat the same conclusion. Their number can give an impression of convergence. Yet if all of them depend on a single study, this multiplication does not create as many independent confirmations. The lineages must be traced back: which observations are genuinely new, which analyses reuse the same data and which texts merely reformulate an interpretation?

1.1.2.2 What a synthesis adds without new observation

1.1.2.2.1 An apparent contradiction in a fictional example

A synthesis can nevertheless contribute knowledge without adding an observation. It puts to the test the relationships established between results. Its work therefore does not consist only in arranging them: it can show that an announced contradiction is not one, or that a shared conclusion rests on different questions. Consider an entirely fictional example. One article reports higher total production of a mediator in a tissue; another finds no increase in production per cell. A synthesis initially presents them as incompatible. Returning to the quantities measured shows that both sentences can be true together: total production and production per cell do not describe the same object. A difference in cell number, in particular, is one possible explanation to be examined; this juxtaposition does not demonstrate it. The synthesis can thus correct the diagnosis of contradiction without having discovered the cause of the difference.

AN APPARENT CONTRADICTION

1.1.2.2.2 A real but delimited contribution

It will be objected that this is a clarification of language, not scientific progress. But calling two results contradictory leads to a search for which one is false. Recognizing their compatibility legitimately shifts the inquiry: it becomes necessary to understand how the quantities combine and what the measurements actually allow to be compared. The distinction changes the problem to be solved and avoids requesting a replication that would not bear on the same proposition. The contribution is therefore real, but delimited. It becomes defensible if the methods and results of the texts allow this juxtaposition to be reconstructed; the elegance of the narrative alone does not guarantee it. A synthesis does not add an independent confirmation to the experiments it brings together. It can add a checkable argument about their shared meaning. It is this argument that should be granted or refused credit, without requiring every fruitful reorganization to announce a revolution.

A REAL BUT DELIMITED CONTRIBUTION

1.1.3 Observation and Its Conditions

Observing presupposes instruments, categories and decisions. It is necessary to choose where to look, at what moment, at what resolution and by what rule a measurement is retained. These conditions can be made explicit and discussed. Recognizing them does not amount to declaring every observation arbitrary. The distinction is important. If observation depends on a method, the consequences of that method can be examined, instruments can be compared and agreement can be sought between different approaches. Objectivity is then built through checks and confrontations that make observation less dependent on a single point of view. Consider a teaching example, not tied to any particular study. Two teams describe different cell populations in the same organ. One analyzes fresh tissue, the other isolated nuclei. Their categories and results may diverge for several reasons: biological differences between samples, cell loss during preparation, measurement resolution or analytical decisions. Asserting at the outset that one of them is wrong avoids precisely the work that would make it possible to understand the disagreement. The relevant question becomes: which differences persist when the conditions of observation are brought closer together? A robust result is not necessarily identical under every method; it is necessary to determine which variations are expected and which variations genuinely put its interpretation in difficulty.

THE CONDITIONS OF OBSERVATION

TWO TEAMS, TWO METHODS

1.2 The Credit of an Explanation and the Measure of Doubt

1.2.1 The Scope of an Explanation

1.2.1.1 An explanation answers a question

AN EXPLANATION ANSWERS A QUESTION

An explanation organizes relationships and answers a question. It therefore selects one aspect of the phenomenon. Explaining why a cell produces a mediator is not enough to explain why a person develops a disease at a given moment. The two questions can communicate without becoming identical. Moving from one level to the other requires intermediate propositions. It must be established that the mechanism exists in the context considered, that it exerts sufficient influence and that the other elements of the system do not neutralize or transform its effect. Each passage adds conditions to the reasoning.

1.2.1.2 Granting credit without knowing everything

CREDIT WITHOUT KNOWING EVERYTHING

It may be objected that such caution makes any conclusion impossible. The objection deserves to be taken seriously. Requiring knowledge of every detail before granting credit to an explanation amounts to confusing reasonable justification with exhaustive knowledge. Research can produce a partial explanation that is useful and solidly supported, even if certain elements remain unknown. The answer also consists in recognizing when an explanation deserves positive credit. In a fictional example, two mechanisms explain an initial observation, but only one correctly predicts a new result obtained by another method. If the alternative accounts for it only by adding a hypothesis without independent support, the reasons are no longer balanced. The first explanation can be provisionally favored and a new experiment devoted to it. It is not necessary to have eliminated every conceivable possibility: what is needed is to have decided between the serious competitors on a defined question. This credit authorizes a research step; a therapeutic decision would still require an examination of its own consequences.

DECIDING BETWEEN SERIOUS RIVALS

1.2.2 Doubt That Clarifies and Doubt That Dissolves

A DOUBT THAT NAMES ITS OBJECT

Doubt can improve reasoning when it names what it calls into question. It can also become a way of avoiding any position. Saying that a study has limitations is not enough to show that its conclusion is false. Not all limitations have the same effect or the same importance for the question asked. A reservation must therefore be translated into a consequence. If the population is highly selected, which generalization becomes fragile? If data are missing, which assumptions are needed to interpret the analysis? If the measurement is imperfect, in which direction and with what plausible magnitude could the error act? An objection gains its force when it describes the link between the limitation and the conclusion. Doubt must also be able to receive an answer. When analyses, new observations or a replication respond to a criticism, it is necessary to accept revising one's judgment. A vigilance that endlessly invents a new requirement after each answer is no longer the same thing as an open examination. This reciprocity also binds whoever formulates the criticism. The critic's analysis may in turn need to be revised. By distinguishing observations from interpretations and making the reasons explicit, the critic allows a precise answer to change that judgment.

TURNING A RESERVATION INTO A CONSEQUENCE

A DOUBT THAT ACCEPTS THE ANSWER

1.3 Concluding, Deciding and Putting One's Conviction to the Test

1.3.1 Knowing and Deciding

CONCLUDING IS NOT DECIDING

A scientific conclusion and a medical decision do not have exactly the same form. The first concerns what the observations allow to be asserted. The second involves a choice between possible actions, with their benefits, risks, constraints and consequences for a person. Knowledge does not disappear at the moment of choice. It becomes one of the reasons for the choice, and its scope must be made explicit. Uncertainty about the effect of an intervention cannot be treated in the same way regardless of the consequences of an error, the available alternatives and the preferences of the person concerned. The action threshold belongs to this deliberation; it is not the same as a statistical threshold. It would nevertheless be dangerous to use this distinction to make any decision independent of the evidence. Deliberation must be able to show how the known results were taken into account, what uncertainties remain and why the chosen option appears defensible. The singularity of a person requires additional reasoning; it does not remove the requirement for reasons. No sequence of rules dispenses with judgment. Evidence makes this judgment better informed when its contribution can be delimited and the point where a personal hypothesis begins can be recognized. Autonomy does not consist in freeing oneself from these reasons, but in being able to discuss them.

THE DECISION SHOWS ITS REASONS

1.3.2 An Example for Testing the Reasoning

1.3.2.1 Three readers, three conclusions

THREE CONCLUSIONS TO BE SPECIFIED

Consider a fictional trial in which a treatment modifies a biological marker, while the functional endpoint shows no statistically significant difference between the groups. Three readers propose three conclusions: the treatment works; the treatment does not work; the biological hypothesis is false. Each of these conclusions requires specification. The first must state which outcome it concerns. The second must distinguish the absence of a demonstrated effect from an estimate precise enough to exclude an effect of clinically important magnitude. The third must explain why the result of the treatment would allow the entire biological hypothesis to be rejected.

1.3.2.2 What each conclusion must specify

ABSENCE OF EVIDENCE, EVIDENCE OF ABSENCE

The principle discussed by Altman and Bland helps to examine the second conclusion: a nonsignificant result is not enough, by itself, to establish the absence of an effect. The estimate and its uncertainty must be considered. This does not mean regarding every nonsignificant trial as devoid of information; a sufficiently narrow interval can rule out certain effects of practical importance.[1]

MARKER AND SURROGATE

The first conclusion calls for another distinction. A marker can replace a clinical outcome only if its use as a surrogate is sufficiently supported in the context considered. Fleming and Powers explain why a relationship with prognosis does not guarantee that the effect of a treatment on the marker predicts its overall clinical effect.[2] The third conclusion, finally, requires specifying what was actually put to the test: a target, an intervention, a dose, a duration, a population and an endpoint. A result can weaken certain versions of the hypothesis while leaving other questions open. Recognizing this limited scope must not, however, become an automatic way of protecting the hypothesis against any refutation.

1.3.3 Peirce: Being Convinced and Being Right

1.3.3.1 Inquiry according to "The Fixation of Belief"

Peirce, 1877

In "The Fixation of Belief", published in November 1877, Charles S. Peirce describes inquiry as an effort prompted by doubt and tending to stabilize a belief. This description does not, however, identify truth with the satisfaction obtained. From the second section onward, he distinguishes the validity of an inference from the impulse that drives acceptance of its conclusion. He then compares four ways of fixing belief: tenacity, authority, preference for what seems agreeable to reason and the scientific method. The last presupposes a reality whose characters do not depend on the opinions held about it. The agreement to which it could lead would depend on sufficient experience and reasoning; mere present agreement does not guarantee it.[3]

1.3.3.2 Felt doubt and the reason to resume inquiry

A tension then deserves discussion. If inquiry ends psychologically when doubt ceases, why continue to check a conviction that is shared and untroubled? From a medical standpoint, the answer cannot be reduced to the wish to be reassured. An error can have consequences for another person even when nothing troubles the one who commits it. The feeling of having searched enough is therefore not, by itself, a sufficient rule for deciding that the examination is over. The distinction proposed here separates felt doubt from the reason to resume an inquiry. A researcher may feel no hesitation about a result; discovering that the instrument was not checked is nonetheless a reason to verify some of its properties. This requirement extends the text critically; it does not appear in it. It does not demand that everything be doubted artificially, but that grounds for checking be recognized that do not depend on anyone's personal unease.

felt doubt, the reason to check

1.3.3.3 A targeted check rather than endless suspicion

an objection opens an operation

The opposite objection is serious: does this not install endless suspicion, since all knowledge involves debatable conditions? It is necessary to specify which error the check would look for and which conclusion that error could modify. In a fictional situation, checking for a measurement drift capable of creating the announced effect has a determinate object. Repeating indefinitely that any measurement might be imperfect has none. A relevant objection opens a possible operation; indefinite skepticism no longer even indicates what would count as an answer. Peirce thus makes it possible to raise a question worth keeping, without claiming to resolve his entire philosophy: how do habits of conviction remain exposed to something other than themselves? For the clinician, the decisive test is not to observe one's own assurance, but to be able to explain why a method gives a real difficulty the possibility of modifying judgment. This requirement does not abolish confidence; it gives it an object that can be examined.

1.4 Texts Put to the Test

1.4.1 Medzhitov: A Fruitful Hypothesis Does Not Become an Established Explanation

1.4.1.1 Situating the Result

PARA-INFLAMMATION

In this 2008 conceptual review, Medzhitov proposes thinking of inflammation beyond infection and overt injury. Para-inflammation designates a response to tissue malfunction or stress, intermediate between basal maintenance and classic inflammation. The text distinguishes inducers, sensors, mediators and effectors; it explicitly leaves open the identity of certain signals and the existence of physiological counterparts to several pathological states. Its evolutionary box distinguishes adaptation, change of context and unfavorable consequences linked to an advantageous trait. It does not present a new experiment validating the model as a whole. The text is read here as a historical organizing proposal, without converting its particular mechanisms into the current state of knowledge.[4]

1.4.1.2 Critical Reading: giving content to the explanation

What is gained by calling a phenomenon para-inflammation? The name becomes useful if it separates questions that were previously conflated. It loses this usefulness if any persistent abnormality can be redescribed after the fact as an attempt at adaptation. The objection is therefore serious: a framework broad enough to accommodate every result might no longer allow a choice among them. The objection calls for a requirement: distinguishing between observing a response and establishing its function. Suppose, in a fictional experiment, that a tissue subjected to stress increases its production of a signal. This observation does not yet make it possible to decide whether the signal compensates for the stress, aggravates its effects or accompanies another response. The word “adaptation” should open this comparison without announcing its result. It would be necessary to specify which capacity would be preserved, over what time frame and under what conditions disrupting the signal would cause this protection to be lost. It will be objected that an exploratory model cannot immediately supply all these answers. That is correct. Its interest can precede its confirmation. But its fruitfulness is then judged by the distinctions and experiments it makes possible, not by the number of diseases that can be filed under its vocabulary. A still incomplete hypothesis remains scientific when it indicates the observations capable of refining or contradicting it.

OBSERVING IS NOT ESTABLISHING A FUNCTION

1.4.1.3 Immunological Reading: a function is not an intention

A FUNCTION IS NOT AN INTENTION

Speaking of a tissue that asks for help makes a relationship intuitive. That does not mean that a cell foresees the interest of the organism. A functional description, a sequence of mechanisms and an evolutionary history answer different questions. Showing a compensation today does not, by itself, reconstruct the conditions of its past selection. This distinction also makes it possible

SAFETY CANNOT BE DEDUCED

to discuss the text. At the end of the box, the author envisages blocking non-adaptive processes as beneficial and free of adverse effects. This expectation is not enough to establish the safety of an intervention. Even if an effect served no purpose, the means used to suppress it could affect other functions. The property attributed to the effect and the actual selectivity of the treatment are two distinct questions. The objection does not abolish the interest of searching for a function; it prevents drawing a therapeutic guarantee directly from it.

1.4.1.4 Cardiological Reading: specifying what the heart would stand to gain

PRESERVED FUNCTION, DEFERRED COST

The contribution to cardioimmunology lies here in the program of comparison. In the fictional example of the signal produced by the tissue, a response could temporarily maintain a function under stress while imposing a later cost. To study it, the preserved function, the possible harm and their time frames would have to be stated separately. Retaining only the decrease in a signal would leave the main question open: does the condition of the organ improve? The reverse deserves the same attention. A response described as compensatory must not become untouchable by definition. A compensation can be insufficient, costly or rendered unnecessary by another intervention. The decision will depend on concrete comparisons, not on the favorable or unfavorable value attached to the name of the process. This text thus shows how to ask more of an explanation: that it specify what intervening might preserve, what might be lost and how to decide between these possibilities.

1.4.1.5 Integrative Clinical Reading

THE COST OF ADAPTATION

For the person, the term "adaptation" can be reassuring even though preserved functioning has a cost. The consultation must be able to distinguish what the organism maintains, what the person still manages to do and the effort this requires. This question extends the model without attributing clinical validation to it: describing a response as compensatory does not decide whether the situation is acceptable. Here, the account of activities that have become difficult adds a question complementary to measuring the signal.

1.4.2 Frantz and the Comparison of Cardiomyopathies

1.4.2.1 Situating the Result

FRANTZ AND THE ESC, 2018

In 2018, Frantz and colleagues published a scientific statement from the European Society of Cardiology (ESC) working group on myocardial function. This review compares the immune mechanisms of several cardiomyopathies, notably ischemic, metabolic, toxic, viral and genetic. It emphasizes a documentary imbalance: animal experiments mainly describe early phases, whereas the treatment of chronic heart failure concerns a different time frame. The authors envisage a partial convergence of late mechanisms, while acknowledging the gaps in human and experimental data. The text proposes an organization of knowledge and research; it reports neither a new cohort nor a trial establishing a treatment common to these diseases.[5]

1.4.2.2 Critical Reading: a comparison requires a common axis

The first question is not to count how many diseases contain the word "inflammation." It is necessary to determine what is being compared: an initial cause, a response to injury, a maintenance mechanism or a functional outcome. Two diseases can share an outcome without sharing their cause; conversely, the same disturbance can produce different consequences depending on the tissue it affects. The common category must therefore help to specify the differences; otherwise it becomes a label that explains everything after the fact.

NOT FILLING THE EMPTY CELLS

The table comparing early and late effects after myocardial infarction retains several cells without data. This presentation invites an important discipline: not mentally filling an empty cell with what seems biologically coherent. An experiment performed before the onset of disease can shed light on its development without answering the question of treating an already established condition. Timing is part of the intervention; it is not a mere detail in the legend.

1.4.2.3 Immunological Reading: common does not mean interchangeable

A COMMON PATHWAY, A TARGET TO DEMONSTRATE

An inflammatory pathway common to several cardiomyopathies first opens a comparison between their mechanisms. To make it a common target, it would be necessary to show that it actually contributes to damage in each situation considered, that its inhibition reaches the relevant compartment and that necessary functions are not lost. These requirements do not disappear when a signal is found in several diseases. Its recurrence increases the interest of the inquiry; it does not determine the direction of the intervention. The hypothesis of late convergence must therefore remain open to refutation. If apparently similar diseases respond differently to a targeted disturbance, exposure, actual engagement of the target and timing of the intervention must be checked before concluding that the mechanisms differ. A discordance becomes instructive when the hypothesis indicates what could genuinely refute it.

1.4.2.4 Cardiological Reading: a clinical unity to be built

A POPULATION TO BE JUSTIFIED

Heart failure is a clinical meeting point; it is not sufficient to constitute a homogeneous population for every immune intervention. To design a trial, the reasoning that links the assumed mechanism to the people included must therefore be made explicit. Selecting an etiology, a stage of progression or a marker does not answer the same question. It must be possible to see which uncertainty each choice reduces and which population it leaves aside. The convergence sought may be therapeutically useful without being universal. A treatment might act on a common maintenance mechanism while leaving different initial causes untouched. This possibility deserves a comparative test; it does not allow the conclusion that all cardiomyopathies call for immunosuppression.

1.4.2.5 Integrative Clinical Reading

THE UNITY OF REASONING

Bringing cardiomyopathies together can facilitate dialogue among specialists, but a person's care pathway retains a history that the mechanisms grouped by the review do not summarize. A shared explanation must keep visible the debated origin of the disease, the treatments already received and current difficulties. The integrative value of this review is to organize this common record: which questions can be addressed together, and which still require distinct expertise? The unity sought is that of reasoning and follow-up, without assuming a universal treatment.

1.4.3 Roy and the Functions That Classification Does Not Contain

1.4.3.1 Situating the Result

ROY, ORECCHIONI AND LEY

Published online in 2021 and then in the 2022 volume, this review by Roy, Orecchioni and Ley organizes immune contributions to atherosclerosis, mainly on the basis of preclinical work. It stresses that several macrophage categories defined by gene expression still await complete functional characterization. For certain preclinical Th17 responses, it reports larger but more stable plaques. Its second figure preserves this tension through red and green indications under the same category. A box specifies the limits of extrapolating from murine models, notably regarding spontaneous atherothrombotic events. The review sets out therapeutic perspectives; it does not constitute a validation trial for them.[6]

1.4.3.2 Critical Reading: a synthesis must preserve several questions

GROWTH AND STABILITY

A classification is often accompanied by adjectives: inflammatory, regulatory, protective. These words can describe a measured activity, summarize a body of work or announce a supposed consequence. Their force changes with the operation. To read the synthesis, it must therefore be possible to return from the name of the category to the proposition it condenses. The problem becomes evident when the same process receives different qualifications depending on the outcome considered. A conclusion about the growth of a lesion and a conclusion about its stability do not necessarily cancel each other out. Sorting them into two columns, favorable and unfavorable, and then counting the studies would make the scientific question disappear. The task of the synthesis is to preserve the dimensions that explain why these conclusions can coexist. It will be objected that such a presentation prohibits any overall view. It does so only if the whole must take the form of a single verdict. A synthesis can be more informative by showing a structured tension: which outcome changes, in which context and which other outcome remains distinct? Whoever reads the synthesis then obtains a map of propositions to be connected, rather than a definitive distribution of roles. This conceptual gain does not depend on the discovery of an additional cell.

1.4.3.3 Immunological Reading: a function is a relation

PROTECTIVE, BUT OF WHAT?

Calling a population protective requires specifying what would be protected. The word is not an additional property that could be read directly among its markers. A signature helps to recognize a set; a function relates this set to a consequence and to conditions. This distinction makes a rigorous use of categories possible without asking them to contain the whole biological story. In an entirely fictional example, two populations share a marker, but only one is associated with a chosen tissue property. Merging them would lose this difference; separating them would not yet demonstrate that the marker explains their role. The right classification depends on the question, while remaining constrained by the observations. It can be relevant for description and insufficient for attributing a function. The difficulty does not disappear with a more detailed image. The favorable color of an arrow must remain attached to the outcome it represents. Extending this color to the whole cell, and then to any intervention that modifies it, adds two inferences. The diagram can be accepted while these extensions are refused. The contribution of the synthesis is thus preserved without treating its vocabulary as an already tested therapeutic program.

THE COLOR OF AN ARROW

1.4.3.4 Cardiological Reading: defining progress before promising it

A NARRATIVE VICTORY

For the heart and for the person, a change in the plaque does not derive its value solely from being measurable. It must be explained what it reveals about a relevant consequence. If several properties change differently, choosing after the fact the one that seems favorable would produce a narrative victory. Examining them together makes it possible, by contrast, to specify the benefit that research has yet to establish. This provides a positive reason to continue: a synthesis that reveals a tension makes it possible to better define the objective of a future study. It helps to choose the question whose answer would be decisive, without predetermining that answer. The value of a model then lies in the problem it makes it possible to solve: deciding between cellular functions in the lesion does not yet amount to establishing a clinical outcome. The organization of immune roles does not summarize the whole epidemiological and therapeutic landscape of atherosclerosis.

1.4.3.5 Integrative Clinical Reading

KEEPING DISCORDANT RESULTS

A favorable result for one property of the plaque can leave another property unfavorable. The same discipline applies to the whole person: an improvement selected by the researcher is not automatically the one that responds to the person's priority. Before announcing overall progress, discordant results must be retained and their respective importance specified. This review thus helps to build a discussion in which the word "protection" remains attached to an object; it does not provide a ready-made clinical weighting of these objects.

1.5 Worked Exercises

1.5.1 Exercise on a Mediation Asserted by an Abstract

1.5.1.1 Problem

two propositions and a "thus"

In the fictional trial (see 1.3.2), an abstract states: "The treatment reduces inflammation and thus improves the patients' condition." Identify the two propositions and the relationship that links them. Indicate which data would be needed to support each of them.

1.5.1.2 Reasoned Answer

a mediation to be demonstrated

The first proposition concerns a biological change: the instrument and the comparison must be specified. The second concerns a benefit for patients, which requires a relevant outcome and an appropriate comparison. The word "thus" asserts a mediation: it attributes the benefit to the inflammatory change. Demonstrating these two effects separately would not prove that the first explains the second. This attribution requires its own data and its own causal reasoning.

Synthesis

Knowing in medicine involves formulating a proposition, examining the reasons that support it and delimiting its scope. Sources make the reasons accessible; method makes it possible to assess their strength; discussion tests the link between observations and conclusions. Decision adds to this examination an appraisal of the consequences and values at stake.

Open Questions

How can a reservation that genuinely changes the conclusion be distinguished from one that merely specifies its conditions? From what set of observations does an explanation become solid enough to guide new research? What must be known in order to extrapolate a result without claiming to erase all differences between situations? These questions will return as the studies are examined.

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2

Defining Health and Disease

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A disease is rarely recognized in a single act. A measurement crosses a threshold, a name groups lesions together, a model orders phases, a person describes what they can no longer do. Each of these operations performs a real service, and each can be mistaken for another. A sharp threshold does not prove that living matter changes its nature at that point; a shared name does not demonstrate a common mechanism; a stage in a model does not become each patient's destiny. Each operation answers to a task: describing a function, forming a comparable group, naming a disease, guiding care. A definition is then judged by what it makes it possible to distinguish and by the situations it actually classifies. It remains revisable, without every biological description becoming arbitrary.

2.1 The Normal between Frequency, Function and Value

2.1.1 A Definition Is Already at Work in the Record

WHAT "NORMAL" CAN DESIGNATE

When a physician writes "normal," several things can be meant: a measurement within a reference interval, a function judged to be preserved, the absence of a disease being sought or a state satisfactory for the person. These propositions can agree. They can also diverge, without a measurement error necessarily explaining their disagreement. Consider an imaginary situation. One person carries out their usual activities, but an investigation discovers an abnormality whose consequences remain uncertain. Another describes a major limitation, while the tests performed do not show the suspected disease. The problem is not to decide which of the two accounts, biological or lived, holds all authority in principle. What each one establishes, and which question remains open, must be understood. The first person does not receive, from the abnormality alone, a complete description of their health. The second does not need a diagnosis to be already available for their limitation to deserve attention. Conversely, feeling well does not guarantee the absence of a pathological process, and suffering does not allow its mechanism to be identified immediately. These distinctions protect both the precision of diagnosis and the recognition of the person. Defining disease is therefore a practical operation in the strong sense: it organizes what is sought, the populations studied and the changes counted as improvements. It calls for reflection on the functions of living beings and on what matters in a life.

TWO ACCOUNTS, NO AUTHORITY IN PRINCIPLE

2.1.2 The Frequent, the Functional and the Desirable

2.1.2.1 The frequent is not enough to define the healthy

THE AVERAGE IS NOT AN IDEAL

An average describes a population according to a variable and a method. It does not become a biological ideal because it is calculated precisely. A widespread characteristic can be unfavorable; a rare particularity can remain without harmful consequence. To move from the frequent to the healthy, an additional argument is therefore needed. This remark does not refute every biological definition of health. It is possible to seek to distinguish the performance of a function from the frequency of a trait. But this approach must explain which functions count, relative to which set of organisms and for which question. The choice of comparator is part of the reasoning; it is not contained in the final figure.

2.1.2.2 Boorse: recognizing a dysfunction without prescribing a life

2.1.2.2.1 A theoretical definition, not praise for the average

Boorse, 1977

In 1977, Boorse sought to distinguish a theoretical concept of disease from practical judgments about treatment. His definition links biological functions, typical efficiency and reference class; it concerns an ability to function in appropriate circumstances, not an activity constantly performed. He acknowledges that frequency alone does not define health and that the precise boundary of a continuous distribution involves a convention. He also notes the difficulties posed to his model by certain conditions common to an entire population and by aging.[1] Reducing this position to "being healthy is being average" would therefore miss both his proposal and its genuine difficulties.

2.1.2.2.2 Controlling the passage from description to prescription

DESCRIBING IS NOT PRESCRIBING

A thought experiment makes it possible to extend this distinction. A person has a diminished biological capacity, but a reliable device enables them to carry out their activities. Describing the diminution and recognizing the success of their life are not two incompatible conclusions. They concern different objects. By contrast, inferring from the diminution that it must necessarily be corrected would turn a description into a prescription without examining the expected benefit, the constraints and the person's preferences. An objection arises: does this separation not simply push values back to the final decision? Boorse could reply that his theoretical definition is precisely not meant to set all the goals of care. The absence of such goals therefore does not refute his proposal. The internal difficulty is more specific: how can functions and reference classes be justified without already choosing what will count as normal? Survival and reproduction offer a biological anchor; they are not sufficient, by themselves, to justify every grouping. The debate that follows should concern this choice, rather than reproach the theoretical concept for not being a complete ethics of care. The decisive difficulty lies in controlling the passage between registers. One can ask whether a capacity is impaired, whether this impairment explains a limitation and then whether modifying it is a desirable goal. None of these questions replaces the other two. This approach preserves a biological requirement while refusing to treat conformity to a type as the obligatory program of a life. The debate on reference classes that follows tests precisely the first of these requirements.

THE DIFFICULTY OF REFERENCE CLASSES

2.1.2.3 Putting reference classes to the test

2.1.2.3.1 Wiśniewski and the criterion of survival or reproduction

Wiśniewski, 2012

A philosophical exchange from 2012 makes it possible to test this requirement. Wiśniewski proposes reconstructing reference classes from the characteristics whose contribution to the prospects of survival or reproduction is not negative. He thereby seeks to defend a definition free of value judgments. His text nevertheless leads him to classify homosexuality among the characteristics that fail this criterion, on the grounds of diminished reproductive prospects.[2]

AN ASSIMILATION TO BE REJECTED

This assimilation must be rejected: a sexual orientation does not become a dysfunction because a line of reasoning has chosen reproduction as its privileged measure. The text does not demonstrate the transition from a supposed consequence at the scale of a group to an individual pathological state. This example matters to the critique: it reveals the classifying power of the criterion and rules out presenting its neutrality as given.

2.1.2.3.2 Kingma's reply and the problem of redescription

Kingma and redescription

Kingma replies in the same issue. Against the rule, she sets groups that differ in survival without having to be declared diseased on that basis alone. She also highlights a difficulty of redescription: if only the characteristic causing the decrease in survival makes a group inadmissible, it would sometimes suffice to name the same group by another sign to make it admissible. The definition would then depend on the way people are designated rather than on their condition.[3] The value of the debate lies in making the consequences of a rule visible. The first proposal seeks an observable criterion; the reply asks whether it actually classifies the situations it claims to cover. A definition is defended not only by the clarity of its terms but also by its resistance to counterexamples. Discussing it does not require assuming that every biological description would be arbitrary.

2.1.2.3.3 One more difficulty and three questions retained

survival does not exhaust the definition

A distinct difficulty must be added; it does not appear in this exchange. Two situations could retain the same survival prospects while differing in pain, possible activity or the constraints imposed. It must then be explained whether, and how, these differences enter the definition. Choosing an observable consequence does not demonstrate that it exhausts everything that was to be defined.

three questions retained

The comparison can nevertheless be checked. Some groupings make it possible to explain and predict differences better than others. They can be confronted with new observations. Three questions thus remain to be kept: what is usually encountered, what can the organism accomplish and what does one seek to preserve or restore? Their answers inform one another without replacing one another.

2.2 Capacity, Lived Experience and Mechanism

2.2.1 A Capacity Must Be Situated

2.2.1.1 A task accomplished does not reveal the whole capacity

a capacity to be situated

Saying that a person can carry out an activity still leaves several questions open. Under what conditions do they carry it out? With what effort, what help and what consequences after the activity? What is called a capacity is not always reducible to the one-off success of a task. The following example is for teaching purposes. Two people make the same journey. The first then retains broad freedom of activity; the second organizes the entire day to make it possible. The observation "journey completed" is accurate for both. It becomes insufficient if it is made to mean that their possibilities are identical.

the same journey, two possibilities

2.2.1.2 The common measure, aids and independence

It might be objected that adding all these circumstances prevents any common measure. But understanding a difference does not require abandoning measurement: it requires specifying what the measure retains and what it leaves aside. The same test can be valuable for tracking a performance and incomplete for describing the effect of the disease on daily life. Another objection concerns aids. If a person succeeds thanks to material or human support, should their capacity be described as nonexistent? That would confuse what they can accomplish in their actual conditions with what they would accomplish in a situation where that support had been withdrawn. Both questions can be useful; neither should erase the other. It is also necessary to avoid turning independence into a measure of the value of a life. Care can improve a

possibility by acting on the body, on the conditions of activity or on both. The evaluation must state which change it observes, without attributing the whole improvement to a single dimension.

2.2.2 Lived Experience Does Not Replace Mechanism

TWO MISREADINGS OF LIVED EXPERIENCE

Giving a place to lived experience invites two opposite misunderstandings. The first would hold that suffering is scientifically negligible as long as it has not received a biological explanation. The second would make the feeling of being ill sufficient proof of the mechanism invoked to explain it. The distinction proposed here concerns the authority of testimony and the justification of interpretation. A person has irreplaceable access to what they experience and to the way their life is affected by it. This testimony sometimes needs to be made more precise in dialogue: circumstances, variations, abandoned activities, expectations. The search for an explanation then draws on other reasons, which can confirm or modify an initial interpretation without canceling the testimony. One objection concerns silent diseases: would a definition attentive to experience not lead to ignoring them? It would if a present complaint were required for recognizing any disease. That requirement is not adopted here. Observations can document a process before any felt limitation. But it remains necessary to explain the significance of this finding, its probable trajectory and what an action could contribute. Thus neither the presence nor the absence of symptoms closes the whole discussion. Medicine must be able to recognize a silent process and take seriously a suffering not yet explained. Both obligations require not using the word "objective" as a synonym for "the only thing that is real."

TESTIMONY AND INTERPRETATION

SILENT DISEASES

2.3 Thresholds, Names and Models

2.3.1 The Boundary a Threshold Draws

2.3.1.1 A sharp classification for neighboring states

A threshold can be indispensable for organizing a study or a decision. It transforms continuous variation into usable categories. This service does not prove that living matter abruptly changes its nature exactly at that point. Suppose a marker M and a cutoff set at 10 units for including participants in a study. One person at 9.9 and another at 10.1 are placed in different groups. To interpret this difference, the precision of the measurement, its variability and the justification for the threshold must be known. The classification can be sharp while the biological states remain close.

A SHARP THRESHOLD, NEIGHBORING STATES

2.3.1.2 The task of the threshold and the place of risk

It will be objected that a line must be drawn and that a debatable boundary is better than permanent indecision. That is correct for certain tasks. The answer is to name the task: selecting a reproducible population, identifying a risk, supporting a diagnosis or determining when an intervention becomes reasonable. Different thresholds can be defensible because they serve different objectives. It does not follow that any threshold will do. It must be examined according to the errors it produces and their consequences. Changing a cutoff can include more people without demonstrating that they will benefit from the same management. The gain in detection and the gain in health are two propositions to be studied separately. Risk is also distinct from current disease. A risk concerns the possibility of an event within a given horizon; a diagnosis asserts the

NAMING THE TASK OF THE THRESHOLD

DETECTION GAIN, HEALTH GAIN

RISK IS NOT DISEASE

presence of a defined condition. Their relationship can be close, but confusing them easily turns a future probability into a present identity.

2.3.2 Defining, Naming and Recognizing

DEFINE, NAME, CLASSIFY, DIAGNOSE

A definition indicates what a category is meant to designate. A nomenclature organizes names and their relations. Classification criteria serve to form comparable groups. Diagnostic reasoning seeks, for a given person, which explanation or category is sufficiently supported by the available findings. These operations support one another but answer to different requirements. The Chapel Hill nomenclature (see 2.4.1) puts this distinction to the test of a concrete document. This difference avoids circular reasoning. Knowing what a name means does not yet make it possible to determine whether that name fits the patient being examined. This requires observations whose value has been studied and an assessment of the other possible explanations. Consensus has a real function here: making communication more precise. It would be wrong to treat it as an underpowered therapeutic experiment, since it does not perform that task. It would be equally wrong to attribute to it automatically the performance of a diagnostic instrument. The quality of a source is also judged by the function it is made to serve. Stable names facilitate comparison between studies. Revising them may become necessary when knowledge changes. A correspondence between old and new categories must then be maintained, so that the same word does not mask different populations, and two words do not suggest two diseases without an established difference.

THE REAL FUNCTION OF CONSENSUS

2.3.3 The Possibilities and Limits of Models

2.3.3.1 A model is worth the precision of its domain

A model offers a selective representation. It can order phases, distinguish mechanisms or bring together observations that were previously scattered. Its value is measured by what it makes intelligible and by the questions it makes it possible to ask. In a fictional example, a model describes a phase of injury, an inflammatory response and a phase of remodeling. This scheme can guide an inquiry without establishing that every person goes through the three stages in the same order, or that their boundaries are always sharp. Turning it into an obligatory story would erase the very trajectories it should make it possible to study. One objection asks what remains of a model if its exceptions are numerous. The answer depends on its ambition. A model limited to certain situations can retain great value. A model presented as universal must answer its counterexamples. The precision of its domain is therefore a scientific property, not a mere drafting precaution.

A THREE-PHASE MODEL

2.3.3.2 The diagnostic category and its subdivisions

THE CATEGORY AND ITS SUBDIVISIONS

The diagnostic category, for its part, groups people according to selected characteristics. It can include several mechanisms. A biological subdivision can be promising, but its name does not yet demonstrate a distinct course or a particular therapeutic response. The reproducibility of the grouping and its usefulness for the clinical question must be examined separately. The texts on myocarditis (see 2.4.3) and on amyloidosis (see 2.4.2) are read in light of this distinction. Syntheses can clarify concepts and organize a literature. When they invoke diagnostic performance or therapeutic benefit, the examination of these claims must go back to the studies on which they are based.

2.4 Texts Put to the Test

2.4.1 Jennette and the Uses of the Disease Name

2.4.1.1 Situating the Result

Chapel Hill, 2012 revision

The Chapel Hill document, revised in 2012 and published in 2013, proposes names and definitions for vasculitides. It states explicitly that it does not provide diagnostic or classification criteria. Its terminology table distinguishes the process defining a disease from the observations that make it possible to identify it in a patient or to form a group for research; the examples given in this table are not validated criteria.[4] It would be insufficient to judge the value of a consensus solely by its adoption. Adoption is an important property of a common language: a term understood in the same way facilitates exchange. Yet it guarantees neither the correctness of each biological division nor the performance of a diagnostic rule. A convention can be useful, widely shared and nonetheless open to improvement.

2.4.1.2 Critical Reading

a task the text does not undertake

Looking for a control group or an effect size in this publication would amount to evaluating a task it does not undertake. The relevant questions concern instead the definition of its object, the deliberation procedure, the coherence of the terms and the handling of situations that resist the proposed categories. These questions must then be distinguished from those raised by a study evaluating criteria: which population, which reference diagnosis, which misclassification errors, which validation in another setting? The consensus, on the interpretation proposed here, organizes the conditions for an empirical discussion. It gives researchers the possibility of asking whether they are talking about the same patients. This contribution does not constitute a demonstration of all the mechanisms suggested by the categories. The precision of the name makes it possible to ask better questions; it does not already contain all their answers. Suppose two teams are studying "cardiac involvement in vasculitides." One selects patients on the basis of documented vascular inflammation; the other uses an old label recorded in the medical files. Even if the titles of the two cohorts appear identical, their comparison may rest on different objects. Before discussing their frequencies, their inclusion rules must be reconstructed. This example is for teaching purposes, with no reference to particular cohorts.

the conditions for empirical discussion

two cohorts under one title

2.4.1.3 Immunological Reading

pauci-immune does not mean without immunity

The difference between pauci-immune lesions and lesions with immune deposits must be preserved without too quickly becoming a complete explanation, and then a therapeutic strategy. The term "pauci-immune" describes in particular the scarcity, or even absence, of observed deposits; it does not mean that immunity is absent. Conversely, recognizing deposits is not enough to reconstruct the whole history of their formation. This contrast between descriptive category and explanation requires specifying three propositions: which property of the lesion is observed, which mechanism is proposed and

which intervention has actually been evaluated? These three sentences require distinct support. The nomenclature helps to avoid a confusion of vocabulary; mechanistic study and therapeutic trials will have to support the subsequent steps. This is how one learns to recognize the value of a category without attributing unlimited explanatory power to it.

2.4.1.4 Cardiological Reading

VESSEL CALIBER AND CARDIAC INVOLVEMENT

Vessel calibers can be related to types of cardiovascular involvement. This connection can help to organize teaching. It becomes excessive if it suggests that an anatomical compartment alone determines all possible manifestations. The category guides questions; the presence, severity and mechanism of involvement in a patient must be established. The philosophical question is that of identity across manifestations: under what conditions do two different presentations belong to the same disease? Answering only by visible resemblance leaves mechanism aside; answering only by a presumed mechanism risks erasing clinical differences. The definition can be read as an explicit commitment, open to confrontation with observations, and not as an essence that has become indisputable because it has a name. What is established here is the declared function of a nomenclature framework. What remains to be tested, for each use, is its translation into operational rules. Neither the diagnostic performance of a score nor the efficacy of a treatment can be inferred from its adoption. Validation studies and observations showing the limits of the categories may modify the use of the framework.

THE IDENTITY OF A DISEASE

2.4.1.5 Integrative Clinical Reading

A NAME SHARED BETWEEN TEAMS

A shared nomenclature has a concrete usefulness when several teams follow the same person: it aims to limit the risk that different names are understood as additional diseases, or that an identical name masks two interpretations. A useful letter must nonetheless distinguish the diagnosis retained, the elements that support it and the manifestations actually observed. This requirement for transmission extends the function of the consensus. It prevents the precision of the vocabulary from lending, by itself, an appearance of certainty to the whole record.

2.4.2 Ruberg and the Boundaries of the Field

2.4.2.1 Situating the Result

RUBERG AND ATTR AMYLOIDOSIS, 2019

The review by Ruberg and colleagues, published in 2019, brings together knowledge of transthyretin amyloid cardiomyopathy. It describes a condition linked to myocardial deposition of misfolded transthyretin, in hereditary or wild-type forms. The review describes a noninvasive diagnostic pathway combining grade 2 or 3 scintigraphic uptake, a compatible context and exclusion of a monoclonal abnormality by serum free light chains and serum and urine immunofixation. These conditions form a whole; scintigraphy taken in isolation does not carry the same conclusion. A monoclonal gammopathy can also coexist with transthyretin amyloidosis (ATTR) and does not, by itself, prove light-chain amyloidosis (AL). The review also presents how therapeutic options had evolved as of that date.[5] Comparing ATTR amyloidosis

with an inflammatory heart disease tests the boundaries of cardioimmunology. Similar cardiac presentations can arise from different mechanisms; their clinical proximity is not enough to attribute a common autoimmune origin to them. Diagnosis must decide precisely between what resemblance brings together.

2.4.2.2 Critical Reading

A REVIEW HAS NO SINGLE LEVEL

Opposing "a review" to "evidence" and assigning the document an overall low level would be too crude. A review can present solid results and connect them with discernment; it can also select or interpret its sources imperfectly. Its lack of a sample of its own does not settle the soundness of all the propositions it contains. Each important claim must therefore be traced back to the work that supports it. A frequency calls for examining the population and the detection method. A diagnostic performance calls for examining the test, the comparator and recruitment. A therapeutic benefit calls for examining the studies of the intervention. Commenting on a synthesis requires making these lineages visible, without counting the review again as an independent replication. This rule applies in particular when a synthesis reports "100% specificity." Such a number must remain attached to a study, its conditions and its uncertainty. It must not become a universal guarantee applicable to every image acquisition or every patient. Nor can the performance of a procedure involving several tests be silently attributed to a single one of its components. It must therefore be checked whether the figure describes an isolated test or the complete diagnostic procedure.

TRACING EACH CLAIM BACK

A SPECIFICITY OF 100%

2.4.2.3 Immunological Reading

A SHORTCUT TO AUTOIMMUNITY

Understanding amyloidosis requires considering protein folding. Its methodological interest lies in forbidding a shortcut: tissue damage and a possible immune response are not enough to designate an autoimmune origin. The role of a process must be discussed according to its place: triggering, accompaniment, repair or consequence. The presence of an immune response does not, by itself, make it possible to choose among these roles. This understanding also guards against another confusion, between ATTR amyloidosis and light-chain amyloidosis. Grouping them under a shared morphological appearance does not allow etiological reasoning or treatments to be extrapolated automatically from one to the other. The general name is useful for bringing together; typing is necessary for distinguishing. A sound nosology must make both movements possible.

2.4.2.4 Cardiological Reading

ONE PHENOTYPE, SEVERAL CAUSES

Consider a fictional case: two people present with thickened ventricular walls and exercise limitation. The resemblance justifies a clinical comparison. It does not establish a common cause. To move from phenotype to disease, findings are needed that decide between competing explanations. A suggestive sign guides the search; it does not necessarily carry, in isolation, the force of the complete procedure. Wall thickening describes a form; typing the amyloidosis identifies what is deposited there. This step changes the clinical question: which findings make it possible to link the observed presentation to a specific disease?

TYPING CHANGES THE QUESTION

Progress does not consist merely in adding a more precise name. It changes which studies are relevant for assessing prognosis and which interventions it becomes reasonable to evaluate. The 2019 review orders these distinctions on the basis of the knowledge then available. Each diagnostic performance and each therapeutic benefit it reports remains attached to its source studies. Its historical interest endures; a current decision requires taking later work into account.

2.4.2.5 Integrative Clinical Reading

NO DIAGNOSIS BY JUXTAPOSITION

For a person facing several hypotheses of amyloidosis, coordination does not consist merely in gathering reports. It must make understandable what has been identified, what remains to be typed and why the tests are not interchangeable. The review makes it possible to organize these questions, without setting out current management here. Maintaining a shared explanation among the professionals involved notably prevents a monoclonal abnormality and a cardiac image from being assembled into a diagnosis by mere juxtaposition, and then transmitted as an acquired certainty.

2.4.3 Tschöpe and the Trajectories of Disease

2.4.3.1 Situating the Result

TSCHÖPE AND MYOCARDITIS

The review by Tschöpe and colleagues, published online in 2020 and in the 2021 volume, examines myocarditis and inflammatory cardiomyopathies. It stresses etiological heterogeneity, differences in course and uncertainties regarding the contribution of infectious agents, the host and the environment. In particular, it distinguishes the association of a virus with the disease from its causal role. Its aim is to synthesize knowledge and to formulate the questions that remain open.[6] It can be tempting to adopt a scheme of initial injury, sustained autoimmunity and then progression to dilated cardiomyopathy without having examined the domain of validity of each of these transitions. The review presents a model of viral myocarditis linking innate response, adaptive response and chronic remodeling; an adaptive response is not necessarily autoimmune. The critical question therefore concerns its domain of validity and the observations supporting each transition, without making it the obligatory trajectory of all forms of myocarditis.

A TRAJECTORY NOT TO BE IMPOSED

2.4.3.2 Critical Reading

The first task is to distinguish what the review reports, what it proposes as a framework and what the commentary adds for teaching purposes. A review does not follow a single cohort from a first event to all its outcomes. A teaching sequence must therefore be linked to the observations that document its transitions. Without this precaution, the continuity of the text becomes an imaginary continuity of the evidence. The review provides a concrete example: some murine models of coxsackievirus B3 also cause pancreatitis and a systemic response. Their trajectory therefore does not describe an isolated involvement of the heart. What emerges is the need to think about time while leaving several trajectories open. Onset of symptoms, onset of the biological process and date of diagnosis may differ. A comparison between patients ob-

THINKING ABOUT TIME

served at different times is not automatically equivalent to following the same person. This logical distinction refutes no particular model; it specifies the data needed to support it.

2.4.3.3 Immunological Reading

A sequence of "injury, response, sequela" does not by itself demonstrate sustained autoimmunity. To support this step, observations suited to the proposed mechanism are needed. Likewise, the persistence of a dysfunction is not enough to establish the persistence of the initial immune process. The interest of the viral question is particularly clear. Detecting an element in a tissue answers a question of presence, within the limits of the method and the sample. Attributing responsibility for the process to it requires additional reasoning. It is necessary to understand what would make a causal interpretation more convincing than an incidental presence.

VIRAL PRESENCE, CAUSAL ROLE

2.4.3.4 Cardiological Reading

DESCRIBING A STATE, PREDICTING ITS COURSE

The clinician needs a current description: symptoms, function, rhythm, structure and available etiological findings. The researcher also wants to explain how this state came about and how it may evolve. The two needs communicate, but their answers are not identical. Describing a severe state does not establish its irreversibility; observing an improvement does not, by itself, allow the assertion that all processes have disappeared. Diagnosis can therefore be read as a reasoned description at a given moment, open to further specification. This does not diminish its value: a category is useful when it correctly organizes what is known and the questions to be resolved. It becomes misleading when it passes off as already known a trajectory that remains to be observed.

A HYPOTHESIS, NOT A DESTINY

A stepwise progression can be fruitful as a hypothesis for certain situations; it must not become the obligatory destiny of all forms of myocarditis. An individual therapeutic indication cannot be deduced from a temporal scheme. Longitudinal description, etiological characterization and direct evaluation of interventions are the elements that could strengthen or modify the judgment.

2.4.3.5 Integrative Clinical Reading

A SHARED CHRONOLOGY

The care pathway benefits from retaining several dates: first reported symptoms, first documented abnormality, diagnosis retained and changes in treatment. This shared chronology allows the physician taking over follow-up to understand what an improvement or a persistence means. It does not force the history into the stages of a model. The integrative contribution of this review is to make current description and evolving history compatible: the person may report feeling better while a biological question remains open, without either invalidating the other.

2.5 Defining in Order to Care without Confining

A DIAGNOSIS THAT CLARIFIES OR CONFINES

A diagnosis can make an experience understandable, open access to care and enable shared research. It can also become too narrow a description if every complaint is attributed to it or if the person is reduced to what is usually ex-

pected of their category. Recognizing these effects does not mean that the truth of a diagnosis depends on how reassuring it is. A justified conclusion can be difficult to receive. Responsibility consists in distinguishing what is established, what is expected and what remains variable. A probable trajectory is not the already written story of a life. Nor should a judgment of health measure conformity to a socially desired life. The ability to work, for example, can matter to a person without becoming the sole definition of their condition. The goals of care must be open to discussion: relieving, preserving a capacity, avoiding a complication or regaining a chosen activity are not always equivalent ways of describing success. The position defended here therefore does not abolish the search for biological functions or for stable categories. It asks that their uses remain intelligible. Before writing that a result is "normal," the comparison must be specified. Before writing that a category is "validated," the task must be specified. Before writing that a person is "better," what has changed for that person must be specified.

HEALTH WITHOUT SOCIAL CONFORMITY

SPECIFYING BEFORE WRITING

2.6 Worked Exercises

2.6.1 Exercise on Three Conclusions to Be Distinguished

2.6.1.1 Problem

THREE INFERENCES TO EXAMINE

In a fictional study, a participant has a marker above the inclusion threshold, meets the selected classification criteria and reports an improvement in daily life after an intervention. Can it be inferred that the participant was ill in the sense of every possible definition, that the proposed mechanism is demonstrated and that the treatment caused the improvement?

2.6.1.2 Reasoned Answer

ELIGIBILITY, INTERPRETATION, EXPERIENCE

The three observations have value, but they do not support the three proposed inferences. The threshold and the criteria indicate how the study constitutes its population; their justification and their relationship to diagnosis must be known. Classification does not demonstrate each mechanism invoked. The report of improvement sheds light on an important dimension of the outcome; causal attribution to the intervention depends on the comparison design. Precise wording keeps eligibility, clinical interpretation and experienced change separate.

Synthesis

Defining health and disease involves several operations that must be kept apart: describing what is frequent, assessing a function, recognizing what matters to a person. A threshold, a name, a category or a model each performs a specific service; none proves by itself a mechanism, a trajectory or a benefit. The texts commented on teach this together. The debate on reference classes shows that a definition is judged by its counterexamples. The Chapel Hill nomenclature organizes a common language without providing diagnostic criteria. The review on transthyretin amyloidosis requires deciding between what resemblance brings together. The review on myocarditis recalls that a

temporal scheme remains a hypothesis, each transition of which requires its own observations.

Open Questions

How can a reference class be justified without choosing in advance what will count as normal? How can pain or possible activity be brought into a definition without giving up the biological requirement? Under what conditions does a biological subdivision of a diagnostic category become a distinct disease, with its own course and treatments? These questions are not settled once and for all; they are taken up again with each case that resists the categories.

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3

Explaining and Establishing a Cause

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The word "cause" promises more than a meeting between two facts. It announces that a difference would appear if one acted otherwise: removing an antibody, lowering a marker, prescribing a treatment or withholding it. No image, however precise, shows this promise, and no calculation fulfills it by itself. Adjusting for a variable can correct a comparison or distort it; a genetic variant can point to a target without fixing the dose or the benefit. Between the observed succession and the successful intervention lie several steps: defining the effect, choosing the comparator, controlling what confounds without blocking what transmits, extrapolating from a model to a person. Each can be crossed for good reasons. None removes the need for the next.

3.1 What Causality Adds to Observed Succession

3.1.1 What the Word Cause Commits To

several true explanations

A disease can receive several true explanations that do not answer the same question. One may ask why it appeared in this person, why a tissue was injured or why an intervention changed its course. The first question concerns a singular history, the second a process, the third a difference produced by an action. Their kinship does not remove the need to distinguish them. Consider an example constructed for reasoning: a mediator rises before a lesion, appears in the affected tissue and falls when the clinical condition improves. Together, these facts sketch a coherent story. Yet several relationships remain possible. The mediator may contribute to the lesion, respond to a process that produces it or accompany an attempt at repair. The chronology eliminates some explanations; it does not automatically designate a single one. The causal problem begins precisely here: several stories may fit the same observations. The right question then becomes: in what situation would their consequences differ? A comparison must be designed that makes these explanations less interchangeable. This approach justifies an inquiry; it does not presuppose that the preferred mechanism is already the true one.

a chronology does not designate the cause

3.1.2 Hume and the Limits of Observed Succession

Hume, the necessary connection

In Section VII of *An Enquiry Concerning Human Understanding*, Hume asks where the idea of a necessary connection between a cause and its effect comes from. An observed succession does not show any additional necessity binding the events together. Repetition leads the mind to expect one when the other appears; Hume locates the origin of the idea of connection in this acquired transition. His problem concerns the foundations of inference, not only the occasional errors of the observer.[1] For medical reading, a requirement follows that extends Hume without appearing in his work: causal vocabulary must make explicit what it adds to the finding. The image of a cell in contact with a lesion can be admirably precise. It does not visibly contain the answer to the question of what would have happened had this cell been absent. That answer calls on reasoning and on other observations. An objection arises: since every conclusion ultimately depends on observations, none could go beyond mere succession. But this objection confuses the absence of direct access to a necessity with the absence of differences between the available reasons. A carefully constructed comparison can decide between explanations that the initial observation left open. There is no need to turn causality into a visible object in order to improve the inference that supports it.

what causal vocabulary adds

3.2 Defining the Effect and Organizing the Possibility of Refutation

3.2.1 Defining the Causal Comparison

3.2.1.1 A comparison between named strategies

potential outcomes

To say that an intervention has an effect implies a comparison: what would have become of the same set of people under one strategy rather than another? The question must name the strategies, the population, the outcome and the time horizon. In the potential outcomes framework, an average effect compares the outcomes of this population under each strategy. A person usually yields only one of these trajectories. Hernán and Robins thus explain why causal inference requires conditions that allow observed groups to be compared: exchangeability, positivity and consistency between the defined intervention and the one actually received.[2] In cardioimmunology, "reducing inflammation" remains too vague to designate a strategy. Should one modify a behavior, administer a molecule, inhibit a receptor or treat a disease that sustains the inflammatory signal? These actions may share a measurable consequence without sharing all their effects. The question is which concrete action corresponds to the conclusion. This question comes before the calculation.

"reducing inflammation," too vague

3.2.1.2 Exchangeability, positivity and consistency

exchangeability, positivity, consistency

These conditions must be made explicit and discussed on the basis of the design and the available data; some cannot be directly verified. Exchangeability requires that the groups be comparable with respect to their potential outcomes, possibly after accounting for baseline characteristics. Positivity requires that each strategy remain possible for the profiles covered by the comparison. If all the most severely ill patients receive A, their outcome without A is not directly represented in this stratum. A model can extrapolate, but extrapolation does not create the missing contrast. Finally, consistency assumes that the intervention is defined precisely enough to link the announced strategy to the one that actually produced the observed outcome.[2]

3.2.2 Necessity and Sufficiency in a Defined Domain

necessary, sufficient, probabilistic

A condition is necessary for an effect, in a defined domain, when the effect does not occur without it. A condition is sufficient, in that domain, when its presence brings about the effect. In biomedical research, an intervention can also change the probability of an effect without producing it in every organism. This probabilistic contribution must be distinguished from unconditional logical sufficiency. The two propositions of necessity and sufficiency are distinct. What can produce a lesion is not necessarily indispensable to every way in which that lesion can arise. The phrase "by itself" is particularly demanding. A molecule injected into an animal never acts outside of all context. The recipient organism, the doses, the timing and the other experimental conditions are part of the situation in which the effect was obtained. Claiming experimental sufficiency means that the addition under consideration suffices in this situation; it does not abolish the conditions that make it possible.

3.2.3 Bernard and the Counterproof

Bernard and the counterproof

Claude Bernard gives the counterproof a precise function: testing the causal conclusion by removing the presumed cause and examining whether the effect disappears. He distinguishes it from the comparison intended to control the circumstances of observation. This requirement belongs to his project of

TWO PATHWAYS, ONLY ONE INHIBITED

determining the conditions of existence of phenomena.[3] The proposition deserves discussion rather than transformation into a universal rule. Suppose, in a teaching example, that two different pathways can each sustain a lesion. Inhibiting the first may leave the second to produce the effect. The failure of the effect to disappear is therefore not enough to declare the first pathway foreign to the process. One must specify whether the aim is an indispensable condition, a contribution or the effect of a particular intervention. The reverse objection is just as important. Invoking compensation whenever a removal fails to produce the expected result risks protecting the hypothesis indefinitely.

A COMPENSATION TO BE DOCUMENTED

A compensation must be documented or lead to a new and risky prediction: which pathway would take over, at what moment, with what measurable consequence? Without this requirement, the explanation becomes available after every result and no longer anticipates any of them. The present fruitfulness of the counterproof thus rests on a discipline: organizing a real possibility of refutation. It does not require a complex disease to have a single cause. It requires the explanation to agree to specify the observations that would weaken it.

3.3 What Adjustment Corrects and What It Can Distort

3.3.1 Confounding, Mediation and Selection

3.3.1.1 The causal diagram and the common cause

THE CAUSAL DIAGRAM AND ITS ARROWS

A causal diagram sets out hypotheses about how relationships are organized. In particular, it helps determine which variables to control for. Greenland, Pearl and Robins show that the usual criteria for selecting confounders can become insufficient when several variables are involved. The diagram is not a proof of its own arrows: its conclusions depend on the postulated relationships, including those assumed to be absent.[4] Consider a fictional case. A team observes patients with a heart disease and an inflammatory syndrome. Some receive the fictional treatment A. The team records baseline severity G, a marker M measured after the start of treatment and hospitalization Y at one year. For this exercise only, G is assumed to influence both the prescription of A and the risk of Y. The path A ← G → Y explains why treated and untreated groups may differ before any effect of treatment. Here, G plays the role of a common cause.

SEVERITY, A COMMON CAUSE

3.3.1.2 The mediator and the total effect

THE MARKER, A MEDIATOR

Now suppose that A lowers M and that this decrease helps reduce Y. In the path A → M → Y, M is a mediator. Entering it into a model intended to measure the total effect of A changes the question: it amounts to holding fixed a step through which the effect might occur. A mediation analysis requires defining the effects of interest and examining its own confounding assumptions; randomizing the treatment does not randomize the mediator. The differences between total, direct and indirect effects do not follow automatically from a comparison of coefficients.[5]

3.3.1.3 Selection on a common consequence

SELECTING ON A COMMON CONSEQUENCE

Finally, suppose that access to a specialist clinic S depends both on inflammatory intensity I and on cardiac involvement K. Restricting the analysis to patients seen in this clinic amounts to selecting on their common consequence: I → S ← K. An association between I and K may then appear, or be altered,

through selection. Hernán, Hernández-Díaz and Robins describe this structure and distinguish it from confounding by a common cause.[6] In this last scenario, a patient with mild inflammation may have been referred mainly because of severe cardiac involvement. Another, with little cardiac involvement, may have been referred mainly for inflammation. Comparing them within that clinic alone can produce a misleading relationship. This example explains why a more complete record does not guarantee a better estimate: one must still understand how that record entered the study.

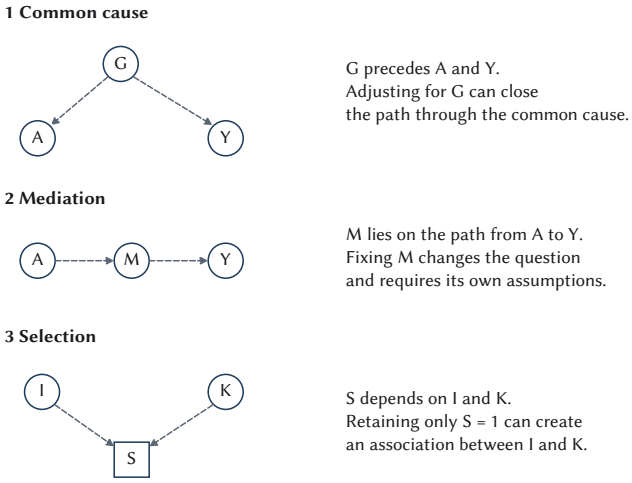


Figure 3.1. Three possible effects of controlling for a variable Distinct fictional models: the arrows express hypotheses, without clinical proof. G: baseline severity; A: treatment; Y: hospitalization at one year; M: marker after treatment; I: inflammatory intensity; K: cardiac involvement; S: specialist clinic. The box indicates restriction to clinic attendees (S = 1). A real situation may require other variables and relationships. Original diagram supporting the discussion.[4,5,6]

3.3.2 A Calculation That Reverses the First Impression

3.3.2.1 The paradox of a crude comparison

The following numbers are entirely synthetic. They describe no drug, no real cohort and no rate expected in practice. They serve only to learn how to read a comparison. Follow-up is assumed to be complete at one year. The table gives the hospitalizations observed in each stratum of baseline severity.

Table 3.1. Hospitalizations observed by stratum of baseline severity, with and without A (fictional data)

Baseline severity	With A, events out of total	Risk with A	Without A, events out of total	Risk without A
High	48 out of 160	30%	16 out of 40	40%
Low	2 out of 40	5%	16 out of 160	10%
Overall	50 out of 200	25%	32 out of 200	16%

THE PARADOX OF A CRUDE COMPARISON

The crude comparison gives 50 hospitalizations out of 200 with A, or 25%, against 32 out of 200 without A, or 16%. The observed difference is +9 percentage points. Taken in isolation, this calculation appears unfavorable to the treatment. Yet the two stratum-specific comparisons point in the opposite direction: −10 points for high severity and −5 points for low severity. The apparent paradox stems from the different proportions of severely ill patients.

3.3.2.2 What standardization corrects and what it assumes

STANDARDIZATION WITH EQUAL WEIGHTS

To compare the strategies in a population made up of half severely ill and half less severely ill patients, give each stratum the same weight. The standardized risk is 0.5 × 30% + 0.5 × 5% = 17.5% with A, and 0.5 × 40% + 0.5 × 10% = 25% without A. The standardized difference is −7.5 points. The change does not result from any modification of the events; it results from a comparison made on the same population composition. This arithmetic correction is nonetheless not enough to prove that A is effective. The calculation assumed that G was measured before prescription, that it was correctly classified and that it sufficed to make the groups comparable for the question studied. In real work, other reasons for prescribing might remain: recent trajectory, contraindications, frailty or access to care. Statistical uncertainty would also need to be assessed. The small numbers in some cells rule out reading the percentages as biological constants. Now change the target population again. If it includes only 20% of severely ill patients, the same stratum-specific risks give 10% with A and 16% without A, or −6 points. Both standardizations can be correctly calculated. They concern different compositions. Presenting an "adjusted effect" without naming the population to which it refers leaves an important part of the reasoning invisible.

WHAT THE CORRECTION ASSUMES

NAMING THE POPULATION OF THE ADJUSTED EFFECT

3.3.3 Adjusting According to the Question Asked

ADJUSTING FOR A MEDIATOR

Return to the marker M measured after treatment. It might be proposed to add it to the standardization "to be more precise." In this scenario, the adjustment no longer simply targets the total effect. It may block a mediated pathway and, depending on uncontrolled relationships, introduce bias. It does not suffice to identify a direct effect. It might also be proposed to keep only the patients still followed at one year. If remaining in follow-up depends on health status and treatment, the restriction would in turn call for an analysis of selection. The teaching rule is therefore a question: why does this variable enter this analysis? An answer based only on its availability, its statistical significance or a larger number of parameters is insufficient. The chronology and the presumed role of the variable must be discussed. The same laboratory name does not carry the same causal information depending on whether it refers to a measurement before or after the intervention.

THE REASON FOR ENTERING THE ANALYSIS

3.4 Mechanism, Genetics and the Convergence of Evidence

3.4.1 Explaining an Effect and Evaluating an Intervention

THREE QUESTIONS ABOUT AN EFFECT

An intervention can change an outcome without all the intermediate steps being known. Conversely, a mechanism can be firmly established without its modification yielding a net benefit. This gap appears as soon as three questions are distinguished: does the effect exist in the comparison studied; through which processes does it occur; what would happen with another intervention or in another situation? This distinction leads to a legitimate objection: why complicate a useful therapeutic result? Because the difficulty returns when another means of obtaining the same biological change is proposed. Two interventions can modify a marker in a similar way while producing other, different effects. Clinical reasoning must retain these differences rather than assume an equivalence that the common measurement does not demonstrate. Mechanistic explanation remains valuable: it suggests where to look for these

differences and which consequences to monitor. But its role is to enrich testing, not to remove the need to study the outcome that matters to the patient.

3.4.2 What Genetics Makes It Possible to Ask

3.4.2.1 The requirements of Mendelian randomization

three requirements of the instrument

Mendelian randomization uses genetic variants as instruments for an exposure. In the presentation by Lawlor and colleagues, three requirements are central. The first is relevance, that is, an association with the exposure. The second is independence from the factors that confound the exposure-outcome relationship. The third is the exclusion restriction, under which the instrument affects the outcome through the exposure studied. Estimating an effect size then requires additional assumptions suited to the method used. None of these requirements becomes true through the mere use of the word "Mendelian."[7] As early as 2003, Davey Smith and Ebrahim discussed several difficulties: variants linked to other variants, the multiplicity of phenotypic effects and compensation during development. Genetics can therefore bring a different test to a causal hypothesis while retaining its own limitations.[8]

Davey Smith and Ebrahim, 2003

3.4.2.2 From a fictional variant to a therapeutic target

the variant V and the mediator X

Suppose, again fictionally, that a variant V is associated with lower activity of the mediator X and with fewer cardiac events. The first task is to specify what X measures: a circulating concentration, an activity, a soluble receptor or a cellular response are not interchangeable descriptions. The second is to examine the other possible paths between V and the event. Consistency of signs does not close this inquiry. Even if the causal interpretation is convincing, a drug started at sixty is not a repetition of a genetic difference present since development. Timing, intensity, the tissues involved and additional effects may differ. This observation can provide a reason to test a target. It cannot, on its own, yield an effective dose or a benefit-risk balance. This consequence belongs to the methodological interpretation proposed here for moving from the etiological question to the therapeutic one.

a drug is not a variant

3.4.3 Building a Convergence of Evidence

what each approach corrects

To assess such a convergence, one criterion is to ask what each approach could correct in the other. A human study describes a relationship in the relevant population but may leave competing explanations standing. An animal perturbation can test a process under controlled conditions but leaves open how far the result can be extrapolated. Bringing them together is instructive if their agreement withstands these different difficulties. It would be excessive to conclude that two limitations neutralize each other simply because they are not identical. The passage between models must itself be argued: which properties are shared, and which differences could change the relationship studied? A convergence without this question risks being an assembly of favorable results. Conversely, requiring each study to resolve every difficulty would make their complementarity impossible. A conclusion can be robust because several comparisons progressively narrow the competing explanations. This robustness concerns a specific proposition, not a global confidence granted to an entire field.

two limitations do not cancel out

3.5 Texts Put to the Test

3.5.1 Xiao and the Experimental Pathogenicity of Anti-MPO Antibodies

3.5.1.1 Situating the Result

witness, consequence or producer

An antibody accompanies a disease. Is it a witness to the lesion, one of its consequences or one of its producers? The question involves more than a dispute over vocabulary. If the antibody is merely an indicator, its disappearance may change nothing in the patient's course. If it takes part in producing the lesion, it becomes a possible target. But between these two propositions lies a third question: will intervening on this target actually bring a clinical benefit? Xiao's work makes it possible to take one step of this reasoning without licensing all of them.[9]

3.5.1.2 Critical Reading

3.5.1.2.1 An antibody transfer within an organized comparison

Xiao, transfer of anti-MPO IgG

The design rests on two interventions: transferring splenocytes from mice immunized against myeloperoxidase (MPO), or transferring purified anti-MPO immunoglobulin G (IgG). The second procedure is particularly instructive: it removes from the transferred material the cells that might have explained the result. All five Rag2-deficient recipients, which lack functional B and T lymphocytes, and all six wild-type recipients exposed to anti-MPO IgG develop renal lesions, against none of the six anti-bovine serum albumin (BSA) controls. Some extrarenal lesions also appear after transfer of IgG alone.[9] The strength of the reasoning comes from the organized comparison. The question is no longer only whether antibodies and lesions occur in the same organisms; it is what happens when the antibodies are introduced under defined conditions. The comparator prevents the effect from being automatically attributed to the experimental manipulation itself. The experiment thus makes an explanation more precise than association alone.

the strength of an organized comparison

3.5.1.2.2 Sufficient in this design, not necessary everywhere

sufficient in this design

The full phrase must nonetheless be kept: the antibodies are sufficient in this design. This restriction does not diminish the result. It gives it its meaning. A cause does not necessarily produce the same effect in every biological setting. The recipient contributes cells, molecules, a vascular architecture and conditions of susceptibility. Removing functional B and T lymphocytes does not remove this whole environment. "Antibodies alone" here describes the material administered, not all the biological actors that make the lesion possible. Nor does the result demonstrate that the antibody is necessary for every disease resembling the one in the model. An introduction experiment can test a factor's capacity to produce a phenomenon in a defined context. A removal experiment can test its role in maintaining that phenomenon at a given time. Neither should be converted, without further argument, into necessity for every route of onset or every stage of the disease. The temptation to fit these two demonstrations into a single verb, "prove," must be resisted.

introduction and removal

3.5.1.3 Immunological Reading

FEW DEPOSITS, IMMUNE ACTION

The apparent paradox of a pauci-immune injury caused by immunoglobulins deserves particular attention. Few deposits do not mean little immune action. They mean that the amount of material visible in the tissue is not enough to describe the function exerted while the lesion was forming. The agent introduced, the cells mobilized and the damage observed must then be distinguished. Identifying the first is not the same as having directly tested each intermediate step. The participation of the neutrophil can be built into the immunological reasoning, but its precise role must be referred to the experiments that assess it. This care avoids a frequent error: considering that a biologically coherent mechanism has been fully demonstrated by the work that established one decisive component of it. The same discipline applies to the antigen. The term antineutrophil cytoplasmic antibodies (ANCA) designates a category; the design studied concerns anti-MPO antibodies. Extending the conclusion without justification to all antibodies in this category would strip the result of its experimental precision.

3.5.1.4 Cardiological Reading

THE TERRITORIES ACTUALLY EXAMINED

The cardiovascular interest of this experiment is that it makes an immune-mediated vascular injury intelligible. It does not remove the need to describe the territories actually examined. A renal and vascular demonstration does not constitute, by verbal extension, a demonstration of myocardial involvement or of large-vessel disease. For the clinician, the work provides a reason to look for relevant interventions. It does not choose the treatment, its dose or the patient likely to benefit. A treatment can reach a real mechanism and nevertheless have an unfavorable clinical balance. The transition to decision-making therefore requires additional evidence concerning the human intervention and its consequences.

A REASON, NOT A DECISION

3.5.1.5 Four Conclusions

Established: in the model studied, transfer of anti-MPO IgG can produce renal lesions and some extrarenal vascular lesions. Plausible: this experimental pathogenicity sheds light on part of the human disease associated with anti-MPO antibodies. Not concluded: complete identity of mechanisms across species, generalization to all ANCA and the efficacy of a specific human treatment. What would change the judgment: reproducible contradictory experiments, a decisive change in the conditions of susceptibility or human data durably dissociating the proposed mechanism from the lesion under consideration. The judgment should evolve by specifying its domain, rather than by classifying the experiment as definitively true or false.

REVISING BY SPECIFYING THE DOMAIN

3.5.1.6 Integrative Clinical Reading

A LINK BETWEEN MANIFESTATIONS

A convincing immune explanation can help link manifestations previously examined separately. It must not make their own description unnecessary: the impact of renal involvement and the consequences of involvement elsewhere remain distinct questions. In a shared record, the proposed mechanism can

serve as a link, while each observation keeps its degree of certainty. The gain for the person would be a coordinated explanation, without suggesting that this experiment already determines all the risks or their treatment.

3.5.2 Jaiswal and the Meeting of Two Types of Evidence

3.5.2.1 Situating the Result

HUMAN AND ANIMAL, TWO QUESTIONS

A human observation and an animal experiment can converge without becoming a single piece of evidence. Their meeting is fruitful precisely because they answer different questions. In humans, the aim is to learn whether a characteristic precedes or accompanies an event. In the experimental model, a component of the system is modified to examine its consequences. Jaiswal's work offers an opportunity to follow this articulation and then to measure its scope.[10]

3.5.2.2 Critical Reading

JAISWAL, CLONAL HEMATOPOIESIS AND TET2

The work brings together human studies of clonal hematopoiesis and a murine component modifying Tet2. For coronary heart disease, the prospective analyses report a hazard ratio (HR) of 1.9, with a 95% confidence interval (95% CI) of 1.4 to 2.7. The retrospective studies of early-onset myocardial infarction report a pooled odds ratio (OR) of 4.0 (95% CI: 2.4 to 6.7). These measures are not interchangeable. The experimental perturbation strengthens the hypothesis of a biological contribution.[10] The lesson concerns the words of risk. An adjusted association is not automatically freed from every competing explanation. Adjustment acts on the variables taken into account and in the ways chosen. It does not retrospectively turn an observation into an assigned intervention. The animal experiment adds an argument of another kind; it does not erase the limitations of the human comparisons. Clone size also invites examination of a graded relationship. But finding a significant association in one category and a nonsignificant one in another is not enough to establish a difference between them. For coronary events, the reported test for heterogeneity gives $p = 0.24$. Caution should therefore apply to the contrast between categories, even if each estimate retains its interest.[10] This distinction matters beyond this example. A gradient hypothesis can be accepted while declining to present it as established. The strength of an argument does not depend on the number of affirmative formulations surrounding it, but on the exact question the comparison answers.

A GRADIENT AND ITS HETEROGENEITY TEST

3.5.2.3 Immunological Reading

A DISEASE AT A DISTANCE

The study brings an important idea to the fore: an acquired change in hematopoiesis might contribute to a disease located at a distance, in the arterial wall. This way of thinking shifts attention from the diseased organ alone to the cells that circulate through it, are recruited into it and act within it. It nevertheless requires several levels to be distinguished. The clone describes a related cell population; the mutation designates a particular alteration; the inflammatory phenotype describes a function; the cardiovascular event is a clinical consequence. Moving from one to the next requires an argument each time. A shared category name, "clonal hematopoiesis," does not guarantee functional

CLONE, MUTATION, PHENOTYPE, EVENT

Tet2 does not explain all clones

identity among all clones. The Tet2 component must therefore not become a universal explanation for all the mutations mentioned. It shows the value of a specific mechanistic hypothesis. Other genes may contribute to risk through partly distinct paths. Clonal hematopoiesis thus designates a situation to be characterized, rather than a uniform inflammatory mechanism. Identifying the gene involved specifies the hypotheses to be examined.

3.5.2.4 Cardiological Reading

screening, another question

A factor associated with a disease can enrich its understanding before becoming a decision tool. Screening raises its own question: what new information is obtained, in whom and what is done with that information? A measurement does not have clinical utility simply because it is technically feasible. The discovery of the factor must be distinguished from the validation of a strategy guided by that factor. This distinction must be kept in its historical context. For contemporary practice, later data would have to be examined separately; a 2017 publication cannot be credited with answers produced after it. Nor does the murine experiment allow the expected benefit of an intervention in a human carrier to be calculated directly. Correcting a mechanism, slowing an experimental lesion and preventing a clinical event remain three distinct objectives.

3.5.2.5 Four Conclusions

Established: the work links a human association with an experimental perturbation of Tet2 in mice. Plausible: some clonal changes may contribute to atherogenesis through a functional modification of hematopoietic cells. Not concluded: a single mechanism for all clones, a definitively demonstrated clinical gradient and the utility of a clone-guided therapeutic strategy. What would change the judgment: better separation of effects by mutation, human observations contradicting the proposed temporal sequence or interventions allowing direct evaluation of the clinical consequences of modifying the process.

3.5.2.6 Integrative Clinical Reading

a new identity as a patient

The discovery of a clone can become, for the person, a new identity as a patient even before its clinical use is defined. Communication must therefore distinguish an acquired characteristic, a statistical association and the consequence expected for the person's own course. The study opens a dialogue between hematology and cardiology. It does not by itself establish that adding tests or surveillance would improve carriers' lives. An integrative question is to ask what new decision would really be informed, before multiplying labels and interlocutors.

3.5.3 Swerdlow and Genetics as an Argument for a Target

3.5.3.1 Situating the Result

THE MARKER OR THE LEVER

When an inflammatory marker accompanies coronary heart disease, a question remains: should one act on this marker, on the pathway that produces it or on another cause that influences both? A circulating measurement can signal a process without being the relevant lever for modifying it. The work of the IL6R Mendelian Randomisation Analysis Consortium (IL6R MR; Swerdlow et al.) takes this difficulty seriously by bringing together genetic variation, biological profile and coronary heart disease.[11]

3.5.3.2 Critical Reading

3.5.3.2.1 A variant as an instrument, and its conditions

SWERDLOW, IL6R AND CORONARY DISEASE

The analysis examines a variant associated with the interleukin 6 (IL-6) receptor (IL6R) and compares some of its biological effects with those of pharmacological blockade. The coronary association is small per allele, with an OR of 0.95 (95% CI: 0.93 to 0.97). The biological concordance supports the relevance of the target; it does not measure the benefit of a cardiovascular treatment. The authors call for a dedicated trial. The signals concerning diabetes and blood pressure are presented only as suggestive.[11] Here a particular type of reasoning is encountered: using genetic variation to shed light on the consequences of a biological modification. The value of the inference rests on conditions. The variant must be linked to the relevant exposure. It must not be associated with the factors that confound the relationship between this exposure and the outcome. Its effect on the outcome must pass through the exposure represented, with no other pathway relevant to this estimate. The term "Mendelian" does not remove the need to examine these conditions.

THE CONDITIONS OF THE INSTRUMENT

3.5.3.2.2 Neither precision nor concordance makes a benefit

PRECISION IS NOT VALIDITY

High statistical precision does not replace this examination. A narrow interval indicates low statistical imprecision under the assumptions of the calculation. It does not measure all uncertainties, notably those related to bias or to the validity of the causal instrument. Sample size and instrument validity answer two complementary questions. The comparison with a drug provides a test of consistency. If the genetic perturbation and the pharmacological intervention show compatible biological effects, the proposed interpretation gains in plausibility. Yet exposure linked to a variant over a long part of life differs from a treatment started in a sick adult. Intensity, duration, context and biological compensation may diverge. The per-allele effect therefore cannot be converted directly into an expected therapeutic benefit.

LIFELONG VARIANT, ADULT TREATMENT

3.5.3.3 Immunological Reading

SIGNALING AND DOWNSTREAM MARKERS

IL-6 signaling must be distinguished from its downstream markers, without turning this distinction into automatic proof that the markers play no causal role. The question concerns the biological intervention represented by the genetic instrument: which pathways does it modify, and which explanations do the data make less likely? The status of a biomarker cannot be reduced to a choice between absolute cause and useless indicator. A marker can serve to

THE STATUS OF A BIOMARKER

recognize biological activity, to select a population or to monitor a response without itself being an effective target. Conversely, a fall in an indicator after intervention does not automatically establish that the whole pathological mechanism has been corrected. Nor should the coherence of a pathway be confused with its exclusivity. Supporting the role of a receptor does not mean explaining the whole disease through that receptor. Useful mechanistic knowledge remains partial; its precision also lies in what it leaves open.

3.5.3.4 Cardiological Reading

A REASON TO TEST

For cardiology, the contribution of the work lies in justifying a research program. It provides a reason to test a strategy. This reason is more substantial than a simple association between a blood concentration and an event, but it is not yet the result of the clinical trial it helps to motivate. The reasoning must keep a specific place for tolerability. The absence of a detected association with an adverse event does not demonstrate that the event is impossible. One needs to know which events were recorded, their frequency, the precision of the estimates and the context of exposure. A genetic result therefore does not guarantee the safety of a drug acting on the same pathway. Treatments and trials that appeared later may extend this story. They must be introduced as new steps and sourced separately. The historical perspective thus retains its interest. It shows how a research decision becomes reasonable before its result is known, without retrospectively reconstructing an inevitable march toward success.

TOLERABILITY IS NOT GUARANTEED

3.5.3.5 Four Conclusions

Established: the analysis provides a genetic argument and biological concordance in favor of exploring the IL6R pathway. Plausible: an appropriate intervention on this pathway could modify coronary risk. Not concluded: the magnitude of benefit of a given drug, its safety and the benefit-risk balance in a specific patient. What would change the judgment: a challenge to the validity of the instrument, better-explained mechanistic discrepancies or clinical trials whose results would specify the effect of the intervention, its population and its conditions of use.

3.5.3.6 Integrative Clinical Reading

SELECTING A PATHWAY, NOT A TREATMENT

A genetic argument for a target is first of interest to research. Presented to a person, however, it risks being heard as the announcement of a drug suited to that person's profile. The difference must be explained between selecting a pathway to study and selecting a treatment for someone who already has a history of care. The integrative question then becomes the additional benefit, the constraints and the other consequences of a concrete intervention. These elements are not calculated from the per-allele effect.

3.6 What an Explanation Must Leave Open

Underdetermination here designates a precise situation: the available data do not fully decide between several explanations. It calls for a reasoned ranking and a program of inquiry. It does not require them to be held equally credible. An explanation may account for more observations, depend on fewer improvised additions or predict a result that was previously unexpected. These reasons carry weight, without guaranteeing that the explanation is definitive. Their articulation must be traceable, and the point at which an additional observation would change the judgment must be recognizable. A rule follows: end every causal discussion by stating its condition for revision. If the hypothesis favors a pathway, which independent perturbation should reproduce the effect? If it relies on a model, which human observation is still missing? If it announces therapeutic utility, which clinical result could contradict it? These questions give doubt a direction and give the explanation a responsibility.

REASONS WITHOUT GUARANTEE

THE CONDITION FOR REVISION

3.7 Worked Exercises

3.7.1 Exercise on Therapeutic Translation

3.7.1.1 Problem

Here is a deliberately excessive conclusion: “Since the antibody causes a lesion in the mouse, removing it will cure the human disease.” Identify the missing intermediate propositions, then propose a proportionate formulation.

3.7.1.2 Reasoned Answer

THE MISSING STEPS OF TRANSLATION

Induction alone does not demonstrate that the antibody is necessary to sustain an established disease. Translation requires checking the relevance of the model to the human disease. It must then be studied whether the intended intervention reaches its target, at what time, with what other effects and by what criterion of benefit. A defensible formulation would be: “This model establishes a pathogenic potential under defined conditions and justifies the study of interventions on this pathway; their clinical utility requires its own demonstration.”

3.7.2 Exercise on the Choice of Adjustments

3.7.2.1 Problem

THREE ADJUSTMENTS TO QUESTION

A cohort reports the association between A and hospitalization. The analysis adjusts for baseline severity, for M measured three months after prescription and for attendance at the twelve-month visit. The authors conclude that the effect obtained represents “the treatment’s own benefit, freed from inflammation.” What clarifications should be requested before accepting this sentence?

3.7.2.2 Reasoned Answer

target effect and timing

First, specify the target effect: total, direct or another defined contrast. Then specify the timing of hospitalization, of the measurements and of selection. Attendance at twelve months is not a baseline characteristic. Adjusting for M could exclude part of the mechanism; selecting on attendance could alter comparability. Baseline severity requires examination of its causal role, its measurement and its sufficiency. "Freed from inflammation" adds an unjustified biological claim: a single assay does not represent all of inflammation. Even a causal direct effect relative to M would not establish independence from all immune pathways. The conclusion must name the quantity estimated, the interpretive assumptions and the mechanisms left open.

"freed from inflammation"

Synthesis

A causal claim adds a comparison to the finding: what would happen under another strategy, in a named population and over a named horizon? This comparison rests on conditions that must be made explicit: exchangeability, positivity and consistency. Adjustment is chosen according to the question asked, not according to the availability of variables. Necessity, sufficiency and contribution remain distinct propositions. Together, the three commented texts show what each type of evidence contributes and where it stops. The transfer of anti-MPO antibodies establishes sufficiency in a defined design. Clonal hematopoiesis brings together a human association and an experimental perturbation without merging them into a single piece of evidence. Mendelian randomization on IL6R justifies the study of a target without measuring the benefit of a treatment. A convergence holds when the agreement withstands different difficulties.

Open Questions

At what point does a compensation invoked after a failed removal stop explaining and merely protect the hypothesis? What properties must an animal model share with the human disease for a convergence of evidence to be justified? How can one move from a genetic difference present since development to a treatment started in a sick adult? Each answer will gain from naming the observation that would lead to its revision.

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4

Reading Sources and Building the Evidence

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BIBLIOGRAPHY, CORPUS, BODY OF EVIDENCE

A bibliography gives access to documents. A corpus gathers them according to an intention. A body of evidence answers a specific question. Moving from one to the next requires decisions that must become visible: why these sources, which versions, which data and in support of which proposition? Choosing one hundred texts imposes a selection. Its quality is not measured only by the presence of the best-known publications. It depends on the corpus's capacity to expose the reasons for a conclusion, the objections raised against it and the work likely to change its scope. A teaching book may choose; it must make its choice intelligible.

4.1 Searching for, Identifying and Reading the Necessary Documents

4.1.1 Defining What Is Being Sought

4.1.1.1 The question guides the selection

THE QUESTION GUIDES THE SELECTION

Searching begins with a question precise enough to guide the selection. Tracing the origins of a concept, estimating the effect of a treatment and examining the performance of a test do not lead to the same documents. A conceptual history may need an old text that provides no estimate usable today. A therapeutic comparison requires searching for the relevant studies, including those that left little mark on the field's narratives. Whether a text is foundational thus depends on its function in a teaching sequence. Some texts change a way of thinking. Others make it possible to decide between hypotheses. Still others expose an error that threatens similar reasoning. These roles must be stated so that it is clear why the publication deserves in-depth analysis.

4.1.1.2 Learning from chosen cases, concluding about the whole

4.1.1.2.1 An anthology is not a sample

TWO USES OF A SELECTION

It may be objected that any selection of this kind favors its author's preferences. Making the criteria explicit and seeking texts that run against the narrative provides a first answer. But two uses of the corpus must also be distinguished: learning to reason about chosen cases, and estimating what the body of research allows one to conclude. A selection that is excellent for the first use

AN ANTHOLOGY IS NOT A SAMPLE

may be poorly suited to the second. Suppose a fictional anthology of ten articles, each chosen because it reveals a different difficulty. Three present a result favorable to an intervention; seven do not establish one. It cannot be inferred from this distribution that only three articles in ten in the literature establish a benefit. The selection was based on teaching interest, not on representation of the available results. Deliberately adding a counterexample may improve the course; it does not, by itself, correct an estimate of the effect. How instructive a text is and how much weight it should carry in a collective conclusion are not the same property.

4.1.1.2.2 A stated and revisable selection

STATING THE CONCLUSION IT PERMITS

The opposite objection would be to demand an exhaustive search before any teaching. Such a demand would make it hard to build a teaching sequence: an article is also chosen for the clarity of its design or the difficulty it makes visible. These motives are legitimate provided the teaching choice is not later turned into an argument about the frequency of results or about a consensus in the field. The answer is therefore not to abolish selection but to state the

conclusion it permits. Supporting a claim about the effect of a treatment requires examining the relevant body of work through a search suited to that question. The mere presence of several examples in a teaching selection does not accomplish that work. The selection remains revisable. A text may give way to another that better illuminates the stated problem; this replacement does not by itself change the state of the evidence. Two means of checking are thus available: discussing the relevance of the example, and separately checking the reasons for a general conclusion. A teaching corpus becomes more robust when it makes both discussions possible.

A REVISABLE SELECTION

4.1.2 Identifying a Document and Its Version

FINDING THE RIGHT TEXT

A bibliographic record must make it possible to find the right text. Title and author may not suffice: a main publication, a secondary analysis and a letter to the editor sometimes carry similar wording. The year, journal, pages or article number and identifiers help remove this ambiguity. An identifier does not, however, remove the need to check the document it leads to. If it has been miscopied, its presence gives a false impression of precision. Good practice consists in checking bibliographic identity against content, particularly when a result plays a central role in the reasoning. Version also matters. A correction may change a value without invalidating all the conclusions; a retraction may radically change the admissible use of a publication. A dated protocol may document an earlier intention without reflecting the latest amendment. The commentary must identify the document it uses and avoid assigning it the function of another. This documentary attention is a condition of scientific discussion. Two readers may seem to dispute the same text while having worked on different versions or analyses. Before their interpretations are discussed, what they actually read must be established.

CORRECTION, RETRACTION, PROTOCOL

4.1.3 Reading What Is Available without Inventing What Is Missing

AN ABSTRACT DOES NOT REPLACE THE TEXT

Access to an abstract makes it possible to examine some information. It does not automatically give access to the detailed methods, the additional analyses or all the limitations discussed by the authors. Trust in a publication cannot materially fill in a section that has not been consulted. The problem is not solved by the length of a file. A large document may be a repeated extraction, a compilation or an assembly of secondary elements. Conversely, a short text may be complete for the editorial genre to which it belongs. The expected structure must be recognized and the continuity of the content checked. The same caution applies to tools that extract, summarize or classify texts. A well-formed response may come from incomplete processing. It must be possible to find the passages on which the synthesis rests. The tool is useful when it facilitates this verification and makes its limitations identifiable. For the one hundred commented texts, free and legal access to the full text is a selection criterion. A record or an abstract can guide the search but cannot stand in for a commented text. If the published version is inaccessible, a legally deposited author manuscript may be used, provided its version is specified. If no admissible full text is found, the commentary is deferred or the text replaced. The restriction is documentary; it does not mean that the excluded publication lacks scientific value.

FULL ACCESS, A SELECTION CRITERION

4.1.4 Linking the Question to the Necessary Documents

4.1.4.1 The documents of a file and their chronology

REGISTRY, PROTOCOL, PUBLICATION

A clinical study may give rise to several documents: registry entry, protocol, analysis plan, main publication, supplement, secondary analyses and later exchanges. Each document answers a different question. The registry helps retrieve what was announced; the protocol specifies the design; the publication presents the results; later documents may complete or correct them. Comparing these documents must not become an automatic hunt for misconduct. A modification may be justified and properly documented. What matters is understanding its nature, date, reason and possible consequences for interpretation. The lack of a sufficient explanation creates uncertainty; it does not authorize inventing an intention. The file of a study must therefore preserve a chronology. This chronology makes it possible to distinguish a decision taken before access to the results from an analysis proposed after they were examined. Both can be informative, but they do not offer the same safeguards against a favorable selection of conclusions. The account must remain proportionate. No one reading the commentary needs to go through every administrative detail to understand the study. It must be possible to identify the elements that change the interpretation and to access the documents if needed. Traceability then serves understanding instead of taking over the whole of the reasoning.

THE CHRONOLOGY OF THE FILE

4.1.4.2 A missing document and the scope of the reservation

A MISSING APPENDIX

Consider a fictional file whose main text is available but one of whose appendices is missing. If this appendix contains the only description of the treatment allocation procedure, its absence prevents a documented judgment on that procedure. It does not prove that the procedure was flawed. If the appendix concerns extended follow-up, the results for the period already described can be examined for that period, while a conclusion on durability must wait. If it contains only an additional illustration with no identified bearing on the argument, reading it may be deferred, provided this is stated. A reservation is thus classified with respect to a specific proposition. A missing document may block its examination, reduce its scope or leave only a secondary point to be explored further. This classification remains revisable: a later correction may make decisive a detail previously judged incidental. Before searching for a new document, the question it must resolve should therefore be written down, together with the interpretations that reading it could decide between.

CLASSIFYING A RESERVATION

4.2 Evaluating and Comparing without Confusing the Objects

4.2.1 Distinguishing the Objects of Evaluation

4.2.1.1 Reporting, risk of bias and certainty of evidence

REPORTING, BIAS, CERTAINTY

Quality of reporting concerns what the publication makes visible. Risk of bias concerns the mechanisms by which the result could be systematically distorted. Certainty of a body of evidence concerns the degree of confidence placed in a conclusion on the basis of the relevant studies. These objects are related but distinct. PRISMA 2020 (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) provides guidance for reporting systematic reviews. Its checklist helps make the procedures understandable; merely citing it does not validate the results. The teaching selection of the one hundred commented

PRISMA 2020

texts will not be described as a systematic review in the absence of a process matching that ambition.[1] For randomized trials, the Cochrane Handbook presents an outcome-specific assessment of risk of bias: randomization, deviations from intended interventions, missing data, outcome measurement and selection of the reported result. A global label assigned on the basis of the study design name alone would lose this precision.[2] The GRADE (Grading of Recommendations Assessment, Development and Evaluation) approach examines the certainty of a body of evidence for an outcome, considering in particular risk of bias, inconsistency, indirectness, imprecision and publication bias. It does not consist in handing out a quality grade to each article because it belongs to a category reputed to be superior.[3]

RISK OF BIAS ACCORDING TO COCHRANE

GRADE AND THE CERTAINTY OF EVIDENCE

4.2.1.2 Choosing the tool and situating the strength of a recommendation

The commentary must choose its tool according to its question. It may assess certain aspects of risk of bias without claiming to have produced a complete rating. It may explain why evidence is indirect without inventing an official grade. A measured conclusion names the operation actually carried out. The strength of a recommendation is yet another judgment: it guides a course of action on the basis of the evidence and its expected consequences. It is identical neither to the certainty of the evidence nor to the percentage of experts who approve its wording. The preamble to the 2025 guidelines of the European Society of Cardiology (ESC) precisely distinguishes level of evidence, strength of recommendation and voting procedure.[4]

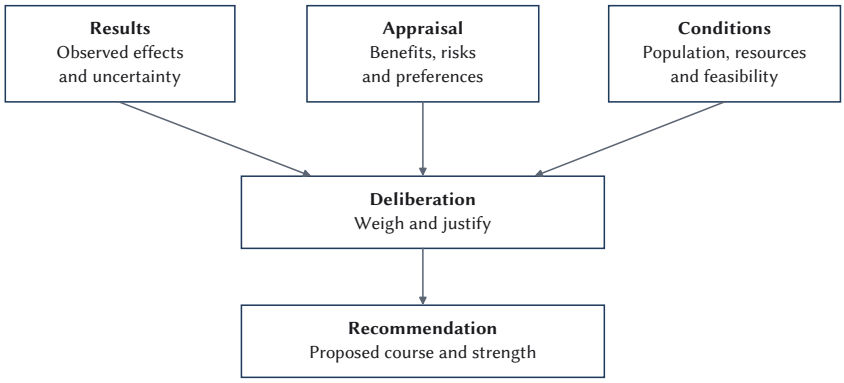


Figure 4.1. Moving from results to a recommendation Original reasoning diagram. The arrows indicate elements to be articulated, not demonstrated causal relationships. It does not reproduce the official procedure of any professional society.

4.2.2 Comparing Results without Erasing Their Questions

4.2.2.1 Five questions before comparing figures

Two publications may concern the same drug and answer different questions. The patients, the timing of treatment, the duration, the comparator and the outcome studied may change. Before speaking of confirmation or contradiction, these differences must be examined. Before comparing figures, what one seeks to learn must be named. The table distinguishes five questions that may run through a single publication. For each, a positive conclusion becomes possible when the corresponding comparison and its conditions are sufficiently supported.

SAME DRUG, DIFFERENT QUESTIONS

FIVE QUESTIONS BEFORE THE FIGURES

Question	Comparison required	Conclusion permitted	Error to avoid
Does the treatment change the outcome?	Defined strategies compared in groups made comparable, over the same horizon.	Estimate an effect for this population and these strategies, with its uncertainty.	Attributing to the treatment the mere improvement observed in its group.
Does the marker transmit part of this effect?	Intervention, marker and outcome linked over time; analysis based on explicit mediation assumptions.	Estimate a component of the effect, if the conditions of this analysis are defensible.	Inferring mediation from two parallel changes.
Does the test distinguish the states sought?	Test result compared with a defined reference, in the population in which it will be used.	Describe its performance relative to this reference, taking the reference's limitations into account.	Equating this performance with the benefit of a strategy using the test.
Does using the test improve management?	Test-guided strategy compared with a reference strategy, on consequences for patients.	Estimate what use of the test contributes, with its adverse effects and constraints.	Taking a changed decision for an improved decision.
Can the result inform another population?	Compare the characteristics and conditions that may modify the effect, check representation of the target profiles, then justify how they are accounted for.	Formulate a transported estimate under explicit assumptions.	Importing an absolute reduction without examining baseline risk and differences in context.

Table 4.1. From the question to the justified conclusion The conditions listed are reading guides; they do not, by themselves, constitute a complete protocol.

Comparison therefore requires a common, sufficiently detailed description. This description must not reduce studies to their p values. It must show which quantity was estimated, with what uncertainty and in what context. A comparative table is useful if it facilitates this reading; it becomes misleading if it lines up, under a single column, results that do not have the same meaning. Discrepancies can then be studied. Some stem from imprecision, others from differences in population or method and others still may weaken an explanation. Two symmetrical reactions must be avoided: declaring all studies contradictory as soon as their conclusions differ, or attributing each disagreement to a secondary difference in order to preserve the initial narrative.

4.2.2.2 Distinguishing measures of effect

ONE RESULT, SEVERAL SCALES

Entirely fictional scenario. At two years, with complete follow-up, 8% of people in the treatment group and 10% of those in the control group experienced at least one adverse event. The odds relate the probability of the event to the probability of its absence. The following measures therefore describe the same results in different ways.[2]

Measure	Calculation in this example	Result	What it expresses
Risk difference	8% – 10%	–2 points	20 fewer people with the event per 1,000
Relative risk RR	0.08 / 0.10	0.80	Estimated risk 20% lower relative to control
Odds ratio OR	(0.08 / 0.92) / (0.10 / 0.90)	About 0.783	Ratio of odds, different from the ratio of risks
Hazard ratio HR	Not calculable from these proportions alone	Not provided	Compares instantaneous occurrence among people still event-free

Table 4.2. The same result does not give the same number on every scale Fictional data, descriptive calculations; no real clinical benefit is established by this example.

OR, RR and HR

The odds ratio (OR) approaches the relative risk (RR) when the event is rare, without being identical to it. The hazard ratio (HR) draws on the timing of events and the duration of observation. Final proportions alone are not enough to estimate it without additional assumptions. Summarizing follow-up by a constant HR requires, in particular, examining the proportional hazards assumption. These distinctions come before any comparison of studies.[2]

4.2.2.3 Reading an interval without turning it into a verdict

WHAT AN INTERVAL DESCRIBES

A confidence interval describes statistical uncertainty conditional on the assumptions of its calculation. The 95% level refers to the coverage of the procedure over hypothetical repetitions. It is not the probability, calculated after the fact, that the fixed parameter lies within this interval. It does not correct biases.[2] Return to the difference of –2 points in Table 4.2. Suppose the analysis reports, for this difference, a 95% confidence interval (95% CI) from –4.5 to +0.5 points. It includes both a reduction and a small increase. The estimate favors a reduction, but these data do not decide between the possibilities that this interval retains. If a minimal relevant reduction had been set at 3 points, the interval would not allow the conclusion that it is reached either.

SIGN, MAGNITUDE, PRECISION, RELEVANCE

Sign, magnitude, precision and clinical relevance are therefore four distinct readings.

A CONCLUSION IN THREE SENTENCES

A useful conclusion can fit in three sentences: "The observed estimate corresponds to 20 fewer people with at least one event per 1,000 at two years. The available precision still leaves open a larger reduction and a small increase. If the threshold of 30 per 1,000 remains relevant to the decision under consideration, a more precise study should better situate the effect relative to it." The research priority thus depends on the decision to be informed, the other effects of the treatment and the information already available.

4.3 Making the Commentary and the Corpus Open to Discussion

4.3.1 The Commentary as a Verifiable Commitment

Commenting on an article commits the person who reads it. The commentary chooses the elements it highlights, formulates the objections and proposes an interpretation. It does not become neutral merely by citing the authors correctly. Three voices must therefore be distinguished: the result reported by the publication, the interpretation of its authors and the analysis proposed

THREE VOICES TO DISTINGUISH

by the commentary. These voices may converge. Keeping them separate remains necessary when they diverge or when the analysis adds a consequence that the authors did not study. The same distinction applies to a recalculation. Recomputing a proportion from a table can check arithmetic consistency. It does not necessarily reproduce an adjusted analysis, a survival method or a procedure that accounts for missing data. The recalculation must state its object, its assumptions and the limited conclusion it allows. This discipline makes precise correction possible. Anyone can dispute a number, an interpretation or a choice without having to reject the whole commentary. The analysis thus gains an important property: its reasons can be discussed separately from its author's authority.

DISPUTING WITHOUT REJECTING EVERYTHING

4.3.2 An Open Corpus That Makes Its Choices Open to Discussion

4.3.2.1 Access as a material condition of critique

ACCESS, A CONDITION OF CRITIQUE

Access to texts is a material condition of critique. A reference may be accurate, its demonstration sound and its commentary well written. If the document cannot be consulted, anyone reading the commentary must nonetheless delegate part of their judgment. An open annotated bibliography seeks to reduce this dependence. It gives everyone the possibility of retracing the reasoning, checking a formulation and proposing another interpretation. An accessible article may remain difficult to read. The four readings make its examination practicable: understanding the design, specifying the immunological proposition, examining its cardiological scope and questioning its place in a person's overall situation.

4.3.2.2 Free access that does not rewrite history

4.3.2.2.1 One hundred points of support rather than one hundred major texts

ONE HUNDRED POINTS OF SUPPORT

A difficulty must be acknowledged. If only texts whose full version is legally accessible without payment are retained, the corpus also depends on conditions of dissemination. It is therefore not enough to assert that the one hundred chosen texts are, without further qualification, the one hundred most important texts in the field. The aim is to gather one hundred points of support for the questions examined, among documents meeting the scientific and documentary criteria adopted. The selection can be ambitious without claiming to erase this condition.

4.3.2.2.2 Replacing a source without shifting a priority

FREE ACCESS AND THE STORY TOLD

The objection is serious: does the choice of free access risk changing the story told? Yes, if an essential publication silently disappears and its replacement is presented as historically equivalent. The answer is to distinguish the function sought from the identity of the document. A recent open-access article may explain a method as clearly as an inaccessible older text; it does not thereby become the text that introduced it. It may serve as teaching support, but it must not receive a historical priority it does not have. Even the original text does not always suffice to establish priority. Priority requires comparing the relevant earlier work and specifying what is meant by introducing the concept. When a source is replaced, two questions must therefore remain separate.

TWO QUESTIONS WHEN REPLACING A SOURCE

Does the new source allow the scientific problem to be examined with sufficient precision? Does the narrative require a historical attribution that this source does not allow to be verified? In the first case, the substitution may be justified. In the second, the historical claim must be reduced or another accessi-

ble document sought. The continuity of a chapter never justifies an attribution reconstructed from memory.

4.3.2.3 Versions, readings and documents shown

When an author version is used, it is named, and references to figures and tables refer to that version. A difference in presentation is not a scientific difference; any difference in content relevant to the reasoning must be examined. Full access, reading, checking a data point and reanalysis must be distinguished. The supplementary documents needed for judgment must also be consulted; their availability does not prove that they were. The commentary shows the documents, names the steps and specifies where another reading could lead elsewhere. Freedom of access takes on its meaning in this freedom of examination.

4.4 Texts Put to the Test

4.4.1 Fiolet and What a Meta-Analysis Adds to Trials

4.4.1.1 Situating the Result

Fiolet and colchicine, 2021

The meta-analysis by Fiolet and colleagues, published in 2021, pools five trials of colchicine in coronary heart disease, totaling 11,816 participants. For the composite primary endpoint (myocardial infarction, stroke or cardiovascular death), the pooled RR is 0.75 (95% CI: 0.61 to 0.92). No statistically established effect on all-cause mortality is shown. The search ends on September 1, 2020; this synthesis therefore predates the later results discussed in Chapter 9.[5]

4.4.1.2 Critical Reading

a meta-analysis creates no patients

The first question concerns the function of pooling. A meta-analysis can make an estimate more precise; it does not create new patients. Counting COLCOT, LoDoCo2 and then their synthesis as three independent confirmations would artificially inflate the evidence. The populations must be followed beneath the references. The bibliography can grow longer without any increase in the number of independent observations. The next question is what the trials actually make comparable. Here, the authors obtain some additional data from the investigators and examine different event definitions.[5] Harmonization can improve the comparison, but its value depends on its traceability. Can the original result be recognized, the transformation understood and the robustness of the conclusion to it checked? The appendix is part of the reasoning, because giving two endpoints the same name does not guarantee that they cover the same events. The central estimate and its precision must also be kept separate from their scope. The absence of a statistically established difference in mortality proves neither certain safety nor equivalence between treatments. To defend equivalence, one would have to define which difference would be small enough to be acceptable, then show that the data allow differences beyond this margin to be excluded. The conclusion “no difference was established” requires less evidence and must retain this modesty.

harmonizing and remaining traceable

no difference, not equivalence

4.4.1.3 Immunological Reading

A benefit without a designated pathway

A reduction in events under treatment does not, by itself, designate the immune pathway that carries the benefit. Consider an intervention that may act on several functions. Even if its clinical effect is firmly estimated, attributing that effect to a single function requires an additional comparison. The statistical precision of a synthesis does not compensate for the absence of this mechanistic comparison. This distinction makes it possible to articulate immunology and trials without conflating them. The former proposes mechanisms and discriminating measurements; the latter evaluate an intervention under defined conditions. To bring the two together, one can ask which observation would make the favored explanation less plausible or which other intervention would distinguish competing pathways. These questions extend the clinical results; the meta-analysis does not demonstrate the mechanism that would answer them.

4.4.1.4 Cardiological Reading

From relative to absolute

A pooled relative benefit does not directly provide the absolute benefit expected for a person. That benefit depends in particular on baseline risk and on the time horizon considered. Before turning an estimate into advice, these two coordinates must be recovered, and it must be examined whether the population concerned sufficiently resembles the populations studied. The shared label of coronary heart disease is not enough to settle this question. Finally, the published correction offers a precise documentary lesson. In the original graphical abstract, the relative reduction in revascularizations was given as 33%; the corrected value is 23%.[6] A memorable image can thus durably transmit an outdated figure. The version used must be dated and the figure tied to its endpoint, without concluding that a graphical correction invalidates the whole study. Reading a synthesis requires understanding its calculation, situating its date and following the corrections that accompany its dissemination.

A corrected graphic, 33% became 23%

4.4.1.5 Integrative Clinical Reading

An average, not a promise

To discuss additional prevention, the relative reduction in a composite endpoint must be expressed as understandable consequences: which events, over what period, with what burden of taking the drug and what uncertainties? The person may give different weights to these consequences, especially when already taking several treatments. This meta-analysis provides a dated collective result; it does not capture these priorities in the person's place. The integrated decision requires relating the expected benefit to the actual care pathway, without converting the pooled average into an individual promise.

4.4.2 Messroghli and the Conditions of Measurement

4.4.2.1 Situating the Result

Messroghli and mapping, 2017

This 2017 consensus concerns cardiac magnetic resonance imaging (MRI) mapping of T1, T2 and T2* relaxation times and of extracellular volume fraction. It combines published data and expert opinion to organize indications, acqui-

sition, quality control and interpretation. It stresses that values depend on technical conditions, recommends local reference values for native T1 and T2 and distinguishes degrees of maturity among applications. A correction published in 2018 specifies its status as a position statement. It also replaces the initial declaration of no conflicts of interest with detailed disclosures. It is thus a dated collective text, to be read together with its correction, and not a trial demonstrating the utility of all the proposed applications.[7,8]

4.4.2.2 Critical Reading: the number is the result of a procedure

4.4.2.2.1 What must be held constant in order to compare

THE NUMBER KEEPS ITS PRODUCTION HISTORY

A map showing values in milliseconds may seem more independent of the observer than a visual assessment. Yet the number retains the history of its production: acquisition, motion, image processing, region measured and reference comparison. A shared physical unit does not guarantee that two procedures yield directly interchangeable values. The preliminary question is therefore: what must be held constant for the observed difference to have the meaning attributed to it? Consider a fictional example. A patient has a higher value at the second examination. Between the two visits, the software and the sequence changed. Without a cross-calibration study, the difference cannot be apportioned between change in the tissue and change in the instrument. Asserting that the disease is progressing would be premature; asserting that it is only the software would be equally so. Here the doubt concerns comparability before it concerns diagnosis.

THE SOFTWARE HAS CHANGED

4.4.2.2.2 Distinct qualities and a production framework

REPEATABLE, ACCURATE, DISCRIMINATING, USEFUL

This requirement leads to distinguishing several qualities that are often confused. Repeating a measurement with little dispersion does not prove its accuracy. Detecting a difference between two groups does not guarantee reliable classification of an individual. Predicting an event does not demonstrate that action guided by the measurement improves its outcome. Each of these propositions requires an appropriate check; one does not automatically follow from another. The document provides precisely a framework for producing and reading results. Its consensus nature does not deprive it of scientific value: harmonizing procedures makes later comparisons possible. But recommending a procedure must not be presented as clinical validation of every decision that might be based on it.

4.4.2.3 Immunological Reading: a tissue property is not the name of a cause

A NONSPECIFIC SIGNAL

In this text, changes in the parameters are not specific to a single disease; interpretation depends on other observations and on context. This limitation matters particularly in cardioimmunology. A signal compatible with a change in tissue water does not, by itself, designate the responsible cell, the recognized antigen or the treatment likely to help the patient. The steps of reasoning must be preserved: noting an abnormality, proposing a tissue explanation, discussing its origin, then assessing its therapeutic relevance. Agreement among several modalities can strengthen an interpretation without turning each measurement into a direct identification of a mechanism. The vocabulary should make it possible to see where each conclusion stands. Another precaution

REGIONAL AVERAGE, LOCALIZED ABNORMALITY

concerns the space studied. An average value in a region and the presence of a localized abnormality do not answer the same question. A relatively unchanged average can mask a heterogeneous distribution; selecting only the most abnormal area can produce a different narrative. The rule for selecting the region must therefore be reported, not only its final value.

4.4.2.4 Cardiological Reading: making the measurement usable without overpromising

The clinician needs a result that can be interpreted in the local setting. A local reference is not an admission that quantification has failed: it makes explicit the conditions under which the number becomes informative. It does not, however, remove the need to check the quality of the examination or to consider other possible causes of the abnormality. Finally, the correction of the conflict-of-interest statement is a reminder that the conditions under which a guidance document is produced are part of reading it. Their disclosure allows discussion of the relationships between expertise, technical development and evaluation. It proves by itself neither that the document is false nor that there was no influence. Reading this document requires keeping the version and its correction together.

THE CORRECTED CONFLICTS OF INTEREST

4.4.2.5 Integrative Clinical Reading

When a person moves to another center, continuity of the record also depends on transmitting the measurement conditions. An earlier value without its method may be hard to compare with the new one and may expose the person to contradictory interpretations. This consensus makes visible an often unnoticed coordination task: preserving the useful technical context and stating whether a comparison is possible. The point is not to request a new examination automatically, but to determine what the examinations already available allow to be understood together.

TRANSMITTING THE MEASUREMENT CONDITIONS

4.4.3 The ESC and the Move from Evidence to Guidelines

4.4.3.1 Situating the Result

Examining how a guideline is developed is not enough to validate all the courses of action it proposes. Each therapeutic proposal retains its own evidence base, population and conditions of use. Diagnostic reasoning does not remove the need for this separate examination. The text distinguishes the strength of recommendations from the level of the evidence supporting them. Its preamble describes two successive votes and a minimum agreement of 75% of voters. It introduces inflammatory myopericardial syndrome as an entry category for diagnostic investigation. It recognizes the possibility of a clinical diagnosis supported by MRI, while specifying that imaging does not provide the histological subtype of the inflammation. These proposals concern different operations: deciding together, organizing an investigation, recognizing an injury and characterizing its nature.[4]

ESC 2025, FOUR OPERATIONS

4.4.3.2 Critical Reading: a vote does not replace an estimate

4.4.3.2.1 A documented agreement and a procedure to be checked

A VOTE DOES NOT MEASURE THE EFFECT

A guideline turns knowledge into practical guidance. This transformation involves relating benefits, risks, a population and conditions of use. The number of experts in favor therefore measures neither the size of the effect nor its precision. It documents agreement on a proposal, after deliberation. Unanimity, if reached, would not make the observations more numerous; a divergence might concern their interpretation or the weight given to consequences. The preamble here forms a scientific part of the document. It makes it possible to ask how texts were selected, how uncertainties were discussed and which procedure produced the final wording. Describing these operations does not prove that each was perfectly carried out. It provides the points of support for a verification.

4.4.3.2.2 Not counting twice, discussing a specific recommendation

NOT COUNTING THE TRIAL TWICE

It is also necessary to avoid treating the whole as one vast independent study. When a trial already commented on feeds into a guideline, citing the guideline afterward does not add a second experiment. The new contribution may be a perspective, a confrontation with other data or an assessment of applicability. Recognizing this work preserves its value without counting the same observations twice. The unit of this critique can be a particular recommendation, with its population, the point at which it applies and its exceptions. One can accept the available results and dispute their application to another population. One can accept this application and discuss the weight of an adverse effect. Or one can judge the course of action reasonable while recognizing that its empirical basis remains fragile. These disagreements call for different answers: data on extrapolation, deliberation on consequences or a better estimate of the effect. Distinguishing them allows a specific recommendation to be discussed without accepting or rejecting the document's authority wholesale.

THE UNIT OF CRITIQUE

4.4.3.3 Immunological Reading: an entry category is not a common cause

AN ENTRY CATEGORY

Grouping several presentations under an initial term helps orient reasoning. But the convenience of a category does not demonstrate that all its members share the same mechanism. A label can organize the search precisely because the etiology remains to be determined. Immediately attributing an immunological unity to it would suppress the question it was meant to help explore. This distinction changes how a guideline is read. A broad term may be appropriate at the start of the pathway, while a later decision requires finer characterization. There is no contradiction in using different levels of description in succession, provided the transition between them is explicit. The difficulty arises when the final vocabulary keeps the generality of the entry point although the intervention depends on a particular mechanism. Imaging thus requires specifying what an observation actually identifies. Recognizing an inflammatory injury does not necessarily designate the dominant cell population; identifying that population does not automatically reveal its causal role. The diagnostic, histological and mechanistic levels communicate, without becoming synonymous.

INJURY, POPULATION, CAUSAL ROLE

4.4.3.4 Cardiological Reading: comparing tests by the decision they inform

CHOOSING THE TEST BY THE DECISION

Two tests may contribute to the same clinical diagnosis without providing interchangeable information. The choice therefore does not come down to electing the best test in the abstract. One must ask what still needs to be known and how that knowledge could change a decision. A more detailed test may be indispensable to one question and superfluous to another. A less invasive answer may suffice for one objective without resolving the whole etiology.

TRUST AND CRITICAL INDEPENDENCE

Comparing tests in this way avoids a false choice between trust and critical independence. The clinician can rely on collective guidance while retaining responsibility for checking that its conditions match the case at hand. The clinician does not redo all of science alone, nor turn a general statement into an automatic personal conclusion.

4.4.3.5 Integrative Clinical Reading

COORDINATING WITHOUT ADDING AN INDICATION

A guideline takes its place in a care pathway already filled with tests, appointments and sometimes several diseases. Integrative reading therefore requires linking each proposal to its objective and to the team responsible for its follow-up. A planned test should answer a question that is still open, rather than repeat one ordered elsewhere under another name. This coordination requirement extends the document methodologically. It does not appear in it, and it adds no therapeutic indication or exception to guidelines whose evidence base remains to be assessed.

4.5 Worked Exercises

4.5.1 Exercise on the Results That Follow-Up Does Not Show

4.5.1.1 Problem

Entirely fictional data. Two randomized groups each include 100 people. The endpoint is the occurrence of at least one adverse event at one year. It remains defined for everyone, but its status is unknown for ten people in each group. Among the 90 people with known status in each group, eight events are observed in the treatment group and twelve in the control group. No known event is erased: only the twenty unknown outcomes vary in the following scenarios.

Table 4.3. Differences according to assumptions about unknown outcomes (fictional data)

Reading or assumption	Risk in treatment group	Risk in control group	Difference in points
People with known status only	8/90 = 8.9%	12/90 = 13.3%	−4.4
No events among people with unknown status	8/100 = 8%	12/100 = 12%	−4
Six unknown events in the treatment group and two in the control group	14/100 = 14%	14/100 = 14%	0
Ten unknown events in the treatment group and none in the control group	18/100 = 18%	12/100 = 12%	+6
No unknown events in the treatment group and ten in the control group	8/100 = 8%	22/100 = 22%	−14

The difference is always calculated as treatment minus control. A negative value favors the treatment for this adverse endpoint. The first row describes only the people with known status; the next four fictionally complete the two randomized groups. They therefore do not have the same denominator or the same documentary status. Questions. Does a 10% rate of unknown outcomes in each group guarantee an unbiased comparison? What imbalance between unknown events is enough to cancel the observed difference? Do the two extreme scenarios form a confidence interval?

4.5.1.2 Reasoned Answer

THE SAME PROPORTION OF UNKNOWN OUTCOMES

The same proportion of unknown outcomes does not guarantee that the people concerned have comparable trajectories. Their reasons for no longer having known status may differ. The Cochrane Handbook encourages examining these reasons and their possible relationship with the outcome, rather than applying a universal threshold for missing data.[2] Let a be the number of events among the ten people with unknown status in the treatment group, and b the corresponding number in the control group. The difference, expressed in percentage points on groups of 100, equals $-4 + a - b$. Four more events in a than in b therefore cancel the descriptive advantage; beyond that, the direction reverses. The pair six and two illustrates this equality. The information provided does not allow a probability to be assigned to any admissible pair, nor the pairs to be ranked by plausibility. The bounds −14 and +6 describe the set of possible completions of these data, under the simplified conditions of the exercise. They do not quantify sampling uncertainty and do not constitute a confidence interval. Nor do they demonstrate that the treatment would be harmful or ineffective. They show what the available observations alone cannot decide. The next task is to document the reasons for the missing data and to examine plausible scenarios with an appropriate analysis. A convenient assumption, such as assigning zero events to each person with unknown status, must remain visible as an assumption. This calculation does not apply as such to a survival analysis or to a measurement that becomes undefinable after a death.

AN IMBALANCE THAT CANCELS THE GAP

BOUNDS, NOT A CONFIDENCE INTERVAL

4.5.2 Exercise on Replacing a Source

4.5.2.1 Problem

AN OLD ARTICLE REPLACED

A chapter attributes the introduction of a concept to an old article. Only its abstract is accessible. A recent full-text review explains this concept and cites the old article. The chapter's author replaces the old reference with the review and keeps the sentence: "This text introduced the concept." What is the problem? What remedy should be proposed?

4.5.2.2 Reasoned Answer

REWRITING THE PROPOSITION

The replacement changes the object to which "this text" refers. The accessible review can provide a teaching explanation; it does not become the historical article it cites. The attribution of priority must be removed if it cannot be verified. If the review explicitly attributes this origin to the old article, that attribution can be reported as such. It is not presented as a priority verified by an independent inquiry. For a commentary devoted to the founding publication itself, access to the abstract remains insufficient under the full-access rule (see 4.1.3). The solution is therefore not merely to change a line of the bibliography. It consists in rewriting the proposition that this line was meant to support.

4.5.3 Exercise on a Review Based on the Most Cited Articles

4.5.3.1 Problem

A review states that it retained only the most cited articles and concludes that a treatment is effective because most of these articles present it favorably. Which steps are missing between this selection and this conclusion?

4.5.3.2 Reasoned Answer

VISIBILITY IS NOT VALIDITY

The number of citations measures a form of visibility, without by itself establishing the relevance or validity of a result. The therapeutic question must be specified, the relevant studies identified, dependencies between publications recognized, their methods examined and the estimates compared. A count of favorable conclusions does not replace this analysis. Results with little visibility that could change the judgment must also be sought, and the limitations of the selection explained.

Synthesis

Building evidence requires specifying the question, identifying the sources and their versions, checking access to the content and comparing results on the appropriate scale. The teaching selection must be justified. Methodological evaluation must name its object. The synthesis must preserve the limitations that change the meaning of the conclusion.

Open Questions

When should an inaccessible source lead to deferring a commentary? How far can a selection of texts represent the controversies of a field? What level of documentary detail genuinely helps reading, and which details can be moved to appendices without making verification impossible?

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PART II

Understanding Mechanisms and Testing Models

A mechanism is readily told in the present tense of the self-evident: a cell releases a signal, a pathway is activated, a tissue repairs itself. These sentences rest on cultures, mice that have received grafts or injuries, reconstructed structures, human samples and syntheses. Each setup makes a relationship visible because it simplifies the living organism. Its value depends on what it preserves, what it actually modifies and what it leaves outside its scope. The common question is that of passage. Does a pathway involved in a rare disease shed light on a common lesion? Does a cell identified in the myocardium perform a function there? Are concordant results independent, or do they share the same weakness? Has a heart that functions better repaired itself in the way that was thought? A model becomes fruitful when its limits lead to a new comparison, not when it promises a treatment in advance.

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5

Inflammation between Protection and Disease

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The same inflammatory response can limit an infection and aggravate tissue damage. It can also subside before the tissue has recovered its function. Yet the words that describe it, "excess," "brake," "resolution" or "danger signal," seem to deliver a verdict before any measurement. Each imports a value that observations have yet to establish. Mutations transmitted in a few families, a cytokine turned therapeutic target, a memory attributed to innate cells, a pore that sorts what it lets out: each discovery captures inflammation at a different scale. None of them, on its own, says whether reducing it will benefit a person. Between a biological pathway and a clinical benefit, each step requires its own evidence.

5.1 The Words That Judge a Response

5.1.1 The Same Word Can Conceal Several Questions

5.1.1.1 Clarifying the word before judging the response

WHAT THE WORD "PROGRAM" SAYS

The Treatise opens its exposition with inflammation viewed as a physiological program. This choice avoids encountering it only when it accompanies disease. It also invites scrutiny of the word "program": does it designate an obligatory sequence, a set of coordinated possibilities or a convenient representation for research? The term gains precision when its use is explained. The opposition between protection and disease must also become a question to put to studies. Is a response protective with respect to the outcome measured, in the organism studied, over the period considered and compared with what other situation? The adjective cannot be detached from these complements without losing part of its meaning.

AN ADJECTIVE AND ITS COMPLEMENTS

5.1.1.2 An imaginary experiment as a guiding thread

WHAT THE IMPROVEMENT LICENSES

Consider an imaginary experiment: compared with an appropriate control, a perturbation reduces an inflammatory signal and improves an early tissue measure. The first finding concerns the effect of this perturbation on this measure, at this time point. To attribute the improvement to the reduction of the inflammatory response, it is also necessary to discriminate between this route and the other pathways through which the perturbation might act. If this link is supported, a harmful contribution becomes defensible under the conditions studied. This does not yet mean that the response has no useful function, that it is unfavorable in the long term or that any way of reducing it will be beneficial. This difference between the propositions serves as a guiding thread. It allows inflammation to be studied without assigning it a single value from the outset, and without giving up the distinction between favorable and unfavorable effects.

5.1.2 A Function Is Not Just Any Effect

MECHANISTIC FUNCTION, EVOLUTIONARY FUNCTION

A process can have several consequences. Calling one of them a "function" requires explaining why it is singled out. In a mechanistic description, the word may designate the contribution of an element to an activity of the system. In an evolutionary explanation, it may designate a consequence that helps explain why a trait has been conserved. These questions are not settled by the same observation. An experiment showing that a cell participates in a response provides information about what it does in that setup. It does not, on its own, demonstrate why this capacity exists in the history of life. Conversely, a possible historical explanation does not automatically predict the current effect of

THE SLIDE TOWARD PURPOSE

every intervention on that cell. Observed contribution, proposed role and evolutionary history remain three propositions to be justified separately. It might be objected that this is a lexical subtlety with no effect on research. Yet the slide changes the conclusion. Writing that a response "is made to protect" can lead to treating any damage as a mere error external to its operation. Writing that it "contributes to a given outcome under given conditions" leaves open the study of its other consequences. This clarification does not prohibit the vocabulary of functions. It prevents that vocabulary from serving as an explanation when it merely renames the outcome. A function becomes informative when it fits within a comparison and within a system whose relations can be tested.

5.1.3 Advantageous for What, and at What Cost?

BENEFICIAL FOR WHICH OUTCOME

The word "beneficial" presupposes a privileged outcome. In a teaching scenario, a response may preserve an activity in the short term while consuming resources needed for another. The statement is not a contradiction. It indicates that the evaluation bears on several consequences whose relations must be described. The reasoning becomes misleading when the scale is shifted unobtrusively. An effect favorable to the survival of a cell is not yet a benefit for the tissue. A favorable tissue outcome does not describe the whole state of the organism. The local measure can be accurate and important while this distance remains. An objection nevertheless confronts a purely global reading: if only the final outcome matters, why study cells? Because understanding the contributions makes it possible to identify the circumstances in which the balance changes. Local study becomes particularly useful when it explains why two apparently similar interventions can have different consequences. Its value does not require that it summarize the whole organism. The object of the balance must therefore be specified. Is the aim to preserve a structure, restore an activity, reduce a symptom or prevent a future event? These objectives may converge. Their convergence must be observed or argued; it does not follow merely from the fact that they all seem desirable.

FROM THE CELL TO THE ORGANISM

WHY STUDY CELLS

5.2 Context, Time and the Outcome of a Response

5.2.1 Context Must Not Become an Excuse

CONTEXT AS NAMEABLE DIFFERENCES

Saying that "everything depends on context" is sometimes accurate, but rarely sufficient. So flexible a proposition can explain all results after the fact without helping to predict any of them. To become scientific, context must be turned into nameable differences: initial conditions, tissue studied, intensity of the perturbation, other processes present, timing of the measurement. Return to the imaginary experiment. The intervention is favorable under condition A and unfavorable under condition B. It must first be established that this contrast is sufficiently supported by the data. If it is, the task is to look for the difference between A and B that could modify the effect, and to determine how to test it. An explanation by context can then be testable. If it attributes the change to a specific initial condition, it must lead to a new comparison in which this condition varies. If it invokes another process, it must indicate which observations would reveal that process's contribution. The word "complex" does not replace this work. The reverse objection deserves as much attention: demanding an identical effect everywhere would erase real relations that exist only under certain conditions. The aim is therefore not to eliminate context,

FAVORABLE IN A, UNFAVORABLE IN B

but to give it an explicit place in the proposition. A limited relation can be solidly known.

5.2.2 Time Is Part of the Mechanism

In a diagram, an arrow often links two states. In an experiment, observations have dates, intervals and sampling conditions. Moving from one to the other requires understanding what was actually followed. Suppose a signal is high at a first time point, then lower at the next. This decrease may accompany several possible stories. Without other evidence, it does not suffice to show that the tissue has returned to its previous state, that the effects of the first phase have disappeared or that the process has simply stopped. An indicator returning toward its initial value does not, by itself, tell everything that happened between the two measurements. The onset of the phenomenon must also be distinguished from the onset of its observation. A so-called "early" sample can be early relative to symptoms and late relative to a biological transformation. This difference changes the question put to an intervention: preventing a step, limiting its progression and modifying its consequences are not the same undertaking. One might think that a better timetable would resolve all uncertainty. Yet accumulating dates is not enough. Measurements must be linked to hypotheses about transitions. Longitudinal observations gain their strength when they make it possible to discriminate between trajectories, rather than adding points to a curve whose meaning remains assumed.

ONE DECLINE, SEVERAL STORIES

FROM DATES TO TRAJECTORIES

5.2.3 The Exit from a Response Deserves Its Own Question

The word "resolution" must be questioned with the same precision as the word "activation." Does it designate the disappearance of a signal, a change in the populations observed, a limitation of damage or the recovery of a function? These events can be linked without being equivalent. Two fictional situations can be conceived in which the same marker decreases. In the first, the functional activity studied is restored. In the second, it remains impaired. The decline in the marker is real in both cases; it does not license the same description of their outcome. The choice of final measures is therefore a decision about what the study calls a completed response. This remark does not require measuring everything in every experiment. It requires limiting the conclusion to the object actually examined. A study can illuminate a cellular transition without assessing the recovery of the organism; a functional study can observe an improvement without describing all the processes that made it possible. The disappearance of a signal does not yet say whether the process has stopped, been transformed or moved elsewhere. Repair adds a question about the reorganization of the tissue; recovery, a question about its functions. The two can diverge. To relate them to resolution, their sequence and consequences must be followed.

WHAT "RESOLUTION" DESIGNATES

ONE FALLING MARKER, TWO OUTCOMES

REPAIR AND RECOVERY

5.3 From Images to Revisable Propositions

5.3.1 Metaphors Also Shape Experiments

Words such as "army," "attack," "brake" or "cleanup" can make a description accessible. Their teaching effectiveness comes from importing a familiar situation. This importation also carries hypotheses: an army has an adversary, a brake slows an action, a cleanup removes what must disappear. A metaphor becomes a problem when these hypotheses take the place of results. Calling a cell

WHAT METAPHORS IMPORT

an enemy does not demonstrate that its removal is desirable. Calling it reparative does not guarantee that all its activities are advantageous at all times. Studies will have to specify the population observed, the state considered and the consequences measured. Should images then be removed from teaching? The answer proposed here is to keep them when their limits can be made explicit. It must be possible to return from the metaphor to a testable proposition. "The brake is missing" must be restated as an identified modification, a relevant comparison and an observed or expected effect. This discipline also helps to avoid intentional explanations. The tissue does not decide on a trade-off the way a physician chooses a treatment. Regulations and consequences can be analyzed without attributing deliberation to them. The teaching purpose of a sentence must not become an assumed biological intention.

KEEP THE IMAGE, STATE ITS LIMITS

A TISSUE DOES NOT DELIBERATE

5.3.2 What the Texts Will Require Revising

The alternative between protection and disease does not mean that inflammation simply divides into a good half and a bad half. It calls for an inquiry into the conditions that give meaning to these evaluations. The same requirement applies to general theories and to experimental setups, with different objects. Of a theoretical proposition, one must ask what it makes intelligible, which situations it claims to cover and which observation would genuinely challenge that ambition. Of an experiment, one will ask what modification was achieved and what it allows to be concluded about the role of the process studied. Of a clinical translation, it remains to ask which concrete intervention and which patient outcome are at stake. The value of this approach is to make disagreements productive. A theory may have to narrow its domain while retaining a fruitful idea. An experiment may modify a model without immediately licensing a treatment. A clinical result may force a reconsideration of the link between a target and the expected effect, even if the target remains biologically interesting. This way of reading is tested on the texts (see 5.4). It must keep together the observed consequences, their interpretation and the conditions that limit their scope. Only then do the words "protection" and "disease" become debatable conclusions, rather than roles assigned in advance in a story.

NEITHER GOOD HALF NOR BAD HALF

THREE QUESTIONS FOR THREE OBJECTS

CONCLUSIONS, NOT ASSIGNED ROLES

5.4 Texts Put to the Test

5.4.1 Hoffman and the Genetic Investigation of an Inflammatory Disease

5.4.1.1 Situating the Result

HOFFMAN, 2001

The authors identify four missense mutations in CIAS1 cosegregating with the disease in three families with familial cold autoinflammatory syndrome and one family with Muckle-Wells syndrome. They are not found in the controls examined. The predicted architecture of cryopyrin suggests an inflammatory signaling function, to be verified experimentally. Neither therapeutic efficacy nor any cardiovascular outcome is evaluated.[1]

5.4.1.2 Critical Reading: strong evidence does not answer every question

TWO SYMMETRICALLY POOR READINGS

Two symmetrically poor readings must be resisted. The first discounts a family study because it does not include thousands of subjects. The second turns a genetic discovery into a complete explanation of the disease. To appraise evidence, the proposition under examination must first be specified. Is the aim a hereditary determinant, a molecular function, a mechanism of injury or an effective treatment? The number of participants takes on meaning only within that question. In a family investigation, the distribution of a variant among relatives can provide information that a simple count of patients without family history would not. To evaluate it, one must look at unaffected individuals, the generations available and the quality of phenotyping. A family, however, is not a random sample of the general population. The strength of a result in explaining transmission therefore does not automatically give it the capacity to estimate a population frequency. Identification and mechanism must then be separated. As a general methodological point, recognizing a defective part does not yet reveal every step by which it disrupts the workings of a whole. Further experiments must establish which interactions change, in which cells, under which conditions and with which consequences. This separation does not weaken a discovery: it makes visible the problem that the discovery now allows to be posed.

A FAMILY IS NOT A SAMPLE

SEPARATING IDENTIFICATION AND MECHANISM

5.4.1.3 Immunological Reading: knowledge that transforms its categories

LOCALIZATION, KINSHIP, FUNCTION

The text detects CIAS1 expression in blood leukocytes and relates the predicted structure of cryopyrin to proteins involved in inflammation. It explicitly leaves the function to be verified. This difference between the localization of a signal, structural kinship and demonstrated function is its immunological foothold. The authors also distinguish the two clinical pictures and leave open the reasons for their differences.[1] A clinical category can thus be reconsidered on the basis of a shared determinant without becoming a uniform disease. Bringing mechanisms together should bring out the differences that the new explanation still leaves to be understood. The discovery of CIAS1 must also be placed within its 2001 horizon. A function proposed from a predicted structure does not retroactively acquire the status of later demonstrations. The historical interest of the article lies precisely in the shift of the question: familial transmission becomes a gateway to a function yet to be established. This discipline also avoids conflating several questions about heredity: why is a disorder transmitted, why does it appear at a particular time, why does it take a given form in this person? A genetic explanation does not preclude studying the circumstances of expression. Nor does it require treating each manifestation as the direct consequence of a single molecular event. Teaching here consists in articulating the levels, without filling the gaps still open with an overly smooth narrative.

THE HORIZON OF 2001

5.4.1.4 Cardiological Reading: welcoming a discovery without inventing a result for it

NO CARDIOVASCULAR OUTCOME

This family investigation includes no cardiovascular outcome. Its contribution to cardiological reasoning lies in the separation it imposes between genetic determinant, inflammatory function and organ manifestation. A discovery can illuminate the first link while leaving the subsequent ones to be demonstrated.

FROM THE RARE TO THE COMMON

Suppose that a pathway involved in a hereditary disease is also found in a vascular lesion. This shared feature would suggest a question; it would not suffice to establish that the pathway plays the same role in both situations. The cellular context, the timing, the intensity of the perturbation and the other determinants of the lesion would need to be examined. Moving from the rare to the common requires a demonstration of its own. Finally, the possibility of intervening remains a distinct question. A target associated with the disease may be inaccessible, act too early or simultaneously participate in a useful function. The discovery therefore opens an evaluation specific to the vascular situation considered; it does not foretell its result.

5.4.1.5 Integrative Clinical Reading

BIOLOGICAL KINSHIP, SINGULAR COURSE

The family investigation brings several people into the same explanation without making their situations interchangeable. Understanding a transmission can change how a family tells the story of the disease; it does not fully reveal what each member experiences or what their priorities are. The integrative contribution consists in holding together biological kinship and the singularity of each person's course. A genetic discovery deserves an intelligible explanation for those concerned, with a clear separation between what it illuminates in their history and the decisions that require other information.

5.4.2 Dinarello and the Passage from Mechanism to Therapeutics

5.4.2.1 Situating the Result

DINARELLO, 2011

The review organizes knowledge about the interleukin 1 (IL-1) family, its receptors, its natural antagonists and the interventions that reduce its activity. It links mechanisms, autoinflammatory syndromes and therapeutic explorations, with a final section on remodeling after myocardial infarction. It also reports disappointing results in other contexts. This juxtaposition constitutes neither a systematic synthesis of all studies nor a new demonstration of cardiovascular benefit.[2]

5.4.2.2 Critical Reading: synthesis as a working instrument

THE NARRATIVE REVIEW AS WORKING TOOL

The value of a narrative review cannot be reduced to the presence or absence of a meta-analysis. It can provide the distinctions needed to formulate a question, connect previously separate fields and show which results should be compared with one another. It should therefore be neither treated as a trial nor declared devoid of evidential value. The evidence it draws on must be examined, then the conclusions it builds from that evidence. Regarding IL-1, three propositions

MECHANISM, TREATMENT, EXTENSION

must remain distinct: one about mechanism, one about treatment and one that extends the treatment effect to other situations. The first describes what takes part in a process. The second asserts that an intervention improves an outcome under certain conditions. The third extends that effect to other people, diseases or time points. Each has its own requirements. A synthesis can link these propositions; their proximity within a paragraph does not make them interchangeable. Response to treatment particularly warrants this vigilance. If a truly specific intervention modifies a manifestation, it can support the causal role of the target in that manifestation. But it does not by itself establish that the target is the sole origin of the disease. A door can allow a journey to be interrupted without being the place where the journey began. This teaching analogy does not replace an analysis of pharmacological effects; it recalls the question to be asked.

WHAT RESPONSE TO TREATMENT PROVES

5.4.2.3 Immunological Reading: thinking in terms of regulation rather than simple excess

A HIGH VALUE TO BE INTERPRETED

The activity of a mediator is also understood through its relations with other processes, rather than through its mere presence. Faced with a high biological value, one must ask which production, clearance, reception and regulation give it meaning. Two identical values can correspond to different situations if these relations differ. The word "excess" also calls for a comparison. Excess relative to a control, to a previous state, to a reference range or to what would preserve a function? These definitions are not equivalent. To be useful in an argument, the term must announce the criterion that makes it verifiable. Without this precision, it readily turns an observed difference into a judgment of harmfulness. The selectivity of an intervention must also be distinguished from the breadth of its consequences. Choosing a precise target describes a property of the means employed. The balance of its effects is judged at another level. Clinical safety therefore cannot be inferred from the adjective "targeted" alone, nor can harm be attributed without having been observed. The passage from mechanism to the whole organism is precisely the space where clinical experimentation becomes necessary.

SELECTIVITY AND BREADTH OF EFFECTS

5.4.2.4 Cardiological Reading: turning a promise into a comparison

A HYPOTHESIS BECOMES ASSESSABLE

For the cardiologist, a hypothesis becomes truly assessable when it names the people concerned, the intervention, its timing of introduction, the comparator and the expected outcome. "Reducing cardiac inflammation" remains too indeterminate: a study must choose whether it targets a manifestation, a lesion, a function or an event relevant to the patient. These objectives may be linked, but their concordance cannot be presumed. Timing matters as much as the target. In terms of reasoning, an intervention may succeed in preventing a process without managing to repair its already established consequences. Conversely, the failure of late administration does not prove that the pathway never contributed to the disease. To interpret a negative result, the failure of the strategy tested must be distinguished from the refutation of any biological involvement. Finally, a historical reading should not turn the review into a prophecy validated by the results that came after it. It is read to understand how a therapeutic question becomes plausible, then how that plausibility must be put to the test. Two requirements must thus be held at once: taking seriously

A REVIEW IS NOT A PROPHECY

a sufficiently supported mechanism, and awaiting an appropriate comparison before attributing a clinical benefit to it.

5.4.2.5 Integrative Clinical Reading

REASON TO PRESCRIBE, CONSEQUENCES TO FOLLOW

A so-called targeted intervention can be understood as an action limited to the single problem one wishes to treat. The distinction between molecular target and consequences in the organism invites a different presentation of a future decision: what benefit is sought, which other effects will need to be assessed and who will monitor them? The review provides a framework for this inquiry, without answering for any given person. Integration consists in holding within the same judgment the reason for prescribing and the full range of consequences to be evaluated.

5.4.3 Netea and the Conditions of an Innate Memory

5.4.3.1 Situating the Result

NETEA, 2016

The review by Netea and colleagues, published in 2016, brings together observations of innate responses modified after a first exposure. It distinguishes this trained immunity from immediate activation and from classical lymphocyte memory, examining epigenetic and metabolic reprogramming. Natural killer (NK) cells, however, complicate a simple opposition between specific and nonspecific responses. The authors discuss protective and potentially deleterious consequences, notably vascular ones. Some explanations remain proposed rather than demonstrated: the persistence of an effect despite the turnover of circulating cells thus raises the question of precursors. This review sets out a conceptual and experimental program; it is not a trial of a cardiovascular strategy.[3]

5.4.3.2 Critical Reading: separating a trace of the past from an ongoing exposure

AN EXPOSURE THAT CONTINUES

Saying that a system retains a memory means that its history modifies its current response. But a difference between two samples does not suffice to show this property. One must first ask whether the initial exposure has actually ceased. A persistent substance could sustain activation, giving the appearance of a memory while still acting. The observation interval, washouts and persistence controls therefore belong to the experimental definition of the phenomenon. Consider a fictional experiment: two cultures receive the same final stimulus, but only one was previously exposed to another agent. If their responses differ, several explanations remain possible. The cells may have changed program; some may have survived better; the composition of the culture may have shifted. Distinguishing these possibilities requires additional observations. The name given to the phenomenon must not prematurely absorb all the causes of the difference. A molecular trace must also not be confused with its function. A mark present after exposure is compatible with several roles: helping to maintain the state, being a consequence of it or simply accompanying it. Removing it could help to discriminate among these possibilities, but an intervention that simultaneously modifies several functions still requires controls. Moving from association to functional contribution remains

A MARK IS NOT A FUNCTION

a precise question, even when epigenetic vocabulary seems highly explanatory.

5.4.3.3 Immunological Reading: a memory is not an intention

TRAINING IS NOT IMPROVING

The term "training" spontaneously evokes improvement. This connotation becomes misleading if it turns a stronger response into a necessarily better one. One can conceive of a cell that better limits an infection while increasing tissue damage in another situation. These are not two incompatible verdicts: they bear on different consequences, in different contexts. The comparison of responses thus depends on the nature of the second challenge. An increased response to a given stimulus does not mean a uniform increase in all functions. To compare two experiments, one must know what was measured, at what time and in which cells. Without these coordinates, the word "memory" becomes a shortcut that conceals precisely what is being investigated. The question of cell turnover adds another difficulty. A property found in a late sample does not imply that the same cells retained it. It may have been transmitted, reinduced or rebuilt by the environment. These scenarios lead to different experiments. Dating the response at the level of the organism therefore does not suffice to date its retention within a particular cell.

5.4.3.4 Cardiological Reading: a biological past does not yet determine a treatment

FROM THE PAST TO THE VASCULAR RESPONSE

For cardiology, the value of the concept is to make a question examinable: how might a past exposure modify a later vascular response? This possibility does not automatically designate the intervention to undertake. It would be necessary to determine which state persists, in which compartment, whether it contributes to damage and whether modifying it improves a consequence that matters to the people concerned. A signature measured in blood could then become an object of inquiry, without directly representing the state of a plaque or the benefit of a treatment. Caution consists in maintaining this distance between sample and organ. It does not prohibit extrapolation; it indicates the observations that extrapolation still lacks. The methodological contribution therefore lies in a requirement for a verifiable account: past exposure, persistent state, new challenge, consequence. Each link must be open to questioning. A convincing biological story becomes scientific when it also specifies what would lead to abandoning its favored explanation.

A VERIFIABLE ACCOUNT, LINK BY LINK

5.4.3.5 Integrative Clinical Reading

THE PAST RECOUNTED, NO IMPOSED CAUSALITY

The concept of innate memory gives a reason to take an interest in the history of exposures, but it does not allow an episode to be retrospectively designated as the explanation of all present disorders. In the clinical interview, the account of the past can guide questions without becoming a causality imposed on the person. A shared chronology must keep reported events, documented traces and still hypothetical links distinct. This distinction notably avoids turning a persistent biological possibility into a foretold individual destiny.

5.4.4 Xia and the Pore as a Selection Device

5.4.4.1 Situating the Result

Xia, 2021

Xia and colleagues bring together gasdermin D structures, transport in liposomes and experiments on murine macrophages. Their results support electrostatic filtering: the pore more strongly hinders certain acidic cargoes and favors the release of the mature forms of interleukin 1β (IL-1β) and interleukin 18 (IL-18) relative to their precursors. The secretion studied can persist without massive cell rupture. This is an experimental explanation of transport, not a demonstration of cardiac protection or damage.[4]

5.4.4.2 Critical Reading: a structure also results from experimental choices

AN IMAGE FROM CHOSEN PREPARATIONS

Seeing a pore seems to give access to the object itself. Yet the reconstruction rests on selected preparations and on forms modified for structural analysis. The form retained does not have an activity strictly identical to that of the comparator, and several architectures were observed. These discrepancies limit the generalization of the image; they must be checked against functional measurements rather than treated as details external to the result. The critical question then becomes: which properties does the preparation preserve well enough for the interpretation to be valid? Demanding complete identity with every natural situation would be unrealistic. Demanding no comparison would be insufficient. The examination must bear on the properties that support the argument: pore formation, structural organization and functional behavior. A sharper image increases the precision of certain descriptions; it does not automatically demonstrate that the chosen object represents all possible states. A reconstruction must also be distinguished from a film. Comparing two states can make a transition plausible without recording each of its steps. An animation based on these states would remain a representation of the link between them, to be evaluated by appropriate experiments. The beauty of a computed movement does not turn its trajectory into a continuous observation.

A RECONSTRUCTION IS NOT A FILM

5.4.4.3 Immunological Reading: an open membrane does not let everything out in the same way

ALTERING CHARGES, OBSERVING PASSAGE

The strength of the reasoning lies in confronting shape, physical properties and functional perturbations. The authors modify charges of the pore or of the transported cargo, then examine passage. This reciprocal comparison gives the interpretation more support than a mere resemblance between colors on an electrostatic potential map. Critical analysis can follow this principle: if a property explains transport, modifying it should change transport without indiscriminately destroying the whole device. The presence of a passage and its preferences must therefore be examined separately. A mutation that abolished all pores would not test the same proposition as a mutation that preserved their formation but altered what crosses them. The quality of the control depends on this difference between the two propositions tested. The word “filter” nevertheless calls for discipline. It sometimes evokes a boundary that permits or forbids absolutely. Here, it rather invites a comparison of rates and quantities. A preference for passage is not a universal exclusion. Likewise, describing a

A PREFERENCE, NOT AN EXCLUSION

SECRETION OR CELL RUPTURE

cargo as mature indicates its state; it does not mean that its release is always advantageous for the organism. The distinction between secretion and cell rupture also changes the interpretation. Measuring what appears outside the cell requires knowing by which route it could have gotten there. Without a control of cell integrity, the same assay could confuse organized transport with the dispersal of contents after destruction. This requirement links a biochemical measurement to the state of the cell that produces it.

5.4.4.4 Cardiological Reading: turning an explanation into a testable question

PRODUCTION, MATURATION, PASSAGE

The value for cardioimmunology is to clarify what an intervention on a release mechanism might change. Blocking the production of a cytokine, its maturation or its passage across the membrane does not pose the same experimental question. Before grouping these actions under a single anti-inflammatory label, their specific effects and the compartments involved would need to be defined. This work therefore supplies comparisons to be designed, without determining which would be desirable in the heart. The relevant cells, the timing of intervention and the tissue consequences would remain to be identified, followed by an assessment of the clinical balance. The mechanistic result is positive and precise: certain properties of the device contribute to the observed transport.

5.4.4.5 Integrative Clinical Reading

THE SHARPNESS OF A MOLECULAR IMAGE

The sharpness of a molecular image can give an impression of therapeutic proximity. For a person awaiting a solution, explaining this discovery requires saying what it makes more understandable: passage across a membrane, not yet an improvement in daily life. This precision preserves the value of the research without manufacturing an immediate clinical expectation. The integrative question remains open: what perceptible outcome, or one important to the person, would one day justify wanting to modify this transport?

5.4.5 Swanson and the Steps of the Mechanism

5.4.5.1 Situating the Result

SWANSON, 2019

This 2019 review describes the assembly and regulation of the NLRP3 inflammasome, distinguishing in particular priming and activation, without making the two-step model a universal rule. The authors weigh several explanations of the events preceding assembly: ion fluxes, organelle disturbances and metabolic changes. They point out contradictions and do not adopt a unifying upstream model. Their pharmacological perspectives rest on this mechanistic synthesis; the review is not a comparative trial of a cardiac strategy. It should be read within its historical horizon.[5]

5.4.5.2 Critical Reading: breaking down what is called an inhibition

PRIMING OR ACTIVATION

The name of the device can lend a misleading unity to different operations. Reducing the quantity of a component, preventing its assembly and reduc-

ing a measured consequence can lead to the same final observation without testing the same proposition. The review provides a precise example: certain inhibitors used to study reactive oxygen species may act on priming rather than on activation. Interpreting their result therefore requires locating the intervention within the sequence, not merely noting that the terminal signal falls.

CHOOSING MEASURES THAT DISCRIMINATE

Construct a fictional experiment. Two interventions reduce a released product equally. The first reduces the availability of its precursor; the second leaves this availability unchanged but blocks a transformation step. A single final assay does not discriminate between these stories. Measuring the precursor and shifting the timing of intervention can make the predictions diverge. The point is not to accumulate measurements: it is to choose those that separate the competing explanations. It will be objected that this breakdown destroys the practical value of a global result. It preserves that value, but assigns it a role. The global result can select an experimental lead; intermediate comparisons make it possible to understand why it works and under which conditions its success should be expected. The object explained remains shared, while the reasons for the explanation become more precise.

5.4.5.3 Immunological Reading: a common device can have several entry routes

UNEQUAL SUPPORT ACROSS LEVELS

The divergences reported in the review do not mean that the existence of the device and all the relations leading to it are equally uncertain. The support available may differ for its composition, for an assembly step and for the path followed from a particular stimulus. Conflating these levels would lead either to presenting the whole map as settled, or to rejecting the whole because one pathway remains disputed. The reverse objection must also be examined: if each discordant result receives its own pathway, does the model become impossible to contest? Plurality must yield distinct expectations. In the fictional scenario, a pathway supposed to bypass the blocked step should preserve a response under conditions where that step has actually been neutralized. A pathway merely renamed after a failure would provide no such anticipation. The mechanism gains content when its variants indicate when they should diverge.

THE STATUS OF AN ARROW

This separation of levels prepares a careful use of diagrams. An arrow can condense an experimental comparison, several compatible studies or a relation that is still proposed. This status must be retrievable from the text. The precision of the drawing does not replace this distinction; it can, however, help in designing the next experiment.

5.4.5.4 Cardiological Reading: choosing the step to be modified

ONE TARGET, SEVERAL OBJECTIVES

For cardioimmunology, this breakdown turns a general intention into a verifiable project. Is the aim to reduce the emergence of a signal, limit a response to an established lesion or modify a late consequence? The same target name does not make these objectives equivalent. Nor does it imply that all of them require the same exposure or the same timetable. A useful comparison would then specify the step targeted, the control showing that it was modified and the tissue or functional outcome sought. If the step changes without improvement in that outcome, something will have been learned about their relation in this context. If the outcome improves, attributing it to that step will require the planned comparisons. The positive gain is a program that distinguishes

success of the molecular intervention, explanation of its effect and value for the organ.

5.4.5.5 Integrative Clinical Reading

THE GOAL OF CARE BEFORE THE TARGET

Breaking down a biological pathway also teaches how to specify the goal of care: is it a current manifestation, a future complication or recovery? These objectives may matter differently to the same person. The Swanson review does not allow a choice among them, but it prevents them from being merged under the single intention of reducing inflammation. Clinical dialogue will need to name the objective sought before explaining what modifying a molecular step might contribute to it; a precise target does not yet define a personal priority.

5.4.6 Schiopu and Cotoi: Situating a Danger Signal

5.4.6.1 Situating the Result

SCHIOPU AND COTOI, 2013

In 2013, Schiopu and Cotoi published a review of S100A8 and S100A9, proteins notably combined in calprotectin. They bring together experimental research, cardiovascular risk factors and clinical associations. Their table distinguishes murine and human observations; their diagram links bone marrow, blood and the vessel wall. The text retains discordant results and context-dependent functions. It examines two perspectives, biomarker and therapeutic target, without demonstrating a human cardiovascular benefit from their use. This synthesis brings together results of different kinds; it does not provide them with independent confirmation.[6]

5.4.6.2 Critical Reading

STRENGTH BORROWED FROM NEIGHBORS

A synthesis can make an explanation more intelligible without adding a new population or a new experiment. The question then becomes one of assembly: is each step supported by an appropriate observation, or does the continuity of the narrative lend it strength borrowed from its neighbors? Several results can be accepted while the conclusion that unites them is contested. The diagram provides a good exercise. For each arrow, ask which proposition would be lost if it were removed, then which observation justifies keeping it. The same graphical appearance does not guarantee the same evidential strength. A human association, an experimental perturbation and a hypothesis about the link between them can be useful together, provided their provenance is kept. The legibility of a figure should facilitate this distinction. Finally, the length of a bibliography does not by itself measure the comprehensiveness of a search. To judge a synthesis, one must also ask how unfavorable results could have entered it. Their explicit presence is informative; it does not establish that all of them were found. This question concerns the robustness of the argument, without requiring a narrative review to silently fulfill the functions of a systematic review.

THE LENGTH OF A BIBLIOGRAPHY

5.4.6.3 Immunological Reading

what a "danger signal" names

The expression "danger signal" calls for a distinction between origin, action and interpretation. A molecule originating from the organism can take part in a response without its every presence signifying a current threat. Danger is not a property that the name suffices to measure. One must specify which relation between the signal and the system studied licenses this qualifier. This reasoning can be tested with a thought experiment. Two samples show the same concentration of a signal but come from different situations. Can the same activity be attributed to them? This equality does not yet describe the site of action, the possible responses or the history that produced the measurement. It constitutes real information, whose interpretation remains to be constructed. No new biological result is assumed in this example. The opposite difficulty would be to make every result incomparable in the name of context. To avoid it, the decisive differences must be named and subjected to scrutiny. A contextual explanation becomes scientific when it specifies what could vary and what should remain comparable. It gains nothing by invoking, after each disagreement, a complexity that cannot be discussed.

equal concentration, meaning to build

5.4.6.4 Cardiological Reading

marker or target: two projects

Recognizing a marker and choosing a target are two projects. For the first, the question may be: what information does the assay add to an already defined decision? For the second: what would an intervention change, with what benefits and what harms? A good answer to one does not remove the need to answer the other. Suppose a signal improves the prediction of an event but modifies no available decision. Its scientific interest could be real, while its immediate usefulness would remain limited. Suppose, conversely, that an intervention improves an outcome without the assay making it possible to select its recipients: its benefit would not automatically validate a selection strategy. These fictional situations show why the targeted decision must be kept in the conclusion. This historical review thus invites the next question to be written precisely. Describing a signal more fully, better predicting an outcome and improving that outcome are three possible advances. Their succession in a research program does not ensure that they follow from one another in reality.

5.4.6.5 Integrative Clinical Reading

the word "inflammatory" for the patient

A patient may receive the word "inflammatory" as a general explanation of scattered symptoms, history and test results. This unity can aid understanding; it can also make problems that require different answers disappear. An integrative reading must be able to bring a record together while preserving the particular decisions it involves. Before giving a place to an additional assay, one should therefore ask what the person and their caregivers would do with it: which uncertainty would be reduced, which decision would become possible, which new burden would appear? Life priorities and coexisting diseases give meaning to these questions; they do not prove the usefulness of the assay. The gain sought is better-justified coordination, whose consequences remain to be assessed, rather than a common biological vocabulary offered as an answer to everything.

5.5 Worked Exercises

5.5.1 Exercise on Biological Effects and Clinical Outcomes

5.5.1.1 Problem

AN INHIBITOR X IN COCULTURE

All the data below are fictional. They represent neither a published trial nor an available treatment. Their sole function is pedagogical. A team conceives an inhibitor X of an inflammatory pathway. In a first experiment, cocultured cardiac and immune cells are subjected to a standardized insult. Twelve independent preparations are randomly allocated to X or vehicle. At twenty-four hours, the mean concentration of a mediator is 100 units with X versus 200 with vehicle; the proportion of dead cells is 15% versus 25%. No whole organism is studied.

THE FICTIONAL TRIAL AND ITS TWO RISKS

A second, independent study randomly allocates 200 patients with the same cardiac syndrome to X or placebo, added to usual care, with 100 patients per group. Follow-up is complete at six months. A prespecified cardiac event occurs in 8 patients with X and 12 with placebo. A serious infection affects 6 patients with X and 3 with placebo. The events may involve the same people. No confidence interval or test result is provided in this problem. Questions. Can the pathway be asserted to be globally harmful? What does the cellular experiment prove? What absolute differences are observed in the fictional trial? Is it justified to subtract the additional infections from the cardiac events avoided? What conclusion would be proportionate?

5.5.1.2 Reasoned Answer

PARALLEL EFFECTS, MEDIATION UNPROVEN

Under the conditions described, the cellular experiment shows results compatible with an effect of X on the mediator and on cell death. It does not demonstrate that the decrease in the mediator causes the decrease in cell death. These two effects could depend on distinct actions of X. Additional experiments would be needed to establish this mediation. The independent preparations are the experimental units here: multiplying measurements within a single preparation would not add as many independent units. In the fictional trial, the observed cardiac risks are 8% and 12%, a difference of −4 percentage points for X minus placebo. For serious infection, the risks are 6% and 3%, or +3 points. These values describe the observations; without their uncertainty, it would be unjustified to announce an established benefit or a demonstrated excess risk. Random allocation supports the comparison of strategies, provided the trial was properly conducted. It does not guarantee that each numerical difference escapes sampling fluctuation. Complete follow-up removes one difficulty here, but it does not suffice to address every dimension of validity: event definitions, measurement procedures, blinding and analysis would still require examination.

AN INADEQUATE SUBTRACTION

The calculation “four events avoided minus three additional infections equals a net benefit” is inadequate. The events do not necessarily have the same severity, may accumulate in one person and do not cover all relevant effects. A benefit-risk assessment requires an explicit definition of the outcomes and of their importance, as well as an account of uncertainty.

A PROPORTIONATE CONCLUSION

The proportionate conclusion would be: “The observations are exploratory signals to be evaluated together with their uncertainty; they allow neither describing this pathway as globally harmful nor recom-

mending the intervention." The next step is not to look for a more assertive slogan, but to explain what additional information would change the decision.

Synthesis

An inflammatory label determines neither the value of a response nor the therapeutic course of action. The genetic investigation and the mechanistic synthesis illuminate different questions; bringing them together becomes useful when the nature of each argument is preserved. A biological contribution, an effect on a measurement and a benefit for a person require linked but distinct demonstrations.

Open Questions

What observation would lead to abandoning the idea that a response is protective in a given situation? How can the role of a pathway in the onset of a disease be separated from its role in maintaining it? Under what conditions could an intervention on the same target have different effects depending on timing and tissue? These questions will guide the studies of the heart as an immune milieu.

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6

The Heart as an Immune Milieu

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The myocardium harbors macrophages, in mice as in human hearts without known heart disease. In mice, they are found even in the vicinity of the cells that conduct the electrical impulse, and some are established there before birth; others may derive from circulating monocytes. Lymphocytes capable of recognizing a cardiac muscle protein do not necessarily cause disease. These findings describe an inhabited organ. They do not yet say what these cells do there, how long they have resided there or through which relations they act. A map of hundreds of thousands of profiles, a molecule deleted in mice, a transplanted human heart: each setup answers its own question. The difficulty lies in the transitions, from presence to contribution and from mouse to person.

6.1 Naming a Milieu and the Cells That Inhabit It

6.1.1 An Organ Is Not Reducible to the List of Its Inhabitants

6.1.1.1 What the word "milieu" commits to

INVENTORY OR RELATIONS

Considering the heart as an immune milieu raises a question about the unity of the object studied. Is the task to study cells present in an organ, or relations for which this organ constitutes a condition? The first question calls for an inventory. The second asks how positions, exchanges and constraints make certain activities possible. An inventory can prepare this inquiry; it does not accomplish it on its own. The word "milieu" takes on a precise methodological sense here: a set of conditions and relations relevant to an activity under study. It designates neither everything that surrounds a cell nor an additional substance that would mysteriously explain its properties. It requires selecting relations and justifying that selection. Depending on the question, a distance, a possibility of contact or a condition of access may count for more than the total number of elements present. Construct an entirely fictional example. Two preparations contain the same numbers of cells A and B. In the first, the A cells lie in immediate proximity to the B cells; in the second, a separation prevents certain exchanges. The preparations yield different results. This scenario shows why an identical composition does not guarantee an equivalent organization. It does not yet prove any mechanism: the separation may alter several conditions, which will have to be disentangled. The argument concerns the question to be asked, not a particular cardiac fact.

A PRECISE SENSE OF THE WORD "MILIEU"

6.1.1.2 Neither the elements alone nor the whole alone

KNOWING A AND B IS NOT ENOUGH

An objection seems natural: since every relation is a relation between elements, is it not enough to know each of them completely? This reply assumes precisely what would need to be established: that the knowledge sought can be obtained independently of their arrangement. Properties of A and of B can be known without knowing the effect of their encounter. Studying the relation does not add an immaterial force to the cells; it adds a comparison absent from their separate examination. The reverse objection guards against another excess. Asserting that only the whole matters risks making any local experiment useless. Yet an analysis of components remains necessary to understand which relations can produce the result. The relational approach therefore does not abolish experimental decomposition. It asks for an explicit statement of what decomposition preserves and what a reconstruction will have to restore. The heart does not become any less an organ when its relations with other compartments are examined. Its anatomical boundary delimits a place; it does not automatically close the explanation. An activity observed in that place may

THE ANATOMICAL BOUNDARY OF EXPLANATION

depend on external conditions. The fruitful question is not to choose between "everything is local" and "everything comes from elsewhere," but to determine which local modification remains explicable once these conditions are taken into account.

6.1.2 Identity, State and Provenance: Three Questions about a Cell

6.1.2.1 Recognizing a cell without confusing signal and entity

RECOGNIZE, DESCRIBE, TRACE

Naming a cell population seems to provide a stable answer. Yet the name can combine different kinds of information: characteristics used to recognize the cells, an activity measured at sampling or a history of provenance. To read studies of the cardiac milieu, these three commitments must be distinguished. Recognizing a cell, describing its state and reconstructing its history are not equivalent operations. Suppose, again fictionally, that a population is defined by markers P and Q. After a perturbation, some cells express less Q. Should one conclude that they have disappeared? That would be possible if other observations established it. But the result could also reflect a change in the characteristic used to count them. An operational definition must therefore be distinguished from the object it makes it possible to detect: losing the identifying signal does not necessarily demonstrate loss of the entity. The alternative is not resolved by declaring all markers misleading. It leads to a search for complementary information that does not repeat exactly the same dependence. A second classification based on Q under another name would not provide independent verification. A measurement bearing on a distinct property could help more, provided its own limits are known. The useful kind of independence here concerns the way an error would arise.

LOSING A SIGNAL, LOSING A CELL

6.1.2.2 Provenance requires a history

ORIGIN AND ACTIVITY

Provenance adds another difficulty. Two cells may share observed characteristics without having followed the same path. Conversely, cells of the same provenance may present different states. Origin and activity are not opposed; they answer distinct questions. One can help explain the other, but this relation must not be silently built into the name of the category. The words "resident" and "recruited" will thus require a definition in each study. "Present at sampling" is not synonymous with "present for a long time." Length of residence, turnover and provenance require observations capable of informing a history. A highly detailed snapshot does not become a follow-up through the precision of its measurements alone. Likewise, a reconstructed history does not dispense with measuring current activity.

"RESIDENT" AND "RECRUITED"

6.1.2.3 What a category must make possible

It might be objected that multiplying distinctions prevents any usable classification. That objection confuses stability of usage with simplicity of the object. A category can be excellent for a limited task: identifying a population, comparing samples or formulating a hypothesis. It need not contain the whole history and all the functions of its members. The requirement at issue concerns the scope of the name, not a prohibition on naming. A classification becomes particularly instructive when it makes it possible to predict a difference that was not used to construct it. If groups are separated according to P, finding afterward that they differ according to P does not suffice to demonstrate their explanatory value. One must examine what this partition helps to understand beyond its own definition, then determine in which situations it ceases to help.

PREDICTING BEYOND THE CLASSIFICATION

6.2 From Map to Perturbation

6.2.1 What a Map Shows and What a Model Adds

6.2.1.1 What a represented distance means

SEVERAL SPACES ON ONE MAP

A tissue map distributes elements in a space of representation. It can locate anatomical positions, group similar profiles or delimit clusters. These operations do not necessarily describe the same space. Two points close together on a plot may have similar measurements without coming from neighboring cells in the tissue. Conversely, anatomical proximity does not suffice to establish a functional interaction. This distinction is decisive for reading. Faced with a figure, the first task is to identify what the represented distance means. Is it a distance in the organ, a similarity between variables or an arrangement chosen to make the data legible? The figure is not misleading because it transforms measurements. It becomes so if interpretation attributes to this transformation a meaning that has not been established.

WHAT A DISTANCE MEANS

6.2.1.2 Describing, then discriminating between models

PROXIMITY, ACTION, COMMON CAUSE

Imagine a fictional map in which A cells lie close to B cells in a region showing high activity. A hypothesis proposes that A controls the activity of B. The map makes this hypothesis conceivable; it may also make certain other localizations less plausible. But it does not, on its own, discriminate between an action of A on B, the reverse action or the common influence of a third condition. The model emerges when a particular organization of these relations is chosen. The experiment therefore does not simply "confirm the image." It must produce a difference that separates models compatible with that image. If two interpretations predict exactly the same result in the chosen setup, obtaining that result will not discriminate between them. This limit persists even with a sharper image or more measurements. The obstacle then lies in the comparison, not only in precision. One objection deserves to be welcomed: a descriptive exploration can discover what no one had thought to look for. It would be absurd to require every inventory to begin with a complete mechanistic hypothesis. Description has value of its own, notably when it reveals regularities, exceptions or overlooked heterogeneities. It becomes the starting point of an explanation once the tests these observations invite are formulated.

DISCRIMINATING, NOT CONFIRMING THE IMAGE

6.2.1.3 Preparing the object and comparing methods

WHAT THE PREPARATION LOSES

The preparation of the mapped object must also be questioned. A method can preserve certain properties and lose others. Asking what is preserved is not denouncing an artifact: every observation requires a setup. The difficulty lies in knowing whether the lost property plays a part in the proposed explanation. A sample may suit an inventory of characteristics while no longer allowing their initial spatial organization to be observed directly. Comparing methods is then more fruitful than an abstract ranking of the best instrument. It makes it possible to look for descriptions that remain compatible when observation conditions change. A disagreement can reveal a technical limit, but also a difference between the objects actually measured. It should lead to reconstructing the comparison before being turned into a rivalry between teams.

6.2.2 Perturbing a Population Does Not Suffice to Isolate Its Role

6.2.2.1 The announced modification and the modification achieved

what "acting on A" means

To move from the map to a functional proposition, one can try to modify a population and observe the consequences. This approach provides experimental leverage on the system. Its strength, however, depends on the relation between the announced modification and the one actually achieved. "Acting on A" can mean removing A, modifying its activity, preventing an interaction or changing a condition on which A depends. These interventions do not answer the same question. In a fictional experiment, a tool reduces population A and modifies a cardiac measure Y. The result could support a contribution of A, but several checks are needed. Does the tool affect other populations? Does it have an effect on Y independently of A? Does the reduction of A lead to intermediate reorganizations? These questions do not cancel the result; they specify the causal interpretation it can bear.

a tool and its other effects

6.2.2.2 Specificity of tools and agreement between tools

the senses of the word "specific"

Specificity has several senses here that must not be confused. A tool can recognize a precise molecular target without that target being exclusive to one population. It can modify a particular population without its consequences remaining confined to the site studied. Finally, it can leave cell numbers intact while changing an activity important for the comparison. Saying "specific" without naming the property concerned postpones the problem instead of solving it. One objection would be to demand a perfect tool before any conclusion. That would make research impracticable. The reasonable requirement is to examine the most relevant competing interpretations, to measure what can be verified and to proportion the conclusion. Several tools whose plausible defects differ can strengthen an argument. Their agreement becomes less instructive if they all share the same uncontrolled action.

6.2.2.3 Restoring a quantity is not restoring an organization

restoring a number, not positions

A restoration experiment can add a valuable comparison: after perturbing the system, one attempts to reestablish the component presumed responsible. But restoring a quantity does not necessarily mean restoring an organization. In this scenario, reintroducing as many A cells would not, by definition, ensure that they regain the same positions or the same relations. A failed restoration would therefore remain ambiguous if these conditions had not been examined.

a possible route back

Conversely, a successful restoration would not demonstrate that the experimental system reproduces the entire initial history. It would show that a route back is possible in this setup. It is important to retain this gain in knowledge without demanding from it a complete identity between the two situations. The comparison can establish a robust contribution while leaving certain steps open. The relational milieu thus becomes a precise experimental object. It is no longer enough to attribute a property to a cell: the task is to find which relation must be present for the modification to produce the effect. This precision makes it possible to distinguish a local responsibility, a dependence at a distance and a reorganization of the whole.

6.3 Extrapolating and Revising an Organization

6.3.1 What Licenses a Comparison between Mice and Humans

6.3.1.1 What a shared name carries over

WHAT IS CARRIED ACROSS SPECIES

Comparing a murine model and a human observation requires specifying what one wants to carry over from one to the other. It may be a population name, a spatial organization, a functional relation or a response to a perturbation. These equivalences are not bound together by principle. Establishing one does not dispense with examining the others. In a teaching example, mouse cells and human cells share several recognition characteristics. The team gives them a common name. This decision can facilitate comparisons, but it does not demonstrate that these cells contribute identically to all the activities of the organ. A shared vocabulary is a hypothesis of correspondence whose domain must be specified. The reverse objection would be to conclude that differences between species make any comparison empty. Yet knowing a relation in one system can make it possible to pose a better-defined question in another. The model need not fully reproduce the human being to be informative. It must preserve, or allow interrogation of, the properties on which the inference sought depends.

A SHARED NAME, A HYPOTHESIS

6.3.1.2 Testing a relation from one species to another

PRESENCE, ORGANIZATION, EFFECT

Suppose the fictional study attributes to a proximity between A and B a role in Y in mice. To discuss its human relevance, searching only for A and B would be a first step. One would also need to examine whether the invoked relation can be observed, then whether the other conditions necessary to the hypothesis are sufficiently comparable. This distinguishes the presence of the elements, the proposed organization and the functional effect. Disagreement between species deserves the same rigor as agreement. It could lead to limiting the model, but could also reflect a difference in sampling, measurement or the situation studied. Immediately attributing the divergence to "the species" confuses a label with an explanation. One must look for the difference that could produce the divergence and design a comparison that gives it a verifiable scope. The teaching strength of the model ultimately lies in this double possibility: suggesting a relation and making its limit visible. An experiment that determines why a correspondence fails can be more instructive than a spectacular resemblance whose conditions remain obscure. Extrapolation then becomes an inquiry in its own right, and not the last optimistic sentence of the commentary.

A DISAGREEMENT BETWEEN SPECIES

WHY A CORRESPONDENCE FAILS

6.3.2 Making Organization a Revisable Proposition

6.3.2.1 Revising categories and relations

A PERSPECTIVE TO BE MADE REVISABLE

Saying that the heart forms an immune milieu will remain a general perspective as long as the relations to which it is applied are not specified. To become a scientific proposition, this perspective must accept limited commitments: which elements, which arrangement, which activity and which expected modifications? Its fruitfulness is measured by the better-posed questions and the comparisons it makes possible. A first ground for revision concerns categories. If a population announced as homogeneous divides into groups whose contributions differ, the relevant unit will have to be reconsidered. This will not mean that the previous classification was devoid of all value. It may have suited a more general description. Revision becomes necessary when this generality conceals precisely the difference one seeks to explain. A second ground concerns relations. A regularly observed proximity can lose its explanatory value

FIRST GROUND: CATEGORIES

SECOND GROUND: RELATIONS

if the relevant perturbation leaves the predicted result intact, under conditions that genuinely test the hypothesis. The relation must then be discussed, rather than indefinitely keeping the same arrow while adding exceptions. A contrary result, however, settles only the proposition that the setup was capable of testing.

6.3.2.2 Expanding a model is not populating it

THIRD GROUND: THE BOUNDARY

A third ground concerns the boundary of the system. A so-called local explanation may have to incorporate an external condition previously omitted. This extension is progress if it brings discriminating observations. It becomes a step backward if "the whole organism" serves as an indefinite answer to every difficulty. Expanding the model must increase its verifiable content, and not merely the number of elements it mentions. These criteria protect against a particular temptation: believing that a more densely populated representation is necessarily a better explanation. A highly detailed network can juxtapose associations without determining the decisive relations. Conversely, a simple model can illuminate a precise comparison without claiming to describe the whole heart. The question is not how many components appear, but what their organization actually helps to understand.

6.4 Texts Put to the Test

6.4.1 Bajpai and the Macrophages of the Human Heart

6.4.1.1 Situating the Result

Bajpai, 2018

The study examines diseased human myocardium, notably in advanced heart failure or after transplantation, distinguishes compartments according to CCR2 and uses, in particular, sex-mismatched transplants to study their maintenance. It also associates macrophage composition with outcome after ventricular assist device support. The authors specify that their design does not demonstrate the embryonic origin of human macrophages.[1]

6.4.1.2 Critical Reading: human tissue does not abolish the conditions of the experiment

BEFORE THE GRAFT, BEFORE BIRTH

A transplant offers a particularly instructive situation for reasoning about provenance. Different characteristics of donor and recipient can serve as markers. However, the question thus made accessible remains situated within the history of the graft. It does not automatically reach back to the donor's embryonic development. A temporal confusion must be resisted: a cell originating from the donor is not necessarily a cell present since the donor's birth. Residence before a transplant and origin before birth designate two different intervals. This situation also calls for vigilance about context. Any exceptional human setup can reveal a mechanism that is hard to observe otherwise; its very exceptionality limits generalization. A property observed in transplanted tissue must be examined in light of treatments, elapsed time and the selection of people who remain observable. These conditions do not constitute a refutation. They define the populations and circumstances to which the result can reasonably be referred. The number of cells analyzed deserves as care-

CONDITIONS OF A TRANSPLANTED TISSUE

MANY CELLS, FEW HEARTS

ful a reading as the number of patients. If several measurements come from the same heart, they share a biological history. Counting them as so many independent hearts would exaggerate the amount of information on variability between people. The analysis can exploit this richness without confusing it with replication: summarizing by patient, using a hierarchical structure or explicitly displaying the sampling levels. Before discussing a probability value, one must understand what was repeated.

6.4.1.3 Immunological Reading: a marker opens an inquiry but does not settle an identity

CAPACITY AND CONTRIBUTION

The distinction proposed here separates capacity from contribution in any functional experiment. An isolated cell may be capable of producing a mediator without being its main source in the organism. Conversely, a cell less active per unit can contribute substantially if it is abundant, persistently stimulated or placed in contact with a sensitive target. The behavior of a cell and the effect of a compartment cannot be deduced from each other without information on number, space and time. Interpreting molecular profiles calls for comparable caution. A signature compatible with repair can motivate an experiment on repair. It does not replace the demonstration that the tissue recovers thanks to these cells. The biological terms associated with gene sets are instruments of interpretation. They are neither clinical criteria nor autonomous proof of benefit.

6.4.1.4 Cardiological Reading: an association can orient without yet predicting or treating

THREE DIRECTIONS OF AN ASSOCIATION

The value of a relation between cellular composition and cardiac recovery is to bring out testable hypotheses. Yet three directions remain possible: the composition participates in recovery, it reflects an injury that also determines recovery or the evolution of the tissue in turn modifies the composition. These directions may coexist. A measurement made before the outcome studied helps to order events, but does not eliminate common causes. A late measurement is particularly exposed to a reversed reading of the causal direction. To evaluate the prognostic scope of the association, one would need to determine what it adds to the variables already available and to verify its performance in other populations. To make it a useful clinical tool, one would further need to specify the decision it can improve and the balance of that use. Access to sampling must also be considered: information obtained during a major procedure does not have the same use as a marker repeatable in the clinic. Biological relevance and practical utility are two complementary evaluations. Turning this same association into a treatment would require a different demonstration. Reducing a population associated with poor outcome does not guarantee improving that outcome. The population may be a bystander, a compensatory response or an actor some of whose functions remain necessary. Selectivity, timing of intervention and effects outside the heart then become central questions. An attractive strategy must be able to fail its tests without its intended claim being redefined after the fact.

PROGNOSIS, UTILITY, ACCESS TO SAMPLING

REDUCING AN ASSOCIATED POPULATION

6.4.1.5 Four Conclusions

Established: within the scope examined, a provenance identified around a transplantation does not, on its own, allow an embryonic origin to be attributed. Plausible: a more precise cellular characterization could help distinguish recovery trajectories that had until now been grouped together. Not concluded: a ready-to-use prognostic classification, a complete clinical causality or the benefit of an intervention targeting a given compartment. What would change the judgment: validation in independent populations, correctly timed longitudinal measurements and perturbations that discriminate the role of the compartment; a clinical evaluation would be needed to judge the benefit of a strategy targeting it.

6.4.1.6 Integrative Clinical Reading

AN EXCEPTIONAL CARE PATHWAY

Knowledge obtained from transplanted or mechanically supported hearts also bears the mark of an exceptional care pathway. For the person and those they consult, the cellular association must not become an individual prognosis detached from this selection. Integration consists in asking how new information would fit with the goals already pursued, rather than assigning it the whole decision. Understanding a biological recovery, estimating the future and choosing a step in the care pathway remain tasks to be coordinated, even when they draw on the same sample.

6.4.2 Hulsmans and the Contribution of Macrophages to Conduction

6.4.2.1 Situating the Result

HULSMANS, 2017

The study combines a description of human tissue with murine experiments. Macrophages are observed in the vicinity of the conducting cells of the atrioventricular node. In mice, the authors combine coculture, electrophysiology, photostimulation, conditional deletion of connexin 43 and macrophage depletion. Together, these support a macrophage contribution to nodal conduction under the conditions studied. The acceleration of repolarization comes from modeling. The human component rests on six autopsies, without known conduction disease but with serious conditions: it does not demonstrate a functional effect in humans. Animal sample sizes are chosen empirically; randomization is reported. Depletion produces greater disturbances than connexin 43 deletion. The article reports a patent application related to the modulation of conduction.[2]

6.4.2.2 Critical Reading: defining what each experiment allows to be asserted

DELETING A POPULATION OR A MOLECULE

Observed proximity does not suffice to establish a functional contribution. The value of the dossier therefore lies in the experimental perturbations that put this contribution to the test, with its murine scope explicitly maintained. This framework also helps in examining a spectacular result. If an intervention eliminates an entire population, it simultaneously modifies all its activities and can

transform its environment. If it deletes a molecule within that population, it poses a more circumscribed question. The difference between the two interventions is therefore not a mere detail of experimental power: the effects sought are not identical. All the damage from the first should not automatically be attributed to the mechanism targeted by the second. A selective restoration of the mechanism would constitute a complementary test, to be designed with its own controls. The same discipline applies to the statistical scale. For each figure, the chain animal-sample-preparation-cell-measurement must be reconstructed. Several cells from the same preparation share conditions that several independent animals do not fully share. Likewise, hundreds of beats recorded in one heart do not become hundreds of hearts. The amount of information increases with repeated measurements, but not in the same way as with new biological units. An appropriate analysis can use this richness; it must preserve the dependence structure.

ANIMAL, PREPARATION, CELL, MEASUREMENT

6.4.2.3 Immunological Reading: a cell's identity does not summarize its activity

MEMBERSHIP, LOCATION, ACTIVITY

The designation "immune cell" is useful for recognizing kinships and organizing knowledge. It should not function as a closed list of what this cell is permitted to do. The commentary proposes distinguishing its cellular membership, its location and its activity in a defined situation. These three descriptions can complement one another without merging. A nondefensive activity does not erase an immune identity; an immune identity does not turn every activity into inflammation. This shift prevents an error central to cardioimmunology: taking the detection of immune cells as a sufficient sign of disease. To conclude that there is a disturbance, one would need to compare a situation with a relevant reference, specify what changes, then link that change to a consequence. The question would no longer be only "are there macrophages?" but "which properties of which cells, in which territory, at what time, take part in which phenomenon?" This formulation makes the experimental program more demanding and more intelligible. It also invites careful use of the words "protection" and "activation." Describing an activity as protective presupposes a threat or damage avoided, a comparator and an outcome. Demonstrating a contribution to basal function answers another question. Likewise, increasing an activity does not guarantee an indefinite increase in its benefit. The model must specify the range within which the relation was tested; its shape beyond that range remains to be established.

IMMUNE PRESENCE AND DISEASE

6.4.2.4 Cardiological Reading: building the passage from mechanism to patient

FOUR DISTINCT CLINICAL OBJECTIVES

Cardiologists must first specify the clinical question they hope to clarify. Explaining a conduction delay, predicting its onset, preventing a complication and proposing a treatment are four distinct objectives. The same mechanism can contribute to the first without improving the other three. The scientific value of a discovery does not require that it immediately yield a prescription; it requires an honest statement of the type of understanding obtained. Territory matters as much as species. The phrase "in the heart" is sometimes too broad to guide a conclusion. A hypothesis concerning a conduction site requires validation at that site; its extension to other structures must be argued. Likewise, a result obtained under basal conditions does not, by itself, describe an estab-

TERRITORY AS MUCH AS SPECIES

lished disease. The clinical context can change the partners, the constraints and the meaning of an electrical variation. Extrapolation is built by specifying these differences, rather than by erasing their existence. The contribution to murine nodal conduction is therefore the result to retain; its diagnostic or therapeutic utility in humans remains to be established. The autopsy observations bring the anatomical territories closer, without providing that functional validation. A future discordance could restrict the mechanism to certain conditions rather than indiscriminately invalidating all the results.

6.4.2.5 Integrative Clinical Reading

two specialties, one organ

The discovery of an immune contribution to an electrical function can bring together two specialties that describe the same organ differently. It does not justify relabeling every conduction disorder as immune. For a person, the shared explanation must preserve the clinical observations and uncertainties instead of substituting a cellular story for the lived problem. The expected integrative gain is a better-formulated question between interlocutors: what could this mechanism explain, and which part of the situation still requires another evaluation?

6.4.3 Lv and Lipes and the Conditions of Tolerance to the Heart

6.4.3.1 Situating the Result

Lv and Lipes, 2012

The review by Lv and Lipes, published in 2012, proposes linking several forms of cardiac autoimmunity to insufficient central tolerance to α-myosin. It brings together, in particular, experiments from their own team and sets out a therapeutic perspective based on antigen-specific tolerance. It is a narrative synthesis committed to a hypothesis, not a new clinical comparison. Its value lies in making visible the passage from an experimental discovery to a broader explanatory framework.[3] To appraise this passage, the 2011 experimental study must also be revisited. In a murine model positive for human leukocyte antigen DQ8 (HLA-DQ8) on a nonobese diabetic (NOD) background, transgenic thymic expression of α-myosin protects against myocarditis during the observation period. The human component comprises three pediatric thymus samples and a comparison of blood responses in three people with heart disease and three controls without clinical heart disease. The authors describe very low, sometimes undetectable, human thymic expression of MYH6.[4]

the 2011 experimental study

6.4.3.2 Critical Reading: the scope of comparisons

6.4.3.2.1 What the thymus experiment establishes

two mentions, one experiment

A first question concerns the form of the reasoning. A review can bring together observations obtained in different contexts; it does not turn this juxtaposition into independent replication. Finding the same hypothesis in the experimental paper and then in its synthesis does not provide two confirmations. One must count experiments and their conditions, rather than occurrences of the conclusion. The intervention on the thymus is of particular logical interest. It shifts the question from the affected tissue to the conditions under which a response against that tissue becomes possible. However, preventing a lesion by modifying a condition early does not mean knowing how to

preventing is not treating

treat that lesion once established. Experimental prevention and clinical treatment differ in their timing, their objective and the population they address. Another distinction concerns the source of the protection. The study leaves open how much the tolerance obtained owes to deletion of reactive lymphocytes and how much to regulatory mechanisms. It therefore does not establish, from the protection result alone, that the entire intermediate chain has been identified.[4] A reading rule follows, extending what the text says: the causality of an intervention must be distinguished from the detailed explanation of its effect. An experiment can make the former credible while leaving several versions of the latter in competition. Even if the preventive effect of the manipulation is established, it does not suffice to show that a defect in central tolerance is a necessary or sufficient cause of all myocarditis. Modifying a condition can prevent a disease in a model without identifying all the conditions of its occurrence or their role in other forms of disease.

A CAUSE WITHOUT THE WHOLE CHAIN

6.4.3.2.2 The human component and its comparators

A SMALL HUMAN SAMPLE

The human component poses a different problem. A small group of people can produce a signal useful for opening a line of research. It does not allow precise estimation of the frequency of the mechanism in a population, nor of the diagnostic performance of a test. Multiplying assay wells from the same sample improves the assessment of certain technical variations; it does not multiply the number of independent people. The apparent precision of a laboratory result must therefore be related to its unit of observation. The question of the comparator is also decisive. "Without clinical heart disease" is not automatically equivalent to "without any other immune particularity." One of the three blood controls in the study has type 1 diabetes.[4] This detail does not make the comparison worthless. It prevents the actual composition of the groups from being hidden behind overly broad labels. To extend the work, comparison populations would need to be defined according to the question: heart disease, autoimmune background or a combination of both.

THE ACTUAL COMPOSITION OF CONTROLS

6.4.3.3 Immunological Reading: the scope of nondetection

6.4.3.3.1 The three levels of a nondetection

METHOD, MECHANISM, DISEASE

A three-level reading is proposed here for a negative measurement result. The first concerns the sample and the method: a signal was not identified under the conditions of the test. The second concerns the mechanism: the level actually present would be insufficient for a certain function. The third concerns the disease: this insufficiency would contribute to damage. These levels can be linked, but moving from one to the next requires an additional argument. A nondetection can constitute an argument against a biologically relevant presence if the sample and the method would have allowed it to be detected. Its scope therefore depends on the sensitivity of the test, the controls and what was sampled. It does not automatically amount to absence throughout the tissue or at all times. Conversely, invoking a signal inaccessible to measurement does not suffice to defend its functional role: an observation capable of testing that hypothesis must be proposed.

6.4.3.3.2 From disposition to pathological process

A DISPOSITION WITHOUT DISEASE

The review also acknowledges that the presence of autoreactive cells is not necessarily equivalent to disease, and discusses the role of peripheral tolerance mechanisms.[3] The conceptual stakes go beyond mere vocabulary. If a disposition can remain without consequence, it must not be described as a disease

already realized. One must look for what turns a possibility into a pathological process: conditions of activation, regulation, access to the tissue and persistence of the response then become questions to be tested. Adding a condition can refine the hypothesis or merely shield it. The difference lies in the new test that the condition makes possible: which situations should it distinguish, and by what independent observation? Even if prompted by an unexpected result, it remains fruitful if it exposes the explanation to a new comparison; invoked without verifiable consequence, it merely preserves the explanation.

REFINING OR SHIELDING THE HYPOTHESIS

6.4.3.4 Cardiological Reading: the passage toward an intervention

LOOKING OUTSIDE THE HEART

The shift of focus is fruitful: a cardiological question may require studying a site outside the heart. But widening the scope of the explanation does not allow the description of the patient, the disease and its course to be dropped. One must always specify which form of heart disease the hypothesis concerns and in which circumstances its examination could be useful. A therapeutic perspective requires a feasible intervention and an evaluation of its clinical and adverse effects. Showing that it engages the targeted mechanism provides a distinct argument, particularly important for the proposed interpretation. A biological action does not suffice to establish a benefit; conversely, a clinical benefit could be established while certain mediators remain unknown. Attributing that benefit to the announced mechanism would require additional evidence.

ANTIGEN, VARIABLE, FEASIBLE TARGET

The vocabulary of the target must remain precise. The biological target of an immune response, the variable manipulated in an experiment and the feasible target of a treatment are not necessarily the same object. Conflating them would suggest that identifying an antigen already provides a clinical strategy. Distinguishing them, on the contrary, makes it possible to construct the necessary comparisons.

6.4.3.5 What is retained and what would revise the judgment

WHAT THE JUDGMENT RETAINS

The judgment retains the value of reasoning that confronts a hypothesis with an intervention and then seeks human relevance. It maintains the difference between a mechanistic argument in a model, an exploratory observation in people and a demonstration of clinical utility. These statuses are neither interchangeable nor final.

WHAT WOULD SHIFT CONFIDENCE

Confidence would increase if independent results reproduced the relevant observations, if the conditions of the response were better delimited and if a prospective study distinguished a factor preceding the disease from a consequence of it. It should decrease or change if controls revealed another explanation of the signal, if the relation were not found in the announced populations or if a properly tested intervention contradicted the main prediction. A sound conclusion preserves both the reasons to continue and the means to be corrected.

6.4.3.6 Integrative Clinical Reading

AUTOREACTIVITY IS NOT ATTACK

Autoreactivity described in the laboratory may be heard as proof that an organism is necessarily attacking its own heart. The distinction between disposition and realized disease here protects the person's understanding. Integrative reasoning then seeks to connect immune observations, the cardiac description and the clinical history without arbitrarily privileging any one of them. In

this exploratory dossier, this linking opens questions; it defines neither family screening nor a preventive treatment program to be offered.

6.4.4 Epelman and the History of Cardiac Macrophages

6.4.4.1 Situating the Result

Epelman, 2014

A cell present in the myocardium may have been born there, have migrated there or descend from a cell that arrived long before. Its current address therefore does not tell its whole history. In mice, the authors combine parabiosis, proliferation labeling, lineage tracing and macrophage depletion. A large proportion of cardiac macrophages is established before birth and maintained locally. After depletion or stimulation with angiotensin II, circulating monocytes can also contribute to tissue populations. Embryonic tracing reveals a yolk sac contribution, without allowing the entire prenatal origin to be attributed to it. The experiments also distinguish several capacities for internalization, antigen presentation and inflammatory response.[5]

6.4.4.2 Critical Reading: observing a legacy requires several clocks

three clocks, three questions

A shared-circulation experiment answers a question about replacement over a given period. It does not suffice to date the birth of the cells that remain in place. A brief DNA synthesis label provides different information: it detects proliferative activity within the observed window. Genetic tracing introduces a third temporality, that of the past activation of the labeling system. The strength

a silent tracer

of this approach lies in articulating answers that are not interchangeable. A negative tracer must nevertheless be interpreted with caution. Absence of signal may mean that the cell did not pass through the lineage sought, but also that the system was insufficiently efficient at the relevant time. Automatically turning "unlabeled" into "derived from the yolk sac" would erase precisely the uncertainty the experiment seeks to resolve. This lesson extends beyond developmental biology: a silent instrument is informative only after its detec-

emptying a compartment

tion conditions have been examined. Perturbation imposes another precaution. Emptying a compartment can open a possibility of colonization that hardly existed while it was occupied. A result obtained after depletion shows what the system can rebuild in that situation; it does not necessarily describe its ordinary functioning. It would therefore be misleading to combine spontaneous maintenance and induced repopulation under a single proportion of "renewal."

6.4.4.3 Immunological Reading: origin, residence and function are three axes

residence, ancestry, activity

Vocabulary benefits from keeping these axes separate. "Resident" designates a relation to the tissue; "embryonic," a developmental history; "inflammatory," an activity or program assessed under certain conditions. Nothing requires these three properties to form categories that permanently coincide. To build a useful classification, one must be able to ask when a cell changes state without changing ancestry, and when a newly recruited cell acquires residence properties.

6.4.4.4 Cardiological Reading: choosing what one would want to preserve

injury, maintenance or repair

Before considering an intervention, the objective must be specified: limiting an initial injury, supporting tissue maintenance or enabling repair. A strategy that alters the abundance of a compartment must be judged on these functions and on their timing. The article thus teaches how to better formulate a target; it does not provide a prescription for depletion or for blocking recruitment in a patient.

6.4.4.5 Four Conclusions

Established: under the murine conditions studied, several mechanisms contribute to the formation and maintenance of the cardiac macrophage compartment. Plausible: distinguishing these mechanisms could help in designing more selective interventions. Not concluded: that an ancestry would determine an immutable function, or that a targeted strategy would already improve human prognosis. What would change the judgment: independent tracing systems with characterized performance would strengthen the conclusions on lineage; their therapeutic exploitation requires other experiments, linking a defined intervention to a cardiac benefit.

6.4.4.6 Integrative Clinical Reading

ancient origin is not fixity

The ancient origin of a cell population is not a complete clinical history. Even if a property dates back to development, its interpretation for a person would require considering the current context in which it operates. This study helps avoid confusing ancient origin with immutability. When the findings are communicated, explaining what is biologically inherited must therefore erase neither the events that have occurred since nor the possibilities for action still to be evaluated. The account of origins illuminates one dimension; it does not close the course.

6.4.5 Litviňuková and the Uses of a Cell Map

6.4.5.1 Situating the Result

Litviňuková, 2020

The precision of a cell map does not prejudge the function of the relations it represents. The team studies six cardiac regions from fourteen deceased donors with no reported notable cardiovascular history. The dataset comprises 487,106 cell or nucleus profiles, obtained through complementary preparations, including enrichment for CD45+ cells. The analyses distinguish populations and transcriptional states; certain localizations are examined by in situ hybridization. Ligand-receptor relations are mainly inferred from gene expression. The article also presents the differences in capture between preparations and their limits.[6]

6.4.5.2 Critical Reading: large numbers must keep their human denominator

PROFILES, NOT PERSONS

The hundreds of thousands of profiles provide considerable observational detail. They do not constitute as many independent people. Cells from the same donor share an organism, a sampling context and part of the technical processing. To generalize a difference between people, the analysis must preserve this nested organization. Multiplying measurements from the same individual does not replace studying other individuals. Comparing proportions also demands a precise question. Is the target the distribution among the elements actually captured, in the sampled fragment or in the whole heart? Experimental enrichment can make a rare population more accessible without making its observed frequency representative of its natural abundance. A nucleus-based preparation and a whole-cell dissociation do not provide identical windows. Comparing them increases information, provided the differences in selection are not erased. Finally, a cluster produced by computational analysis remains a classification proposal. Its stability across donors, its recognition by independent measurements and its functional relevance answer different tests. A clean graphical separation does not settle all these questions. An excellent exploratory map can be available while certain boundaries are kept open to revision.

CAPTURED PROPORTIONS, TRUE ABUNDANCE

6.4.5.3 Immunological Reading: a predicted conversation has not yet been heard

A LIGAND-RECEPTOR COMPATIBILITY

A compatibility between ligand and receptor indicates where to look for communication. It does not demonstrate that the proteins are present, accessible, active and necessary for the supposed effect. Spatial proximity reduces certain uncertainties, but still leaves the direction of action and its consequences to be examined. To move from the map to a mechanism, one would need to interrupt or modify a defined relation, then measure what changes and what can be restored.

6.4.5.4 Cardiological Reading: building a reference, not a diagnostic threshold

A REFERENCE, NOT A NORM

A descriptive reference helps to formulate what appears altered in a diseased heart. It is not automatically an individual norm, a disease threshold or a model of aging. Before attributing a contrast between diseased and reference tissue to the disease, differences in sampling, preparation, age and clinical context must be examined. This requirement protects the comparison without diminishing the value of the atlas.

6.4.5.5 Four Conclusions

Established: the sample studied reveals regional and cellular diversity that bulk tissue analysis does not capture with the same precision. Plausible: this mapping can guide mechanistic experiments and better-targeted comparisons of disease states. Not concluded: that each cluster would be a stable lineage, that each predicted interaction would be active or that the map would describe

all human diversity. What would change the judgment: replication in other donors and contexts, followed by spatial and functional validation of the specifically claimed relations.

6.4.5.6 Integrative Clinical Reading

NO IDEAL COMPOSITION

A reference map can suggest that a healthy heart has a single composition to which everyone should conform. To convey the value of this atlas, it must be explained that it describes the tissues studied and provides comparisons, without establishing an individual ideal. The whole person cannot be reduced to proximity to a cellular profile. A new clinical question becomes relevant when one can say what this comparison would add to the history, difficulties and goals already known for that person.

6.5 Worked Exercises

6.5.1 Worked Exercise: A Map and Twelve Animals

6.5.1.1 Problem

All data, group names and results in this exercise are fictional. They reproduce no result from Litviňuková, Hulsmans or any clinical trial. The exercise distinguishes a descriptive observation, an experimental perturbation and a clinical extrapolation. A team builds a transcriptomic map from eight human hearts obtained post mortem, with no documented history of conduction disorder. It analyzes 80,000 nuclei. It identifies a population named M expressing transcript G, as well as a population of conducting cells expressing transcript R. Software proposes a G-R interaction. The frequency of M varies between donors. Neither electrical current nor the effect of a perturbation of G is measured in these tissues. A second team randomly allocates twelve mice to two groups: an intervention reducing G in M cells, or a control procedure. Each animal provides twenty electrocardiographic sequences. The endpoint announced before the experiment is, for each animal, the mean change in a conduction delay between before the intervention and 48 hours after. The individual values, in milliseconds, are as follows: Control group: 0; 1; −1; 0; 1; −1. Intervention group: 2; 4; 3; 5; 2; 2. The team describes a mean increase of 3 milliseconds and writes this conclusion: “The analysis of 240 sequences demonstrates that G protects humans from conduction disorders; increasing G will make it possible to treat them.” 1. What proposition does the human map directly support? 2. What is the unit of assignment, and what mean difference should be calculated? 3. Which parts of the conclusion must be rewritten? 4. What complementary experiment would strengthen the attribution of the result to G, without already claiming to demonstrate clinical efficacy?

EIGHT HEARTS, ONE PREDICTED INTERACTION

TWELVE MICE, TWENTY SEQUENCES

6.5.1.2 Reasoned Answer

AN INTERACTION THAT REMAINS PREDICTED

The human map supports the presence of expression profiles in the samples analyzed and a hypothesis of interaction. A prediction based on transcripts measures neither the availability of the proteins, nor their actual encounter, nor the electrical consequence of that encounter. The word “interaction” must

therefore remain accompanied by "predicted." Nor does the absence of a documented disorder suffice to define the samples as a universal reference for health. The animal is the unit of assignment: six mice per group, twelve in total. The sequences are repeated measurements within these units. Their multiplication can improve the precision of each mouse's mean; it does not multiply the number of independent animals. Here, the values provided are already the individual means of the announced endpoint. The control mean is 0 milliseconds; the intervention mean is 3 milliseconds. The intervention minus control contrast is therefore +3 milliseconds. This calculation describes the data; it replaces neither an estimate of their uncertainty nor a justification of the biological relevance of this magnitude. No relevance threshold is provided in the problem. Random allocation helps to interpret the difference as an effect of the intervention in this setup, subject notably to its proper conduct, any losses and measurement quality. It does not guarantee that all consequences pass exclusively through G in M cells. Target engagement, the specificity of the intervention and the relevant comparabilities must therefore be verified. It would be premature to conclude that the observations demonstrate the mechanism in every detail. The proposed sentence then merges species and reverses the direction of the intervention. Showing a consequence of reducing G in mice does not establish that increasing G in a sick human would have the opposite effect, with a net benefit. One possible rewrite: "In this fictional murine setup, the intervention is accompanied by a mean difference in delay change of +3 milliseconds relative to control. The human map motivates study of the plausibility of a related mechanism; it demonstrates neither its functional role nor its therapeutic value." A complementary experiment could restore G in M cells after its reduction, with appropriate controls and an endpoint defined before analysis. A return of the delay toward baseline would strengthen attribution to the target, while requiring verification that the restoration does not act through an additional pathway. This experiment would remain preclinical. For the human follow-up, a setup that genuinely allows study of a function would need to be chosen, rather than simply increasing the number of molecular profiles. The right next experiment depends on the proposition that remains uncertain.

THE ANIMAL, UNIT OF ASSIGNMENT

TWO SPECIES, A REVERSED DIRECTION

RESTORING G AFTER ITS REDUCTION

Synthesis

Studying the heart as an immune milieu requires linking the composition of the organ to the organization of the activities examined there. The list of cells provides indispensable knowledge; it does not, on its own, describe their positions, exchanges or dependencies. The milieu becomes explanatory only when these relations are named and subjected to comparisons capable of testing them. This inquiry begins with a discipline of categories. The characteristics that make it possible to recognize a cell are not to be confused with its current state or its history of provenance. A change in classification may reflect a transformation of the object, but also a limit of the means used to recognize it. The scope of each conclusion therefore depends on what the method actually tracks. It continues with a discipline of representations. An anatomical map, a grouping of profiles and a causal model can concern the same samples while providing different information. Their articulation becomes solid when experiment discriminates between interpretations that description left open. Perturbation must in turn be examined in its actual effects, without turning its announced precision into a general guarantee. Finally, the passage between

systems requires its own justification. A resemblance between human and murine cells prepares a comparison; it does not establish all the functional equivalences. The study of differences can help specify the properties needed for extrapolation. This work avoids both rejecting every model and granting it unlimited scope. The central question is therefore the following: which relation does this study make it possible to know better and through which comparison? The answer must distinguish description, proposed explanation and contribution actually tested. This is how the heart as a milieu becomes an object of research rather than a general formula.

Open Questions

A population changes markers after a perturbation: what observation would distinguish its disappearance from a change of state? A map shows neighboring cells: what experiment would discriminate between a direct interaction and a common response to another condition? When two experimental tools agree, what information makes it possible to judge whether this agreement genuinely corrects their limits? Finally, what property would need to be conserved between mice and humans for the specific extrapolation envisaged to be justified?

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7

Atherosclerosis and the Convergence of Evidence

Chapter contents

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Cholesterol crystals, poorly cleared dead cells, lymphocytes that recognize a lipoprotein protein, a blood clone carrying a mutation: each has been linked to inflammation of the atherosclerotic plaque. Lipid and inflammation appear here less as two rival explanations than as processes to be articulated. This accumulation gives an impression of solidity. Yet it brings together transplanted mice, cell cultures, human plaques and reviews whose authors sometimes produced the work under discussion. Before treating it as evidence, it is necessary to know what makes these results comparable, what makes them dependent and which result would require their alignment to be undone.

7.1 The Conditions of a Convergence

7.1.1 Several Results, a Single Question?

SEVERAL PATHS, ONE CONCLUSION

A field becomes convincing when several paths seem to lead to the same conclusion. This impression nevertheless deserves analysis. What does "the same conclusion" mean? Studies may refer to the same biological pathway while examining its presence, its activity, its role in a process or the effect of an intervention. Taken together, they outline a coherent program; they do not yet form a shared demonstration. To examine convergence, the starting point is a limited proposition. In an entirely fictional example, a team posits that the activity of pathway V contributes to a lesional property L and that reducing this activity might improve a clinical outcome Y. V, L and Y designate no particular molecule, lesion or outcome. This teaching device makes it possible to follow the shifts in question without attributing any biological validation to them. Three imaginary studies appear to support the program. The first finds a signal of V in samples associated with L. The second modifies V in an experimental system and observes a change in L. The third compares an intervention targeting V with a control and measures Y. To speak of convergence, it will be necessary to explain what each contributes to the others and what links them. The results do not add up like three votes cast in favor of an opinion.

THREE STUDIES, NOT THREE VOTES

7.1.2 Counting Dependencies before Counting Confirmations

ONE REAGENT, TWO POPULATIONS

Suppose that the first study measured V with a reagent R. Another team reproduces the association in new samples using the same reagent. This repetition is informative: the observation is no longer confined to the first set studied. It does not, however, correct on its own a possible inability of R to distinguish V from another signal. Two independent populations can leave a shared instrumental dependency in place. It would be excessive to conclude that the second study adds nothing. Independence is not an indivisible quality that one either has or lacks. A study can be independent in its recruitment and dependent in its measurement, independent in its analysis and dependent in the hypothesis linking the signal to the mechanism. The task is to identify the type of error that each new study renders less plausible. In this example, a measurement based on a different property of V could test the interpretation of the R signal. But changing technology does not guarantee this independence: the two techniques could be calibrated against the same inadequate reference. The question is therefore not how many devices were used, but which reasons could lead them to reach the same wrong conclusion. Another dependency concerns the choice of comparisons. If all the studies select situations in which L is already present, they may share a limitation in distinguishing what precedes L from what accompanies it. The diversity of laboratories does not erase

INDEPENDENCE BY DEGREES

A SHARED OBSERVATIONAL STRUCTURE

a shared observational structure. Triangulation becomes useful when it genuinely changes a condition of the inquiry.

7.1.3 What Converges Must Be Linkable

ONE NAME, DIFFERENT MEASUREMENTS

A diversity of methods is not sufficient either. For an experimental result to shed light on the initial association, it must be established that the two studies examine a shared relevant property. The name V can conceal different measurements: the quantity of a signal, an activity attributed to a pathway or a downstream consequence. Their relationship must be explained rather than presumed. Imagine that the first study measures an abundance, while the second reduces an activity without changing that abundance. The results are not contradictory by definition. But the second experiment does not automatically test the interpretation of the first. It is necessary to know what relationship the model proposes between abundance and activity, then to ask whether the

THE LINK BETWEEN RESULTS

data can support that relationship. The term "link" will refer here to the argument that warrants setting results against one another. It may concern the measurement, the system studied, the chronology or the outcome selected. It is not a mere detail of presentation: if the link fails, the studies may remain sound individually while ceasing to support the announced shared conclusion. The objection would be that this requirement prevents any pooling of

HETEROGENEITY IS NOT A GUARANTEE

heterogeneous work. It prevents only treating heterogeneity as a guarantee. Differences may be precisely what makes a comparison instructive, provided it is understood what remains comparable. The aim is to keep the useful differences without fabricating a nonexistent identity between the objects.

7.2 A Convergence Open to Revision

7.2.1 A Contradiction Can Refine the Model

TWO PERTURBATIONS, TWO RESULTS

Continue the scenario. Two perturbations assumed to reduce V yield different results on L. The temptation would be to keep the experiment that agrees with the hypothesis and to explain the other by a technical flaw. Yet the disagreement may reveal a distinction that the initial model had erased. It must first be established that the results genuinely differ. A clearly estimated effect in one study and an imprecise estimate in the other do not necessarily constitute two opposite effects. If the divergence is supported, the next step is to examine what the interventions actually modified: the same activity, the same location, the same intensity, the same starting conditions? A shared target name settles none of these questions. One possible explanation would be that the second intervention acts on an additional condition. This idea deserves study if it leads to a new prediction. A comparison separating the action on V from this other action would then have to be designed. Without this step, the explanation remains a possibility; it does not yet allow the discordant result to be set aside.

A GENUINELY SUPPORTED DIVERGENCE

ABANDONING THE LINK

Conversely, a contradiction may require abandoning the link between V and L rather than adjusting the interventions indefinitely. Revision must remain available. A convergence that accepts only the results compatible with its narrative reduces to a selection of confirmations.

7.2.2 Treatment Adds a Proposition

A STRATEGY IS NOT A MECHANISM

The third fictional study measures Y after an intervention targeting V. Its arrival does not automatically close the program. It introduces a proposition

about a strategy, in a defined population and over a defined period. This proposition bears a relationship to the mechanism under study, but it is not identical to it. Two possible results do not test the same links. If Y improves, the intervention gains clinical support; the path from V to L and then to Y remains to be examined in order to understand this improvement. If a precise estimate rules out the expected benefit despite an adequate modification of V, the link between target and clinical outcome becomes more directly open to question. This distinction (see 3.4.1) has a consequence specific to convergence: the trial can strengthen one part of the program and weaken another, without contradiction. Why keep a model in which one link fails? Because it may still explain observations for which it has separate support. But maintaining it requires explicitly abandoning the extension that has been called into question. Keeping a local proposition does not permit silently keeping the global promise that had been attached to it.

STRENGTHENING HERE, WEAKENING THERE

7.2.3 Organizing a Triangulation That Takes a Risk

THREE STATED COMMITMENTS

An informative triangulation requires a plan for testing. In this scenario, the first commitment concerns measurement: if R does signal the property attributed to V, an independent comparison should find a defined relationship. The second concerns the experiment: if the activity of V contributes to L under the stated conditions, an adequate modification should change L according to a specified prediction. The third concerns the clinical step: the strategy tested should produce an effect on Y whose size and uncertainty can be assessed. One objection remains: investigators cannot foresee every way in which a program will run into difficulty. This is true; revision does not reduce to executing a contract written before the experiment. Yet keeping the initial expectations makes it possible to distinguish a successful anticipation from a new accommodation. The latter may be fruitful, but it must earn its own support against other data. The risk taken does not lie in a prohibition on changing one's mind: it lies in the obligation to acknowledge what has changed. Finally, it is important to keep the discordant results within the representation of the program. A diagram showing only the supported arrows gives an incomplete picture of the reasoning. The links that remain uncertain, the missing comparisons and the relationships whose domain has had to be narrowed must also be made visible. Triangulation thus ceases to be an emblem of solidity; it becomes an organized way of finding where an explanation holds and where it must change.

ANTICIPATION OR ACCOMMODATION

SHOWING THE UNCERTAIN ARROWS

7.3 Texts Put to the Test

7.3.1 Duewell and the Material Form of Cholesterol

7.3.1.1 Situating the Result

DUEWELL, 2010

The authors combine observations of crystals, cellular experiments and murine models. In the atherogenesis experiment, mice deficient in the low-density lipoprotein (LDL) receptor receive normal bone marrow or marrow deficient in NLRP3, ASC or interleukin 1α/β (IL-1α/β). After the feeding period, lesion burden decreases in the experimental groups. The cellular experiments distinguish priming from activation and support a lysosomal disturbance. The human observations do not constitute a clinical trial.[1] A 2010 correction amends an

affiliation and completes the funding information of one author; it does not alter the scientific results presented.[2]

7.3.1.2 Critical Reading: the apparent chronology

SEEING BEFORE DATING

The most original methodological contribution is a question of visibility. Before classifying an event as late, it is necessary to know at what point the instrument allows it to be recognized. If a preparation removes part of the object sought, an absence of image no longer amounts to an absence of object. Changing the method of observation can then shift the apparent chronology of the disease without the disease having changed. This is a precise lesson for the reading of histological work: the properties of the sample and of its preparation belong to the definition of the result. Bone marrow transplantation also calls for exact wording. An intervention on a hematopoietic compartment does not become, for narrative convenience, an intervention specific to the macrophage. Attributing the effect exclusively to one lineage would require a design that separates its contribution from those of the other cells involved. The word "compartment" here protects an essential piece of experimental information. Irradiation, reconstitution and diet also define a specific system; they do not reproduce all the routes by which a disease develops spontaneously.

A COMPARTMENT, NOT A LINEAGE

7.3.1.3 Immunological Reading: the conditions of activation

The term "required" in the title must be related to the result obtained: a reduction in lesions after perturbation supports a contribution in this setting; it does not demonstrate that any lesion is impossible without this pathway. The distinction between priming and activation sharpens this reading. In an experimental setup, a factor may produce an effect after the cells have been prepared, yet fail without this preparation. It does not thereby become inactive: its effect is conditional. To carry the explanation over to a tissue, it is then necessary to identify what supplies the missing condition, at what intensity and for how long. The result on primed cells thus opens an inquiry into the environment; it does not exempt anyone from that inquiry. An intermediate mechanism must also be distinguished from its ultimate molecular link. A series of experiments can support a lysosomal disturbance and a dependence on a pathway without demonstrating a direct binding between a crystal and a protein. Filling this gap graphically with a single arrow makes the diagram simpler, but may make the claim stronger than the data. The right arrow must name what has been tested and leave visible what remains to be explained.

A CONDITIONAL EFFECT

AN ARROW THAT FILLS THE GAP

7.3.1.4 Cardiological Reading: the relationships between lipid and inflammation

LIPID AND INFLAMMATION ARTICULATED

The merit of this program is that it allows a question that goes beyond the choice between a lipid disease and an inflammatory disease. A physical property of accumulated material may contribute to a cellular response; this response may, in turn, modify the lesion. It then becomes more relevant to study how the processes are articulated than to ask which one alone would own the disease. This way of reasoning does not yet fix their relative weight in a given patient. The experimental endpoint must remain named. A lesion

A LESION AREA, NOT AN INFARCTION

area provides information on one component of atherogenesis; it does not directly measure the infarction prevented, the stability of each plaque or the net benefit of a treatment. It would also be wrong to interpret a future clinical result on IL-1 as retroactive validation of every upstream relay proposed. A target can receive several signals. The efficacy of a downstream intervention, if established separately, does not automatically identify the dominant source of the signal in each patient.

7.3.1.5 Four Conclusions

Established: within the field examined, the experimental setup supports a contribution of the axis studied to murine atherogenesis, without immediate equivalence to a clinical effect. Plausible: a characterization of the lipid material and its environment could complement a measure of quantity in explaining a lesion. Not concluded: the universal necessity of NLRP3, an exclusively macrophagic role or a human therapeutic efficacy drawn from this study alone. What would change the judgment: more specific perturbations, replications in other contexts and complementary disease endpoints capable of separating a local mechanism from a general property.

7.3.1.6 Integrative Clinical Reading

SEVERAL PROCESSES, A SINGLE HISTORY

The articulation between lipid material and inflammation avoids telling a person that a single explanation of the disease must be chosen. Several processes may belong to the same history without imposing several additional treatments. The move to care requires knowing what each option would actually add to the existing care pathway. This experiment provides a biological link to be studied; the integrated discussion must still distinguish an enriched explanation from a useful change in management, with its constraints for the person.

7.3.2 Fuster and the Arterial Consequences of a Blood Clone

7.3.2.1 Situating the Result

FUSTER, 2017

In a murine model predisposed to atherosclerosis, the authors transplant bone marrow initially containing 10% Tet2-deficient cells. This population expands; on a high-fat, high-cholesterol diet, lesions increase without any difference in blood cholesterol detected between the groups. The cellular experiments point toward IL-1β and NLRP3. Under NLRP3 inhibition by MCC950, the contrast in plaque size between the groups is no longer found in this experiment. These results are preclinical.[3]

7.3.2.2 Critical Reading: the clone and its measurements

10% AT BASELINE, THEN EXPANSION

The first methodological interest is to vary one component of the hematopoietic system within an organism. The proportion introduced at baseline, its evolution and the activity of the cells derived from it must be distinguished. An experiment that begins with a minority of modified cells does not necessarily describe an organism durably exposed to that same minority. The phrase

"effect of 10% mutated cells" would make this dynamic disappear. To distinguish clonal expansion from cellular activity, both dimensions would need to be followed and perturbations designed that can vary them separately. Simply comparing animals with similar final clonal burden would not suffice: this burden may itself depend on processes that influence the lesion. A second point of vigilance concerns denominators. The fraction of donor-derived cells is not the same as an allele fraction measured by sequencing. Converting one into the other requires information on genotype, copy number and the populations actually sampled. The exact name of each measurement must therefore be kept, rather than merging them under an undifferentiated notion of "mutational burden."

CELL FRACTION, ALLELE FRACTION

7.3.2.3 Immunological Reading: the damage and its origin

REDUCING LESION WITHOUT REDUCING CLONE

Modifying the lesion does not imply reducing the clone. Interpretation therefore requires two forms of follow-up, one of the cell population, the other of the damage. A further question, distinct from the results reported here, would be whether the effect persists after the intervention is stopped. The lipid context must also be kept within the reasoning. A difference observed between two groups with comparable blood cholesterol allows an additional mechanism to be studied; it does not demonstrate that the lipid environment has become secondary. The most fruitful interpretation looks for articulations between processes instead of staging a competition between "lipids" and "inflammation."

THE LIPID CONTEXT REMAINS

7.3.2.4 Cardiological Reading: risk and benefit

RISK, BENEFIT, TOLERABILITY

Considering clinical use raises three distinct questions: does the marker identify a risk? Does it predict a different therapeutic benefit? Does the intervention improve an outcome that matters to the patient with acceptable tolerability? Answering the first does not answer the others. An experiment on Tet2 does not constitute a common verdict on all the alterations that can give rise to clonal hematopoiesis.

7.3.2.5 Four Conclusions

Established: an experimental contribution of Tet2 deficiency to the atherosclerotic phenotype under the conditions studied. Plausible: a research strategy targeting its inflammatory consequences. Not concluded: a preventive treatment indicated for every person carrying a clone. What would change the judgment: future clinical results, which would specify the scope of this strategy according to the alteration, the context and the duration.

7.3.2.6 Integrative Clinical Reading

TWO MEASUREMENTS, TWO MISSIONS

The clone and the lesion evolve according to different measurements. For a person whose record contains both, improving one would therefore not allow the other to be declared resolved without its own observation. This distinction invites coordination of the objectives of the different clinicians: monitoring a blood population and preventing a cardiovascular event are not the same

mission. The murine study helps formulate this division of questions; it sets neither clinical surveillance nor an intervention threshold for people carrying a clone.

7.3.3 Kojima and the Clearance of Dead Cells

7.3.3.1 Situating the Result

Kojima, 2016

The authors describe increased expression of CD47 in human plaques and test its blockade in mice deficient in apolipoprotein E (apoE). The antibody reduces lesion burden in several experimental setups. The cellular assays and tissue observations support improved efferocytosis, the clearance of apoptotic cells. The work also explores regulation of CD47 by tumor necrosis factor alpha (TNF-α). Anemia appears in the acute model at full dose. Adding an anti-TNF agent does not demonstrate, in the animal comparison reported, superiority over anti-CD47 alone.[4]

7.3.3.2 Critical Reading: measuring a stock does not describe its flows

LESS DEATH OR MORE CLEARANCE

A plaque contains less debris after an intervention. Two explanations come immediately to mind: fewer cells die, or more dead cells are cleared. The final image does not separate them. This is why the reading must bring together the experiments measuring cell death and those measuring ingestion, then check whether the conditions are comparable. A negative result on one form of apoptosis in culture does not become a universal exclusion of any change in cell death in the animal. The proximity between an apoptotic body and a macrophage likewise provides a clue, not a film of its clearance. A colocalization, an ultrastructural image and a functional assay carry different

SLOWING IS NOT REVERSING

information. Timing is another essential control. Administering an intervention before a lesion is established and beginning it afterward address two different problems. Even if both results point the same way, slowing progression does not automatically mean causing an already formed plaque to regress. To claim regression, a reference point allowing a decrease to be established is required, not merely a final lesion smaller than that of the control.

7.3.3.3 Immunological Reading: the brake is part of the function

A SIGNAL, NOT AN INTENTION

The phrase “do not eat me” serves teaching purposes, but it risks turning a contextual signal into an intention of the cell. A mechanism that protects certain cells against their clearance may become unfavorable in another environment. Calling it pathological therefore requires specifying which cell should be pre-

TWO DIFFERENT SELECTIVITIES

served, which should be removed, at what time and by what criterion. This reading explains why the lesional benefit and the hematological consequence belong to the same reflection. A molecular target is not necessarily confined to the tissue one wishes to treat. The possibility of modifying its activity does not ensure that every exposed cell will benefit from that modification. A drug’s selectivity for a molecule and its selectivity for a pathological situation are two different properties. Finally, the connection with TNF-α requires distinguishing coexpression from experimental regulation. A network analysis proposes

a candidate; it does not on its own measure its responsibility. Likewise, adding two interventions and observing an improvement is not enough to name a synergy: the effect expected from their combination, and the contrast that tests it, must be defined.

7.3.3.4 Cardiological Reading: an accessible target is not yet a validated strategy

DEFINING THE BENEFIT SOUGHT

Observations on human tissue motivate examining this target in humans; they do not demonstrate that the experimental mechanism explains the effect of a treatment in patients. Moving toward that comparison must also define the benefit sought: modifying a histological feature, preventing a rupture or reducing a cardiovascular event. These objectives are not equivalent. Rereading Kojima after Duewell brings out two distinct operations: the production of a signal from accumulated material and the clearance of dead cells. This juxtaposition suggests a new comparison: does reducing the first operation restore the second, or must it be addressed separately? The two studies do not jointly test this alternative. Their shared value is to make it formulable: the same decrease in inflammatory activity could accompany different clearance capacities. Resolution then becomes a question about the work accomplished by the tissue, and not only about the signal that decreases.

PRODUCING A SIGNAL, CLEARING THE DEAD

7.3.3.5 Integrative Clinical Reading

LOCAL SUCCESS, GLOBAL BALANCE

The hematological damage observed in the model is a reminder that a local success and a global balance can diverge. For a future clinical evaluation, a consequence outside the artery would not be a detail to be handed afterward to another specialty. It would be part of the same judgment on the intervention. The integrative value of this work is that it requires defining together the benefit sought and the functions to be preserved, without converting an animal signal into a human frequency or announcing in advance which balance would be acceptable.

7.3.4 Wolf and the Fate of an Autoreactivity

7.3.4.1 Situating the Result

WOLF, 2020

The authors characterize, in mice, CD_4 lymphocytes recognizing an apolipoprotein B (apoB) peptide, with profiles combining regulatory and effector markers. Lineage tracing reveals a loss of $FoxP_3$ in a fraction of these cells. After transfer, the populations studied do not reduce total lesion area, despite a difference concerning small lesions. In people with coronary artery disease, stimulation of blood with apoB peptides elicits a greater cellular response than in the controls studied.[5]

7.3.4.2 Critical Reading: a resemblance is not a descent

RECOGNIZING, EXPRESSING, ACTING

Three questions must remain distinct: which cell recognizes the antigen, which program it expresses and what it does to the tissue. Binding to a reagent that recognizes an antigenic specificity answers the first. Expression of genes or

proteins addresses the second. The third requires measurements of function; attributing that function causally to the cells further requires comparisons able to adjudicate between competing explanations. The elegance of a molecular map does not exempt anyone from these tests. The coexistence of regulatory and inflammatory signatures within a cell does not reconstruct its history. It could correspond to a stable state, a transition or a temporary response. Tracing past expression therefore provides different information from comparing young and old animals. But establishing that a fraction has changed does not allow this origin to be attributed to the whole population observed subsequently. A trajectory demonstrated locally does not become a universal genealogy. Cell transfer adds a particularly instructive test. A population announced as protective must retain this property in the environment into which it is placed. However, failing to reduce a lesion area does not prove that the cells worsened the disease. That latter claim would require a contrast showing worsening. Likewise, a difference in the number of small lesions does not replace the result on total area: counting and measuring an extent answer different questions.

FAILING TO PROTECT IS NOT WORSENING

7.3.4.3 Immunological Reading: an identity with several dimensions

THE THREE SENSES OF "REGULATORY"

The word "regulatory" may designate a marker, a transcriptional kinship or a suppressive function actually measured. Moving from one sense to another without saying so produces an artificial certainty. To read this article, the antigenic specificity, the molecular state, the history of the cell and its function in the experimental setup must be kept side by side. This discipline also changes the use of the word "autoimmunity." Recognizing a component of the self is not enough to establish that this recognition causes damage. An autoreactivity may contribute to a balance; its pathogenic character must be demonstrated in a defined situation. Conversely, finding an autoreactive component does not necessarily place a whole complex disease in a single etiological category. Stability then becomes a therapeutic question as important as the initial identity. If transferring a protective population is contemplated, its persistence, its fate and the effects of the recipient environment must be examined. Correctly selecting a cell before administration does not guarantee that its function will be preserved afterward.

RECOGNIZING THE SELF WITHOUT DAMAGE

7.3.4.4 Cardiological Reading: from the cellular test to individual risk

A RESPONSE ELICITED IN THE LABORATORY

A response obtained after stimulation in the laboratory depends on the setup that reveals it. It is equivalent neither to spontaneous secretion within a plaque nor to a direct measure of vascular damage. Comparing people with and without disease constitutes a first clinical check; it does not demonstrate that the test predicts a future event or adds information useful to the criteria already available.

7.3.4.5 Four Conclusions

Established: the observations support a diversity and an instability of the cellular states studied. Plausible: preserving certain regulatory states could modify

the disease. Not concluded: a regulatory origin for all effectors, worsening caused by the transferred cells or the efficacy of a vaccination in humans. What would change the judgment: functional measurements and prospective clinical follow-up, which could specify which properties remain protective, in whom and for how long.

7.3.4.6 Integrative Clinical Reading

A RESPONSE TO SELF, NOT A DIAGNOSIS

A test showing a response to the self may carry strong symbolic weight for the person who receives it. In the clinical exchange, autoreactivity must not become a global diagnosis supposed to explain every symptom. The integrative question is that of the information added to a situation already described: does the result shed light on documented involvement, on a course of disease or only on a lead? Keeping this distinction makes it possible to discuss the test without reducing the person to an immunological identity whose consequences would remain presumed.

7.3.5 Bäck and Resolution as Work to Be Accomplished

7.3.5.1 Situating the Result

Bäck, 2019

This 2019 review organizes the relationships between lipid accumulation, inflammation and resolution in atherosclerosis. It describes an active resolution, including in particular the clearance of dead cells, and examines mediators together with their receptors. Its experimental table gathers interventions on ligands and receptor knockouts; their effects do not all follow a uniform direction. The authors also discuss the physiological concentration of the mediators and their local exposure. They propose a therapeutic perspective, without reporting a clinical trial demonstrating the cardiovascular benefit of a pro-resolving strategy. This synthesis is to be read within its 2019 horizon.[6]

7.3.5.2 Critical Reading: defining a different success

NEUTRALIZING OR RESTORING

Neutralizing an activity and restoring a function do not designate the same result. This difference changes the experimental question. In a fictional example, two interventions lower an inflammatory signal to the same extent. With the first, the tissue remains cluttered with dead cells; with the second, their clearance resumes. If the only criterion retained were the fall in the signal, the two strategies would appear equivalent. The concept of resolution becomes useful by requiring the measurement of an operation that this criterion leaves invisible. The objection will be that a successful anti-inflammatory treatment is simply being renamed. That would be the case if "resolution" designated only the final improvement. To avoid this circularity, it is necessary to state what the intervention is to restore, how that restoration will be observed and which dissociation would place the explanation in difficulty. A tissue improvement without resumption of the proposed process would require another pathway to be sought; a resumption without improvement would show that this process was not sufficient under these conditions. This requirement also provides a positive reason to continue. An intervention that restores a deficient operation, before the expected improvement and in comparisons able to rule out the main competing explanations, contributes more than an agreement between

A CIRCULARITY TO BE AVOIDED

two markers. The gain in knowledge lies in the link that has been tested, not in the favorable character of the word "resolution."

7.3.5.3 Immunological Reading: the receptor does not impose the direction alone

ALX/FPR2 and its agonists

The review sets out in particular the possibility, for ALX/FPR2, of transmitting different effects depending on the agonist. Naming a "resolution receptor" therefore does not confer on it a favorable function independent of its interactions. Deleting this receptor and supplying it with a chosen ligand are not two exactly opposite interventions: deletion also removes the receptor's other possible uses. The table of experiments here takes on argumentative value. Effects that differ across models must neither be erased to preserve a simple narrative nor automatically declared incompatible. The conditions that could explain their direction must be identified, and it must then be asked whether these conditions allow a new prediction. An explanation adjusted after each result, without a new test, would protect the vocabulary. Even when born of a disagreement, it can advance the mechanism if it leads to a discriminating comparison. The question of concentrations requires specifying where the reasoning applies. In a fictional example, a low circulating measurement would be compatible with sufficient local exposure as well as with insufficient exposure. Choosing the first interpretation because it saves the model is not an answer. The sample, the compartment and the duration must make it possible to distinguish these possibilities.

circulating concentration, local exposure

7.3.5.4 Cardiological Reading: shifting the endpoint, not only the target

removing or making possible

For the heart and the arteries, this reading invites a distinction between what is removed and what is made possible again. A research program may aim at less damaging activity, better clearance or a more resistant structure. These objectives can complement one another without becoming interchangeable. The clinical comparison to be built should preserve this ambition: to check whether the restoration sought leads to a relevant benefit and at what cost. A presumed reparative property is not a dispensation from studying harms. The text thus opens a substantial therapeutic hypothesis: acting on the conditions under which a response terminates, rather than measuring only the intensity that has been taken away from it.

the termination of a response

7.3.5.5 Integrative Clinical Reading

what the person recovers

Resolution invites the question of what the person recovers, beyond what decreases on a report. A lower biological activity and the resumption of a daily activity are different results, which should be related without presuming their agreement. This question gives functional objectives a place in a future clinical program. It does not turn a mechanistic review into evidence of experienced recovery: integration consists precisely in planning both levels of assessment and in making any discordance between them interpretable.

7.3.6 Döring and Disagreements within a Promising Mechanism

7.3.6.1 Situating the Result

Döring, Soehnlein and Weber, 2017

This 2017 narrative review brings neutrophil extracellular traps (NETs) together with atherosclerosis and arterial thrombosis. It sets experimental results, observations of human lesions and therapeutic proposals against one another. The difficulties of identifying NETs within lesions are discussed. Their figure assembles relationships several of which remain hypothetical. The synthesis organizes these results; it does not constitute a replication of the studies that produced them.[7]

7.3.6.2 Critical Reading

two competing responses

The divergences set out concern in particular the respective place of a type I interferon response and of an IL-1β/Th17 response in atherogenesis. As regards the degradation of NETs, the review distinguishes the prospect of promoting the lysis of a thrombus from the more contested results on lesion growth. These are different questions, which the general formula "role of NETs" risks merging too quickly. This comparison is all the more instructive because the authors of the review contributed to some of the research discussed. They possess knowledge of the problem, but their synthesis does not constitute an independent replication. The force of an argument must be distinguished from the closeness of its author to the work concerned. This closeness does not invalidate the analysis; it makes the presentation of objections and discordant results particularly useful. A fruitful critique should lead to a narrower proposition that is better exposed to error. If every divergence were absorbed into the formula "it all depends on the context," the explanation would become impossible to test. It is necessary to specify which difference in context matters, why it matters and which result would contradict this interpretation. Disagreement then becomes a resource for restricting the hypothesis, rather than noise to be suppressed.

the closeness of the authors

7.3.6.3 Immunological Reading

what detection identifies

Before discussing the role of NETs, it is necessary to know what their detection makes it possible to identify. A component found within the lesion does not necessarily suffice to recognize the structure from which it may derive, nor its activity at the time of sampling. Uncertainty about the object measured then precedes uncertainty about its function. Presence and contribution must then be separated. An element associated with a lesion may participate in its formation, appear after it or accompany another process. A localized observation specifies where to look for the explanation; it does not, on its own, fix the direction of the relationship. Likewise, a result attributed to a pathway does not permit all the other functions of its components to be ignored. The figure in this review links inflammation, plaque instability and thrombosis in a continuous visual narrative. Reading it must preserve the reservations stated in its legend. The useful exercise would be to assign each relationship an explicit status: observation, interpretation or hypothesis. One would thus learn to use a representation without lending it a uniform certainty. The coherence of a diagram can guide an inquiry; it does not replace its tests.

observation, interpretation, hypothesis

7.3.6.4 Cardiological Reading

PLAQUE GROWTH, THROMBUS LYSIS

These distinctions change the cardiological question. An intervention intended to modify the growth of a plaque and an intervention aimed at the lysis of a thrombus do not pursue the same result. Gathering them under the promise of an "anti-NET" treatment would make precisely what has to be evaluated disappear: the timing of the intervention, its intended effect and its clinical consequences. Temporal order also matters. Information collected during an acute event does not necessarily describe the state that preceded that event. To build a preventive strategy, it would be necessary to know what was measurable beforehand and which choice would have been changed. To judge a treatment strategy, its consequences beyond the biological signal would have to be examined. This historical review provides questions to be organized; it does not remove the need to check current knowledge and guidelines before any clinical use.

BEFORE THE ACUTE EVENT

7.3.6.5 Integrative Clinical Reading

THE LIFE AROUND THE EVENT

A molecular explanation can make a cardiovascular event more intelligible without becoming the whole explanation of the life surrounding it. General practice retains the other diseases, the treatments, the material conditions of follow-up and the person's priorities. This perspective adds no evidence to the biological hypotheses; it specifies the outcomes an innovation would have to improve in order to take its place in care. The teaching issue is to be able to say simultaneously: this is what the research helps to understand, and this is what it does not yet permit anyone to promise. A patient should not receive the most appealing prospect in a review as a personal certainty. An accurate account can preserve the interest of a discovery while leaving open the decisions that this text does not settle.

7.4 Worked Exercises

7.4.1 Exercise on the Scope of a Convergence

7.4.1.1 Problem

FORTY-EIGHT MICE, TWO TRANSPLANTS

All the data below are fictional, created for this exercise. They reproduce neither Fuster nor any real therapeutic trial. Forty-eight mice with an identical predisposition to atherosclerosis are allocated between two types of transplant; within each, they randomly receive an inhibitor X or its vehicle. Each group comprises twelve animals. The diet is identical, histological analysis is performed blind and each mouse provides a mean plaque value. The cell fraction is the percentage of live blood leukocytes carrying the marker of the donor studied. This marker identifies the same initial subpopulation in both types of transplant; it does not designate all the cells of the graft. Group: control transplant + vehicle; mean plaque area: 100 units; cell fraction of the donor studied at the end of follow-up: 10%. Group: control transplant + X; area: 90 units; fraction: 11%. Group: transplant with modified clone + vehicle; area: 150 units; fraction: 45%. Group: transplant with modified clone + X; area: 105 units; fraction: 44%. No confidence interval or interaction test is provided. A second

ONE MEDIATOR MEASURED TWICE

laboratory observes, in culture, a decrease in an inflammatory mediator after X. A third finds no difference in the concentration of this mediator in the blood of the animals at the end of follow-up. Questions. What is the observed effect of X on plaque in each type of transplant? Does the effect of X on plaque differ according to the type of transplant? Does X reduce the modified population itself? Do the cellular and blood results necessarily contradict each other? Which experiment or analysis should be requested before concluding?

7.4.1.2 Reasoned Answer

AN INTERACTION ON THE ADDITIVE SCALE

In the control transplant, the difference in means is −10 units; in the transplant with the modified clone, it is −45 units. The difference between these two effects is −35 units. This is a descriptive estimate of interaction on the additive scale. Without dispersion or a confidence interval, how precisely it is estimated remains unknown. A difference between two separate p values would not replace this analysis. The final fractions of 45% and 44% do not document elimination by X of the population identified by the marker. They demonstrate neither equivalence between the groups nor stability of the absolute count: a fraction can remain similar if the numerator and the denominator change together. To study the expansion or reduction of the clone, its course and the corresponding counts would have to be known. A conclusion of equivalence would further require an appropriate margin and precision. A culture and a blood concentration do not measure the same object. Before speaking of contradiction, the times, the compartments, the stimulation conditions and the sensitivity of the measurements must be compared. This possible explanation does not by itself prove a local activity: that activity has to be studied. What should be requested is the uncertainty of the interaction, relevant temporal and tissue measurements and an independent verification of the mechanism. These observations constitute experimental signals; they permit no prescription in humans.

SIMILAR FRACTIONS, UNKNOWN COUNTS

CULTURE AND BLOOD CONCENTRATION

Synthesis

Two distinctions guide these comparisons: an observation depends on the technique that makes its object visible and affects its apparent chronology; the effect on damage does not describe the whole fate of the cells involved. A convincing convergence preserves the differences between experiments. It specifies the reasons for bringing them together, the possible shared errors and the results that would require the conclusion to be revised. These requirements also apply to clinical trials (see Chapter 9).

Open Questions

Does a replication with the same instrument test all the competing explanations? What makes it possible to bring together two results obtained in different compartments? Does a downstream intervention validate the whole mechanism assumed upstream? Finally, which precise result would lead to abandoning the proposed alignment?

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8

Repair and the Limits of Experimental Models

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After an infarction, the adult heart forms a scar, whereas that of the newborn mouse can rebuild organized muscle. Between the injury and its outcome, monocytes follow one another, neutrophils orient macrophages, regulatory T lymphocytes release factors and lipid mediators accompany the exit from inflammation. Each of these observations comes from a constructed model: permanent ligation or reperfusion, clodronate depletion, animals of a chosen age. Dead cells can produce the benefit expected from living cells; removing a population can take away a useful function. It must then be asked what an experimental success really measures, and to which situation it can be related.

8.1 What Is to Be Recovered

8.1.1 Repairing: Recovering What?

FORM, ACTIVITY OR CAPACITY

The word "repair" seems to designate an immediately recognizable result: something was damaged, then restored. This obviousness becomes uncertain as soon as what should be recovered is specified. Is it a previous form, an activity, a capacity to withstand a load or a possibility of living with the consequences of the injury? These objectives may coincide, but their identity must not be presumed. Three verbs serve here to specify the questions, without imposing a universal nomenclature. Repairing designates the response to damage whose result is being assessed; restoring specifies the property one seeks to recover; rebuilding concerns the organization one seeks to remake. These ambitions may overlap without being equivalent. A restored activity does not yet describe the organization that makes it possible; an organization resembling the one before the injury does not guarantee the return of that activity. The word "regeneration" likewise requires specifying what would have been rebuilt. These distinctions establish no automatic ranking: for a given objective, a durable compensation could be more useful than a structural resemblance without recovered capacity. One might object that so cautious a definition deprives the word of its usefulness. On the contrary, it gives the word a controllable use. A team can set itself a limited ambition without giving up the study of the other dimensions later. The problem arises when the conclusion silently broadens the objective: a morphological success becomes a functional restoration, then a promise of benefit, with no change in the data.

REPAIRING, RESTORING, REBUILDING

A CONTROLLABLE USE OF THE WORD

8.1.2 The Same Apparent Progress Can Correspond to Two Different Stories

8.1.2.1 Two organizations for the same value

TWO INTERVENTIONS, THE SAME VALUE OF F

Follow an entirely fictional experiment. After a standardized injury, the experimental units are randomly allocated among A, B and a control group subjected to the same procedures, without active intervention. At a set date, A and B give the same mean value of the global measure F, higher than that of the control; all units remain assessable in this first scenario. A complementary measurement nevertheless suggests two different organizations: A preserves more of the initial structure, whereas B is accompanied by a new arrangement. These letters refer to no real treatment or result. Can A be declared superior because it comes closer to the previous state? This preference requires justification. If the new organization durably ensures the activity sought, its difference is not in itself a failure. Conversely, equality of F does not establish that the two tissues possess the same future possibilities. The restoration of a measured result is not the complete identity of the systems that produce it.

A PREFERENCE TO BE JUSTIFIED

8.1.2.2 Restoration and compensation without a prior hierarchy

A further test could ask how each responds to a defined load. This proposal adds a question about a capacity not measured until then; it does not assume that B would be less robust. The experiment must leave both directions of the result open. Without this openness, the distinction between restoration and compensation would become a hierarchy decided before the observation.

SUCCESS ACCORDING TO THE PROJECT

The relevant comparison thus depends on the project. To understand how a structure rebuilds itself, its organization matters directly. To judge a practical capacity, a different organization may suffice. The temptation to turn a legitimate scientific objective into the sole definition of every therapeutic success must be resisted.

8.2 Choosing the Reference and the Model

8.2.1 The Reference Is Not a Past Accessible without Change

8.2.1.1 The initial state is not the expected state

INITIAL STATE, EXPECTED STATE

Comparing the injured tissue with its initial state appears to provide a natural reference. Yet time would have continued to pass in the absence of injury. The initial state and the state expected at the final date are not necessarily the same thing. The first is an observation; the second requires reasoning and

A SPONTANEOUS RISE IN F

a suitable comparison. In the fictional scenario, F rises spontaneously in the group subjected to injury without intervention. A rise in F after A is therefore not enough to attribute that rise to A. The question becomes the difference between comparable trajectories, and not the distance traveled by the treated tissues alone. This distinction is particularly important when the outcome studied can change during the observation.

8.2.1.2 What each control allows one to say

But the control without intervention does not answer every question. It may help to assess the effect of A after the injury without indicating on its own what would have been observed without that injury. Depending on the objective, the effect of the insult, the subsequent changes and the effect of the experimental strategy must be distinguished. Naming them separately avoids

THE DECISIVE COMPARISONS

turning every favorable change into induced repair. One might call for a control for every conceivable circumstance. That would forget that a design must remain feasible. The reasonable choice is to protect the comparisons that are decisive for the conclusion announced. An unresolved limitation can be stated; it must not be erased by a global term such as “recovery.” Entirely fictional scenario. The useful control changes with the property sought. The comparisons in Table 8.1 assume comparable units, the same time points and identical measurement procedures; they are not all necessary in every experiment.

Table 8.1. Choosing the control according to the repair question

Precise question	Comparison to organize	What it informs
Does A modify F after the injury?	Injury + A versus injury + procedural control, with random allocation.	Effect of strategy A on F at the time point chosen.
Is the level expected without injury recovered?	Add a contemporaneous group without injury, with an appropriate sham procedure.	Difference from the contemporaneous reference level; to establish equivalence, set a margin and examine the uncertainty.
Do A and B preserve the same functional reserve?	Subject the groups to a predefined, identical load after the measurement of F.	Response to this load, distinct from the measurement at rest alone.
Does the result persist after a delay in administration?	Compare delay strategies defined from the same initial time, with suitable allocation.	Effect of timing from that initial time, also accounting for losses before administration. F is measured only where it remains defined; a result restricted to the units still eligible answers a different question.

8.2.2 A Model Can Be Instructive Thanks to What It Simplifies

A CONSTRUCTED SITUATION

A standardized injury is a constructed situation. It may allow interventions to be compared while limiting certain sources of variation; it does not necessarily reproduce the whole history of a disease. This construction becomes a problem only if the properties omitted are necessary to the inference sought. Nor does a general resemblance to clinical practice guarantee that the comparison is well controlled. In this example, suppose that A is introduced immediately after the injury. This condition allows an early action to be studied. If the project aims at administration after a delay, the time of introduction becomes a property to be varied, not a detail that can be omitted from the report. The initial model was not wrong; the next question goes beyond its domain. The quality of a model must therefore be related to a use. A design may help to isolate one step and be less suited to the evaluation of a complete strategy. This judgment establishes no definitive hierarchy between models. It organizes their succession: which uncertainty does the first reduce, and which must the next address?

A MODEL RELATED TO A USE

8.3 The Scope of a Success

8.3.1 A Local Success Can Conceal a Narrowing of the Result

8.3.1.1 The units that are no longer assessable

THE UNITS THAT REMAINED ASSESSABLE

Return to F, the global measure improved by A and B. In a variant where some units are no longer assessable, retaining for analysis only the tissues still assessable at the final date risks comparing groups selected differently by their course. A better value among the remaining units does not necessarily describe a better outcome for all the units initially studied. The problem is not always corrected by arbitrarily assigning the worst value to the missing units. It is first necessary to understand why the measurement is missing and whether F remains definable in those situations. When a system can no longer function, the absence of F is not necessarily a very low value of the same quantity. It may reveal that the chosen endpoint describes only part of the relevant outcomes.

A MISSING MEASUREMENT

8.3.1.2 What is the success a success of?

THE SUCCESS OF WHAT

This observation calls for a concrete philosophical question: what is a success a success of? Of the mean of the available observations, of the preservation of the units or of a capacity attributable to the initial set? A measurement can be accurate while answering a narrower question than the one asked by whoever reads the study. The transparency of the denominator therefore belongs to the meaning of the result. The objection that would render every partial indicator useless must also be resisted. A limited measurement may be excellent for tracking one determined dimension. It becomes misleading when it collects, for convenience, the authority of a complete assessment. Complementary results serve to illuminate these differences; they must not be added only when they make the picture more favorable.

8.3.2 Acknowledging a Limitation without Losing the Result

ASSERTING WITHIN ITS SCOPE

A final objection would be to see in these reservations a way of avoiding any commitment. The answer is to make the conclusion as assertive as its scope allows. "A improves F in the comparison studied" can be an important proposition. There is no need to inflate it into "A repairs the heart" to give it value. The limitation must, however, remain open to discussion: which new comparison would genuinely license the more ambitious statement?

8.4 Texts Put to the Test

8.4.1 Serhan and the Physiology of Resolution

8.4.1.1 Situating the Result

Serhan, 2014

This single-author review articulates the specialized lipid mediators of resolution: lipoxins, resolvins, protectins and maresins. It links experimental results to still limited human observations and distinguishes their identification from the demonstration of their physiological importance. It opens a pharmacological strategy without establishing a clinical cardiovascular benefit.[1]

8.4.1.2 Critical Reading

DETECTED, ACTIVE, NECESSARY, USEFUL

The phrase "pro-resolving mediator" contains several claims that must be separated. A molecule may be detected in a sample, exert an action when added to a preparation, normally take part in that action within the organism and become useful when administered as a treatment. These four propositions call for different experiments. Conflating them would give a chemical observation the scope of a therapeutic trial, or a pharmacological response that of proof of endogenous function. To test a physiological mediation, one would ask in particular whether the compound is present at the relevant place and time, in a quantity compatible with the proposed effect. Further questions are whether reducing its availability modifies that effect, and whether a controlled restoration reinstates it. Each step has its difficulties. An intervention on an enzyme may affect several products; a concentration in plasma may poorly represent the immediate vicinity of a cell; a synthetic product may reach an exposure foreign to the spontaneous situation. This reasoning refutes no result as a matter of principle: it specifies the conclusion that a result can support.

8.4.1.3 Immunological Reading

PREVENTING ARRIVAL OR EASING DEPARTURE

A thought experiment can be proposed: two interventions reduce the number of inflammatory cells in a tissue to the same extent. The first prevents their arrival; the second eases their clearance once their function has been accomplished. An identical final photograph would not necessarily show the same course, nor the same consequences. The flows, the fate of the cells and the state of the tissue would have to be observed to distinguish these situations. The final result should then specify what has ceased, what has been cleared and which activity has been recovered. The rigor of the measurement then precedes that of the interpretation. To study a candidate lipid, its identity, the sampling conditions, its stability and the performance of the assay should be documented. An analytical uncertainty would not be offset by the elegance of the biological narrative. Conversely, a flawless chemical identification would still leave the question of function open.

THE RIGOR OF THE ASSAY

8.4.1.4 Cardiological Reading

PRECURSOR, MEDIATOR, DRUG

The nutritional precursor, the mediator formed within a tissue and the pharmacological agent administered should also be distinguished. Assuming their equivalence would amount to erasing the steps of transformation, distribution and exposure. A hypothesis about resolution is therefore not enough to choose a supplement, a dose or a patient. The review thus provides a framework for interpreting experimental work on the exit from the inflammatory reaction, with a human scope still limited at its date. It makes plausible an intervention on several coordinated functions, without establishing a universal resolution deficit in heart disease. To specify the therapeutic proposal, robust measurements of endogenous mediation would first be needed, then a comparison demonstrating a benefit relevant to the heart and its function.

8.4.1.5 Integrative Clinical Reading

THE REQUEST FOR A SUPPLEMENT

Bringing together the nutritional precursor, the local mediator and the pharmacological agent may prompt an immediate request for supplementation. This review requires these objects to be distinguished before a personal expectation is discussed. An integrative approach does not consist in accumulating products in the name of a shared mechanism; it examines what each proposal claims to modify and which data support that effect. The history of what is actually being taken, and the reasons for continuing, then become elements of the dialogue, without this text determining any nutritional prescription.

8.4.2 Nahrendorf and the Timetable of Healing

8.4.2.1 Situating the Result

NAHRENDORF, 2007

After permanent coronary ligation in the mouse, the authors describe a succession of Ly6C-high and then Ly6C-low monocyte populations. They combine kinetics, cell transfers, receptor deficiencies and depletions at distinct times. Histological analysis supports a complementarity in healing. The follow-up

used is not a genetic tracing demonstrating the impossibility of a cellular transition.[2]

8.4.2.2 Critical Reading

CLODRONATE AND ITS DATES

The succession observed is not enough to define what each population contributes to healing. Here, clodronate administered at different dates does not achieve a perfectly specific suppression of a single subpopulation. The intervention says more about a temporal participation of phagocytes than about the exclusivity of a lineage. The difficulty can be put as a general example: preventing a first operation can impair the result of a second without acting directly on its agents. A late defect observed after an early intervention therefore does not automatically locate the mechanism at the time of measurement. The chain of consequences must be reconstructed, possibly with an intermediate restoration planned, and then what belongs to the initial damage must be compared with what belongs to its repair. The comparison with atherosclerotic animals constitutes another test, but it modifies several properties at once. It does not allow monocytosis to be isolated as the sole cause. In this protocol, age also differs between the atherosclerotic animals and the ordinary mice studied. Presenting this contrast as a pure experiment on monocyte number would erase part of the design.

A LATE DEFECT, AN EARLY CAUSE

MONOCYTOSIS AND AGE CONFOUNDED

8.4.2.3 Immunological Reading

ENTRY, SURVIVAL OR EXIT

A classification by markers describes the cells accessible to one method at one moment. To know whether a cell subsequently changes category, a follow-up suited to that question is required. Differences in accumulation may support differential recruitment without excluding any local maturation. Likewise, fewer cells in a tissue may correspond to reduced entry, diminished survival or increased exit. Choosing between these explanations requires further observations. Interpreting a cell transfer also requires consideration of the environment receiving the cells. Injecting the same number of labeled cells in two situations does not mean that they encounter the same recruitment possibilities. The endogenous cells and the state of the tissue take part in the result. A relative frequency and an absolute quantity thus answer distinct questions; reading them together avoids calling a population dominant when only its proportion increases.

RELATIVE FREQUENCY, ABSOLUTE QUANTITY

8.4.2.4 Cardiological Reading

A TIMETABLE THAT CANNOT BE EXTRAPOLATED

For the clinician, the issue is to think through the succession of the tissue's needs before imagining an intervention. A research objective should specify the lesion considered, the time of treatment and the benefit expected. The timetable of a permanent ligation model is not a therapeutic schedule applicable to a person managed with reperfusion. The temporal perturbations therefore show that the monocyte system takes part in healing in this model. The possibility that the coordination of functions matters as much as the abundance of cells deserves to be tested by a follow-up of cell fates and by more specific interventions. This extension would make it possible to distinguish recruitment from maturation, before any extrapolation of the timetable to other lesions or to clinical practice.

8.4.2.5 Integrative Clinical Reading

DATING DECISIONS

The experimental timetable draws attention to a difficulty in the care pathway: information that is accurate at one moment may be taken up later as if the situation had remained the same. Handover between teams should therefore preserve the date, the objective and the context of a decision. This study indicates no human care schedule; it provides a reason to ask whether the same phase is still being discussed. Clinical continuity requires a memory of changes, as much as a preservation of reports.

8.4.3 Aurora and the Conditions of a Myocardial Reconstruction

8.4.3.1 Situating the Result

AURORA, 2014

After coronary ligation in mice on the first postnatal day, depletion with clodronate liposomes impairs myocardial reconstruction, increases fibrosis and decreases the function and vascularization measured. Mitotic labeling of cardiomyocytes studied at seven days is nevertheless not significantly different. Comparison with mice operated on at fourteen days reveals differences in the population and transcriptional profile of mononuclear phagocytes. The study demonstrates neither the abolition of all vessels nor a general absence of regulation of the cardiomyocyte cycle by these cells.[3]

8.4.3.2 Critical Reading

TISSUE AND MITOTIC LABELING IN CONTRAST

The contrast between the tissue results and the single mitotic labeling delimits the question here. The perturbation impairs several dimensions of reconstruction, whereas no significant difference is found on this marker at seven days. Their relationship must therefore be studied without thereby declaring proliferation unchanged. A nonsignificant result on one marker at one moment does not demonstrate an invariance of the whole process. To explore this question, several observation times could be proposed, together with measurements able to distinguish progression through the cycle, completed division and survival

AGE VARIES EVERYTHING

of the cells produced. The comparison between two ages raises another difficulty. No difference in repair can be attributed to immunity on the sole ground that immunity also differs. Age varies many properties of the organism at once. To separate those of the cells from those of the environment, controlled exchanges of populations or more targeted perturbations could be designed, while checking the specific effects of these manipulations. The relevant question becomes: what share of the difference persists when the environment is made comparable?

8.4.3.3 Immunological Reading

A DEPLETION THAT EXTENDS BEYOND THE HEART

Clodronate depletion is not a manipulation confined to the macrophages of the heart alone. This systemic reach must remain visible when the effect is named. An action within the organ and a change in its cellular supply or its environment may both take part in the same result. They would have to be distinguished before an exclusive role is attributed to a local subpopulation. The expression profile proposes candidates; it is not enough to identify the

mediators of reconstruction. Each candidate would have to be examined at the level of the protein and of its activity, then through an appropriate perturbation. Determining whether it contributes to the effect, whether it is an indispensable condition in this setting or whether adding it can produce part of the response are three distinct questions.

8.4.3.4 Cardiological Reading

TOWARD THE ADULT HEART

Moving to the adult human heart does not consist in seeking only the same cellular signature. The reconstruction aimed at must be defined, and it must then be checked whether the conditions on which the neonatal result depends are present or can be modified. The data in this article do not show that reproducing an expression profile would suffice to obtain that reconstruction. The perturbation therefore impairs the neonatal recovery studied, without explaining on its own why this capacity differs with age. A direct or indirect contribution to vessel formation remains plausible. To adjudicate, the age of the cells, their environment and the various components of repair would have to be distinguished. It is this dependence on precise conditions, rather than an isolated neonatal signature, that would have to be understood before any extrapolation to the adult heart is contemplated.

8.4.3.5 Integrative Clinical Reading

AGE, A CONDITION AND NOT A WORTH

The age of the model is a condition of the result, not a ranking of the worth of persons according to their supposed potential for repair. The neonatal finding opens a precise biological question; its communication must announce neither absolute impossibility in the adult nor available reconstruction. To relate this research to lived experience, the return of a structure, the improvement of a function and the resumption of an important activity must be distinguished. A person might value one of these gains without requiring that it be described as complete regeneration.

8.4.4 Vagnozzi and the Revision of an Explanation

8.4.4.1 Situating the Result

VAGNOZZI, 2020

In the murine therapeutic arm after ischemia-reperfusion, the injections are performed one week after the injury. Bone marrow mononuclear cells, cardiac cells selected for c-Kit and zymosan improve function measured by echocardiography. Bone marrow cells rendered nonviable also produce a benefit. Cyclosporine abolishes the gain associated with the marrow cells or with zymosan; depletion with clodronate liposomes abolishes the gain associated with the marrow cells. The authors propose an immune and matrix explanation rather than a replacement of cardiomyocytes by the injected cells.[4]

8.4.4.2 Critical Reading: an experiment must adjudicate between explanations

8.4.4.2.1 Controls that shift the active element

DEAD CELLS THAT WORK

The dead-cell control is particularly instructive. If the effect required their survival and then their transformation into muscle, their destruction should block that explanatory route. Finding a benefit therefore requires reconsidering what is taken to be the active element of the intervention. This does not mean that living and dead cells necessarily act in identical ways; two interventions may reach the same result by different routes. The experiment narrows the domain of an exclusive explanation, and then calls for the comparisons that adjudicate between the others. Zymosan adds a noncellular intervention to the reasoning. Its value does not lie in one more vote in favor of an appealing story. It lies in the possibility of eliciting part of the phenomenon without the substrate to which the initial explanation tied it. To conclude that there is a shared mediation, the perturbations, their local consequences and the function measured must still be connected, and the competing explanations examined.

8.4.4.2.2 Homogeneity is not random allocation

HOMOGENEOUS IS NOT RANDOMIZED

One design limitation deserves to be discussed directly: the animals are not randomized, the authors invoking their relatedness and genetic homogeneity; several analyses are performed blind, but not the whole set of interventions and echocardiographic acquisition. Yet homogeneity does not mean guaranteed interchangeability at the moment of allocation. The order of the interventions, the preparation or unmeasured differences may still be distributed unevenly. Blinding protects certain observations from the influence of expectation; it does not replace random allocation. This reservation does not destroy all the comparisons, but it forbids treating their structure as equivalent to that of a randomized experiment.

8.4.4.3 Immunological Reading: removing a condition does not identify the whole chain

A MORE DISCRIMINATING RESTORATION

The loss of a benefit after an immune perturbation strengthens the hypothesis that the immune environment takes part in producing it. It is not enough to isolate a single population or its precise action. An extensive intervention may modify several conditions necessary to the result. The next question therefore becomes: which element must be restored, where and when, for the effect to return? A targeted restoration would be more discriminating than a second general perturbation producing the same failure. The distinction between marker and function also remains decisive. Deleting a receptor throughout the organism is not equivalent to removing only the cells that express it in one region of the heart. The number of cells, their recruitment and their activity may then raise different questions. The temptation to convert every genetic experiment into a perfectly specific depletion experiment must be resisted.

8.4.4.4 Cardiological Reading: an improved function does not name its mechanism

AN IMPROVEMENT, NOT A REGENERATION

The study brings together measurements of function, the matrix and tissue mechanics. This combination makes it possible to interrogate an improvement

without translating it immediately into "regeneration." Conversely, a negative search for proliferation under the conditions examined is not a general demonstration that producing new cardiomyocytes is impossible. The issue is to determine which contribution explains the result observed, at the scale at which it is measured. Finally, clinical inference must preserve the timing and the route of administration. A local stimulation at a precise experimental moment does not justify a general inflammatory activation in a person with heart failure. The gain in knowledge is already considerable without this extension: a result can outlive the abandonment of its initial explanation. What would advance the case would be a replication better protected against allocation bias, then interventions allowing the immune and mechanical contributions proposed to be separated.

A RESULT THAT OUTLIVES ITS EXPLANATION

8.4.4.5 Integrative Clinical Reading

AN EXPLICIT CORRECTION

Changing the explanation of an effect does not mean that the hope placed in the research was meaningless. But, for a person to whom a cell therapy was presented as a replacement of muscle, the correction must be explicit. This work teaches how to separate what was observed, the biological story that explained it and what remains expected in humans. The quality of scientific guidance then lies in the continuity of that explanation, rather than in the silent replacement of one promise by another.

8.4.5 Horckmans and the Price of a Cellular Suppression

8.4.5.1 Situating the Result

HORCKMANS, 2017

In mice after permanent coronary ligation, Horckmans and colleagues observe a deterioration of cardiac function after neutrophil depletion. The macrophage profile changes; the increase in so-called M2 markers does not correspond to improved repair. Supplementation and secretome immunodepletion experiments implicate neutrophil gelatinase-associated lipocalin (NGAL) in the expression of MerTK and the clearance of apoptotic cells. An exploration after ischemia-reperfusion finds a change in MerTK, without reproducing the whole functional demonstration of the permanent model. The text, published online in 2016 and then in the 2017 volume, thus invites examination of the functions suppressed together with the population targeted.[5]

8.4.5.2 Critical Reading

A depletion indicates what becomes of the system when the population targeted is reduced. It may modify the recruitment, survival or multiplication of other cells. Attributing each consequence to a precise action therefore requires adjudicating between these mediations. Reintroducing a secreted product narrows this inquiry. If a feature is restored, that product may take part in maintaining it. But the scope of the rescue depends on what was measured: restoring a receptor or a phagocytic capacity is not the same as normalizing the whole of remodeling. The same links would have to be followed, through to the function of the organ, to support that broader conclusion. The dose and the site of administration also matter: an experimental compensation does not necessarily reproduce the physiological distribution of the mediator.

NGAL AND THE SCOPE OF THE RESCUE

8.4.5.3 Immunological Reading

A LABEL VERSUS A FUNCTION

The conceptual interest lies in the disagreement between a label and a function. Calling a profile anti-inflammatory or reparative is not enough to demonstrate what it accomplishes. A cell may express certain expected markers and fail at a task indispensable to the tissue. Classification becomes fruitful when it organizes functional tests; it becomes misleading when it replaces them. To specify the mechanism, the cell that releases the signal, the cell that receives it and the action modified would have to be identified. A perturbation restricted to one lineage would adjudicate between the possible sources; an intervention on the receiving cell would test the relay. These proposed experiments are not presented as already performed in this article.

8.4.5.4 Cardiological Reading

THE TIMETABLE BEFORE THE VALUE

The teaching consequence is to build the timetable of the intervention before assigning it a favorable or unfavorable value. Limiting an early lesion and allowing a robust scar to form are two objectives that may require different conditions. Comparing them requires specifying the delay, the reperfusion, the territory involved and the horizon of assessment. One and the same cell population has no therapeutic value independent of these conditions. These observations justify neither increasing neutrophils nor administering NGAL to patients. They help to ask whether an intervention could reduce a damaging action while preserving a useful function. Assessment will have to distinguish a change in mechanism, a transient functional improvement and a durable benefit for the person.

8.4.5.5 Integrative Clinical Reading

STRENGTHENING OR WEAKENING IMMUNITY

Suppressing a population may remove a damaging activity and a useful contribution together. For the person, this tension invites going beyond the spontaneous sorting of treatments into those that strengthen and those that weaken immunity. An intelligible explanation must specify the function sought and those that are to be preserved. This preclinical work does not allow that human balance to be calculated. It does, however, help to formulate a care objective more intelligible than a global decrease: improving a situation without confusing reduced intensity with recovered function.

8.4.6 Zacchigna and the Transmission of a Reparative Activity

8.4.6.1 Situating the Result

ZACCHIGNA, 2018

This study combines cardiomyocyte cultures, murine pregnancy and experimental infarction. The authors examine the depletion or administration of regulatory T lymphocytes, their conditioned media and the expression of a set of secreted factors. They report markers of cell cycle progression and favorable functional results under certain conditions. The presence of Aurora B at the intercellular bridge is studied notably in neonatal culture; it does not constitute a longitudinal count of new adult cardiomyocytes after infarction. The authors acknowledge that the necessity of the factors proposed is not demonstrated.[6]

8.4.6.2 Critical Reading: keeping the address of each piece of evidence

cell, age, environment, intervention

The methodological originality lies in the circulation of a hypothesis across several experimental setups. This circulation strengthens its plausibility if each experiment answers a precise objection. It becomes misleading if a result obtained in one setup is silently attributed to all the others. Each piece of evidence must therefore carry its address: which cell, at which age, in which environment and after which intervention? The word "proliferation" requires particular attention here. A signal of DNA synthesis, an index of mitosis, a cell separation and a durable increase in cell number do not answer exactly the same question. Their convergence matters, but it does not allow the last observation to be replaced by the first. Critical reading avoids two excesses: ignoring observations that go beyond DNA synthesis, or carrying those observations over to a situation where they do not have the same demonstrative force. In the neonatal cultures, the control receives the usual medium of regulatory T cells, containing interleukin 2 (IL-2): this comparison avoids attributing to the conditioning, without a control, an effect of the medium itself. This discipline also concerns replicates. Counting more cells refines the description of one preparation; it does not create as many independent preparations. The precision of the observation and the diversity of the experimental units contribute to robustness in different ways. A convincing photograph must thus remain tied to the whole set of samples and to the quantification rule, rather than becoming the autonomous proof of the mechanism.

the degrees of proliferation

a control containing IL-2

8.4.6.3 Immunological Reading: reproducing an effect does not identify all of its mediation

acting without contact

The experiment with a conditioned medium raises a fruitful question: can a transmissible activity act without direct contact with the cell that produces it? A positive answer narrows the inquiry, without determining on its own which substance acts, on which initial target or with which accessory contributions. Moving next to selected factors constitutes another intervention, whose composition and concentrations do not necessarily reproduce natural secretion. The active mixture retained combines six factors: Cst7, Tnfsf11, interleukin 33 (IL-33), Fgl2, Matn2 and Igf2. The cardiac experiment has them expressed by vectors derived from adeno-associated virus serotype 9 (AAV9), compared with a vector encoding no protein. This plurality invites examination of several coordinated functions, rather than necessarily searching for a single reparative cytokine. The decisive distinction is between experimental sufficiency and physiological necessity. Producing an effect with a set of molecules shows a possibility under the conditions tested. To establish that these molecules explain the natural action of regulatory T cells, it would be necessary in particular to interrupt their contribution, preserve the other functions as far as possible and observe what disappears. Conversely, too broad a perturbation could suppress several pathways at once and make attribution undecidable. A mixture more active than each of its components does not automatically demonstrate a synergy. Synergy requires an explicit reference to the effect expected from their combination. This precaution preserves the full exploratory value of the mixture: it distinguishes a promising strategy from an already quantified mechanistic interaction.

six factors, sufficiency and necessity

8.4.6.4 Cardiological Reading: separating the functional benefit from its explanation

number, quality, coordination

A heart can function better by several routes. A functional improvement is therefore not enough to measure the specific contribution of new cardiomyocytes. To link these two propositions, it would have to be shown that cell production precedes the benefit, actually contributes to the benefit and results in a useful integration. An additional cell must still take part in an organized tissue; number, quality and coordination are different properties. Pregnancy, neonatal tissue and the injured adult heart are not interchangeable versions of one and the same state. Bringing them together may uncover a biological possibility; it does not directly yield the dose, the timing or the safety of a human treatment. The clinical question then becomes precise: which components of this possibility remain active in the heart to which it would be applied? The contribution of the text is to make this inquiry testable. It licenses the search for a reparative function of regulatory T cells beyond their immunological designation alone, while requiring three propositions to be separated: transmitting an activity, explaining a benefit and rebuilding an organ.

pregnancy, newborn, injured adult

8.4.6.5 Integrative Clinical Reading

the word "regeneration" for the patient

Moving from particular cultures or tissues to an adult person requires recovering the context in which the promise is heard. In a patient's imagination, the word "regeneration" may announce a complete return to the previous state; the study describes much more situated observations. The integrated explanation must preserve their interest while asking which capacity would be recovered, for how long and under which constraints. The person's own pathway supplies these questions of value; cell cycle markers do not answer them alone.

8.4.7 Prabhu and Frangogiannis: Repairing Is Not Regenerating

8.4.7.1 Situating the Result

Prabhu and Frangogiannis, 2016

This 2016 review examines repair after infarction, from inflammation through to maturation of the scar. It distinguishes replacement by a scar matrix from myocardial regeneration. It gathers chiefly experimental arguments, with human observations and trials. The authors stress effects that depend on time, compartment and context; they discuss failures of clinical translation. Their fifth figure proposes two explanations for late inflammation: incomplete resolution or new activation. The durations given for the early phases of this scheme concern the mouse. The text demonstrates neither a neonatal regeneration strategy nor a new therapeutic benefit.[7]

8.4.7.2 Critical Reading

a reconstructed history

A synthesis can give the impression of a continuous film when it is in fact bringing together different observations. The difficulty is to recognize the links: were the steps followed within the same system, or does their succession result from a reconstruction? This question does not cancel the value of the model. It distinguishes a directly observed history from an argued history,

MAINTENANCE OR RESUMPTION

each of whose transitions must keep its own support. Imagine two fictional biological trajectories. In one, an inflammatory activity declines and then rises again; in the other, it remains durably high. If only an identical late assay is available, these two trajectories remain compatible with the same observation. The name "persistent inflammation" would not adjudicate between them. The right question concerns the information missing between the dates, and then the observations capable of differentiating maintenance from resumption. The model becomes fruitful when it organizes this difference instead of concealing it under a continuous curve. A second difficulty lies in the meaning of success. If one study seeks to preserve a structure and another to restore the cells lost, their results are not judged on the same scale. A repair may meet the first objective without attaining the second. Presenting this difference as an absolute failure would devalue an achievement; erasing it would turn a limited achievement into a promise larger than its demonstration.

8.4.7.3 Immunological Reading

BEYOND M1 AND M2

The review declines to reduce the macrophages of repairing tissue to two fixed populations, M1 and M2. It also acknowledges that certain general mechanisms of resolution remain little studied in the infarcted heart. These two qualifications limit the reading of its diagrams: a category summarizes properties, and an arrow may represent a link whose cardiac support remains to be specified.[7] Describing a cell as reparative does not yet make it possible to know which operation it accomplishes, which result depends on it or at what moment its presence becomes useful. A classification should help to pose these questions, without incorporating their answers into the label. This requirement illuminates the link with resolution as work to be accomplished (see 7.3.5). Decreasing a damaging activity and restoring an insufficient operation are distinct objectives. They may be compatible, but they may also require different comparisons. The idea of resolution therefore deserves better than a favorable value attached to the word: it must specify what the tissue begins to accomplish again and how that change is recognized.

DECREASING OR RESTORING

8.4.7.4 Cardiological Reading

"LESS" IS NOT "BETTER"

Consider a fictional result: an intervention reduces a quantity of scar tissue, with no information on the strength of the wall or on subsequent function. The figure may be accurate and still leave its clinical meaning open. "Less" does not automatically become "better" when the quantity measured takes part in several functions. It must be asked which objective justified this measurement and which complementary consequences are needed to interpret it. This distinction avoids gathering all results under the term "recovery." Preserving a geometry, improving a function and avoiding a hospitalization are not three interchangeable ways of measuring the same event. A research program gains from announcing how they are articulated before the results. If these dimensions diverge, that divergence becomes a lesson of the program, rather than a detail removed from its narrative.

8.4.7.5 Integrative Clinical Reading

WHAT IMPROVES, WHAT REMAINS

For the person, the word "repair" may evoke a return to the previous state. The dialogue must therefore specify what improves and what remains transformed, without deciding on their behalf the value of that improvement. A favorable measurement exhausts neither the capacities recovered nor the constraints that persist. Conversely, a daily activity that has become possible again remains an achievement, even if it does not erase every abnormality. The integrative question is that of a shared objective: what does one hope to preserve, to recover or to make possible again? The different participants must be able to relate their observations to that objective, while keeping their units and their limits. A distinction between biological processes does not fix, for each person, what would count as sufficient recovery. That appraisal requires relating the capacity recovered to the shared objective.

8.5 Worked Exercises

8.5.1 Exercise on Depletion and Restoration

8.5.1.1 Problem

Entirely fictional situation. The following experiments and their values are invented in order to learn how to interpret a perturbation; they reproduce none of the studies discussed in 8.4. In a model of neonatal cardiac injury, a team studies an immune population P. At thirty days, it measures the fraction of the initially injured zone occupied by organized contractile tissue: Control: 80%. Depletion of P: 35%. Depletion then reintroduction of P: 70%. In adult animals subjected to an injury comparable in its experimental definition, another experiment gives: control, 15%; supply of P, 18%. Sample sizes, dispersions and confidence intervals are not provided. The investigators announce: "P is the cell that regenerates the heart; transferring it is sufficient to restore this capacity." Questions. Which part of this claim goes beyond the observations? What does the reintroduction teach? Which additional result would help to interpret the adult experiment?

A CLAIM THAT GOES BEYOND THE DATA

8.5.1.2 Reasoned Answer

8.5.1.2.1 The contribution of P in the newborn

A NEONATAL CONTRIBUTION

The decrease after depletion suggests a contribution of P in the neonatal context, subject to the specificity and validity of the design. It does not demonstrate that P itself produces the contractile tissue. P could support another population or modify a local condition necessary to its reconstruction. The reintroduction is an interesting restoration argument: it descriptively narrows the gap with the control. However, 70% and 80% are neither identical nor shown to be equivalent. The precision of these estimates must be known, and it must be checked that the supply does restore P without simultaneously modifying other major factors. This experiment strengthens one interpretation; it does not automatically cancel all the competing explanations.

70% IS NOT 80%

8.5.1.2.2 The adult experiment and the defensible wording

REACHING THE TARGET, PERSISTING

In the adult, only a descriptive gap of three percentage points is presented here. It cannot be taken to show an established effect, an absence of effect or an equivalence. Even a precisely estimated null effect would not necessarily refute the neonatal role: a condition may be useful when other conditions are met and insufficient when they are lacking. Before explaining this contrast by a difference in environment, it would have to be checked that the cells supplied reach the target, persist and exert the expected activity. An absence of effective exposure and a resistance of the recipient tissue are not the same failure. Functional measurements distinct from the tissue endpoint alone, and a follow-up suited to durability, would also be necessary. The defensible wording is therefore: these observations explore the contextual contribution of P. They establish neither an autonomous regenerative capacity nor a clinical transfer strategy.

Synthesis

Repair is not a simple measurable return to an unchanged past. It involves a choice of properties to be restored, a horizon and a comparison. A different structure may support a useful activity; a recovered activity may leave other capacities open. The oppositions between restoration and compensation must therefore not become automatic judgments on the value of the result. The fictional A/B/F example shows why the initial comparison calls for several extensions: the course without intervention, a reference without injury when the question requires it, the response to a load and the fate of all the units involved. The point is not to measure everything in every study, but to recognize the exact question each result answers. A model becomes fruitful when its simplifications are understood and when its major limitations lead to new comparisons. This discipline makes it possible to welcome an experimental advance without crediting it in advance with a clinical success. A sound conclusion keeps together what has changed and what remains to be tested.

Open Questions

In what situation could a different organization constitute a repair sufficient for the objective pursued? Which comparison distinguishes a spontaneous improvement from an effect of the intervention? Is an equality of the measure F enough to declare capacities restored? Finally, what becomes of the judgment of success when some units no longer allow F to be measured?

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PART III

Testing Hypotheses through Treatments

An immunological hypothesis changes status when the pathway it designates is modified in patients. Treatment then becomes a test: a benefit makes the hypothesis more credible, while an absence of gain or a harm must take something away from it. Yet this test never bears on a bare idea. It passes through a chosen population, a timing of intervention, a comparator, an endpoint and a duration; its result remains attached to these conditions. The same requirement applies to the favorable trial whose conclusion one would like to extend and to the negative trial for which the weakened proposition must be named. It also applies to the result obtained in another disease that one would like to apply to the heart, and to the diagnosis expected to designate the treatment. Everywhere, a biological response, a clinical result and a decision for a person require evidence that is connected, but distinct.

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9

Coronary Inflammation Put to the Test in Trials

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The idea that inflammation contributes to coronary events has led to trials in patients: a pathway is modified, then events are counted. The answers do not agree. A targeted intervention reduces a composite endpoint, with more fatal infections. Colchicine appears effective in two trials and remains without demonstrated benefit in a third. Low-dose methotrexate provides no demonstrated benefit. Faced with this disagreement, two shortcuts are tempting: to regard the hypothesis as confirmed by the successes or refuted by the failures, then to rank the treatments by their presumed strength. Yet each result belongs to a population, a time of initiation, an endpoint and a duration. Before explaining a divergence, it is necessary to know what each trial actually compared.

9.1 Reading a Trial under Its Own Conditions

9.1.1 What the Comparison Entails

STRATEGIES, NOT A VERDICT

Coronary trials (see 9.3) do not directly submit "inflammation" to a verdict. They compare strategies: an intervention, a defined group of people, associated care, a duration and outcomes. The biological hypothesis explains why this comparison is worth undertaking. The result must then retain the conditions under which it was obtained. An objection is legitimate: if each conclusion remains attached to its design, how can anything be learned beyond the participants studied? Generalizing consists precisely in deciding which properties of the design matter for the effect and which may vary. The conclusion must not remain a prisoner of the trial; its extension must become explicit reasoning. The formula "same disease, same treatment" is not enough to replace that reasoning. Consider two entirely fictional trials. A introduces drug X after a recent event. B introduces it in a stable situation, after a preliminary phase that excludes certain people. The shared name of the drug erases neither the timing nor the selection. B may answer a particular decision precisely without answering the decision addressed by A. Calling B less representative overall teaches less than naming the population and the strategy to which its result applies.

TWO FICTIONAL TRIALS, A AND B

9.1.2 Reading What Makes Up the Result

DEFINITIONS AND COUNTING RULES

Suppose that A and B combine several events in their primary endpoint. Before comparing their estimates, the definitions and counting rules must be set side by side. The same label may conceal different events or procedures. An analysis of the first event and an analysis of recurrent events do not bear on exactly the same outcome. Imagine that a favorable overall result is driven mainly by a less serious component, while a major component remains imprecisely estimated. The conclusion on the primary endpoint may remain justified; it does not establish a reduction in each component. It must be possible to see what makes up the result, without reconstructing an artificial total by adding events that may concern the same people. Conversely, demanding separate significance for each component would lose the meaning of the primary analysis. The data do not necessarily offer that precision. The reading therefore keeps two levels: the overall result specified by the trial and the information that allows its clinical significance to be assessed. The second reading refines the first; it should neither arbitrarily replace it nor be hidden behind it.

9.1.3 Benefit and Harm in the Decision

CONSEQUENCES WITHOUT A COMMON CURRENCY

FACTS, VALUES AND WEIGHTING

If X prevents certain events but increases another type of complication, subtracting the numbers is not enough to decide. The consequences may affect the same people, differ in severity or occur at different times. They do not become a common currency merely because they can be counted. This evaluation could be criticized for introducing preferences into knowledge that ought to be objective. But distinguishing facts from values does not mean suppressing values. The available estimates, their uncertainty and the importance given to each consequence must be made visible. Two people may agree on the estimated effects without judging their balance in the same way. This discussion does not justify an improvised balance sheet. It calls for defined consequences, comparable time horizons and a presentation that does not relegate harms to an incidental note. Clinical judgment relies on the results; it is not reduced to the sign of the main contrast.

9.2 Comparing Trials That Do Not Ask the Same Question

9.2.1 A Disagreement Deserves an Investigation

A DIFFERENCE IN SIGNIFICANCE

AN EXPLANATION THAT PREPARES THE TEST

Suppose that A is judged favorable and B less convincing. The first step is to specify what differs: estimates, uncertainties, endpoints or thresholds for drawing conclusions. A difference in significance does not demonstrate a difference in effects. Explaining an apparent disagreement before defining it can manufacture a problem that the data do not pose. If a relevant divergence persists, timing or selection may supply hypotheses. They do not automatically become its causes. A and B may differ on several points at once; choosing after the fact the point that makes the most satisfying story does not isolate it causally. Must interpretation then be abandoned until a new trial is done? No: an explanation can be proposed, together with what makes it plausible. It is also necessary to state which comparison would distinguish it from its competitors. A hypothesis arising from a disagreement is fruitful when it prepares a new test; it merely protects the narrative if no result could weaken it.

9.2.2 A Research Step Is Not an Incomplete Prescription

FICTIONAL TRIAL C AND SIGNAL M

Some trials first examine a biological response or a property of a lesion. Their value does not require them to answer the entire therapeutic question at once. In a third fictional trial, C, drug Z modifies signal M. Depending on the design, this result may shed light on a dose, an exposure or target engagement. The research decision thereby better informed must be named. “Promising” then means promising for a specific step. A change in M does not by itself establish the relationship with the clinical events that are to be prevented. It may reduce one uncertainty while leaving the uncertainty about benefit entirely open. Recognizing this limited contribution also avoids devaluing a trial because it lacked the mission attributed to it after the fact. A useful comparison therefore begins with the questions specific to each trial. It retains the differences in strategy, then examines the decision informed by each result and the comparison that is still missing. An indirect ranking of treatments would erase these differences before they had been understood.

9.3 Texts Put to the Test

9.3.1 CANTOS and the Conditions of Success

9.3.1.1 Situating the Result

CANTOS, canakinumab and elevated CRP

CANTOS randomizes 10,061 patients with a history of myocardial infarction and a high-sensitivity C-reactive protein (CRP) level ≥ 2 mg/L. Three doses of canakinumab are compared with placebo. The primary endpoint combines nonfatal myocardial infarction, nonfatal stroke and cardiovascular death. For 150 mg every three months, it yields a hazard ratio (HR) of 0.85, with a 95% confidence interval (95% CI) of 0.74 to 0.98, over a median follow-up of 3.7 years. Only this dose crosses the multiplicity-adjusted threshold. The rates are 3.86 versus 4.50 events per 100 person-years. No benefit in all-cause mortality is demonstrated. Fatal infections or sepsis are more frequent with all doses pooled: 0.31 versus 0.18 per 100 person-years. Low-density lipoprotein (LDL) cholesterol does not decrease with canakinumab.[1]

9.3.1.2 Critical Reading

reading the testing plan

The primary result must remain attached to the planned comparison. To understand why two similar estimates can receive different labels, the testing plan must be read, not only the p values. The 150 and 300 mg doses here have close estimates but distinct formal thresholds. The statistical success of one and the absence of formal success of the other do not demonstrate that the first is biologically superior. Such superiority would require a comparison directly addressing that question. The event rate calls for another precaution.

a rate is not an individual risk

The arithmetic difference is 0.64 events per 100 person-years. Its unit relates events to cumulative follow-up time; it is not a 0.64-point reduction in individual three-year risk. Obtaining a number needed to treat at a given time point would require cumulative probabilities at that time point and their uncertainties. Precise vocabulary prevents a valid result from becoming a numerically false promise.

9.3.1.3 Immunological Reading

an LDL level that does not fall

The absence of a fall in LDL makes it possible to resist an explanation that would simply attribute the result to a reduction in this parameter. It does not mean that lipid and immune processes become independent in the disease. A measurement that does not change does not abolish the biological milieu

a function that also defends

in which the intervention acts. The infection signal belongs to the same reasoning. A function may contribute to a disease while also contributing to the body's defense. The value of a target is therefore not judged solely by the direction of an effect on the disease studied. It is necessary to look for which other functions are modified and in which patients they may become decisive.

9.3.1.4 Cardiological Reading

The result on the composite must not be rewritten as a demonstrated reduction in all-cause mortality. The observed cardiovascular benefit and the infectious

CRP, an entry condition

harm call for a parallel presentation. Elevated CRP here defines an entry condition. It does not demonstrate that this threshold alone distinguishes the people who benefit from treatment from those who do not. This predictive property would require a specific evaluation, distinct from the result obtained in the selected population. The benefit therefore concerns the composite, within the defined population and comparison. It demonstrates neither the efficacy of all anti-inflammatory strategies nor the superiority of 150 mg over the other doses. Specifying uses would require direct comparisons and a better characterization of the absolute benefit, its variations and the harms.

9.3.1.5 Integrative Clinical Reading

A balance lived after infarction

CANTOS invites discussion of what the person wishes to preserve after a myocardial infarction, without limiting this conversation to coronary risk. The cardiovascular result and the infection signal belong to a single lived balance. The integrative task is to make these dimensions understandable together, with their uncertainty, rather than letting each specialty comment on its own event alone. The trial does not establish how individual priorities should be weighted; it provides the terms of a discussion, not its personal answer.

9.3.2 COLCOT and the Composition of Benefit

9.3.2.1 Situating the Result

COLCOT, colchicine after infarction

COLCOT randomizes 4,745 patients in double-blind fashion within thirty days after a myocardial infarction: colchicine 0.5 mg/day or placebo. The primary endpoint combines cardiovascular death, resuscitated cardiac arrest, myocardial infarction, stroke and urgent hospitalization for angina leading to revascularization. At a median follow-up of 22.6 months, it occurs in 5.5% versus 7.1% of participants: HR 0.77 (95% CI: 0.61 to 0.96). The primary test accounts for an interim analysis; the intervals of the secondary analyses are not adjusted for multiplicity. CRP is studied before and after in only 207 selected participants, with no significant between-group difference in change. Serious pneumonia is reported in 0.9% versus 0.4% of exposed patients.[2]

9.3.2.2 Critical Reading

Naming the whole composite

A composite endpoint asks a precise question: does the strategy modify the occurrence of the first event belonging to a defined set? It does not promise a separate answer of equal strength for each event. The first discipline of reading is therefore to keep the name of this set in the conclusion. Writing only "prevents infarctions" or "reduces mortality" would shift the question after seeing the answer. The opposite criticism would be equally excessive: the clinical heterogeneity of the components does not automatically invalidate the endpoint. An avoided hospitalization may matter to the patient. It is nevertheless necessary to ask what importance is attributed to it, what margin of judgment enters into its recognition and whether the overall result masks divergent directions. The endpoint can be judged relevant while refusing to treat its components as interchangeable. Multiplicity calls for the same precision. Adjusting the threshold of the primary test because the data were examined during the trial does not, by itself, resolve the multiplication of secondary conclusions. These are

Two families of decisions

two families of decisions. A component whose interval appears particularly favorable may orient a new hypothesis; it does not thereby become the assured explanation of the overall success. The planned conclusion on the composite must be distinguished from the hypotheses proposed on the basis of its components.

9.3.2.3 Immunological Reading

A BENEFIT WITHOUT A CLOSED NARRATIVE

The intellectual interest of the case lies in the possibility of supporting a therapeutic inference without a fully closed mechanistic narrative. An intervention may improve an outcome while the chosen marker cannot identify the path of that effect. Conversely, a marker may miss a biologically relevant change because its site, timing or precision does not match the mechanism studied. These possibilities are questions to be examined, not demonstrated explanations of COLCOT. Two symmetrical shortcuts must be resisted: declaring immunology useless if CRP does not separate the groups, or invoking an invisible "local" inflammation to shield any hypothesis from contradiction. An explanation gains strength when it proposes a measurement that could prove it wrong: in which compartment, at what time and with what contrast is the change expected? The observed benefit does not spontaneously designate a single mediator.

9.3.2.4 Cardiological Reading

The infection signal calls for a discussion parallel to that of the benefit: its importance does not depend exclusively on crossing a statistical threshold. Its precision, severity and reproducibility must be considered. The favorable result on the composite thus retains its value without demonstrating each separate benefit or designating a single mediation. Its scope becomes clearer through comparison with other trials and through a more reliable estimate of harms. This difference between an overall result and its components also governs the way it is explained to the patient.

9.3.2.5 Integrative Clinical Reading

URGENT INTERVENTION AND PROLONGED LIFE

For a recently hospitalized person, avoiding a new urgent intervention and prolonging life do not represent the same promise. COLCOT therefore requires translating the composite into understandable experiences: acute event, return to hospital, new procedure. Tolerability and the possibility of continuing care have their place in this translation. An integrative reading asks what the result changes in the complete care pathway, without attributing to each of its steps a benefit that the trial did not separately establish.

9.3.3 LoDoCo2 and the Population That Reaches Randomization

9.3.3.1 Situating the Result

LoDoCo2, AN OPEN-LABEL RUN-IN MONTH

LoDoCo2 enrolls patients with coronary disease who have been clinically stable for at least six months. After one open-label month on colchicine, 5,522 of the 6,528 entrants are randomized: colchicine 0.5 mg/day or placebo. Among the 1,006 not randomized, 611 report intolerance. The primary composite

(cardiovascular death, spontaneous nonprocedural myocardial infarction, ischemic stroke or ischemia-driven revascularization) occurs in 6.8% versus 9.6% of participants; HR 0.69 (95% CI: 0.57 to 0.83), median follow-up 28.6 months. Noncardiovascular mortality yields an HR of 1.51 (95% CI: 0.99 to 2.31). The endpoints and their hierarchy were modified before unblinding, for the last time in January 2020. Inflammatory markers were not measured systematically.[3]

9.3.3.2 Critical Reading

WHERE RANDOMIZATION TAKES PLACE

Randomization must be placed at the right point. It compares strategies among the people who passed through the preliminary phase; it does not compare two initiation policies among all entrants. Selecting before assignment does not, in principle, destroy the validity of the comparison that follows. It restricts its scope. The question becomes: what effect is observed in people able and willing to continue after this initial exposure? This selection cannot be corrected simply by replacing the denominator of randomized patients with that of the entrants. For those not randomized, what would then be missing is their outcome under the compared strategies and an allocation allowing their difference to be interpreted. A broader question requires a broader design. In particular, presenting tolerability after randomization as the tolerability of any new initiation would erase precisely the people whose prior experience prevented their entry into the comparison. It is also necessary to distinguish what "planned" means. An endpoint registered before the first enrollment, an amendment during the trial and a decision taken once the group results are known do not have the same history. An amendment made before unblinding is not, by itself, proof of manipulation; it nevertheless deserves explicit traceability. Judging it requires knowing its date, its justification and the information available at the time. Transparency allows the choice to be discussed without reducing the case to suspicion or to a blank check.

THE HISTORY OF AN AMENDMENT

9.3.3.3 Immunological Reading

PROGNOSIS, EFFECT, SELECTION

The absence of systematic measurements precludes turning this case into a demonstration of a strategy guided by an inflammatory profile. Three questions remain distinct: can an intervention modify the outcome; does a marker describe prognosis; does this marker make it possible to choose who will benefit more from the intervention? A favorable answer to the first does not contain the other two. To make progress, a precise hypothesis about a marker could be formulated, specifying before the analysis how it modifies the expected effect, and selection strategies could be compared. It would then be necessary to avoid confusing a higher baseline risk, which can increase absolute benefit for a comparable relative effect, with a particular pharmacological sensitivity. This distinction changes the meaning of the "inflammatory" patient: a patient more exposed to events, a patient in whom the treatment acts more or both?

9.3.3.4 Cardiological Reading

AN INTERVAL DOES NOT CERTIFY SAFETY

An interval that includes the null value of the effect does not certify safety. The noncardiovascular mortality signal requires an investigation proportionate to its severity and its uncertainty. It would be as excessive to assert an established

fatal toxicity as to write that no problem remains because the conventional threshold has not been crossed. The comparison with COLCOT must therefore keep two dimensions: the definition of events and the construction of the population. A shared molecule and dose allow the trials to be set side by side; they do not automatically make the questions identical. Convergent clinical results may strengthen a research program while leaving open the decision to initiate, to continue or to select differently. The benefit on the composite thus concerns the people who passed through the preliminary phase. Better characterizing those who benefit could make the strategy more relevant, but this trial validates neither selection by CRP nor identical tolerability in all new users. A comparison from initiation onward would allow benefit, selection and safety to be examined jointly over this broader pathway.

9.3.3.5 Integrative Clinical Reading

INITIATING OR CONTINUING

The question asked at the start of a treatment differs from the one asked after a tolerability phase. LoDoCo2 makes this difference visible: the experience of people who do not reach randomization also belongs to the initiation pathway. From a general practice perspective, explaining the benefit therefore requires locating the stage at which it was estimated. Difficulties in continuing are not personal failures; acknowledging them allows the strategy to be discussed without promising every new user the experience of the participants who were ultimately randomized.

9.3.4 CLEAR SYNERGY and the Scope of the Disagreement

9.3.4.1 Situating the Result

CLEAR SYNERGY, A FACTORIAL DESIGN

In this randomized factorial trial, 7,062 patients after myocardial infarction are assigned to colchicine or placebo, alongside a spironolactone/placebo comparison. The composite endpoint combines cardiovascular death, new myocardial infarction, stroke and unplanned ischemia-driven coronary revascularization. After a median follow-up of three years, it occurs in 322/3,528 versus 327/3,534 patients: HR 0.99 (95% CI: 0.85 to 1.16; p = 0.93). CRP measured in a subset is lower at three months: adjusted difference −1.28 mg/L (95% CI: −1.81 to −0.75). Diarrhea is more frequent: 10.2% versus 6.6%. Treatment discontinuations affect about a quarter of participants in each group.[4]

9.3.4.2 Critical Reading

9.3.4.2.1 The interval does not demonstrate equivalence

BENEFITS MADE LESS COMPATIBLE

The result does not demonstrate equivalence. It does, however, make certain large benefits that might have been expected less compatible with the data. On the HR scale, the bounds correspond approximately to a relative instantaneous rate 15% lower or 16% higher. They are not differences in cumulative three-year risk. This translation describes values compatible with the estimate and its model; it does not assign each of them an identical probability, nor does it point to a hidden effect.

9.3.4.2.2 Interruptions, protocol and factorial design

INTERRUPTIONS WITHIN THE STRATEGY

Analysis by assigned group answers the question of the strategy offered under the conditions of the trial. Interruptions are part of that strategy. Removing them from the reasoning because they disappoint the hypothesis would change the question. An analysis limited to the on-treatment period can shed light on robustness, but the decision to stop is not necessarily independent of prognosis. Such an analysis therefore does not automatically enjoy all the guarantees of the initial randomization. The protocol also evolved: the colchicine regimen, initially twice daily for three months in participants weighing at least 70 kg, was changed to a single daily dose of 0.5 mg for all from September 2020. This change must appear in any comparison of strategies; it does not, by itself, establish the cause of the result. The factorial design also calls for an explicit reading. It is necessary to know the comparison evaluated in parallel and to examine whether the effects might depend on their combination. An undetected interaction does not certify that none exists. Conversely, the mere fact that another intervention is being studied does not permit attributing a disappointing result to it: an explanation must be tested against the estimates and their precision.

AN UNDETECTED INTERACTION

9.3.4.3 Immunological Reading

MEDIATOR OR MERE CHANGE

In the measured subset, the fall in CRP attests to a biological change. It does not, however, allow the conclusion that all relevant immune functions were modified, at the right site and during the decisive period. The question becomes more precise: which measured changes accompany the intervention, and which ones truly constitute mediators of the clinical result sought? Measurement in a subset deserves attention of its own. Its interpretation requires knowing why some participants were measured and others were not. A randomized comparison across the whole trial does not spontaneously guarantee that all subsets defined after entry are interchangeable. This reservation does not destroy the observation; it bounds its mechanistic use.

9.3.4.4 Cardiological Reading

ACUTE VERSUS CHRONIC, A CONTRAST TO TEST

Enrollment after myocardial infarction and early initiation define the strategy studied; they do not reduce the observation to the first hours of the event. It would therefore be excessive to summarize the entire disagreement with earlier trials as an already demonstrated opposition between “acute” and “chronic.” Establishing an effect of the timing of initiation would require a comparison designed for that contrast. The gastrointestinal signal must remain visible alongside the cardiovascular result. Tolerability influences the acceptability of a strategy and its durable application; it is not a detail to be subtracted from the commentary on efficacy. The absence of demonstrated benefit on the composite restricts the promise of this strategy; it establishes neither equivalence nor the absence of any inflammatory contribution. The disagreement with other trials remains to be explained. A precise comparison of effects must precede experiments devoted to the differences in strategy that might modify them.

9.3.4.5 Integrative Clinical Reading

THE REASONS FOR STOPPING

In CLEAR SYNERGY, interruptions invite a distinction between the result of the strategy offered and the reasons why continuing it proved difficult. These reasons cannot be deduced from the number of discontinuations alone: tolerability, practical constraints and personal choice need to be documented. The integrative reading seeks to hear them without attributing each interruption in advance to poor adherence or excessive burden. This inquiry sheds light on the possibility of continuing care; it does not restore after the fact a cardiovascular benefit that the primary comparison did not demonstrate.

9.3.5 Kleveland and the Meaning of a Biological Response after Myocardial Infarction

9.3.5.1 Situating the Result

KLEVELAND, A BIOLOGICAL WINDOW

In the trial by Kleveland and colleagues,[5] 117 patients with non-ST-segment elevation myocardial infarction are analyzed: 58 on tocilizumab and 59 on placebo. An intravenous dose of 280 mg is given before coronary angiography, at a median of two days after symptom onset. The primary endpoint is the area under the curve of high-sensitivity CRP during hospitalization; troponin T is the main secondary endpoint. Both measures are reduced with treatment. The question asked therefore concerns, first of all, a biological response within a precise window.

9.3.5.2 Critical Reading

A SAMPLE SIZE BASED ON CRP

The sample size calculation is based on CRP. Clinical events must be read within this frame: the trial is not powered to demonstrate their reduction. A repeated measurement remains the announced biological endpoint; even a prognostic association established elsewhere would not make it interchangeable with a clinical event. Here it indicates what the comparison directly observed. The step from 121 patients enrolled to 117 analyzed leaves a question about the participants' flow. The main report is not sufficient to reconstruct the exclusions, which are detailed in the supplement. Describing this analysis as a complete intention-to-treat analysis therefore requires this additional verification. The difference in numbers does not, by itself, demonstrate that the exclusions explain the result.

FROM 121 ENROLLED TO 117 ANALYZED

9.3.5.3 Immunological Reading

Circulating interleukin 6 (IL-6) increases under blockade of its receptor, while CRP decreases. The authors propose a reduction in receptor-mediated clearance of IL-6 to explain this apparent discordance. This interpretation must keep its status as a proposal. This example invites a distinction between the measured quantity of a mediator and the activity of its pathway. Under intervention, their relationship may change. Before calling a high concentration a pharmacological failure, it is necessary to ask what the drug modifies: production, elimination, binding or signal transmission. The meaning of the assay

MEASURED QUANTITY, PATHWAY ACTIVITY

depends on the system perturbed. The same biological value therefore does not automatically receive the same interpretation before and after treatment.

9.3.5.4 Cardiological Reading

RECONSTRUCTING THE SEQUENCE

Timing is decisive for formulating the next question. Administration after the onset of infarction and before coronary angiography does not evaluate indiscriminately every moment of the injury. The sequence must be reconstructed before speaking of prevention, limitation or repair of damage. The signal obtained thus points toward comparisons concerning the timing of intervention and directly relevant cardiac outcomes.

9.3.5.5 Integrative Clinical Reading

BIOLOGICAL FALL, HOPED-FOR RECOVERY

A biological fall obtained in hospital can easily become, when passed on to the patient or the primary care physician, the announcement of a better recovery. Kleveland shows why these statements must remain distinct. The care pathway continues after the measurement, with expectations concerning activities and future events. An integrative reading preserves this horizon without claiming it is already documented: it explains what changed during the study and what remains unknown for the rest of life.

9.3.6 CIRT and the Difference between a Class of Treatments and a Tested Hypothesis

9.3.6.1 Situating the Result

CIRT, LOW-DOSE METHOTREXATE

CIRT compares low-dose methotrexate with placebo in 4,786 patients with stable coronary disease and diabetes or metabolic syndrome, after a preliminary on-treatment phase. The trial is stopped for futility. After a median follow-up of 2.3 years, the expanded primary endpoint (nonfatal myocardial infarction, nonfatal stroke, cardiovascular death or hospitalization for unstable angina leading to urgent revascularization) occurs in 201 versus 207 participants. The HR is 0.96 (95% CI: 0.79 to 1.16). The original endpoint gives an estimate close to unity. Concentrations of interleukin 1β (IL-1β), IL-6 and CRP do not decrease more with methotrexate.[6]

9.3.6.2 Critical Reading

AN ENDPOINT EXPANDED DURING THE TRIAL

A change in the primary endpoint requires the timeline to be reconstructed. According to the article, those who chose the expansion decided on it without access to the results; the monitoring board then recommended stopping. The article presents both endpoints. This transparency shows that the overall interpretation does not depend here on a single composite. The history of the

STOPPING FOR FUTILITY

endpoints remains an element distinct from the reading of the results. Stopping for futility is a monitoring decision based on the trial's prospects; it does not demonstrate that every possible effect is exactly zero. The interval remains necessary to assess the effects compatible with the data and the assumptions of the analysis. A rigorous conclusion concerns the absence of demonstrated benefit for this strategy, with this precision; it does not turn a monitoring deci-

sion into a biological law. The preliminary phase also limits the population to which tolerability observations after randomization apply; difficulties arising before randomization must remain visible.

9.3.6.3 Immunological Reading

THE IDENTITY OF AN INTERVENTION

The philosophical question is that of the identity of an intervention. Classifying two drugs as anti-inflammatory agents is not enough to support the claim that they test the same causal chain. A class name groups properties together; an experiment must specify which of them it actually modifies, in what milieu and at what time. It could be objected that this precision indefinitely shields a theory from negative results: each failure would be attributed to the "wrong" intervention. The objection is serious. To avoid this, the mechanistic answer must become a testable proposition before the next experiment: which pathway should be modified, how should this be verified and what result would lead to revising the explanation? A justification added after the fact calls for independent testing. CIRT and CANTOS thus invite a distinction between a hypothesis about the targeted pathway and an already demonstrated explanation of the contrast between trials. Their differences are not randomized against one another. They suggest research; they do not allow the divergence of results to be attributed to a single characteristic. Conversely, the absence of a fall in the three markers measured does not demonstrate an absence of any immune action.

CIRT AND CANTOS, NOT RANDOMIZED TOGETHER

9.3.6.4 Cardiological Reading

TALLIES NOT TO BE SUMMED

The expected benefit must be weighed against the observed harms. The safety table notably describes more hepatic abnormalities, leukopenias and non-basal-cell skin cancers with methotrexate. It presents several methods of ascertaining cancers, with differing counts.[6] Citing a figure without specifying how it was obtained would create misleading precision; it is better to retain the concordant signal, without summing these tallies. The negative trial thus narrows the space of defensible conclusions. It requires specifying what should change in the next experiment for that experiment to truly test another proposition.

9.3.6.5 Integrative Clinical Reading

ACCUMULATION IS NOT PROGRESS

The combination, in CIRT, of coronary disease and metabolic disorders is a reminder that a patient cannot be reduced to the biological category chosen for a hypothesis. Adding an intervention to a care pathway presupposes seeking a benefit that also justifies the additional burden and the possible harms. Here, the negative result prevents presenting the accumulation of treatments as progress in itself. It invites a distinction between an attractive explanatory possibility and an addition that is actually useful for the person.

9.3.7 ASSAIL-MI and What Salvaged Myocardium Measures

9.3.7.1 Situating the Result

ASSAIL-MI, the salvage index

ASSAIL-MI is a blinded trial comparing a tocilizumab infusion with placebo around the coronary intervention. The reported comparison concerns 199 people treated for ST-segment elevation myocardial infarction within six hours. The primary endpoint is a myocardial salvage index measured by magnetic resonance imaging (MRI) at three to seven days. Data are available for 195 people. The adjusted difference is 5.6 points (95% CI: 0.2 to 11.3). The work evaluates an effect on this measure; its clinical significance remains to be established.[7]

9.3.7.2 Critical Reading: a ratio retains the history of its components

9.3.7.2.1 An index related to an estimated territory

a denominator that changes the meaning

The index relates the difference between an estimated area at risk and infarct size to that same area at risk. This choice seeks to take account of the territory initially threatened. The points gained must therefore not be read as so many points of ejection fraction, nor as an additional proportion of the whole ventricle that has become viable. The same number can change meaning when its denominator changes. The difficulty runs deeper than a conversion of units. The threatened territory is estimated from the images after the intervention; it is not an anatomical measurement taken before any exposure and then preserved unchanged. Interpreting the ratio requires examining how each component is obtained, at what time and with what sensitivity to tissue changes. This does not demonstrate that the result is an artifact. It prevents the denominator from being regarded as an immutable reference by mere definition.

9.3.7.2.2 Attribution, uncertainty and missing participants

200 enrolled, 195 analyzed

Randomization strengthens the attribution of a difference between strategies on the chosen endpoint. It does not automatically validate every biological interpretation of that endpoint. Moreover, the stated uncertainty covers a range of effects of very different sizes. Crossing a statistical threshold completes neither the assessment of their importance nor that of their reproducibility. The article describes 200 enrollments, followed by the withdrawal of one person who refused the use of their data. Among the other 199, four have no MRI at three to seven days: 195 therefore contribute to the primary endpoint. No imputation was performed. These steps must not disappear behind the label "intention to treat." A useful experimental contrast can be recognized while examining what information is missing and how its absence might weigh on the estimate.

9.3.7.3 Immunological Reading: modifying the pathway does not localize the whole injury

ischemia, reperfusion, inflammation

An intervention on a receptor brings the inquiry closer to a defined pathway. However, an imaging difference after its administration does not, by itself, allow the damage to be apportioned among ischemia, reperfusion and subsequent inflammatory evolution. These stages follow one another; their mea-

surements do not necessarily constitute independent counters. Concordance with an inflammatory marker may support pharmacological coherence without demonstrating that this marker transmits the tissue effect. Conversely, a less clear-cut result on another measure does not automatically cancel the first. It requires asking whether the measures concern the same phenomenon, at the same time and with comparable precision. Choosing only those that support the mechanism would manufacture a coherence that the experiment did not provide. The time to intervention deserves the same discipline. A biological possibility of a different response depending on the delay may justify a new comparison. It does not turn a subgroup exploration into a rule permitting care to be delayed. The proposed interaction, its stability and the conditions under which it was observed would need to be verified.

9.3.7.4 Cardiological Reading: an intermediate result prepares a decision

FROM PROGNOSIS TO STRATEGY

To become a clinical strategy, the hypothesis must connect with consequences relevant to patients. A measure associated with prognosis can help design this step; it does not guarantee that every intervention that modifies it will change that prognosis. The favorable and unfavorable pathways of a treatment may extend beyond what the image summarizes. Likewise, a brief exposure in a selected population is not enough to define the safety of other doses, durations or clinical presentations. The next question is therefore not only to reproduce a favorable index. It is to determine whether the result extends into benefit, for which people and with what possible harms. Subsequent correspondence specifies that clinical heart failure was not a prespecified endpoint and corrects the claim of its absence.[8,9] The result on the salvage index therefore does not demonstrate clinical prevention of heart failure. The experimental attribution remains attached to the measurement actually obtained.

A CORRECTIVE CORRESPONDENCE

9.3.7.5 Integrative Clinical Reading

THE WORD "SALVAGE"

The word "salvage" spontaneously evokes a life or a function preserved. In ASSAIL-MI, it designates a measure constructed by imaging. The clinical conversation benefits from making this difference explicit before discussing the future: what abilities does the person hope to regain, what constraints will the person accept, which outcomes remain to be studied? This shift toward lived experience does not invalidate MRI. It prevents the evocative force of a scientific name from turning, on return home, into an unproven individual promise.

9.4 Comparing Trials without Manufacturing a Hierarchy of Treatments

9.4.1 Five Trials and Three Distinct Operations

FIVE OUTCOME TRIALS

The five outcome trials do not deliver a single verdict on inflammation. CANTOS examines a targeted intervention in people selected in particular by elevated CRP. COLCOT, LoDoCo2 and CLEAR SYNERGY evaluate colchicine in different populations and sequences. CIRT examines another strategy, low-dose methotrexate. Comparing them requires preserving the conditions

DESCRIBING, SEARCHING, EXPLAINING

that make each result interpretable. Three operations must remain distinct. Describing the results consists in presenting the estimates with their intervals. Searching for heterogeneity consists in examining whether the differences in effects exceed what their uncertainty explains, within an appropriate comparative framework. Explaining this heterogeneity further requires an argument about its causes. Moving from the first operation to the third is a frequent source of hasty narratives.

9.4.2 Timing and Markers Do Not Yet Explain the Disagreement

TIMING, A HYPOTHESIS TO EXAMINE

Timing provides a concrete example. A difference between early initiation and treatment of stable disease is a hypothesis to be examined. It does not become a demonstrated cause of the disagreement because it seems biologically coherent. The trials may also differ in selection, associated care, endpoints or actual continuation of treatment. An experiment directly comparing times of initiation would answer a different question from their mere juxtaposition. The same discipline applies to markers. A biological fall and a benefit on events are not two interchangeable confirmations. Nor do CRP measurements in subsets permit ranking treatments by their clinical anti-inflammatory potency. Establishing mediation would require specifying the timing and conditions of measurement, then examining the assumptions needed for causal reasoning.

A DISAGREEMENT THAT SHARPENS THE THEORY

Disagreements become instructive when each conclusion keeps its object. A favorable trial requires defining the limits of extension; a trial without demonstrated benefit requires specifying what was tested. Both can improve a theory, provided the theory agrees to become more precise and to leave certain propositions open.

9.5 Worked Exercises

9.5.1 Exercise on Two Composite Endpoints

9.5.1.1 Problem

THE PROBLEM, TWO RECRUITMENT SETTINGS

All the following data are fictional. They come from no colchicine trial and do not represent real clinical results. Two randomized trials compare drug X with placebo. Follow-up is complete at two years, with 1,000 people per group. Trial A enrolls after a recent coronary event; B enrolls clinically stable people, after a preliminary phase selecting those who tolerate X. The diagnostic and adjudication rules are identical for the shared events. A chooses cardiovascular death, myocardial infarction or stroke as its primary endpoint. At least one of these events occurs in 80 participants on X and 100 on placebo. The two-year risk ratio is 0.80 (approximate 95% CI: 0.60 to 1.06). B adds ischemia-driven revascularization to these three events. Its composite involves 120 participants on X and 160 on placebo: risk ratio 0.75 (approximate 95% CI: 0.60 to 0.93). In B, the composite restricted to the three shared events involves 80 versus 100 participants. In addition, at least one revascularization involves 60 versus 90 participants: some of them also had an event in the restricted composite. No test directly comparing effects between trials is provided. 1. Does B show that X is more effective in stable people? 2. Why can its expanded composite not be obtained by adding the two numbers of participants involved? 3. What does the comparison on a shared endpoint teach? 4. Which clinical conclusion would remain unjustified, even if all the calculations were correct?

9.5.1.2 Reasoned Answer

SIGNIFICANCE AND INTERACTION

The first answer is no. A and B do not give the same statistical result on their primary endpoints, but these endpoints do not measure exactly the same event. The difference in significance does not demonstrate a difference in effects. An interaction must be estimated within an appropriate comparison; clinical stability is linked here to other protocol differences, notably the prior selection. In B, on X, 80 + 60 = 140 would count some people twice; the distinct total is 120. The intersection therefore comprises 20 people. On placebo, 100 + 90 − 160 = 30 people belong to both sets. This calculation is possible because the problem describes participants who had at least one event over a complete and common duration. It reconstructs neither the chronology of events nor a survival analysis from simple margins. On the shared composite, the two trials here have the same observed risks: 8% versus 10%, a difference of −2 points. This brings the description of the results closer; it proves neither a biological equality of effects in the two populations nor their interchangeability. Uncertainty persists, and no equivalence margin was defined. B's contrast on the expanded composite is −4 points, but these two additional points of difference must not be recounted as two points of avoided mortality. Finally, choosing a prescription requires examining separately the components, the harms and the people to whom the result applies. Prior selection on tolerability does not allow the safety of initiation to be described in all patients who might receive X. The comparison becomes useful when it preserves the differences that define each question.

COUNTING THE SAME PEOPLE TWICE

KEEPING WHAT DEFINES THE QUESTION

Synthesis

A trial answers a comparison of strategies, in a population, at a time and according to an endpoint that are all defined. Its result becomes neither a verdict on inflammation nor a rank in a hierarchy of treatments. Reading a composite endpoint requires keeping the name of the set measured and presenting harms alongside benefit. Together, the trials of canakinumab, colchicine and methotrexate show that the same target or molecule does not fix the question asked: prior selection, timing and the markers measured change what the result can teach. Trials with a biological or imaging endpoint illuminate a research step without establishing clinical benefit. A disagreement becomes instructive when it is first described, then explained by a hypothesis that accepts a new test.

Open Questions

Which comparison would directly test the effect of the timing of initiation, rather than inferring it from the juxtaposition of several trials? Under what conditions could an inflammatory marker identify the people who will benefit more from a treatment, and not only describe their risk? How can a benefit on a composite and an infection signal be presented together so that a person can judge their balance? These questions require trials designed for them, not new juxtapositions.

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10

Pericarditis and Myocarditis as Testing Grounds for Clinical Experimentation

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Pericarditis that recurs despite colchicine, myocarditis recognized by imaging, giant cell myocarditis confirmed by biopsy: these situations are rare, and some are serious. Randomized comparisons are few and take particular forms. Randomization takes place after a response to treatment, a registry is described, a series is followed whose protocol had to change. Clinical words easily shift their meaning here. An improvement maintained under treatment becomes a cure, a low observed risk becomes a guarantee, a variant classified in another disease becomes a cause, a detected arrhythmia becomes a life protected. Each of these shifts requires evidence that the initial study did not provide.

10.1 A Population Selected for a Precise Question

10.1.1 A Common Question, Distinct Diseases

ONE QUESTION, TWO DISEASES

How do the conditions of entry into a study determine what its result can teach? This question unites the studies examined in 10.3; it does not assume that pericarditis and myocarditis constitute a single disease. The pedagogical juxtaposition preserves their own objects, endpoints and clinical decisions. Selecting participants is not merely a matter of finding people who carry a diagnosis. A study may retain a particular course, a prior response or a test result. It thereby constructs a population suited to its question. This choice may strengthen the interpretation of the result while limiting its extension. The objection is serious: by dint of selection, does research not end up studying a disease that exists only in the protocol? But selection is not a distortion in principle. It can isolate an important clinical situation. The responsibility lies in naming this situation and showing its relationship to the people who are then to be treated.

A DISEASE OF THE PROTOCOL ALONE?

10.1.2 The Starting Point Belongs to the Question

STARTING, CONTINUING, WITHDRAWING

Starting, continuing and withdrawing a treatment are different decisions. When randomization takes place after a period of exposure, the prior response belongs to the history of the people being compared. It cannot be erased from the conclusion. A group that then receives placebo does not thereby become a group that was never treated. This does not diminish the value of a maintenance question. For a person who has already improved, knowing what happens with continuation or discontinuation may be decisive. The error lies in making this contrast answer another decision, that of initiating treatment in all people who might be considered for it. The scope of the attribution must then be distinguished from the pathway leading to it, rather than labeling the study globally as representative or biased.

10.2 What a Test or a Registry Can Establish

10.2.1 A Test Defines Information, Sometimes an Entry Condition

A NEGATIVE TEST AND ITS CONDITIONS

A negative test can specify a population without explaining its entire disease. The absence of a signal depends on the sample, the method and the timing. It must not surreptitiously become the certain absence of a process everywhere and at all times. Yet it would be excessive to render every nondetection useless. If the procedure should have revealed a relevant presence, its negative result provides information. The question is which absence it allows to be assessed. Keeping the procedure in the conclusion does not mean giving up on

knowledge; it gives that knowledge its conditions of validity. Symmetrically, detecting a signal may justify an inquiry without automatically making the phenomenon responsible for the damage. Presence, activity, contribution to disease and therapeutic relevance remain distinct propositions. An enrollment criterion does not establish them all together.

10.2.2 From Prognosis to Decision

THE FUNCTIONS OF SELECTION

Selection does not have the same function in every study. In a therapeutic comparison, it delimits the people about whom a causal attribution can provide an answer. In a prognostic registry, it delimits the trajectories made observable. Asking the registry to prove the efficacy of each treatment received would miss its own contribution. One difficulty remains: identifying a low-risk group does not yet demonstrate that reducing investigations will preserve its outcome. The observed risk exists within a context of care. Nor does identifying a high-risk group demonstrate that therapeutic intensification will benefit it. Stratification provides a reason to design a comparison; it does not take its place. This distinction also makes it possible to criticize a conclusion without supporting its opposite. If the data do not demonstrate that a test is useless, neither do they prove its necessity for everyone. It is necessary to specify what each policy would change, for whom and according to which outcome. The question then becomes: which decision, based on the available information, improves what matters for these people?

STRATIFYING IS NOT COMPARING

10.3 Texts Put to the Test

10.3.1 AIRTRIP and the Two Phases of the Comparison

10.3.1.1 Situating the Result

AIRTRIP, RANDOMIZED WITHDRAWAL OF ANAKINRA

AIRTRIP enrolls 21 patients with recurrent idiopathic pericarditis, elevated C-reactive protein (CRP), corticosteroid dependence and colchicine resistance. All respond to a first open-label period on anakinra. At 60 days, they are randomized between continuation and replacement with placebo, for a maximum of six months. Recurrence occurs in 2 of the 11 patients continuing anakinra and 9 of the 10 on placebo. Corticosteroids were withdrawn before randomization; colchicine could be continued. Local reactions are frequent during the open-label phase.[1]

10.3.1.2 Critical Reading

TWO CLOCKS

The first clock starts when the drug is introduced. It describes a course under treatment, with associated care and organized monitoring. The second starts when chance decides whether to continue or withdraw this treatment. It is this second clock that carries the randomized comparison. Confusing the two would amount to attributing to randomization events that precede it. The contrast answers the decision to continue in a person who has already improved; it does not compare two initiation strategies. Here, all the patients enrolled reach randomization. A massive selection of nonresponders during the run-in phase must therefore not be narrated as if it had actually occurred. The response condition remains inscribed in the experimental design; its concrete

"ENRICHED TRIAL," AN ARCHITECTURE

effect on exclusions must be read in the participants' flow. "Enriched trial" designates an architecture, and never exempts anyone from counting those who pass through each of its stages. Finally, two measures of the same recurrence, its occurrence and its timing, are not two independent confirmations. The small denominator invites a distinction between the existence of a marked contrast and the precision of its magnitude. It also makes a reassuring conclusion about rare complications hard to defend. The absence of an event in a small group does not give that event a zero probability in the population.

10.3.1.3 Immunological Reading

A USEFUL PATHWAY, NOT A FIRST CAUSE

The maintenance contrast thus leads to distinguishing a pathway worth modifying from a first cause. If recurrence becomes more frequent after withdrawal of the antagonist, the hypothesis that this pathway plays a role in maintaining control gains credibility. It remains to be determined where this dependence lies in the biological history: triggering, amplification or perpetuation. These positions are not interchangeable. A favorable response should not be turned into a retrospective test establishing, on its own, an autoimmune disease, a mutation or a specific local cytokine production. The diagnostic category and the therapeutic target may overlap without merging. This distinction preserves the explanatory value of the intervention: it teaches what can control a process, without claiming to have reconstructed all its origins.

10.3.1.4 Cardiological Reading

MAINTENANCE AND DURABLE REMISSION

An improvement maintained during exposure and a durable remission after discontinuation are two different objectives. The advantage of continuation makes a dependence of control on the intervention credible. It fixes neither the optimal duration nor a cure after discontinuation; it does not compare another active treatment and does not allow a precise estimate of rare risks. A comparison of discontinuation strategies would therefore extend the question that became experimental with AIRTRIP: maintaining an improvement, then seeking the conditions under which it might persist.

10.3.1.5 Integrative Clinical Reading

DEPENDENCE ON TREATMENT

AIRTRIP places dependence on treatment at the center of a question that goes beyond the recurrences counted. How can a real improvement be explained without implying that it will necessarily persist after discontinuation? This distinction matters for the account the person gives of the illness and for their plans. The integrative reading examines what continuing or interrupting a strategy would mean for that person. It recognizes that the trial compares specific pathways and determines neither individual preference nor a universal way of stopping.

10.3.2 RHAPSODY and the Pathway before Randomization

10.3.2.1 Situating the Result

RHAPSODY, 86 entrants, 61 randomized

In RHAPSODY, 86 patients enter a run-in phase on rilonacept, a trap for interleukin 1α (IL-1α) and interleukin 1β (IL-1β). After response and withdrawal of concomitant treatments, 61 are randomized between continued monotherapy and placebo. The first recurrence occurs in 2 of 30 and 23 of 31; the hazard ratio (HR) is 0.04, with a 95% confidence interval (95% CI) of 0.01 to 0.18. Among the 25 not randomized, 15 were still in the run-in phase at the planned closure; 3 did not meet the response criteria, 4 had stopped because of an adverse event and 3 for other reasons. Recurrences are adjudicated blindly.[2]

10.3.2.2 Critical Reading

A count is not an explanation

Subtracting the randomized from the entrants yields a count, not an explanation. An exclusion related to response, an intolerance and an administrative interruption do not mean the same thing. Grouping them under "treatment failure" would manufacture a result that was never observed. Conversely, showing only the randomized people would erase the experience of those who started without being able to continue. The randomized population answers the comparative question of maintenance. The whole exposed group also provides information on the difficulties of the preliminary phase, which do not disappear from the tolerability assessment. The HR relates the instantaneous rates of first recurrence in the model used; it is neither a proportion of people cured nor a difference in risks at a fixed date.

10.3.2.3 Immunological Reading

Two targets, a single comparison

The presence of two targets in the drug's mechanism raises an additional experimental question: what share should be attributed to each? An overall comparison cannot distribute the benefit between the two cytokines. That would require a design able to distinguish the interventions, under sufficiently comparable conditions. The plausibility of biological cooperation does not take the place of this comparison. A second question concerns remission. A process that has become silent under inhibition may be contained, resolved or transformed; these words entail different predictions after discontinuation. Recurrence is informative about the control exerted, but it does not directly map the cells responsible. The mechanistic inquiry remains open even when clinical efficacy becomes convincing.

10.3.2.4 Cardiological Reading

What "placebo" designates

In a randomized withdrawal, "placebo" describes what is administered after a common sequence. It represents neither a person never treated nor necessarily an ordinary care pathway. This clarification prevents naively comparing its recurrence rate with that of another trial with a different run-in phase. Judging the real decision requires naming the options: continue, stop according to the protocol, taper or switch. Only the options actually compared benefit from randomization. The advantage of continuation therefore supports recurrence

control in the randomized population. It does not apportion this benefit between the two targeted cytokines, does not compare rilonacept with anakinra and does not extend to myocarditis. The next question would concern the conditions for remission after discontinuation: a comparison of exit strategies would examine what maintenance left open.

10.3.2.5 Integrative Clinical Reading

remission, a lived period

RHAPSODY leads to considering remission as a period lived under certain conditions, rather than as the closure of the care pathway. For the person, being able to plan activities and knowing how a possible exit from treatment will be discussed may be two different expectations. The trial informs continuation after response; it does not settle this whole conversation. An integrative reading therefore holds together the control obtained, the burden of maintenance and the uncertainty about what follows, without converting a withdrawal comparison into a mandatory individual duration.

10.3.3 Ammirati and the Population Made Visible by Diagnosis

10.3.3.1 Situating the Result

Ammirati, the Lombardy registry

The retrospective Lombardy registry retains 443 patients among 684 records reviewed in 19 hospitals. Symptoms had to have begun within the thirty days preceding admission. Inclusion was based on histology or on necrosis markers associated with two magnetic resonance imaging (MRI) criteria. The presentation was termed complicated in the presence of an echocardiographic ejection fraction below 50%, sustained ventricular arrhythmia or a low-output syndrome requiring inotropes or circulatory support. This category comprised 118 patients, versus 325 without these complications. At five years, the estimate of cardiac death or transplantation was 14.7% in the first group and 0% in the second.[3]

10.3.3.2 Critical Reading

10.3.3.2.1 What the registry makes visible

a shift in the denominator

The first methodological result is a shift in the denominator. A biopsy-based series and a population also recognized by imaging do not necessarily make the same people visible. If a better prognosis is observed in the latter, two explanations must remain distinct: trajectories may have changed; enrollment may have included more of the less severe cases. A crude historical comparison does not decide between these possibilities. Conversely, this difficulty does not make the registry useless: it specifies which population its estimate describes.

10.3.3.2.2 A zero and a subset to be read exactly

a zero at five years

The observed zero then deserves an exact reading. A five-year estimate does not mean that every person was followed for five years: its precision also depends on the number still under observation at that horizon and on censoring. This result constitutes neither an individual guarantee nor the absence of any consequence of the disease. The objection that “nothing is ever reassuring, then” would be excessive. A favorable observation can genuinely modify the assessment of risk. It becomes misleading when its vocabulary exceeds

200 follow-up scans out of 415

the event sought, the duration observed or the people enrolled. These three limits must be kept together. Another question concerns how outcome is measured. Follow-up MRI involved 200 of the 415 patients initially examined by this method. This subset must not silently replace the entire cohort in a conclusion about ventricular evolution. The problem is not only sample size: the reasons for having a second scan may be linked to the evolution itself. The description remains informative; extending it requires characterizing those who are missing.

10.3.3.3 Immunological Reading

prognosis is not mechanism

A prognostic category is not automatically a mechanistic category. Grouping patients because they present a complication provides an instrument for describing risk. It demonstrates neither that a single immune pathway explains all their situations nor that they would respond to the same treatment. Moving from "group at higher risk" to "group sensitive to an intervention" requires an additional comparison. It is also necessary to distinguish information that happens to be available from information sought in a comparable way. When tissue examination depends on the clinical presentation, directly comparing lesion frequencies between groups may mix a biological difference with a difference in investigation. The answer is not to abandon histology, but to make explicit to whom it was applied before interpreting its distribution.

10.3.3.4 Cardiological Reading

a surveillance question

The registry helps organize a surveillance question; it does not, by itself, demonstrate which testing or treatment policy improves outcome. A good prognosis in people managed according to certain practices is not enough to prove that each of these practices is necessary, useless or risk-free. The possible decisions would need to be compared in people to whom they could genuinely apply. Stratification therefore describes a prognosis in the observed population; it does not demonstrate the causal efficacy of a treatment or biopsy strategy. A prospective validation retaining the definitions would test its stability. Evaluating decisions based on this stratification would then show whether knowledge of risk improves the care pathway.

10.3.3.5 Integrative Clinical Reading

the "good prognosis" label

Announcing a "good prognosis" category may reassure, but it may also make present difficulties less audible. Ammirati invites a distinction between the statistical outcome studied and the recovery experienced by each person. The registry's classification does not measure every dimension of the return to ordinary life. Its integrative use is to inform a conversation about the rest of the care pathway, keeping the limits of enrollment visible, rather than summarizing all needs after myocarditis in a single adjective.

10.3.4 Cooper and the Transformation of the Protocol

10.3.4.1 Situating the Result

Cooper, giant cell myocarditis

In 2008, Cooper and colleagues report a prospective multicenter study of biopsy-confirmed giant cell myocarditis. The initial plan included a usual care group; recruitment difficulties led to a change in design. Twelve people were enrolled, eleven treated and analyzed, of whom four were randomized between two immunosuppressive combinations. The work describes a reduction in inflammatory lesions on follow-up biopsies, without significant short-term improvement in ejection fraction. Complications occur; a fatal recurrence is reported after discontinuation of immunosuppression in one participant. These observations call for a reading of the design actually carried out, beyond its initial intention.[4]

10.3.4.2 Critical Reading: following the transformation of the question

"randomized," for whom?

The word "randomized" does not describe all the evidence contained in this article. The people concerned, the interventions compared and the analysis supporting each conclusion must be identified. Random allocation limited to part of the enrollment does not turn the entire series into a randomized comparison. Nor does a comparison between therapeutic combinations answer the initial question of treatment versus usual care. The adaptation of the project reveals a concrete difficulty of research in a rare disease: patients may refuse an option that the team wishes to compare. This refusal does not establish the efficacy of the preferred option. It explains why certain hoped-for data will not be produced. Transparency about this change then becomes an essential documentary quality, without restoring the strength of the lost contrast. It might be tempting to recover this contrast from an older series. But different periods, places and diagnostic circumstances may select different patients. A disease identified early enough to allow enrollment does not necessarily have the same severity distribution as a disease discovered at transplantation or after death. A better observed result therefore does not, by itself, measure the benefit attributable to treatment. The historical comparison retains a function of putting things in perspective, with an uncertainty that cannot be reduced to its p value.

the temptation of an older series

10.3.4.3 Immunological Reading: not asking the same thing of every measure

biopsy and function, two observations

Biopsy brings observation closer to the targeted tissue process. It provides, however, neither an exhaustive map of the heart nor a direct measure of all its functions. If inflammatory lesions decrease without discernible functional change, both observations must be maintained. Declaring them incompatible would assume that inflammatory activity alone determines the measured function, immediately and reversibly. Circulating antibodies raise another question. A weak signal directed against a few antigens does not demonstrate the absence of any humoral contribution; nor does it identify, by elimination, the sole responsible cellular mechanism. A hypothesis concerning lymphocytes requires appropriate positive evidence. The coherence of an immunological

story does not permit turning what was not found into proof of what remains proposed.

10.3.4.4 Cardiological Reading: a recurrence informs without fixing the whole course of action

a relapse after discontinuation

A relapse of the disease after discontinuation of treatment is an important observation. It makes the automatic equation of improvement with lasting disappearance of risk hard to defend. Yet it determines neither the frequency of recurrences, nor the optimal duration of treatment, nor the risk the same person would have run had treatment been continued. The severity of the event increases its clinical importance, not its capacity to resolve all these comparisons. The balance must also keep the possible complications of immunosuppression visible. Monitoring disease, toxicity and function together makes it possible to formulate the next questions: what result is to be maintained, with what exposure and in which patients?

10.3.4.5 Integrative Clinical Reading

a family and a histological result

In a serious and rare disease, the family may expect a histological result to answer questions about the entire future. Cooper requires these levels to be separated without stripping the observed improvement of all value. Coordination should make understandable the goals pursued, the complications encountered and the questions still open about duration. This is a requirement of communication, not a care protocol deduced from the series. A tragic trajectory after discontinuation deserves attention without becoming, by itself, the fate announced to everyone.

10.3.5 Lota and the Scope of a Genetic Classification

10.3.5.1 Situating the Result

Lota, two cohorts, 336 people

Lota and colleagues study the overlap between acute myocarditis and cardiomyopathy genetics in two cohorts totaling 336 people. Their recruitment and clinical profiles differ. In 27 people, they identify variants that would be classified as pathogenic or likely pathogenic in a cardiomyopathy context. This classification does not establish as many certain causes of myocarditis. The study brings diagnostic categories together; it does not compare the clinical consequences of two genetic screening policies.[5]

10.3.5.2 Critical Reading: the diagnosis accompanies the classification

rare, protein-altering, pathogenic

A genetic result cannot be reduced to the presence of a difference in a sequence. Its meaning still has to be decided. “Rare,” “protein-altering” and “pathogenic” do not designate interchangeable categories. Moving from one to another requires additional elements, including the clinical setting to which the classification refers. A property recognized in one disease does not become, by a mere shift of the word, an explanation of every disease in which the same variant is found. This distinction avoids two opposite errors. The first would

make each classified result a sufficient cause of the observed episode. The second would dismiss the result as uninteresting because it does not resolve this episode. Between the two lies a precise contribution: identifying a reason to reconsider the relationship between categories, then formulating the observations likely to decide between explanations. The scope of a test is not measured solely by its capacity to give an immediately definitive answer. Recruitment thus becomes an element of biological interpretation. Two centers may observe different proportions because they receive different people, investigated at different times or by different procedures. Calling these populations by the same diagnosis does not eliminate these gaps. Before attributing a difference to geography or mechanism, the routes that made the patients visible must be reconstructed. This does not make every comparison impossible; it indicates which comparisons would be most instructive.

RECRUITMENT WITHIN THE INTERPRETATION

10.3.5.3 Immunological Reading: predisposition and episode require articulation

PREDISPOSITION AND EPISODE

Finding a variant and observing inflammation does not automatically specify the relationship between the two. A predisposition might modify the response to an insult; a structural disease might include an inflammatory presentation; several determinants might also coexist without their cooperation being demonstrated in the study. These propositions may guide an inquiry, but they do not have the same experimental consequences. To distinguish them, a simple enrichment of variants among cases is not enough. It would be necessary to bring together the sequence of events, the tissue manifestations and comparisons suited to the proposed mechanisms. An interaction requires more than the simultaneous presence of two factors: the expected joint effect must be specified, as well as the reference against which it is compared. The vocabulary of an encounter between genetics and immunity is useful only if it makes this test formulable.

10.3.5.4 Cardiological Reading: additional information must find its use

THE USE OF A PIECE OF INFORMATION

Moving to a decision involves a step of its own. Even if a piece of information modifies the interpretation of a case, it remains to be known what using it would change for the person concerned. The options may concern surveillance, further investigation or a family discussion. Their value does not follow automatically from the frequency of the variants observed. A useful evaluation should therefore name the decision triggered, the people to whom it applies and the expected consequences, favorable as well as unfavorable. It should also distinguish an uninformative result from a result that would actually exclude a possibility. Not finding an explanation within the scope of a test does not close off all other routes to knowledge. Genetic classification is therefore sufficient neither to explain the episode nor to demonstrate the usefulness of a decision. An additional prognostic conclusion would require concordant estimates and an examination of the detailed genomic methods. Classifying a variation, explaining an episode and improving a decision are three related but distinct tasks.

CLASSIFYING, EXPLAINING, DECIDING

10.3.5.5 Integrative Clinical Reading

THE PERSON AND THEIR RELATIVES

Genetic information may concern the person who came for an acute episode and immediately raise questions about their relatives. Lota does not measure the consequences of all these decisions. An integrative reading therefore asks what concrete use is envisaged before attributing personal value to the result: understanding the episode, organizing follow-up or opening a family discussion are distinct projects. The possibility of obtaining no explanation must also remain intelligible, without turning a classification uncertainty into a failure of understanding on the patient's part.

10.3.6 Tscholl and the Difference between Detection and Protection

10.3.6.1 Situating the Result

TSCHOLL, WEARABLE DEFIBRILLATOR

Tscholl and colleagues retrospectively describe, in a single center, 59 people with myocarditis who were prescribed a wearable defibrillator. While the device is worn, four episodes of sustained ventricular tachycardia are recorded in two people. No shock is delivered. The rhythm findings contribute to changing their management. This result documents surveillance in selected people; it does not compare the device with its absence or with another surveillance strategy.[6]

10.3.6.2 Critical Reading: identifying the benefit in question

10.3.6.2.1 The benefit and the missing comparator

THE WORD "BENEFIT" IN A TITLE

The title uses the word "benefit." This word can designate several propositions: detecting an event, changing a decision, terminating an arrhythmia or improving survival. These propositions may form a coherent chain, but they are not equivalent. An observation of the first step does not establish all the following steps. Recognizing the value of information therefore does not require immediately attributing a therapeutic effect to it. The missing comparator must be named precisely. The alternative is not necessarily a person without any care or surveillance. It might include other means of recognizing an arrhythmia, raising the alarm and deciding. Asserting that the event would have remained unknown without this device requires more than noting that it was discovered with it. The difference concerns two possible pathways, only one of which was observed here.

THE MISSING COMPARATOR

10.3.6.2.2 The absence of shocks and the prior prescription

THE ABSENCE OF SHOCKS

The absence of shocks calls for the same discipline. It proves neither that the device is useless nor that it prevented a fatal outcome. A technical capability may not have had the opportunity to be used in the series. Evaluating its usefulness requires distinguishing this possibility of rescue, the events actually treated and the comparative clinical consequences. Adding these three registers under the same word would make the estimate of benefit impossible to recover. Finally, enrollment starts from a prescription already decided. The people equipped do not automatically represent all those with myocarditis. The reasons for this decision may be linked to severity, to the ability to

wear the device and to the physician's judgment. A frequency observed in this group therefore does not become the average risk of the disease. This selection does not remove the descriptive value; it indicates which question that value answers.

10.3.6.3 Immunological Reading: linking observations without erasing their timeline

ELECTRICAL TRACING AND IMMUNE ACTIVITY

The inflammatory diagnosis locates the problem, but it does not make an electrical recording interchangeable with a measure of immune activity. Linking the two requires keeping the date, site and nature of each observation. A tissue assessment and a later arrhythmia do not necessarily describe the same state of the heart. The methodological consequence is important: a slight abnormality in a sample is not enough to guarantee a low future risk; conversely, a rhythm event does not by itself demonstrate intense or persistent inflammation. Neither piece of information can be converted into a universal standard for the other. Their relationship would need to be verified in a population and according to a timeline able to distinguish activity, sequelae and other determinants. This study does not evaluate the efficacy of immunosuppression. The cardioimmunological interest of the juxtaposition lies precisely in this boundary: a disease characterized by inflammation may pose a protection decision that requires its own evidence. Treating the presumed cause and preventing a complication do not answer the same contrast.

PROTECTING IS NOT TREATING THE CAUSE

10.3.6.4 Cardiological Reading: surveillance is part of the strategy

WEARING THE DEVICE, BEING PROTECTED

The time during which a person actually wears the device belongs to the observed exposure. A prescription and continuous protection are not synonymous. Nor should the total duration of follow-up automatically become an identical duration of surveillance for everyone. Comparing strategies would require describing the periods covered, the events sought and the responses triggered by their detection. Changing a decision may be a genuine clinical contribution, but its usefulness remains to be assessed by its consequences. A comparative study could examine outcome, complications, constraints and decisions avoided or prompted. The series prepares these questions; it does not provide their complete balance. The common diagnosis must therefore not conceal the difference between the interventions, the comparators and the outcomes that each allows to be studied. Evidence concerning rhythm surveillance does not replace evidence concerning immunosuppression.

10.3.6.5 Integrative Clinical Reading

LIVING WITH A DEVICE

Wearing a surveillance device also involves a way of sleeping, moving about and understanding one's sensations. These dimensions would need documenting in an overall evaluation; the results of the series do not provide an account of them. Coordination also concerns the meaning given to an alert and the person responsible for responding to it. Detecting more is fully intelligible only when the pathway that follows is described, with its decisions, constraints and consequences, without confusing available surveillance with continuously guaranteed protection.

10.4 Worked Exercises

10.4.1 Exercise on Randomized Withdrawal

10.4.1.1 Problem

SIXTY FICTIONAL ENTRANTS

All the data below are fictional. They reproduce neither AIRTRIP nor RHAPSODY and describe no real drug. A trial admits 60 people with recurrent inflammatory episodes. All receive treatment X for eight weeks. At the end of this period, 42 respond and tolerate the treatment; 8 do not meet the response criteria, 6 stop because of intolerance and 4 withdraw consent for personal reasons. The categories are mutually exclusive. The 42 eligible people are randomly allocated, in blinded fashion: 21 continue X and 21 receive placebo. Follow-up after randomization lasts six months for each, with no missing data. A first recurrence occurs in 3 people on X and 12 on placebo. The endpoint and its adjudication are identical in both groups. After a recurrence, rescue treatment is allowed; only the first recurrence is counted here. 1. What is time zero of the randomized comparison, and which population does it concern? 2. Calculate the risks, their difference and their ratio. Can the difference be applied to the 60 entrants? 3. Can it be asserted that X prevents recurrence in 75% of all the people to whom it is offered, or that the 18 not randomized are all resistant to X? 4. What trial would be needed to compare two strategies from their initiation?

10.4.1.2 Reasoned Answer

Time zero is randomization after eight weeks, among the people then eligible after response and tolerance. Initial enrollment precedes a common intervention; it does not begin a randomized comparison between X and its absence.

RISKS, DIFFERENCE AND RATIO

The observed risks are 3/21 = 14.3% and 12/21 = 57.1%. The difference, continuation minus withdrawal, is −42.9 points. Their ratio, 0.25, corresponds to an observed relative reduction of 75%. No interval is provided; the uncertainty of the small sample remains. This reduction is neither an individual probability of benefit nor a proportion of cures. Multiplying 42.9 points by 42/60 does not estimate the effect of initiation. All received X; the comparative outcome of the other participants is unknown. The 18 not randomized comprise three distinct situations: insufficient response, intolerance and withdrawal of consent.

ONE RANDOMIZATION PER DECISION

To compare initiation, randomization would here have to take place before the strategies begin, with follow-up of changes and rescue treatments. Each decision requires its own contrast: initiation, maintenance and discontinuation strategies may be studied in separate trials or in a protocol with several randomizations.

Synthesis

The population of a study is constructed by its question. A randomized withdrawal after response informs the continuation of a treatment in people who have already improved; it does not compare two initiations and does not describe remission after discontinuation. A registry describes the trajectories made visible by its diagnostic criteria; it does not demonstrate the usefulness

of the care received. A series with a transformed protocol, a genetic classification or device-based surveillance each provides real information, whose use for decision-making requires another comparison. Together, these texts teach how to count those who pass each stage, to name the missing comparator and to distinguish detecting, explaining and protecting.

Open Questions

How can discontinuation strategies be compared to learn under what conditions an improvement obtained under treatment can persist? What prospective validation would show that a prognostic stratification genuinely improves the care pathway of the people it classifies? What comparative study would measure what rhythm surveillance changes in outcome, beyond the events it detects? Each of these questions calls for a starting point chosen for it.

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11

Systemic Diseases and the Question of Extrapolation to the Heart

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In vasculitides, genetically determined recurrent fevers or systemic sclerosis, therapeutic trials most often measure the disease as a whole: a remission, a flare avoided, a relapse delayed, a walking distance. Cardiac involvement is rarely the primary endpoint in these trials. Life-threatening forms are often excluded, and cardiac involvement then concerns only a few participants. The cardiologist must nevertheless decide, often without a comparison devoted to the situation at hand. Borrowing a result from another disease then becomes tempting, in the name of a shared target or drug. The borrowing is not illegitimate; it must state what it carries. A definition of remission, a corticosteroid tapering rule or a marker that the treatment itself modifies travels with the result and changes its meaning.

11.1 What a Result Obtained Elsewhere Can Teach about the Heart

11.1.1 The Same Patient, Several Questions

SEVERAL ORGANS, ONE DIAGNOSIS

A systemic disease can bring together, under a single diagnosis, involvement of several organs. This grouping gives coherence to the inquiry; it does not make the results measured in each organ interchangeable. Improving the overall activity of a disease, reducing a concomitant treatment and preserving cardiac function are three propositions. A study can shed light on the first without having directly tested the third. The unity of the person must not be lost in this distinction. A general benefit can matter deeply to someone whose heart is affected, even if the study does not demonstrate a cardiac effect of its own. It must be possible to state simultaneously what the treatment brings to the person and what remains unknown about its action on an organ. The error begins when the importance of the first benefit serves as proof of the second. A symmetrical difficulty consists in declaring useless, for cardiology, any study in which the heart is not the primary endpoint. Such a study can provide information on an intervention, its adverse effects or the conditions of its use. Extrapolation therefore requires an examination of the propositions being carried over, without confining each specialty to its own measures.

THE SYMMETRICAL DIFFICULTY

11.1.2 A Resemblance Must Concern What Makes the Effect Vary

IDENTITY OF NAME, IDENTITY OF DEPENDENCE

Why might a result obtained elsewhere shed light on a cardiac question? A frequent answer invokes a shared target. But recognizing the same biological element is not enough: its role in producing damage, its modifiability and the timing of intervention may differ. An identity of name does not establish an identity of dependence. The objection is immediate: if all these relationships must be known before acting, no generalization will be possible. Yet the point is not to demand complete identity between situations. The task is to specify which differences could change the conclusion, then to seek the observations that inform them. Reasoning gains from naming a small number of decisive hypotheses rather than accumulating favorable resemblances. In particular, two expectations can be distinguished: preventing further damage and restoring a function already lost. The second does not automatically follow from the first. An intervention can modify an active process without erasing all its earlier consequences. This distinction predicts the result of no drug; it specifies what would need to be measured so as not to call “failure” the absence of an effect that the initial hypothesis gave no reason to expect.

PREVENTING OR RESTORING

11.2 From the Global Endpoint to a Controllable Extrapolation

11.2.1 The Global Result Can Hide the Question of the Organ

HOW THE HEART ENTERS THE SCORE

An endpoint aggregating several dimensions can suit a systemic question. To know what it teaches about the heart, one must examine how the heart enters it: does it contribute to the score, can it be offset by other improvements or does it fall under a separate measure? The problem is not only statistical. An identical word, "remission," can designate different operational rules depending on the protocol. The presence of a few participants with cardiac involvement does not, by itself, establish an answer for that involvement. Their number, the events observed and the prespecified question must be known. However, their small number does not prove either that the effect would be different. Insufficient information must be turned neither into general permission nor into absolute prohibition. The comparison must therefore keep its exact object: controlling general activity, avoiding a complication or improving established involvement. A shared treatment and a shared diagnosis do not make these expectations equivalent. Reading them separately then makes it possible to bring them together in a decision for the same person.

11.2.2 Distinguishing the Justification of a Decision from the Proof of Its Mechanism

DECIDING BEFORE THE COMPARISON

A clinical decision must sometimes be made before a comparison devoted to each situation exists. This does not turn the extrapolation argument into direct observation. The benefit sought, the possible harms, the competing options and the missing information must be made explicit. The soundness of a decision also depends on the options available; it is not measured solely by the number of studies addressing exactly the same diagnosis. This responsibility includes a requirement of revision. If a result is extrapolated because a property appears shared, the observations that would lead to reexamining this analogy must be announced: a persistent discordance between general activity and organ function, or a particular harm. An absent individual response may justify clinical reassessment; it does not, by itself, refute a hoped-for average effect. Extrapolation thus becomes a controllable line of reasoning. It preserves available knowledge while giving a precise form to what remains to be learned.

11.3 Texts Put to the Test

11.3.1 MIRRA and the Definition of a Remission

11.3.1.1 Situating the Result

MIRRA, MEPOLIZUMAB VERSUS PLACEBO

MIRRA randomizes 136 adults with relapsing or refractory eosinophilic granulomatosis with polyangiitis to mepolizumab 300 mg subcutaneously every four weeks or placebo, added to usual care for 52 weeks. Remission combines a Birmingham Vasculitis Activity Score (BVAS) of zero and prednisone or prednisolone ≤ 4 mg/day. The two primary endpoints are the accrued duration of remission, analyzed by categories, and remission at weeks 36 and 48. Both favor treatment; for the second, 32% versus 3%. However, 47% of treated participants never achieve this remission. Organ- or life-threatening forms in the

three months before screening are excluded; histories of cardiomyopathy are nevertheless present.[1]

11.3.1.2 Critical Reading: treatment enters the definition of success

THE CORTICOSTEROID WITHIN REMISSION

This remission combines a clinical state and low corticosteroid exposure. This choice answers a relevant question: can the disease be controlled while lightening a burdensome treatment? It also requires specifying what failure means. Not crossing the threshold does not prove the absence of any improvement; conversely, reaching the threshold does not mean cure or the disappearance of all earlier damage. The timeline deserves the same precision. Accruing favorable weeks, being in remission at two visits and remaining continuously in remission are not three formulations of the same result. A careful reading must reconstruct the rule before comparing percentages. It must also avoid turning an ordinal indicator, built from several categories, into a simple multiplication of the probability of a binary event. Corticosteroid tapering is part of the strategy evaluated. Its organization and blinding help make the comparison interpretable; they do not automatically separate a direct effect of the drug from all the consequences of changes in concomitant treatment. The proper unit of interpretation remains the strategy actually compared.

ACCRUING, REPEATING, LASTING

11.3.1.3 Immunological Reading: a targeted intervention does not exhaust the disease

INTERLEUKIN 5, WITHOUT EXCLUSIVITY

Mepolizumab targets interleukin 5 (IL-5). Improvement under this intervention supports a therapeutic interest without making this pathway an exclusive explanation of the disease. Above all, MIRRA does not allow each absence of remission to be attributed to a still-active target: the endpoint also requires corticosteroid reduction. Residual activity, established damage and an obstacle to treatment reduction would call for distinct observations.

11.3.1.4 Cardiological Reading: an included history does not represent an excluded emergency

A HISTORY IS NOT AN EMERGENCY

The presence of participants with a cardiac history does not turn this trial into a study of the treatment of acute, life-threatening cardiac involvement. The exclusion concerns a clinical and temporal situation; it cannot be reduced to the name of an organ. To carry over a result, severity, disease phase, concomitant treatments and the benefit sought must therefore be compared. A favorable global endpoint does not, by itself, provide a reliable estimate for each organ. A cardiac outcome would need to be defined, with sufficient events and sample size, and its precision then examined. The absence of a statistical difference on a few safety events would not demonstrate equivalent cardiac safety either. Established: more remissions combining activity control and low-dose corticosteroids, in the population studied. Not concluded: specific cardiac protection or efficacy in recently life-threatening forms. A dedicated comparison could clarify this scope.

11.3.1.5 Integrative Clinical Reading

what "remission" covers

MIRRA compels the question of what the word "remission" covers for someone in whom several dimensions of health remain at stake. An endpoint combining measured activity and low-dose corticosteroids does not constitute an inventory of all the difficulties experienced. Integrative reading keeps the value of the result while leaving room for symptoms, possible sequelae and personal goals not described by this single score. It does not carry the trial over to the recently life-threatening situations it had excluded.

11.3.2 CLUSTER and the Limits of a Shared Target

11.3.2.1 Situating the Result

CLUSTER, three cohorts

Grouping rare diseases can make research feasible without declaring them identical. CLUSTER raises this question in a particularly fruitful way: how far does a response to the same intervention allow diseases with different genetic causes to be brought together? Three randomized cohorts compare canakinumab with placebo: colchicine-resistant familial Mediterranean fever, mevalonate kinase deficiency and tumor necrosis factor receptor-associated periodic syndrome (TRAPS). They include 63, 72 and 46 participants, respectively. Complete response combines resolution of the index flare by day 15 and no new flare through week 16, according to a clinical and biological definition. It concerns 19/31 versus 2/32 participants, 13/37 versus 2/35 and 10/22 versus 2/24. A dose increase classifies the participant as a nonresponder for the primary analysis. After this phase, a new randomization involves some responders; the other pathways include open-label treatments.[2]

11.3.2.2 Critical Reading: keeping successive questions separate

11.3.2.2.1 Three cohorts and one response rule

one protocol, three populations

A shared protocol facilitates the comparison of methods. It does not create a homogeneous population. Reading the three contrasts within their cohorts prevents a global result from masking a difference in response or in uncertainty. Nor are the observed percentages sufficient to rank the diseases by sensitivity: a comparison between diagnostic groups requires accounting for their characteristics and for its own uncertainty. The rule concerning dose increase alters the precise meaning of the word "response." It distinguishes efficacy obtained under the conditions initially assigned from improvement obtained after adjustment. A patient classified as a nonresponder may have experienced a real benefit; conversely, an analysis allowing adjustment must not be presented as if it answered exactly the primary question. The name of the treatment pathway must be kept together with its result.

the meaning of the word "response"

11.3.2.2.2 The second randomization and tolerability across pathways

the second randomization

The second randomization selects on an already observed history. Its interest is to pose a maintenance question in those who have passed this stage. It does not simply reestimate initial efficacy in the whole population. Merging the phases would erase this selection and give follow-up duration a demonstrative value it does not possess on its own. Safety requires the same attention

to pathways. When participants switch treatment, the exposure denominator and the attribution of events become essential. The tolerability table classifies under canakinumab the events occurring after at least one dose. Raw numbers are therefore not enough to reconstruct an unaltered randomized comparison. Even a rate adjusted for exposure time does not automatically restore all the properties of the initial randomization.

11.3.2.3 Immunological Reading: one accessible pathway among several causes

interleukin 1 beta, a shared pathway

Interleukin 1β (IL-1β) can occupy a useful position in several causal chains. Its inhibition does not require assuming that everything upstream proceeds identically. Reasoning thereby gains an organization: distinguishing the cause of the dysregulation, the pathways that sustain the manifestations and the points where an intervention can modify them. This connection remains revisable. A nonresponse does not exclude all involvement of the pathway; a response does not prove its exclusivity. To make progress, the contrasts in response would need to be explained with measurements able to distinguish insufficient exposure, biological diversity and limits of the chosen endpoint.

11.3.2.4 Cardiological Reading: carrying over a question before carrying over a conclusion

from flares to the heart

A trial devoted to systemic flares does not become a trial of prevention of cardiac lesions. To extend a conclusion to the heart, the targeted involvement, its presumed mechanism, its timeline and a measurable cardiac outcome must be defined. A shared therapeutic logic opens a line of research; it does not remove the need to conduct it. The favorable result in each of the three cohorts makes the therapeutic interest of a shared pathway credible, without establishing that the diseases are identical or correcting their genetic cause. Specific cardiac protection remains a different proposition, which would require comparisons suited to the targeted involvement and trajectories.

11.3.2.5 Integrative Clinical Reading

one target, several diagnoses

CLUSTER brings together rare diseases without erasing their differences. For a person whose care pathway crosses several specialists, sharing a therapeutic target can facilitate a common language, but does not make diagnoses or expectations interchangeable. Integrative reading asks what "no longer having flares" concretely changes and what the endpoint leaves out of scope. It keeps room for the individual history and the burden of follow-up, without drawing from the cohorts an identical benefit for each organ.

11.3.3 TAKT and the Definition of Relapse

11.3.3.1 Situating the Result

TAKT, Takayasu arteritis

In TAKT, 36 Japanese patients with Takayasu arteritis who had recently relapsed, then returned to remission with corticosteroids, are randomly assigned to tocilizumab or placebo, double-blind. Corticosteroid reduction is mandated.

For time to relapse, the primary intention-to-treat endpoint does not establish superiority: hazard ratio (HR) 0.41, 95.41% confidence interval 0.15 to 1.10, p = 0.0596. The per-protocol analysis is more favorable. C-reactive protein (CRP) is masked; the assessment limited to clinical signs gives an HR of 0.70, with wide uncertainty. These are several readings of the same participants, not independent replications.[3]

11.3.3.2 Critical Reading

MAINTAINING, INDUCING, ADJUSTING

The first lesson is to respect the question actually posed. Maintaining remission during a scheduled reduction of background therapy differs from inducing that remission in a patient with a flare. It also differs from comparing two strategies in which the physician would freely adjust doses. In the first case, the mandated reduction creates a common challenge; in the second, adjustments are part of the strategy studied. A result travels from one to the other only if this change is discussed. The second lesson concerns the statistical threshold. The unusual level of the interval stems from the planned handling of the interim analysis. This framework must not be replaced by a boundary chosen after seeing the result. The absence of demonstrated superiority nevertheless leaves a range of effects open: it establishes neither equivalence nor futility of the treatment. The right sentence keeps the favorable estimate and its imprecision together; removing either would distort the conclusion. The per-protocol analysis does not automatically repair the primary result. As soon as participants are selected according to what happened after their allocation, the groups may lose part of their initial comparability. This analysis sheds light on the sensitivity of the finding; it does not provide a second randomization. To give it more weight, the reasons for exclusion, their possible relation to prognosis and the results of appropriate analyses would need to be made explicit, rather than retaining only the most favorable p value.

THE PER-PROTOCOL ANALYSIS

11.3.3.3 Immunological Reading

THE TREATMENT ACTS ON THE INSTRUMENT

A treatment can act on the disease and on the instrument used to recognize it. This possibility becomes particularly important when the biological signal of the endpoint depends on the targeted pathway. Making a marker disappear and making the process that produces it disappear are not interchangeable propositions. Conversely, requesting an independent measure does not mean that any fall in the marker is artificial: the same intervention can simultaneously modify several levels of the phenomenon. Masking CRP protects against certain interpretations influenced by knowledge of it; it does not cancel its biological dependence on the treatment. Observer blinding must therefore be distinguished from the independence of the endpoint. Comparison with a clinical definition of relapse is illuminating, but it also changes the rule for classifying events. A divergence between definitions is not enough to measure the exact share of a biological effect, a difference in sensitivity or chance. Above all, it indicates where the demonstration must become more precise.

11.3.3.4 Cardiological Reading

ACTIVITY, LESION, SEQUELA

The ultimate stake is preserving an arterial network capable of perfusing the organs. Yet inflammatory activity, mural lesion, anatomical sequela and

circulatory impact do not necessarily evolve at the same pace. Symptomatic improvement may precede anatomical stabilization; a stenosis may persist after activity has subsided. Reading the trial as a cardiologist requires keeping these timescales distinct instead of immediately converting an avoided relapse into an avoided myocardial infarction or avoided heart failure. To strengthen this translation, a future design could combine a clinical definition fixed in advance, repeated imaging assessed independently and sufficiently long follow-up of complications. This would be an extension of the question, not merely an increase in the number of participants. TAKT thus teaches how to keep a therapeutic hypothesis open without announcing that all its vascular consequences are already established.

11.3.3.5 Integrative Clinical Reading

THE MANDATED SCHEDULE

The mandated corticosteroid tapering schedule belongs to TAKT as much as the drug compared. A person may nevertheless receive the result as an answer concerning the drug alone. The integrative task is to restore the whole strategy, with its pace and relapse criteria, before relating it to an individual care pathway. Follow-up of several organs cannot be reduced to a single signal; this remark opens a question of coordination without turning the trial into a demonstration of each expected benefit.

11.3.4 GiACTA and the Measurement of a Remission under Treatment

11.3.4.1 Situating the Result

GIACTA, FOUR STRATEGIES

GiACTA randomizes 251 patients with giant cell arteritis among four strategies: two tocilizumab dosing schedules combined with a 26-week prednisone taper, or placebo with a 26-week or 52-week taper. The population called "intention to treat" comprises 250. At 52 weeks, protocol-defined sustained prednisone-free remission reaches 56% and 53% with tocilizumab, versus 14% and 18% with placebo. CRP is part of its definition, although the drug lowers it directly; it is masked from assessors. The analysis without this biological requirement remains favorable, but no longer establishes superiority of the every-two-weeks schedule over the long taper at the prespecified threshold.[4]

11.3.4.2 Critical Reading

11.3.4.2.1 Separating the requirements of remission

SEPARATING THE REQUIREMENTS OF SUCCESS

A definition of success often combines several requirements. They must be separated before their combination is interpreted: clinical state, biological indicator, adherence to the treatment schedule. A strategy may act on these three dimensions in different ways. The question is then which component carries the result and what experiment would test its robustness. The sensitivity analysis here answers a precise objection: does part of the advantage come from the change in the marker used to declare remission? Removing this marker allows an instructive comparison; it does not construct a new independent trial. The same care regarding thresholds and multiple comparisons must be maintained after this removal. Noninferiority established from a prespecified margin and

NONINFERIORITY AND SUPERIORITY

demonstrated superiority answer two different questions. The mere failure of a superiority test is not enough to conclude the former.

11.3.4.2.2 Compared schedules and numbers analyzed

THE TAPER WITHIN THE OBJECT EVALUATED

Comparing the schedules raises a second question. With identical tapering, what is studied is the addition of the drug within a defined strategy. Against a longer taper, two ways of organizing treatment are compared. Differences in corticosteroid exposure are then not simply a defect to be corrected: they belong to the object evaluated. To interpret an adverse event, this entire strategy must be considered, without automatically attributing its cause to a single molecule. Finally, the name of the analysis population does not dispense with checking the numbers. The gap between those randomized and those analyzed must be made explicit, even when it seems small. A demanding reading examines its reason and its possible consequences; it does not mechanically turn every exclusion into an invalidation of the trial.

11.3.4.3 Immunological Reading

A MEASURE TRANSFORMED BY THE INTERVENTION

The difficulty is fruitful: intervening on a mechanism can simultaneously transform the disease and the way it is observed. A marker relevant before intervention could thus lose part of its informative power during it. The stability of a measuring instrument must be discussed in the context in which it is to be used. It would nevertheless be excessive to conclude that any biological fall is devoid of meaning. It provides information on a response to treatment. The next question is what it allows to be inferred about the state of the tissue. To test this step, one would seek complementary observations whose relation to vascular involvement does not depend solely on the targeted mechanism. This research proposal does not describe a result already obtained in GiACTA.

11.3.4.4 Cardiological Reading

CONTROL, WALL, EVENTS

The cardiologist distinguishes inflammatory control, integrity of the vessel wall and prevention of events. A remission defined during a trial does not, by itself, give the probability of a late structural complication. Answering that other question would require appropriate follow-up, defined events and enough information to estimate differences in them. GiACTA and TAKT can therefore be placed in dialogue without being conflated. Their ways of defining relapse, reducing corticosteroids and preserving blinding can be compared. The effect size from one disease cannot automatically be imported into the other. The shared lesson is a discipline of formulation: stating the population, the strategy compared, the outcome measured and the duration before calling the treatment effective.

11.3.4.5 Integrative Clinical Reading

"PREDNISONE-FREE PER PROTOCOL"

In GiACTA, remission is tied to a corticosteroid reduction strategy; the care pathway therefore cannot be reduced to the addition of a molecule. For the person, controlling manifestations and reducing the treatment burden may represent associated, but not identical, goals. Integrative reading makes this dual expectation explicit, along with the limits of what was measured. It avoids

translating "prednisone-free per protocol" into a general absence of treatment, constraint or need for follow-up.

11.3.5 Zamanian and the Result the Primary Analysis Does Not Confirm

11.3.5.1 Situating the Result

Zamanian, rituximab and walking

This trial randomizes 57 people with systemic sclerosis-associated pulmonary arterial hypertension to rituximab or placebo. At 24 weeks, the primary analysis estimates a difference in the change in walking distance of 23.1 meters, with a standard error of 14.7 meters and p = 0.12. A secondary analysis incorporating measurements up to 48 weeks estimates, still for the change at 24 weeks, a difference of 25.1 meters, with p = 0.03. The safety table counts serious infections in 7 of the 29 participants in the rituximab group and 2 of the 28 in the placebo group.[5]

11.3.5.2 Critical Reading

11.3.5.2.1 Two analyses of the same data

two p values, a single trial

The two p values do not describe two independent trials. The secondary analyses are not corrected for multiple comparisons. In the longitudinal model, later data modify the estimate of a contrast at an earlier time point. Presenting the second result as confirmation of the first would erase their dependence and their different status. The proper object of reading is the link between the question, the analysis rule and the observations it uses. The model assumes data missing at random conditional on the information it takes into account. The imbalance in early withdrawals therefore calls for a sensitivity analysis; it does not, by itself, demonstrate bias. A nonsignificant result does not demonstrate an absence of effect. Nor does it allow the benefit suggested by the point estimate to be retained as established. Uncertainty must remain attached to this estimate. Adding analyses can clarify the hypotheses to be tested; it does not retrospectively create a confirmatory demonstration.

11.3.5.2.2 A modified endpoint, two conclusions of unequal strength

an endpoint changed during the study

The main article reports a change of the primary endpoint during the study, followed by a late registry update. The authors state that they did not use the trial's walking data for this change. This calls for a chronological examination, without sufficing to establish opportunistic selection. The date on which a decision is made, the data then accessible and the date on which it becomes publicly verifiable must be distinguished. Comparing the protocol with its successive versions would allow this chronology to be checked independently of the authors' account. Two conclusions of different strength can thus be kept: the primary analysis does not confirm the announced benefit; certain observations still justify a research question. Conflating them artificially forces a choice between a therapeutic promise and the abandonment of any investigation.

11.3.5.3 Immunological Reading

WHICH PROPOSITION IS CHALLENGED

The uncertain functional result raises the question of disagreement between levels of evidence. If an intervention reaches its biological objective but the functional result remains uncertain, which proposition is really challenged? The efficacy of the intervention under the conditions studied is directly questioned. The general involvement of immunity in the disease is not, by itself, settled by this result. Conversely, invoking this involvement does not allow the intervention to be declared effective. The same discipline must be applied to the search for response profiles. A rule built to recognize the people who improved must be tested in other data, with a comparison that allows this improvement to be attributed to the treatment. Recognizing a favorable trajectory is not enough to recognize an individual benefit caused by the treatment. No selection tool can be regarded as validated by this exploration alone.

11.3.5.4 Cardiological Reading

WALKING IS NOT CIRCULATION

Walking performance deserves attention for what it represents in a person's life. It must not, however, be turned without argument into an exclusive measure of the pulmonary circulation, right ventricular function or survival. The name of the disease does not make each of the endpoints used to study it specific. Safety requires an equally precise reading. The people affected, the number of events, their severity and their attribution must be distinguished. A small trial can document problems without precisely estimating all risks; labeling certain events as unrelated does not demonstrate general safety. Safety observations must contribute to the judgment without being converted, in the other direction, into certainty about a causal harm that the study does not demonstrate. Systemic sclerosis-associated pulmonary arterial hypertension is the situation studied here. Kinship between connective tissue diseases alone does not allow the estimate to be extended to lupus or mixed connective tissue disease. The primary analysis and the safety assessment leave open the therapeutic value of the intervention in its own context.

11.3.5.5 Integrative Clinical Reading

WALKING FARTHER

In a systemic disease, walking farther is an understandable goal beyond the cardiology office. But the measured distance does not, by itself, tell why an activity becomes possible or difficult. Zamanian invites attention to this gap between standardized performance and daily life, without using it to circumvent the uncertain primary result. An integrative reading sets the functional expectation, the constraints of treatment and the safety observations side by side; it turns neither an individual hope nor a secondary analysis into sufficient proof.

11.3.6 Broderick and the Certainty Specific to Each Result

11.3.6.1 Situating the Result

Broderick, a Cochrane review

The Cochrane review by Broderick and colleagues makes it possible to examine how coronary evidence is built in Kawasaki disease. For initial treatment, the comparison of intravenous immunoglobulin, generally combined with aspirin, with aspirin alone reports fewer coronary artery abnormalities up to thirty days. The odds ratio (OR) is 0.60, with a 95% confidence interval (95% CI) of 0.41 to 0.87 and moderate certainty. Certainty regarding adverse effects is very low. An analysis excluding studies at high risk of bias in the detection of coronary abnormalities yields an interval that includes no effect. These findings concern this initial comparison; they do not settle the rescue strategy after immunoglobulin resistance.[6]

11.3.6.2 Critical Reading: giving confidence a precise address

where the evidence lies

The strength of this text lies in the possibility of asking exactly where the evidence lies. Calling a review "positive" flattens several questions: which contrast was studied, at what time point, for which manifestation and with what capacity for detection? A conclusion on arterial involvement does not automatically pass its certainty on to a conclusion on treatment complications. It must be possible to hold simultaneously a favorable judgment on one result and a strong reservation about another. The shift of the interval in the sensitivity analysis likewise calls for a graded reading. It would be improper to treat this analysis as a demonstration that the benefit disappears; it changes the set of available studies and its precision. It would be equally improper to neglect it because the main analysis crosses the conventional threshold. The useful question becomes: what part of the confidence depends on how the lesion was sought and recognized? The sensitivity analysis makes this dependence visible without perfectly measuring its cause. The reassuring wording used by the authors regarding safety deserves, in this context, additional restraint. The absence of a detected difference does not carry the same weight when the search for adverse effects is precise and homogeneous as when its uncertainty remains substantial. The reasoning does not consist in assuming a hidden danger: it consists in leaving open a question that the data do not close. Certainty must accompany each conclusion, rather than becoming an overall rating attached to the title of the article.

a reassuring formula

11.3.6.3 Immunological Reading: a therapeutic effect does not by itself name the cause

response and etiology

Responding to an immunological intervention does not, in itself, constitute a test defining an autoimmune disease. The clinical effect concerns a change obtained in a given context; an etiological classification or attribution to a particular mediator requires additional arguments. The review can therefore support the therapeutic contrast studied without selecting a single cause among the possible explanations.

11.3.6.4 Cardiological Reading: preserving the observed object

CORONARY ABNORMALITY AND OUTCOME

A coronary abnormality is a relevant cardiac object, but it does not summarize all cardiovascular outcomes. Additional reasoning is needed to move from its frequency to the persistence of lesions, then to events experienced later on. This step does not become automatic because the organ concerned remains the same. The time point and definition attached to the measure must be kept, and it must then be asked which observations document the subsequent steps. In particular, preventing a new lesion and the course of an existing lesion raise different questions. An aggregated result must not allow anyone reading it to choose freely the one that matches their expectation. Precision of language here has a clinical function: it prevents promising, in the name of real evidence, a result that was not that of the comparison. The initial comparison therefore does not settle the choice of a rescue treatment after immunoglobulin resistance. This new decision requires a dedicated comparison. The confidence attached to the initial prevention of a coronary abnormality must not silently carry over to another phase of the disease.

11.3.6.5 Integrative Clinical Reading

THOSE CLOSE TO A CHILD

For those close to a child, a coronary abnormality, an adverse effect and a hospitalization can occupy different places in their worry. Broderick's review makes it possible to give each result its degree of certainty, without requiring families to regard them as equivalent. Integrative reading here concerns the quality of explanation: distinguishing initial treatment from later rescue, specifying the time horizon studied and recognizing what the synthesis does not measure of the family experience. It derives no individual course of action from it.

11.4 Worked Exercises

11.4.1 Exercise on a Global Benefit and a Cardiac Result

11.4.1.1 Problem

TWO HUNDRED FICTIONAL ADULTS

All data are fictional. They reproduce neither MIRRA nor any other real trial. A trial randomizes 200 adults with a systemic inflammatory disease to X added to usual care or placebo added to the same care. Each group includes 100 people. Organ- or life-threatening involvement in the past three months is excluded. A few participants nevertheless have a history of stabilized cardiac involvement. The primary endpoint simultaneously requires an activity score of zero and a daily corticosteroid dose below a defined threshold, at two spaced visits. It is met in 40 people on X and 20 on placebo. The 95% CI of the absolute difference is, by assumption, +8 to +32 points. Between these two visits, no data on continuous remission are provided. New cardiac complications affect two people on X and three on placebo. This exploratory analysis was not powered for this endpoint and no confidence interval is provided. Among the people not meeting the primary endpoint, some have a score of zero but remain above the corticosteroid threshold; their number is not specified. A reader concludes:

"X doubles the cure rate, nearly halves cardiac complications and should work in life-threatening cardiac forms." Which propositions need correcting? What should be asked to extend the interpretation?

11.4.1.2 Reasoned Answer

SUCCESS IS NOT CURE

Protocol-defined success concerns 40% versus 20% of people: a difference of 20 points, relative risk (RR) of 2. The interval provided for the absolute difference runs from +8 to +32 points. The word "cure" adds a durability and a disappearance of the disease that were not measured. Nor do two favorable visits demonstrate continuity between them. The endpoint combines clinical activity and treatment exposure. Not all failures can therefore be classified as persistent inflammatory activity. The components and their course would need to be known, without retrospectively replacing the primary endpoint with whichever appears most favorable. For cardiac complications, the observed risks are 2% and 3%: a difference of −1 point and a descriptive RR of 2/3, that is about 0.67. "Nearly halves" is already numerically inaccurate. Above all, five events do not allow an established cardiac benefit to be declared without an estimate of uncertainty. The significance of the global endpoint does not carry over to this comparison. Finally, a stabilized history does not represent currently life-threatening involvement. The exclusion leaves precisely this latter situation outside the randomized contrast. It demonstrates neither efficacy nor inefficacy of X in this population. To move forward, an appropriate comparison, a defined cardiac endpoint, suitable monitoring and sufficient precision would be needed. The defensible conclusion remains a benefit on the defined clinical and therapeutic success, in the population actually studied.

"NEARLY HALVES" IS INACCURATE

THE HISTORY AND THE EXCLUSION

Synthesis

Extrapolating to the heart a result obtained in a systemic disease requires naming what is being carried over. A favorable global endpoint justifies a conclusion about the disease; it does not, by itself, provide an estimate for the organ. The vasculitis and arteritis trials show that remission depends on a rule: corticosteroid threshold, tapering schedule, a marker sometimes directly modified by the treatment. CLUSTER shows that a shared target can bring diseases together without making them identical. Zamanian and Broderick teach how to attach certainty to each result, instead of making it an overall quality of the study. An extrapolation becomes controllable when it announces the observations that would lead to its revision.

Open Questions

What cardiac outcome, defined in advance and measured with sufficient precision, would allow an extrapolation to be tested rather than assumed? How should disease activity be judged when the treatment directly modifies the marker used to recognize it? What observation would force revision of an analogy based on a shared target between two diseases? These questions accompany any decision that precedes its own comparison.

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12

Heart Failure and the Lessons of Negative Trials

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In some people with heart failure, an elevated C-reactive protein (CRP) signals measurable inflammation, and the idea of reducing it to improve exercise capacity seems reasonable. The trials that have tested this idea are often small, brief and designed to prepare a next step. They readily report a fall in CRP, more rarely a functional improvement demonstrated by the comparison between groups. These results are easy to misname. An absence of difference becomes a failure of immunology, or on the contrary a benefit hidden by a lack of power. An improvement within one group becomes an effect of the treatment; a function returned to normal becomes a cure. Each of these formulas says more than the trial, or less.

12.1 What a Negative Result Really Weakens

12.1.1 Which Statement Has Just Been Put to the Test?

THREE CONCLUSIONS UNDER THE WORD "FAILURE"

A negative result calls for a precise sentence before it calls for an explanation. Did the trial fail to establish superiority, did it render implausible a benefit judged important or did it reveal harm? These conclusions are not equivalent. Grouping them under the word "failure" masks their scope. Heart failure gives this distinction particular weight for cardioimmunological inquiry. The proposition that an immune process participates in a disease is not identical to the proposition that modifying it, through a given intervention, improves a clinical outcome. Between the two lie a direction of action, an intensity, a timeline and a disease state. A biological explanation does not, by itself, contain the prescription that would succeed. This distinction does not, however, permit shielding the explanation from therapeutic results. If a hypothesis served to predict a benefit, the absence of that benefit must change something in the confidence it is granted. The weakened proposition must be identified, instead of declaring by turns that the whole theory is dead or that it was never concerned.

IDENTIFYING THE WEAKENED PROPOSITION

12.1.2 When Does an Explanation of Failure Become Instructive?

AN INDICATOR WITHOUT FUNCTIONAL GAIN

Suppose, in an entirely fictional example, that a drug reduces an inflammatory indicator without improving functional capacity. It is conceivable that the indicator accompanies the damage without driving it, that the modified process plays several roles or that an established lesion is no longer reversible through this pathway. The initial observations do not automatically choose between these possibilities. The objection is serious: with enough possibilities, can the hypothesis not always be protected? Yes, if no explanation exposes itself to another observation. Asserting after each result that the treatment would have succeeded "earlier" preserves every possible outcome. By contrast, defining the relevant stage by a measurable criterion, independent of the result one wishes to save, makes a new comparison possible. The difference lies in the risk of refutation that the explanation accepts. Revision can also lower the ambition of the proposition: moving from a supposedly dominant mechanism to a limited contribution. This correction is legitimate if it acknowledges the loss of scope. It becomes misleading when a discourse keeps the initial promise while silently reducing what was supposed to verify it.

THE RISK OF REFUTATION

12.2 From Observed Harm to the Next Research Decision

12.2.1 Harm Can Be More Instructive than an Absence of Gain

WORSENING AND ABSENCE OF GAIN

Failing to demonstrate a benefit and observing a worsening do not present the same difficulty. In the second case, the question also concerns what the intervention disrupts beyond the intended effect. Clinical harm can coexist with the expected modification of a marker. The success of an intermediate measure then does not neutralize the unfavorable result for people. It might be objected that such harm concerns only the molecule. That is possible, but it remains a proposition to be examined. Conversely, immediately attributing it to an entire class or to any immune intervention would also go beyond the result. To choose the scope of revision, what the treatments actually share must be distinguished: target, mode of modification, exposure and associated effects. The practical lesson can be firm while the explanation remains open. Abandoning a strategy in the situation studied does not require having resolved all the reasons for its result. Mechanistic doubt suffices neither to establish an undemonstrated benefit nor to erase an observed harm.

THE MOLECULE OR THE CLASS

12.2.2 Putting Words to the Test as Well

A withdrawal trial raises a complementary question: does an improvement maintained under treatment mean that the treatment has become unnecessary? A difference observed after withdrawal can refute this expectation without constituting a "negative trial." Reading a trial without demonstrated benefit begins, for its part, by holding estimate and uncertainty together. A very imprecise result may leave room for substantial effects; a precise result may rule out the promise that justified the trial, without establishing an absolute absence of effect. This assessment requires defining clinical importance before looking only at the statistical threshold. It continues with a research decision. Repeating, changing population, modifying the intervention or abandoning a line of inquiry have different costs. A new study is better justified by an identified uncertainty than by the mere wish to finally obtain a favorable result. At each modification, the question becomes: what new observation would allow the still plausible explanations to be separated? Negative trials thus contribute to knowledge when they restrict promises, correct conditions of application and prevent an interesting biology from being confused with a useful therapy. Their value presupposes no hidden success. The size and objective of a study change what is expected of its result. An exploration can justify a more demanding comparison without already supplying its conclusion. When moving from a pilot to a subsequent study, the exact question must be kept: have population, duration and endpoint changed? A succession of publications does not constitute replication when these elements differ.

ESTIMATE AND UNCERTAINTY

FROM PILOT TO THE NEXT STUDY

12.3 Texts Put to the Test

12.3.1 The D-HART Pilot and the Conditions of a Crossover Comparison

12.3.1.1 Situating the Result

D-HART pilot, fourteen days

The D-HART pilot studies stable heart failure with preserved ejection fraction, with CRP above 2 mg/L. Fourteen people are enrolled; twelve receive treatment and complete both periods. Each person successively receives anakinra and placebo for fourteen days each, in randomized order, with no treatment-free interval. The reported contrast in peak oxygen consumption favors anakinra by 1.2 mL/kg/min (p = 0.009). CRP also decreases. The twelve treated participants are all obese and hypertensive; eleven are women.[1]

12.3.1.2 Critical Reading: the subject remains the same, its history changes

12.3.1.2.1 The economy of the crossover design and carryover

THE ECONOMY OF THE CROSSOVER DESIGN

The crossover design is attractive when people differ greatly from one another: each participant provides an observation under each treatment. It thereby reduces certain differences between individuals. But this economy rests on a difficulty of its own: the second treatment arrives in someone who has already lived through the first period. The body does not necessarily return to its initial state on the date the syringe changes. The absence of a washout period therefore requires examining a possible carryover effect. A benefit persisting during placebo could modify the contrast; spontaneous change or familiarization with the test could also play a role. Randomizing the order helps distribute calendar effects, without demonstrating their absence in a small sample. The repeated-measures model described by the authors accounts for time and allocation. The main article does not, however, report a specific assessment able to rule out a residual effect. This uncertainty must accompany the result, without becoming an assertion that the result is necessarily biased.

WITHOUT A WASHOUT PERIOD

12.3.1.2.2 Periods are not additional patients

PERIODS, NOT PATIENTS

A second trap would be to take periods for additional patients. Two measurements in the same person do not provide two independent histories. The comparison must preserve their pairing. Likewise, the twelve people analyzed must not obscure the fourteen initially enrolled: exclusions before treatment and follow-up after treatment answer different questions. Saying that all treated participants completed the trial does not describe the entire recruitment. Finally, a result on the primary endpoint does not permit turning each secondary or subgroup analysis into confirmation. Multiplying views of the same observations can give rise to hypotheses; it does not multiply independent experiments.

12.3.1.3 Immunological Reading: a biological response does not localize its effect

COHERENCE IS NOT A CAUSAL PATH

An immune intervention, a decrease in an inflammatory marker and a functional improvement constitute coherent observations. Their coherence is

nevertheless not enough to reconstruct the entire causal path. CRP reveals neither the responsible cell nor the compartment where any benefit takes place. Its change does not directly measure myocardial repair. Even if two changes are correlated, this finding alone does not reveal whether one transmits the effect of the intervention to the other. They may share a cause, reflect parallel responses or depend on other characteristics of the participant. Demonstrating mediation requires more than an association between changes. The fruitful question then becomes: what measurements would separate a cardiac action from a vascular or muscular contribution to exercise capacity? Choosing an elevated CRP for enrollment also delimits the experiment. It does not prove that people below this threshold could not respond. An entry criterion is not automatically a predictive test: to support this second function, effects would need to be compared according to the marker in an appropriate design.

AN ENTRY CRITERION, NOT A TEST

12.3.1.4 Cardiological Reading: measuring a capacity without promising a fate

AN INTEGRATED EXERCISE CAPACITY

Peak oxygen consumption describes an integrated exercise capacity. Its functional interest deserves to be kept, without renaming it an anatomical improvement of the heart. It does not, by itself, provide information on a reduction in hospitalizations or mortality. The prognostic value of a measure and the clinical value of its induced modification are two distinct propositions. The participants' profile and the brevity of the periods also limit how far the result can be carried over. An observation in stable, selected people able to perform the tests does not directly address the situation of acute decompensation. Nor does it provide a precise assessment of rare or late adverse effects. This small crossover trial therefore provides a functional signal compatible with a modifiable inflammatory contribution to certain exercise limitations. It demonstrates neither myocardial repair, nor durable benefit, nor the ability of CRP to select responders. A confirmation would need to better control period effects and extend observation toward clinical consequences: it would test the scope of the signal, instead of presuming that the signal will be reproduced.

12.3.1.5 Integrative Clinical Reading

ONE'S OWN CONTROL

Being one's own control does not mean living through two interchangeable periods. In the D-HART pilot, integrative reading draws attention to what the exercise test represents for the person, beyond the figure: perceived capacity, desired activity, constraints of repeated visits to the center. Not all these dimensions are assessed by the trial. Recognizing them serves to formulate the next clinical question, without converting a short-term change into general recovery or forgetting the highly selected profile of the participants studied.

12.3.2 D-HART2 and the Distance between Biological Response and Functional Benefit

12.3.2.1 Situating the Result

12.3.2.1.1 The question and the participants analyzed

D-HART2, 31 randomized, 28 analyzed

The question of this study is precise: does inhibition of interleukin 1 (IL-1) with anakinra improve cardiorespiratory fitness in people with heart failure with preserved ejection fraction and CRP above 2 mg/L? The publication reports 31 randomized participants, 21 on anakinra and 10 on placebo, for twelve weeks. The efficacy analysis is based on 28 participants with at least one follow-up visit, namely 20 on anakinra and 8 on placebo. The authors call it intention-to-treat, but it does not include all 31 randomized people.[2]

12.3.2.1.2 Biological changes without demonstrated functional benefit

A biological fall without exercise gain

At twelve weeks, the publication reports, for the comparison of changes in peak oxygen consumption under anakinra and under placebo, a difference of −0.4 mL/kg/min. The 95% confidence interval (95% CI) runs from −2.2 to +1.4, with p = 0.64. The sign of this difference must be read with the convention of the contrast and its interval. Nor does the publication show a significant difference in the slope of the relation between minute ventilation and carbon dioxide production (VE/VCO_2). It does, however, report decreases in CRP and in N-terminal pro-B-type natriuretic peptide (NT-proBNP) at four weeks compared with placebo. The trial thus reports biological changes without demonstrating the functional benefit sought. The difference in significance, at measurement times that must be kept in view, does not by itself demonstrate a difference between biological and functional effects.[2] For NT-proBNP, the comparative signal concerns four weeks; at twelve weeks, the fall in the anakinra group is not accompanied by a significant difference in changes between groups.[2] An improvement within one group and a difference between strategies answer two distinct questions.

12.3.2.2 Critical Reading

12.3.2.2.1 The planned comparison, its interval and its imputed data

The planned comparison first

The comparison between groups answers the question of the treatment effect. A change observed in a single group does not separate the effect of the intervention from that of time, concomitant care or other changes. Interpretation must therefore begin with the planned comparison, even when some secondary results appear more encouraging.

The limits of a p value

The p value gives neither the probability that the treatment is ineffective nor the size of its effect. The American Statistical Association statement stresses that scientific conclusions should not depend solely on whether a threshold is crossed. The question here is to understand together the estimate, its precision, the study design and the clinical relevance of the effects still compatible with the results.[3] The interval reported for peak oxygen consumption crosses zero and covers effects of different sizes. To conclude that there is no clinically important benefit, that benefit would need to have been defined and the data shown to rule it out.

Last observation carried forward

The trial cannot be retrospectively turned into an equivalence trial merely because its result is not significant. The authors carry the last available observation forward to impute missing data at twelve and twenty-four weeks. This

method assumes stability between the last observation and the missing time point; it does not, by itself, restore the lost information. Sensitivity analyses are announced in the supplement. The threshold for the two coprimary endpoints is set at $p < 0.025$; the secondary signals retain their exploratory status.[2] Those randomized, those contributing to the analysis and the values actually observed or imputed must therefore be distinguished. An overall number does not summarize these choices.

12.3.2.2.2 The protocol and the question of sample size

The protocol registered on ClinicalTrials.gov, in its April 2016 version, plans 60 participants and an allocation of two patients on anakinra for every one on placebo. This document confirms the intention to study exercise capacity and ventilatory efficiency.[4] The design publication and the methods of the final report describe a revision of the sample size, in October 2016, from sixty to thirty participants, prompted by slow recruitment and a new variability assumption. Yet the discussion in the same final report speaks of recruitment stopped after thirty-one participants out of the sixty initially planned.[2,5] The tension therefore also belongs to the final report itself. The reading must neither ignore the revised target nor take this revision as a guarantee of precision. The complete chronology of the amendments and the information accessible at the time of decision remain to be reconstructed. The precision finally obtained remains to be assessed in light of the effect sought, the variability assumed, the analysis method and the available data. Reaching a recruitment target does not automatically validate the assumptions used to set it. The methodological scope extends beyond this case. A prospective calculation helps design a study; after the results, the estimate and its confidence interval directly indicate what the data can distinguish. The expression "lack of power" must neither turn an absence of demonstration into a hidden benefit nor make the assumptions of the initial calculation disappear. It must be possible to trace the decision to modify the study and to judge the information finally produced.

sixty, then thirty

"lack of power"

12.3.2.3 Immunological Reading

"target engaged"

The fall in an inflammatory marker is a biological observation. It does not by itself demonstrate target occupancy in cardiac tissue, complete interruption of the pathological pathway or the exclusive responsibility of this pathway for the symptoms. These distinctions must be kept when the expression "target engaged" is used. The coexistence of the reported results invites several questions. Does the marker provide information on the determining mechanism in this population? Does the duration of intervention allow the functional result to change? Can other limitations persist when the inflammatory signal decreases? These questions organize future research; they are not demonstrated explanations of the trial's outcome. This distinction also guards against a favorable interpretation by default. It would be too easy to argue that any absence of benefit merely proves that the wrong patient, the wrong duration or the wrong endpoint was chosen. A hypothesis retains its scientific value when these conditions can be specified and tested, with a real possibility of producing a result that weakens it.

12.3.2.4 Cardiological Reading

FUNCTION AT THE CENTER

Clinical reading must keep the functional result at the center of the initial question. If the goal is to improve exercise capacity, a biological change does not fully meet this expectation. It may justify further investigation without becoming a demonstration of the benefit sought. Peak oxygen consumption probes an integrated performance. The fact that it does not exclusively measure a cardiac mechanism is a property to be considered in the design and interpretation of the trial. This property is not enough to disqualify the endpoint after the fact. The endpoint had precisely the merit of addressing a function, beyond an isolated biological measure. Two legitimate projects can be distinguished: showing that a biological mechanism is modifiable and showing that an intervention improves an important function. The instruments and sample sizes required will not necessarily be identical. Which of these projects supports each conclusion must be identified, without allowing a silent passage from one to the other. The historical trial is not enough to define the current place of the treatment. That assessment would require a synthesis of subsequent data and guidelines.

12.3.2.5 Integrative Clinical Reading

A DIFFICULT CONVERSATION

D-HART2 makes possible a difficult conversation: a biological result can improve without the expected function improving demonstrably. For the patient, this divergence should be neither concealed nor presented as a personal failure. Integrative reading returns to the initial life goal and to the other dimensions that the endpoint does not summarize. It can help name the remaining questions, but it does not provide, on the basis of this historical trial, a decision to continue or stop a current treatment.

12.3.3 REDHART and the Comparison of Treatment Durations

12.3.3.1 Situating the Result

REDHART, THREE STRATEGIES

REDHART randomizes 60 people recently hospitalized for heart failure, clinically stabilized, with ejection fraction < 50% and CRP > 2 mg/L. Three strategies are compared: anakinra for two weeks then placebo, anakinra for twelve weeks or placebo. The two primary endpoints compare the pooled anakinra groups with placebo at two weeks: peak oxygen consumption and ventilatory efficiency. No significant advantage is demonstrated. At twelve weeks, peak oxygen consumption improves in the long-treatment group, but the contrast between groups is not significant. CRP decreases; clinical outcomes remain exploratory. Eight people have no initial follow-up visit.[6]

12.3.3.2 Critical Reading: a group cannot be its own randomized control

12.3.3.2.1 The primary endpoint versus within-group tests

WITHIN-GROUP TESTS

The formula “improvement in the extended group, no improvement in the others” draws attention to duration. Yet it does not demonstrate that extending treatment produces a greater improvement. Each within-group test sets

a change against zero; the therapeutic question sets changes against one another. A difference in significance may depend on dispersion, available data or group size, even when the estimated effects are close. The primary result at two weeks therefore keeps its place. Later observations may give rise to a hypothesis about time to effect; they do not retrospectively shift the primary time point. A subsequent study could choose a later comparison, announce its endpoint and plan the corresponding sample size. That would be a new test of the hypothesis, not the simple repetition of a success already achieved.

12.3.3.2.2 Comparing durations despite missing data

The design nevertheless raises an interesting question: what does maintaining an intervention after a common initial period change? To answer it, the assigned comparison between short and long strategies must be kept, rather than selecting the most encouraging trajectories after the fact. A duration is compared together with its schedule, its successive treatments and its follow-up conditions. Missing data are part of this reading. The primary analysis omits absent observations; for some later analyses, the last known value is carried forward, with another imputation method examined by the authors. Neither the term "intention to treat" nor the sophistication of an imputation recreates the missing measurements. The questions are how many people contribute, why values are missing and how a different course in these people would change the judgment. A sensitivity analysis tests a dependence on assumptions; it does not erase that dependence.

THE MISSING MEASUREMENTS

12.3.3.3 Immunological Reading: modifying a marker and restoring a capacity are two tests

A TEMPORAL DISSOCIATION

The temporal dissociation between biological response and exercise measurement prevents their link from being regarded as automatic. A fall in CRP provides information on a measured inflammatory consequence; it does not by itself localize the action in the myocardium and does not identify the mechanism limiting exercise. A correlation between biological and functional changes can guide inquiry. It does not show that modifying the first produces the second: general recovery, concomitant treatments or other changes may contribute to both. The value of the dissociation is to make these competing explanations visible, without declaring the marker useless when it does not immediately predict the full benefit hoped for.

12.3.3.4 Cardiological Reading: preparing a trial without announcing its result

ATTRIBUTING AN IMPROVEMENT

After decompensation, the clinical trajectory can change under the combined effect of all care received. Functional improvement must be attributed through the appropriate comparison. It does not automatically convert into fewer hospitalizations or longer life. Likewise, a few clinical events favoring one strategy can help design research without constituting a sufficiently stable estimate of its benefit. The observed inflammatory modification is therefore not accompanied by demonstrated success on the early primary endpoints. The hypothesis of a later functional response remains to be tested; it suffices to establish neither the superiority of twelve weeks over two, nor a reduction in serious events, nor general safety. A confirmatory comparison of durations, with more com-

plete follow-up and prespecified endpoints, would allow this hypothesis to be distinguished from a favorable course under overall care.

12.3.3.5 Integrative Clinical Reading

THE MOST RECENT CHANGE

After a hospitalization, an improvement may be spontaneously attributed to the most recent change. REDHART shows why the account of recovery must also keep the entire care pathway and the comparison between groups in view. Integrative reading does not set this account against measurement: it brings them into dialogue. Resuming an activity, tolerating an additional duration of treatment and avoiding a new hospitalization are different expectations. The late functional signal does not allow all of them to be announced as demonstrated consequences of prolonged exposure.

12.3.4 TRED-HF and the Difference between Recovery and Cure

12.3.4.1 Situating the Result

TRED-HF, PHASED WITHDRAWAL

TRED-HF raises a question that improvement does not resolve: would the good function observed persist without the treatments? This open-label pilot trial assigns 51 people with recovered dilated cardiomyopathy to phased withdrawal or continuation of therapy. In the first six months, the relapse criterion is met in 11 of the 25 participants assigned to withdrawal, versus none of the 26 controls. Relapse requires crossing predefined thresholds of ejection fraction, ventricular volume or NT-proBNP, or clinical signs of heart failure. These components are not equivalent to death or hospitalization.[7]

12.3.4.2 Critical Reading

AFTER THE RANDOMIZED COMPARISON

The randomized comparison must be separated from the subsequent phase, in which controls are offered withdrawal. Pooling all the people who stopped their treatment makes it possible to describe their outcome; it does not create a second randomized comparison against concurrent continuation. The timeline changes the strength of the inference, even when the observations seem to converge. The trial is open-label, but analysis of cardiac volumes is performed blind. These two properties can coexist. Blinding of the measurement protects one step; it does not erase the possible effects of knowing the group on reported symptoms or decisions. Monitoring is also more frequent during withdrawal. A precise reading must ask which events might be detected differently, without equating all measurements with subjective judgments. Withdrawal concerns a strategy comprising several drugs, according to a defined order. The result therefore does not allow all the protection to be attributed to one class, nor does it determine which drug could be removed on its own. A change in heart rate may suggest a lead; it is not equivalent to a separate randomization of the withdrawal of each drug. Finally, small size does not preclude a clear contrast on the primary endpoint. Here it mainly limits precision, the study of rare situations and the search for individual predictors. The retrospective power calculation presented by the authors does not replace a prospective justification of the sample size. The useful question remains what the estimate can distinguish, for the population and duration studied.

OPEN-LABEL, BUT BLINDED MEASUREMENT

POWER CALCULATED AFTER THE FACT

12.3.4.3 Immunological Reading

dependence without a designated mechanism

The study does not test an immune intervention and does not demonstrate persistent inflammation. A relapse after withdrawal can reveal a dependence on treatment without identifying the process that produces it. Translating "disease still liable to manifest itself" as "immune mechanism still active" would add a conclusion absent from the experiment. This restriction is fruitful for cardioimmunology. A functional measure returned to a reference range does not necessarily describe all the biological states of the tissue. But their possible persistence must be sought through appropriate measurements and comparisons. The result opens this inquiry; it does not choose its answer in advance.

12.3.4.4 Cardiological Reading

monitoring and reinstatement

Monitoring and the reinstatement of treatment at the first sign of deterioration are part of the design. The absence of death or unplanned hospitalization for heart failure therefore does not demonstrate that stopping without monitoring would have been safe. Conversely, reaching a biological or imaging threshold does not mean having suffered a serious clinical decompensation.

cure as a tested stability

The specific severity of each event must be kept in view. The notion of cure thus becomes a proposition about the stability of a state when its conditions change. It might be objected that no health is independent of every external condition. The objection is fair: the study does not seek absolute autonomy, but the persistence of the result after a precisely defined modification. This limited test suffices to correct an overly hasty equation between a normalized measure and therapeutic independence. What is established is the increased risk of protocol-defined relapse after this withdrawal strategy. What remains open concerns the mechanisms, the indispensable components of treatment and the reliable identification of people able to maintain recovery. This historical study does not, by itself, constitute a guide to stopping contemporary treatments.

12.3.4.5 Integrative Clinical Reading

returning to a normal life

TRED-HF also concerns the meaning each person gives to returning to a life considered normal. Continuing treatments once measurements have improved may seem contradictory if their possible role in that improvement has not been explained. Integrative reading takes this expectation of relief seriously, without confusing it with a demonstration of therapeutic independence. It distinguishes the reasons for wanting to reduce the burden, the comparison actually studied and the monitoring conditions of the protocol; it does not provide a personal rule for withdrawal.

12.3.5 Wohlford and the First Exposure in Patients

12.3.5.1 Situating the Result

Wohlford, dapansutrile in phase 1B

Wohlford and colleagues study dapansutrile, an NLRP3 inhibitor, in a single-center phase 1B study in 30 patients with stable systolic heart failure. Three successive cohorts receive escalating doses for up to fourteen days: each com-

prises eight participants on treatment and two on placebo. Functional analyses address within-group changes, without statistical comparison with placebo or correction for multiple comparisons. No serious adverse event is reported; nonserious events occur. Some functional parameters improve in the cohort receiving the highest dose. CRP does not decrease significantly. These observations constitute an early exploration of tolerability and pharmacodynamics, not a demonstration of clinical efficacy.[8]

12.3.5.2 Critical Reading: a randomization whose use must be examined

A CONTROL THE ANALYSIS DOES NOT USE

This work could be classified among controlled trials and immediately credited with the strength of a therapeutic comparison. An additional question must nevertheless be asked: what difference does the analysis estimate? The presence of a control group in the protocol and its use to support a conclusion are two distinct properties. The design can help maintain blinding without providing, in the published analysis, the comparative estimate expected for efficacy. An improvement between two visits leaves several explanations open: drug effect, spontaneous change, variation in measurement conditions or familiarization with a test. These are possibilities to be separated, not biases whose presence would be demonstrated here. Randomization does not eliminate them from a comparison limited to treated participants. Likewise, a significant change in one cohort and a nonsignificant one in another do not constitute a test of difference between doses. To speak of a clinical dose-effect relationship, an analysis designed for this question and an estimate of its uncertainty would be required. Tolerability calls for different reasoning. The absence of an observed serious event may make an additional research step conceivable; it does not provide sufficient information on complications that are rare, late or specific to frailer patients. The time horizon and the number of people exposed are part of the conclusion. Removing them turns a limited observation into a general assurance.

THE ABSENCE OF A SERIOUS EVENT

12.3.5.3 Immunological Reading: distinguishing exposure, target and response

EXPOSURE, TARGET, RESPONSE

Three propositions must remain separate: the drug reaches the body, it modifies the targeted biological pathway and this modification benefits the patient. Their sequence is an experimental hypothesis to be documented. A circulating concentration is not enough to establish relevant inhibition in all compartments; an inflammatory marker does not by itself summarize the entire activity of a target. Conversely, the absence of a statistically discernible change in a marker does not automatically refute the mechanism. It requires specifying what the marker measured, at what time and with what sensitivity. It would, however, be equally incorrect to systematically resolve a disappointing result by invoking an insufficient dose or a parallel pathway: these possible explanations require their own tests. The mechanistic argument retains its value when it accepts the conditions of being put to the test.

12.3.5.4 Cardiological Reading: defining the next research decision

AN EARLY TRIAL, ONE STEP

The role of an early trial is to reduce certain unknowns before exposing more people. It should therefore be judged neither as a failed large morbidity and mortality trial nor as its miniature equivalent. The useful question is that of moving to the next step: what population, duration, comparator and endpoint would turn the signal into an interpretable estimate? For cardiac function, exercise capacity or a symptom, improvement must also be assessed according to its relevance to the patient. A short observation does not settle durability, hospitalization or survival.

12.3.5.5 Integrative Clinical Reading

TAKING PART WITHOUT ESTABLISHED BENEFIT

Taking part in a first study in patients does not mean receiving an already established benefit. Wohlford invites clarification of this difference without reducing participation to abstract risk-taking. Time spent, assessment constraints and hopes of relief belong to the experience that would need describing. Integrative reading distinguishes what research learns about an exposure from what a person can reasonably hope for from it; it keeps open the possibility of contributing to knowledge without an individual therapeutic promise.

12.3.6 Kim and the Passage from Metabolism to Immune Response

12.3.6.1 Situating the Result

KIM, EMPAGLIFLOZIN AND GLIMEPIRIDE

Kim and colleagues randomize 71 people with diabetes at high cardiovascular risk to empagliflozin or glimepiride for thirty days; 61 complete the study. The trial is open-label, but the examiner performing the cellular experiments is unaware of the assignment. The primary endpoint concerns interleukin 1β (IL-1β) secretion by blood-derived macrophages after experimental stimulation. Its decrease is greater with empagliflozin. Ex vivo manipulations of β-hydroxybutyrate, glucose and insulin explore possible mechanisms. The work does not measure whether this immune modification explains a reduction in cardiovascular events.[9]

12.3.6.2 Critical Reading: an active comparator does not neutralize all differences

12.3.6.2.1 What an active comparator leaves free to vary

AN ACTIVE COMPARATOR

Choosing another diabetes treatment allows a more precise question than comparison with no treatment alone: do the two strategies have the same inflammatory consequences? But an active comparator is not a biologically neutral control. An observed difference expresses the contrast between two interventions, each liable to modify several pathways. It does not directly yield the share attributable to an isolated mechanism. Likewise, a glycemic course without a statistically established difference does not demonstrate complete glycemic equivalence. To support that latter proposition, an acceptable margin would need to be defined, relevant measures chosen and sufficient precision

obtained. The contrast can be regarded as informative without turning it into an experiment in which all metabolic determinants other than the one of interest were held constant.

12.3.6.2.2 The participants retained and the repeated measurements

COMPLETING THE STUDY, A LATER FACT

The reading must then follow people through to the results. Completing the study is not a characteristic fixed before randomization. Examining withdrawals and the available information would make it possible to judge whether their exclusion could alter the comparison. The initial number of participants must therefore not be used as the automatic denominator of each biological experiment. Another point of vigilance concerns repetitions. The figure legends distinguish biological samples from repeated experiments. The data file reflects this structure, with several measurements per sample, but does not give the statistical model used. Repeating a measurement improves its characterization; it does not create as many independent people. To assess the stated precision, it remains to be checked how the analysis accounts for these dependencies. A comparison based on people can exploit their repeated measurements, provided this organization is respected. Without knowing this statistical handling of dependencies, neither invalidating all the tests nor validating their precision is justified.

12.3.6.3 Immunological Reading: an induced response is not spontaneous secretion

AN INDUCED RESPONSE

A cellular assay can reveal a difference that the blood measurement does not summarize. Still, the name of what it observes must be kept: a capacity for response under conditions prepared in the laboratory. Culture, priming and stimulus define the experiment. Its result is not interchangeable with the circulating concentration of a cytokine, nor with the ongoing activity of cells in the myocardium. The ex vivo manipulations strengthen the interpretation when they separately test elements of the proposed chain. They do not, however, abolish the question of concentrations and milieu. A response obtained at an experimental exposure must be related to the exposures actually reached, before it is made into a sufficient physiological explanation. A pathway can be capable of producing an effect without carrying most of that effect in patients.

12.3.6.4 Cardiological Reading: what the clinical comparison must add

THE FRACTION CARRIED BY THE PATHWAY

Even if other research establishes a clinical benefit, juxtaposing that benefit with an immune modification does not measure the fraction of the effect transmitted through this pathway. That question would require data linking intervention, mediator and clinical outcome, with appropriate timing and explicit assumptions about their other determinants. The scope of this text is therefore to make an explanation testable. It helps design comparisons that could separate several mechanisms, without providing a new therapeutic indication.

12.3.6.5 Integrative Clinical Reading

SOPHISTICATION AND CERTAINTY

In a person with diabetes already exposed to several therapeutic decisions, explaining an immunological result requires specifying what it really adds to the discussion. Kim brings a question about possible mechanisms, not a new measure of lived benefit. Integrative reading prevents biological sophistication from artificially increasing the certainty of a choice. It distinguishes the proposed explanation, the clinical results established by other work and personal priorities; bringing these levels together requires argument, not their mere presence in the same file.

12.4 Worked Exercises

12.4.1 Exercise on Rewriting a Dissociated Result

12.4.1.1 Problem

Rewrite the sentence "The treatment lowers CRP, but the patients are no better" so that it distinguishes the observed result, the undemonstrated result and the scope of the study.

12.4.1.2 Reasoned Answer

A DEFENSIBLE SENTENCE

A defensible wording would be: "The trial reports a biological response, but does not demonstrate improvement in the functional endpoints studied compared with placebo." This sentence avoids equating the absence of a significant difference with proof that each patient remained unchanged. It also does not extend the result to every possible dimension of clinical status.

12.4.2 Exercise on Comparing Durations

12.4.2.1 Problem

NINETY FICTIONAL PEOPLE

Entirely fictional data, not reproducing REDHART. A trial randomly assigns 90 people, after stabilization of a cardiac episode, to three strategies: X for two weeks then placebo, X for twelve weeks or placebo for twelve weeks. Each group comprises thirty people; all final measurements are available. The primary endpoint planned at two weeks shows no demonstrated advantage of X over placebo. At twelve weeks, the mean change in an exercise capacity is +1.2 units in the short group, +2.0 in the extended group and +1.4 on placebo. Within-group tests against zero change give p = 0.08, p = 0.01 and p = 0.06, respectively. The exploratory comparison extended minus short gives +0.8 units (95% CI: −0.9 to +2.5). No minimal threshold of clinical relevance was set. An investigator asserts: "Twelve weeks work; two weeks and placebo do not." Correct this sentence. What would be the relevant contrast for studying extension? What should be planned next?

12.4.2.2 Reasoned Answer

All three means change favorably. Only the extended group crosses the threshold chosen for its within-group test; this difference in significance does not demonstrate a difference in efficacy. The placebo group also improves, by +1.4 units: spontaneous change and other care must remain part of the reasoning. For extension, the relevant contrast compares the extended and short strategies: +2.0 − 1.2 = +0.8 units. Its interval includes zero and values in both directions. Superiority is not demonstrated; nor is equivalence. Extended minus placebo is descriptively +0.6 units, but no interval is provided for this contrast. The late result does not replace the initial primary endpoint. It may motivate a study in which the comparison between durations and the schedule would be announced in advance. A clinically important difference would need to be specified, the study sized to estimate it and multiple analyses organized. Finally, better exercise capacity would not, by itself, answer the question of serious events or additional adverse effects of prolonged exposure.

EXTENDED MINUS SHORT

A STUDY ANNOUNCED IN ADVANCE

Synthesis

A trial without demonstrated benefit weakens a precise proposition: the one linking a given intervention, in defined people and over a defined duration, to an expected result. It does not refute all immune involvement, and it does not hide a success. The anakinra trials show that a biological response and a functional benefit require two tests, and that a within-group test does not replace the comparison between strategies. A crossover design, imputed data, a revised sample size or an active comparator changes what the estimate can distinguish. TRED-HF shows that a recovered function may still depend on treatment. An explanation of failure becomes instructive when it accepts a risk of refutation and prepares a new comparison.

Open Questions

What measurable criterion, defined independently of the result, would make it possible to identify the stage at which an inflammatory intervention could still improve function? What measurements would separate a cardiac action from a vascular or muscular contribution to exercise capacity? How can the people able to maintain recovery without treatment be recognized before withdrawal? These questions give negative trials their most useful continuation.

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13

Infiltrative Cardiomyopathies and the Specificity of Treatments

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Transthyretin amyloidosis, cardiac sarcoidosis and giant cell myocarditis can produce similar images and presentations. Their treatments bear no resemblance to one another. A transthyretin stabilizer is not an immunotherapy; immunosuppression targets an entirely different process. Diagnosis thus carries a particular weight: it opens or closes access to an intervention. Scintigraphy, positron emission tomography (PET) and magnetic resonance imaging (MRI) make it possible to recognize some of these diseases without biopsy, to stratify a risk or to monitor a response. Each of these uses requires its own validation. Recognizing a substance, measuring an activity, predicting an event and guiding a treatment are distinct tasks, even when a single image seems to accomplish them all.

13.1 What Validates a Diagnostic Distinction

13.1.1 A Resemblance Organizes Inquiry without Settling Its Conclusion

GROUPING IN ORDER TO INVESTIGATE

Grouping diseases according to a shared tissue or cardiac presentation makes it possible to begin an inquiry. Such grouping does not guarantee that they call for the same intervention. The reading of Ruberg's review (see 2.4.2) already distinguished recognizing a pattern from identifying what produces it. The question here becomes more demanding: how can it be established that a diagnostic distinction deserves to guide a therapeutic choice? A morphological diagnosis describes an observable organization. An etiological diagnosis seeks to identify the responsible process. These two descriptions do not simply replace each other. Knowing a cause may leave the current severity open; describing a lesion precisely may leave several causes possible. Information is better served by preserving these dimensions than by making the most specialized name a summary of everything that matters for the person. It might be objected that this caution deprives diagnosis of its practical function. The reverse is true: specifying what a diagnosis establishes makes it possible to know which decision it can support. A classification robust enough to organize follow-up is not necessarily robust enough to justify an irreversible intervention. The certainty required also depends on the consequences of error, although these consequences cannot make an insufficiently supported diagnosis true.

THE PRACTICAL FUNCTION OF DIAGNOSIS

13.1.2 A Classification Is Not Validated by Repeating Its Definition

A SIGNAL THAT CLASSIFIES AND CONFIRMS

Consider an entirely fictional example. Two categories, A and B, are separated by an imaging signal. It is then observed that A more often displays this same signal. The observation is accurate but does not validate the distinction: it reproduces the classification rule. If the reference diagnosis incorporates the result of the test under evaluation, their agreement risks overestimating what the test contributes independently. The question must therefore be asked: which observations could contradict the classification? An independent tissue identification, a different trajectory or a distinct therapeutic response can provide complementary tests. They are not interchangeable. A classification can correctly describe a substance without predicting the entire course; it can predict a risk without identifying the mechanism responsible for that risk. The objection is that a perfect and independent reference does not always exist. This does not condemn diagnosis to circularity. The dependencies between observations must be made visible, the disagreements specified and the cases that

THE IMPERFECT REFERENCE

the first rule misclassifies put to the test. An imperfect reference does not become infallible because it is called the reference. Knowing its conditions and limits makes it possible to evaluate it; this does not yet prove its superiority. The preference granted to it must rest on documented advantages for the task at hand, for example a more reproducible comparison or a better distinction between relevant situations.

13.2 Therapeutic Specificity and the Test of Classification

13.2.1 Identifying a Disease and Recognizing an Activity

recognizing, monitoring, estimating

A marker can serve to recognize a condition, to monitor an activity or to estimate a risk. The same instrument can contribute to several tasks, but validation for one does not automatically hold for the others. A category may remain relevant while a signal varies over time. Conversely, a persistent signal does not imply that all of the damage is still modifiable. This distinction sheds light on therapeutic specificity. One intervention may target a constituent element of a process; another, its consequences; a third, the body's tolerance of those consequences. The treatment closest to the biological definition is not, by that proximity alone, the one with the greatest clinical benefit. Comparing outcomes remains necessary. Calling a treatment "specific" must therefore state the relation intended: specificity for a target, for a disease or for a benefit in a given population. These properties can support one another without merging. The molecular precision of an action contains neither the measure of its benefit nor that of its harms.

"specific" to what?

13.2.2 When Treatment Tests the Classification

comparing effects across categories

A difference in response between categories can strengthen the case for a classification. The effects must still be compared, rather than merely noting a statistically significant result in one group and a nonsignificant one in the other. The treatment modalities, the recruitment and the precision must allow this comparison. Otherwise, differences in information risk being turned into differences in disease. Nor can therapeutic response alone define the "true" diagnosis retrospectively. If every responder is declared correctly classified and every nonresponder reclassified, the treatment necessarily validates the category rebuilt around its result. A group defined before the intervention, then examined in another population, exposes the proposition more fully to a genuine test. Finally, the benefit of a strategy does not by itself demonstrate the usefulness of every procedure that might select its recipients. A new diagnostic pathway must also be judged by the errors and delays it introduces or avoids. The chain remains open at each step: recognizing, distinguishing, choosing and verifying. Progress consists in bringing these operations into closer agreement, while making it possible to locate the disagreement when one of them fails.

recognizing, distinguishing, choosing

13.3 Texts Put to the Test

13.3.1 Gillmore and the Conditions for a Diagnosis without Biopsy

13.3.1.1 Situating the Result

Gillmore, a diagnosis without biopsy

Gillmore and colleagues study a rule combining bone scintigraphy with testing for a monoclonal protein. Among 1,498 patients referred to specialized centers, 1,217 are analyzed; 281 cases without histological confirmation and with imaging suggestive of amyloidosis are excluded because of diagnostic uncertainty. Only 374 participants have an endomyocardial biopsy. In this subgroup, grade 2 or 3 cardiac uptake combined with the absence of a monoclonal protein shows an observed specificity of 100% and a sensitivity of 70% for transthyretin amyloid (ATTR) deposits. The positive predictive value is 100%, with a confidence interval of 98 to 100%. The authors propose this rule in a setting of heart failure with echocardiography or MRI consistent with or suggestive of amyloidosis.[1]

13.3.1.2 Critical Reading

a rule with several conditions

The proposition evaluated is not "a positive image is enough." It brings together several observations, some of which help to rule out another explanation. Omitting a condition amounts to changing the test. The uptake threshold, the laboratory tests and the setting in which the rule was constructed must be kept together. The reference standard deserves as much attention as the new test. A cardiac biopsy and an extracardiac biopsy combined with imaging do not provide exactly the same observation. Pooling them increases the sample size but does not turn every diagnosis into a cardiac histological verification. The analysis must therefore begin with the directly verified subgroup, then examine what the whole set contributes, without treating these two nested analyses as two independent replications. Excluding uncertain cases raises a particular question: does performance in classifiable situations adequately describe performance in the difficult situations where the test will be used? It does not make the study useless. It limits the population actually evaluated and calls for a validation that retains the ambiguities encountered in practice. The success rate must remain attached to the pathway that produced it. Finally, an observed 100% does not mean that a future error is impossible. The confidence interval conveys sampling uncertainty; it does not resolve the differences between expert centers and the general population. The positive predictive value also depends on the population examined. A series oriented toward one disease cannot therefore, on its own, provide a guarantee valid for all screening.

uncertain cases excluded

100% observed

13.3.1.3 Immunological Reading

immunoglobulins without autoimmunity

Testing for a monoclonal protein relies on serum and urine immunofixation and on measurement of serum free light chains. Its role here is to distinguish processes capable of producing a similar cardiac presentation. A signal related to immunoglobulins must not become, through a slide in vocabulary, proof of autoimmunity. The nature of the protein sought, its origin and its role in the damage remain distinct questions. The reasoning also shows why an as-

sociated abnormality should not automatically be designated as a cause. A monoclonal protein can coexist with deposits of another kind. Its detection changes the path of evidence; it does not dispense with identifying what the deposits are made of. Conversely, its absence is not sufficient, in isolation, to name every cardiomyopathy. The strength of the diagnosis lies in the defined combination of clues.

13.3.1.4 Cardiological Reading

DISEASE EXCLUDED OR PATHWAY INAPPLICABLE

A rule constructed to affirm certain diagnoses may miss some patients. Here, failing to meet the conditions for diagnosis without biopsy therefore does not mean the absence of transthyretin amyloidosis. It means that this particular shortcut is not established. The distinction between "disease excluded" and "diagnostic pathway not applicable" is essential to avoid ending the inquiry prematurely. It might be objected that any restriction reduces the value of a noninvasive diagnosis. Yet avoiding a biopsy in a defined group already constitutes a possible advance. There is no need to promise its disappearance for all patients. The right question is which people can benefit from this simplification and which situations require another form of proof. This study examines diagnostic performance; it neither randomizes a diagnostic strategy nor demonstrates its effect on survival. Identifying a category more accurately can make a treatment relevant without yet proving that each use of the test improves outcome. The lesson is a demand for precision: a useful rule remains tied to its conditions, its reference standard and the decision it actually enables. The 2016 rule must be placed within the evolution of the evidence before guiding a current decision.

THE 2016 RULE

13.3.1.5 Integrative Clinical Reading

A NAME AND HOW IT WAS ESTABLISHED

Avoiding a biopsy can simplify a care pathway, but only if the conditions of the diagnostic rule are preserved. Gillmore thus points to a requirement for coordination: an image, a laboratory workup and a clinical context must be interpreted together. For the person, receiving the name of a disease and understanding how it was established are two linked needs. An integrative reading also explains why a result that does not satisfy the rule does not necessarily close the inquiry, without multiplying tests as a matter of principle.

13.3.2 ATTR-ACT and the Definition of Therapeutic Benefit

13.3.2.1 Situating the Result

ATTR-ACT, TAFAMIDIS VERSUS PLACEBO

ATTR-ACT randomizes 441 people with transthyretin amyloid cardiomyopathy to tafamidis 80 mg, tafamidis 20 mg or placebo for thirty months. The primary analysis pools the two doses: 264 participants versus 177. It first compares mortality, then the frequency of cardiovascular hospitalizations when the first comparison does not separate the people compared. This hierarchical ranking favors tafamidis; the ratio of comparisons won to comparisons lost, or win ratio, is 1.70, with a 95% confidence interval (95% CI) of 1.26 to 2.29.[2]

13.3.2.2 Critical Reading: a clinical hierarchy in the calculation

13.3.2.2.1 A clinical priority and its counting conventions

A CLINICAL HIERARCHY

The method does not give equal weight to all events. It expresses a priority: survival precedes hospitalization in the comparison. This choice makes a clinical judgment explicit, instead of hiding it behind the word "composite." It provides neither an individual probability of being cured nor a multiplication of life expectancy. Two favorable results calculated differently may answer two different questions. A decisive detail concerns the events counted as deaths: heart transplantation, combined heart-liver transplantation and implantation of a mechanical cardiac assist device. The published hazard ratio (HR) of 0.70 must be read with this convention. A sensitivity analysis that does not count these interventions as deaths gives 0.67, with an interval of 0.49 to 0.94.[2] The conclusion therefore does not simply depend on a change of vocabulary; the sensitivity analysis supports its stability under this other definition. But the numbers must keep their own meaning: an event counted as a death for the analysis does not become an actually observed death.

EVENTS COUNTED AS DEATHS

13.3.2.2.2 Pooled doses and missing data

POOLED DOSES

Pooling the doses demands a different discipline. Demonstrating the efficacy of a pooled strategy does not demonstrate the equivalence of the doses. Answering this second question would require a comparison designed for that purpose and sufficiently precise. Likewise, the label "modified intention-to-treat" is not enough to judge the analysis: here, all randomized participants meet its conditions and are included.[2] What matters is the path of the participants, not the label in isolation. Including all randomized participants in an analysis does not, however, mean that all their walking or quality-of-life measurements are available. Deaths and withdrawals change who can still be measured; longitudinal estimates must be read with their assumptions about missing data.

13.3.2.3 Immunological Reading: a boundary useful to the field

A BOUNDARY OF CARDIOIMMUNOLOGY

Tafamidis is presented as a transthyretin stabilizer, not as an immunotherapy.[2] This action clarifies a boundary of cardioimmunology: explaining a cardiomyopathy by a protein process and modifying that process is not equivalent to treating every cellular reaction that might accompany it. The trial supports a clinical proposition concerning one intervention and one selected disease. It does not measure each step between molecular stabilization and the event avoided. The benefit therefore proves neither that every tissue consequence has disappeared nor that every patient responds through an identical biological chain.

13.3.2.4 Cardiological Reading: slowing a trajectory is not returning to the starting point

SLOWING, NOT RESTORING

Walking capacity and quality-of-life score decline less with tafamidis than with placebo.[2] A favorable difference between groups does not mean that the treated group recovers its previous state. Speaking of "slowing" preserves a

NYHA class III, more hospitalizations

real benefit while avoiding the promise of a restoration that the result does not contain. Hospitalizations pose an additional difficulty. In New York Heart Association (NYHA) class III, their frequency is higher with treatment; the authors propose prolonged survival during a severe phase as an explanation. They present it as a conjecture.[2] Dying and being hospitalized are indeed not two independent possibilities: remaining alive leaves time to experience a hospitalization. But this general reasoning does not demonstrate that this mechanism fully explains the observed contrast. The subgroup result and its interpretation must be kept separate. The absence of NYHA class IV and the exclusion of light-chain amyloidosis also delimit the direct scope of the trial.[2] Identifying another disease through a morphological resemblance is not enough to extrapolate the benefit to it. Conversely, this limit does not mean that each excluded situation would respond unfavorably: it indicates where the comparison is missing.

13.3.2.5 Integrative Clinical Reading

13.3.2.5.1 From established diagnosis to expected benefit

From diagnosis to outcome

Bringing Gillmore and ATTR-ACT together makes it possible to follow two successive questions without conflating them. Gillmore examines the conditions under which a combination of clues can support a diagnosis; ATTR-ACT compares outcomes under therapeutic strategies in a selected population. A better-established diagnostic category makes the second question relevant. It does not, by itself, decide what a given person should choose.[1,2] Moving to judgment begins by specifying which proposition each text contributes to the case: justification of the diagnosis, then expected benefit under the conditions studied. It remains to be examined whether these conditions sufficiently illuminate the present situation. This step must not be replaced by enthusiasm at finally having a name for the disease. The certainty gained in identifying a condition does not carry over fully to the prediction of individual benefit.

13.3.2.5.2 The person's priorities in the decision

What matters to the person

Then comes a different question: which consequences matter most to the person? ATTR-ACT ranks mortality and hospitalizations in its analysis. This hierarchy makes the calculation interpretable; it does not by itself capture an individual preference. To discuss the result, it is possible to ask what preserving an activity, limiting hospital stays or accepting a burden of care means for the person. These dimensions must be presented together with what the study actually shows and what remains unknown, without fabricating a personal estimate absent from the data. It will be objected that introducing priorities risks relativizing evidence of efficacy. That would be the case if a preference retrospectively altered the trial result. Here, it comes in elsewhere: in appraising the expected consequences and the options actually available. A justified decision must therefore be able to state which evidence supports it, which extrapolation it assumes and which priority it respects. This juxtaposition teaches a continuity of reasoning, without turning two historical publications into current clinical practice.

13.3.3 Blankstein and the Distinction between Prognosis and Utility

13.3.3.1 Situating the Result

Blankstein, PET and prognosis

This single-center cohort[3] analyzes 118 patients referred for known or suspected cardiac sarcoidosis, without known coronary artery disease. PET combines rubidium perfusion with fluorodeoxyglucose (FDG) metabolism. Thirty-one patients experience death or sustained ventricular tachycardia during a median follow-up of 1.5 years. The combination of perfusion and metabolic abnormalities retains a prognostic relationship after adjustment: HR of 2.87 (95% CI: 1.05 to 7.85). Imaging results were available for care; arrhythmic events were largely detected by implanted devices. The technical and statistical details referred to the supplement remain necessary to appraise the methods in full.

13.3.3.2 Critical Reading: specifying what a test adds

adjustment and calibration

The prognostic relationship validates neither the diagnostic specificity of the test nor the benefit of a care strategy guided by it. Adjustment calls for equally precise interpretation. An association that persists in a model is not a complete measure of predictive improvement. Building an individual tool would require, in particular, checking calibration: do the predicted risks match the observed frequencies? Separation between groups can be clear while the probabilities assigned to each person remain poorly estimated. The tool would also need to be tested in another population, with a sufficiently reproducible reading procedure. Knowledge of the result also transforms the question. When the test enters into decisions, the observed outcome is that of patients who were actually managed. It does not automatically represent the natural history associated with each image. Treatment intensification could attenuate a contrast; closer surveillance could increase detection. Without an appropriate comparison, choosing one of these explanations as certain would replace an unknown with a narrative.

13.3.3.3 Immunological Reading: keeping the distance between signal and mechanism

a localized signal

The image prompts thinking in terms of inflammatory activity; a methodological reading must ask which proposition this expression authorizes. A spatially localized signal does not, by itself, name the responsible cell population, the dominant molecular pathway or the initial cause. Moving directly from a visible territory to a particular immunotherapy would add several untested inferences.

13.3.3.4 Cardiological Reading: from classification to choice

comparing actions

The cardiologist does not merely choose a degree of concern, but compares actions, their expected benefits, their complications and the consequences of abstaining. A higher risk does not guarantee a greater therapeutic benefit: that

benefit also depends on the share of risk the intervention can modify. A study of an imaging-guided strategy should therefore prespecify a decision rule, a comparator, relevant events and the harms of the diagnostic or therapeutic pathway. Finally, an implicit equivalence between a detected arrhythmic event, a device intervention and a death avoided must not be assumed. The clinical relevance of each event must be preserved without attributing to it an inaccessible individual counterfactual. The prognostic association therefore justifies examining a rule that incorporates imaging, without demonstrating the benefit of a treatment chosen on that result alone. Validation must address the consequences of this rule in its context of use. Only under this condition can risk classification contribute to a useful care pathway.

13.3.3.5 Integrative Clinical Reading

AN ADDITIONAL DEGREE OF WORRY

An image associated with higher risk can durably change the way a person envisages the future. Blankstein invites caution against letting this classification act alone: what additional decision would be considered, with what consequences and what uncertainty? The integrative reading also considers the tests, explanations and interventions prompted by the information. It does not dispute the prognostic value of the signal; it asks for verification that the pathway opened by this signal offers something other than an additional degree of worry.

13.3.4 Nordenswan and the Comparison of Diagnoses

13.3.4.1 Situating the Result

NORDENSWAN, TWO DIAGNOSES COMPARED

The Finnish cohorts compared by Nordenswan comprise 351 cases of cardiac sarcoidosis and 28 of giant cell myocarditis, after a second diagnostic review. The survival analysis excludes sudden deaths that were the sole manifestation. At five years, survival free from cardiac death, aborted sudden death or transplantation is estimated at 77% for sarcoidosis and 27% for giant cell myocarditis. The association between diagnosis and events weakens after adjustment for markers of myocardial injury and dysfunction. But the successive models do not include the same numbers of people: the last rests on 190 people, compared with 324 for the first multivariable model.[4]

13.3.4.2 Critical Reading: the registry also preserves the history of classification

13.3.4.2.1 A revised classification changes the groups compared

A HISTOLOGICAL REVISION

A national cohort broadens the view of a rare disease. It does not turn diagnosis into immutable data. Here, second histological review moved many cases initially classified as giant cell myocarditis to sarcoidosis. This operation is not a simple correction of vocabulary: it changes the composition of the groups whose trajectories will later be compared. Two objects must therefore be distinguished. The first is the outcome of people classified according to the criteria adopted after revision. The second would be the outcome attached to the labels as they had been used at the time of care. Both questions are legitimate; their answers are not interchangeable. Comparison with an older series requires

checking its histological rule before attributing any difference in prognosis to therapeutic progress.

13.3.4.2.2 What adjustment isolates and what it does not

ADJUSTING FOR THE LESION

Adjustment raises a further difficulty. If the diagnosis characterizes a process that produces the lesion, holding that lesion constant no longer answers the same question as comparing the groups as they present. What is then examined is what the label adds, conditional on the measures introduced. This does not, by itself, decompose the causal contribution of the biological process. A prognostic model can thus show a limited additional contribution without making the classification useless for understanding or treatment. Added to this is the change in population between models. When tests are missing, adding a variable can also remove people from the analysis. The attenuation of a coefficient may therefore reflect both the new information and the observations retained. To separate these contributions, a comparison of the models on a common set, followed by suitable sensitivity analyses, would be informative. The study does not allow the decrease in the coefficient to be read as an experiment isolating a cause.

13.3.4.3 Immunological Reading: shared pathways do not demonstrate a single disease

WHAT THE COHORT MEASURES

An inflammatory kinship is compatible with several classifications: it does not, by itself, decide between two diseases and two variants of a single condition. The reasoning must also respect what this study measures: presentations, revised diagnoses and outcomes. It does not experimentally follow the transformation of one disease into the other and does not identify a single mechanism by intervention. Supporting a biological continuity would require evidence independent of the clinical resemblance: mechanisms, timing or tissue trajectories capable of deciding between competing explanations.

13.3.4.4 Cardiological Reading: the start of the curve is part of the result

SUDDEN DEATH AS FIRST MANIFESTATION

A survival estimate calculated among those who had a clinical presentation that was then followed does not describe all affected people, notably those whose first manifestation was fatal. This exclusion must remain visible when explaining to which population the estimate belongs. The composite endpoint also brings together outcomes of different meaning. A transplantation expresses severity but also access to an intervention; an aborted sudden death remains distinct from a death. The comparison describes outcome under the care actually received. It does not measure, by randomization, the relative efficacy of immunosuppressive regimens or devices. The reconstructed groups thus show different profiles and outcomes. Injury markers can provide prognostic information beyond the name, without making histology useless or establishing that the diseases are identical. Independently reproduced classifications and comparisons retaining the same patients would clarify the respective roles of classification and severity; a causal attribution to treatments would require yet another comparison.

13.3.4.5 Integrative Clinical Reading

two diagnoses, not two destinies

Comparing two diagnoses does not amount to announcing two individual destinies. Nordenswan shows how much the name adopted, the initial severity and the care pathway shape the population described. The integrative reading calls for conveying the reasons for any diagnostic revision without suggesting that everything preceding it becomes useless. It preserves continuity for the person: which elements remain observed, which interpretations change and which questions must now be discussed? The collective curve does not abolish this situated explanation.

13.3.5 Coulden and the Monitoring of Images

13.3.5.1 Situating the Result

Coulden, two scans per patient

In 2020, Coulden and colleagues report a single-center retrospective study of 31 patients with extracardiac sarcoidosis and suspected cardiac involvement. All have two assessments combining FDG PET and MRI; 22 receive individually decided immunosuppression, and nine do not. In treated patients, the myocardial volume with abnormal FDG uptake decreases and mean ejection fraction increases, without a statistically significant change in the volume of late gadolinium enhancement. In this series, however, the decrease in metabolic volume does not predict individual improvement in ejection fraction. The study therefore observes several measurement trajectories; it does not randomly compare two therapeutic strategies or two ways of guiding treatment by imaging.[5]

13.3.5.2 Critical Reading: the same people, but not all patients

the patients reexamined

Requiring two assessments makes a longitudinal comparison possible. It also selects the patients able to complete this pathway. To interpret the result, a change among the reexamined participants must be distinguished from a change in all the patients to whom the test would initially have been offered. This is not an accusation of demonstrated bias: it is a restriction of the population to which the conclusion can directly apply. The presence of an untreated group does not automatically solve the causal problem. The reasons for treating help to form the groups; they may also be related to the expected course. The authors do not know the reasons for withholding treatment in several patients who nevertheless met the diagnostic criteria. An adjustment could not then use these missing reasons. Comparing the observed changes remains instructive, but does not recreate random assignment of a common intervention.

13.3.5.3 Immunological Reading: a discordance does not yet tell a story

activity and tissue trace

The gap between a decreasing metabolic signal and a persistent MRI abnormality invites a distinction between activity and tissue trace. It is nevertheless insufficient to attribute to each patient an obligatory succession of immune stages. Such a reconstruction would require verifying what each signal rep-

resents, then establishing the temporal relationship between these properties. Observations consistent with a model do not demonstrate all of its transitions. No patient in this series has cardiac confirmation by biopsy. The imaging categories cannot therefore be presented as a histological validation performed in this cohort. This does not invalidate their possible uses: it prevents the plausible mechanism of the signal from being confused with direct identification of the cells present. Nor does a response to immunosuppression, on its own, possess the specificity of an etiological test. If the disease is recognized because a signal responds to treatment, and that response is then invoked to demonstrate both the diagnosis and the efficacy, the same observations bear two insufficiently independent conclusions. The solution is to specify what serves to classify, what serves to measure and what will serve to test the hypothesis.

CLASSIFYING, MEASURING, TESTING

13.3.5.4 Cardiological Reading: from the average to individual benefit

AVERAGES AND PERSONS

The most instructive result is the distance between two average improvements and their relationship within individuals. The patients who contribute most to the fall in a marker are not necessarily those who contribute most to the functional gain. The coexistence of the two movements at group level therefore does not validate an individual rule: "if this image improves, this person will benefit more from the treatment." To use the signal as a guide, the decisions that a value or its change would trigger would need to be specified, and the consequences of this strategy then compared with those of another course of action. This test should consider the harms linked to the decisions as much as the favorable measurements. An image that responds to treatment is a valuable observation; its ability to guide care reliably remains a distinct proposition.

13.3.5.5 Integrative Clinical Reading

AN IMAGE THAT IMPROVES

When an image improves, the person may legitimately ask whether they are better in a broader sense. Coulden helps to keep this question open: average changes in imaging and function do not necessarily describe the same improvement in each individual. The integrative reading brings measurements closer to lived experience without merging them. It asks what decision the follow-up imaging was meant to inform and what repeated tests add to the pathway, rather than treating any favorable change as sufficient reason to continue unchanged.

13.4 Worked Exercises

13.4.1 Exercise on Risk and Testing Strategy

13.4.1.1 Problem

two hundred fictional people

All the data below are fictional. They illustrate a line of reasoning and reproduce neither the Blankstein study nor any care guideline. A cohort of 200 people referred to a specialized center undergoes an imaging test. The report classifies 50 people in group A and 150 in group B. At two years, the same predefined clinical event affects 15 people in group A and 15 in group B. Follow-up is complete. The physicians know the results; 35 people in group A and 15 in group B subsequently receive treatment X. No randomization takes place, and detailed baseline risks are not provided. The person in charge concludes: "The test identifies those who should receive X. The risk is three times greater: treating all A subjects will therefore reduce this risk by two-thirds." Questions. Calculate the observed risks. Which part of this conclusion is supported? Which study would allow the proposed rule to be examined?

13.4.1.2 Reasoned Answer

30% versus 10%

The observed risk is 30% in A and 10% in B: a difference of 20 percentage points, a ratio of 3. These estimates describe a prognostic separation in this cohort; no confidence interval is presented in this exercise. They do not measure the effect of treatment X, and the ratio of 3 allows no conversion into a two-thirds therapeutic reduction. Treatment was given to 70% of A subjects versus 10% of B subjects. What would have been observed under another care policy is unknown. Treated subjects may also differ from the others in severity or contraindications. A crude comparison between treated and untreated subjects would therefore not resolve the question. To test the rule, a strategy explicitly guided by the test could be compared by randomization with a defined reference strategy, with the same follow-up of events and complications. What each result triggers would have to be specified. Randomizing X against its comparator only among A subjects would instead address its efficacy in that group; it would not by itself demonstrate the usefulness of testing everyone. The expected answer thus distinguishes three objects: the observed risk, the effect of the treatment and the benefit of the testing strategy.

randomizing the testing strategy

Synthesis

A diagnostic distinction is validated by observations that could contradict it, not by repeating its classification rule. Gillmore shows that a diagnosis without biopsy remains tied to a combination of clues, to a reference standard and to a population. ATTR-ACT shows that a benefit is defined by a hierarchy of events and that slowing is not restoring. Blankstein, Nordenswan and Coulden separate prognosis, classification and response to treatment: an image associated with risk does not demonstrate the usefulness of a strategy guided by it, and a diagnostic revision changes the groups compared. The specificity of a treatment must state its relation: to a target, to a disease or to a benefit in a population.

Open Questions

How can a diagnostic rule be validated in the ambiguous situations that a study had to exclude? Which randomized comparison would show that an imaging-guided strategy improves outcome, and not merely the separation of risks? Which observations, independent of the clinical resemblance, would make it possible to decide between two diseases and two variants of a single condition? The classification of infiltrative cardiomyopathies remains a hypothesis that treatment helps to test, without being able to define it alone.

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PART IV

Connecting Systems and Interpreting Measurements

A signal measured in the blood, a complication arising during treatment, an abnormal image and a histological grade have one thing in common: they connect the heart to something other than itself. A microbial community, a past infection, an immune intervention, an instrument or an organ from another organism then enter the explanation. This opening enriches knowledge, provided that not everything is connected to everything. Each passage requires its own justification. An identified origin does not yet designate a cause; an association does not say where to intervene; a precise measurement does not say which property it measures. A reference, finally, can be justified without being infallible. The same requirement returns in different forms: naming what each observation connects, under what conditions and at what moment. A class, a threshold or a monitoring rule earns its value through the tests it accepts, and then through the decisions it actually improves for the people receiving care.

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14

The Heart in Its Relations with the Microbiota and Infection

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The heart does not by itself create all the conditions of its diseases. What a person eats, what a microbial community transforms and what an infection has left behind can weigh on a cardiac event occurring later. The difficulty lies in the distance: between food and lesion, between infection and diagnosis, several places and several periods intervene. A molecule measurable in the blood, a more abundant species or a date of exposure offers a point of entry. None of them alone designates the center of the system. Each passage, from one organ to another as from one moment to another, must remain open to discussion: it requires its own comparison, its controls and its own test before becoming an explanation.

14.1 Following What Is Transformed between Places

14.1.1 Shifting the Boundary of Explanation

14.1.1.1 Seeking the decisive relation beyond the organ

EXPLANATION BEYOND THE ORGAN

A disease may be defined by the organ it affects without its explanation stopping at the boundaries of that organ. Studying a relation between the microbiota and the heart therefore requires distinguishing the site of damage from the sites where its conditions are built. But extending the inquiry to the whole organism is not enough to explain better: an explanation that links everything to everything no longer allows the decisive relation to be recognized.

ONE VALUE, SEVERAL HISTORIES

The difficulty becomes concrete in this entirely fictional example. A food contains a precursor P; a microbial community can transform it; the body then produces a molecule M, measurable in the blood. A team observes an association between M and a cardiac event. It has a point of entry into a system, but this point does not yet designate its center. The concentration could vary because intake changes, because transformation accelerates or because elimination slows. The same value can thus correspond to several histories.

14.1.1.2 A predictive signal does not indicate where to intervene

PREDICTING IS NOT LOCATING WHERE TO ACT

It will be objected that the essential thing is to recognize a useful signal, whatever its origin. That may suffice for a properly validated predictive function. However, the observed association does not, by itself, indicate where to intervene: reducing intake, modifying a transformation or acting on elimination are not the same operation. Deciding between these histories can justify the choice of a line of inquiry. The effect of an intervention can also be established by an appropriate comparison before its mechanism is fully explained; such a result would not automatically validate the other means of modifying the same signal.

14.1.2 A Community Is Not a Collection of Culprits

COMPOSITION AND FUNCTION

In the fictional example, suppose that a microbial species is more abundant in subjects with high M. Naming it does not resolve the question of its role. Is it responsible for the transformation, a beneficiary of the same environment or associated with another population that performs it? A description of composition and a description of function answer different questions. The objection seems immediate: removing this species and observing what follows should settle the matter. The experiment would be instructive, but its interpretation would depend on what the removal actually changes. If it also transforms the resources available to other organisms, the result concerns a reorganized community. Attributing the whole effect to an activity specific to the removed

THE TEST OF REMOVAL

species would require an additional comparison capable of isolating that activity. Conversely, reintroducing the species would not guarantee the return of the previous situation. In the fictional scenario, other populations might have taken its place; the necessary resource might be lacking. A failure of restoration would therefore not automatically cancel the initial observation. This possibility must not, however, become an indefinite excuse: the changes invoked must be measurable and subjected to a new test. Thinking in ecological terms adds relations to verify, not reasons to shield the hypothesis from contradiction.

14.2 Determining What Changes between Periods

14.2.1 What a Past Infection Can Explain

AN ABSENT CAUSE, LASTING EFFECTS

Infection introduces another question: must the cause remain present for its effects to persist? Logically, no. In a hypothetical scenario, an agent could trigger an alteration later sustained by other processes. Its absence at the time of assessment would be compatible with this initial role. But this compatibility would demonstrate neither the triggering nor the proposed maintenance process. A possible history would have to be distinguished from a supported one.

PRESENCE IS NOT ACTIVITY

The reverse objection deserves the same attention: identifying the agent today would not, by itself, establish that it sustains the observed cardiac involvement. The question concerns an activity and its consequences, not merely a presence. A coherent inquiry thus specifies whether it is looking for a past trigger, a current activity or a sequela. These objects may lead to different measurements and comparisons. This distinction has an important consequence for therapeutic reasoning: an intervention directed against a trigger does not necessarily test the whole explanation when that trigger has already disappeared. It must be stated which step is still assumed to be modifiable at the time of intervention.

14.2.2 Building a Comparison over Time

WHAT "AFTER" MEANS

Studying an event occurring after an infection requires, finally, defining what "after" means. Symptom onset, the positive test and entry into a medical record are different reference points. Confusing them can shift the window during

MORE DISEASE OR MORE TESTING

which an event is attributed to the exposure. Suppose, still fictionally, that an infected group subsequently undergoes more frequent testing. An increase in diagnoses could combine two phenomena: more disease and more opportunities to detect it. Mentioning this difficulty does not refute the association; it indicates the additional comparison its interpretation requires. Comparing each person with themselves might seem preferable. This approach removes certain stable differences between individuals, but the periods compared may differ in the care received, the circulation of the agent or the possibilities of diagnosis. The unity of the person does not ensure the equivalence of the moments. Linking the heart to its biological environment therefore requires two complementary clarifications: following what is transformed between places and determining what changes between periods. A good explanation keeps these passages open to discussion, instead of concealing them behind the name of a microbe or a simple succession of dates.

14.3 Texts Put to the Test

14.3.1 Wang and the Construction of an Explanation across Several Scales

14.3.1.1 Situating the Result

Wang, choline and TMAO

After a search for plasma molecules in two independent sets of cases and controls defined by events at three years, Wang and colleagues identify choline, betaine and trimethylamine N-oxide (TMAO). A third analysis, in 1,876 subjects evaluated in cardiology, associates their concentrations with prevalent cardiovascular disease: it does not constitute a prospective validation in 1,876 people. In mice, isotopic labeling tracks the conversion of oral phosphatidylcholine into TMAO; production disappears after antibiotics and returns after recolonization. Germ-free experiments reinforce the role of the gut flora. In apolipoprotein E-deficient mice, choline or TMAO supplementation increases aortic lesions; antibiotics prevent the increase linked to choline. The study also examines macrophage cholesterol and the macrophage receptors CD36 and SR-A1. These experimental results do not concern human clinical events.[1]

14.3.1.2 Critical Reading: comparisons with different scopes

discovery, validation, recruitment

Separating discovery from validation reduces the risk of retaining only peculiarities of the first sample. It does not dispense with examining the population: subjects are recruited at a referral center, on the occasion of cardiac evaluations. Validation in a distinct set may therefore retain shared recruitment limitations. It does not automatically amount to generalization to the whole population.

what isotopic labeling proves

Labeling provides a particularly instructive argument: recovering a labeled product links it to the substrate administered. The same reasoning is not enough to establish the exact share of this product in a lesion. A chemical transformation and a pathological contribution are two propositions, even when they belong to the same explanation. Antibiotics, germ-free mice and recolonization support the microbial dependence of the production studied. For atherosclerosis, the intervention is less selective: antibiotic therapy changes far more than a single reaction. The observed protection therefore does not establish that each effect passes exclusively through TMAO. Direct administration of TMAO adds an experimental piece; it does not retrospectively turn antibiotic therapy into specific inhibition.

precautions not described

The main methods detail the measurements, without specifying randomization of the animals or blinded quantification of lesions. This observation concerns the main text: the absence of a description does not establish the absence of these precautions, which the supplementary materials may specify.

14.3.1.3 Immunological Reading: an activity rather than a single culprit

an activity, not a pathogen

The approach follows a microbial transformation through its products and relates it to a macrophage phenotype. It does not require finding an infectious agent in the lesion. Participating in possible damage does not, however, confer

on an entire community the identity of a pathogen: the proposition concerns an activity under defined dietary and experimental conditions. One objection remains: without an identified responsible organism, is the explanation complete? It is partial, but informative. The demonstrated dependence makes it possible to search for which activities are necessary; it does not yet designate the species to eliminate. Likewise, the increase in macrophage receptors and cholesterol does not prove that this step alone explains the entire increase in lesions. Isolating this contribution would require perturbing it while preserving the other relevant steps. This distinction avoids two excesses: rejecting the experiment because it does not explain everything, or presenting a coherent set as a fully closed mechanism. The value of the proposed chain is that it makes each passage accessible to distinct tests.

14.3.1.4 Cardiological Reading: defining the intervention before promising its benefit

THREE WAYS TO LOWER A METABOLITE

The clinical choice cannot be inferred from the lowering of a metabolite alone. Modifying its intake, its production or its elimination would correspond to different interventions, each with its own effects. The results do not establish therapeutic equivalence between these possibilities. Nor do they demonstrate

PREVENTING IS NOT REVERSING

that general suppression of the flora would be an acceptable strategy. The timing of intervention would also need to be specified: preventing an experimental increase in lesions is not equivalent to reversing established disease. This difference concerns the object of the treatment, not merely its intensity. The article opens hypotheses for intervention; it establishes neither the cardiovascular benefit of antibiotics in humans, nor that of a probiotic, nor a dietary prescription.

14.3.1.5 Integrative Clinical Reading

A PRECURSOR WITHIN A DIET

A person's diet cannot be reduced to the intake of an isolated precursor. Any intervention would have to be examined with its nutritional consequences, its day-to-day feasibility and the other goals of prevention. The article does not compare these pathways. For general practice, its value lies in opening a question about the relations between habits, biological transformations and health, without making a metabolite a total explanation of lifestyle or a sufficient reason for restriction.

14.3.2 Tang and the Link between Metabolism and Prognosis

14.3.2.1 Situating the Result

TANG, A TRACER AND A COHORT

Tang and colleagues combine two protocols. After ingestion of labeled phosphatidylcholine, a sequence with antibiotics in six volunteers shows near-suppression and then reappearance of TMAO after the challenge. Separately, 4,007 people referred for elective coronary angiography are followed for three years; 513 experience death, myocardial infarction or stroke. The hazard ratio (HR) between extreme quartiles of TMAO falls from 2.54 without adjustment to 1.43, 95% confidence interval (95% CI) 1.05 to 1.94, in the most adjusted model.[2]

14.3.2.2 Critical Reading: two experiments do not close the whole chain

14.3.2.2.1 An experiment on origin is not a clinical trial

ORIGIN OF A MOLECULE, WHAT IT PREDICTS

The teaching value of this work lies in the meeting of two questions. Where does a molecule measured in the blood come from? What does its concentration predict? The first calls for a perturbation of the system; the second, a prospective observation. Combining them makes the hypothesis more coherent, without turning the two answers into proof of all the intermediate steps.

SIX VOLUNTEERS DO NOT MAKE A TRIAL

The tracer helps to link an ingested precursor to a recovered product. The antibiotic perturbation examines the dependence of this transformation on microorganisms. The return after discontinuation strengthens this interpretation. But a sequence in six people does not constitute a cardiovascular prevention trial. Neither its duration, nor its population, nor its endpoint answers that question. The biochemical precision of an experiment does not compensate for the absence of the clinical result sought.

14.3.2.2.2 What an adjusted concentration supports

FROM CRUDE TO ADJUSTED COEFFICIENT

In the cohort, the move from the crude to the adjusted coefficient is a reminder that the number presented depends on the model. Reading only the first could amplify the impression of an autonomous effect. Reading only the last would conceal the role of adjustment choices. The persistence of an association indicates information conditional on the variables retained; it guarantees neither the absence of residual confounding nor the identification of the effect that an induced reduction in TMAO would have. The circulating concentration also raises a question of mass balance: it may vary with intake, production and elimination. Even when a production pathway is demonstrated, each difference between people is not automatically attributable to it. A measure of renal function introduced into a model does not make all biological measurements perfect and does not, by itself, settle the causal place of each variable.

14.3.2.3 Immunological Reading: situating the mechanism not yet measured

THE UNMEASURED IMMUNE MECHANISM

The vocabulary of the microbiota could give the impression that the immune mechanism is already established. This temptation must be resisted. A microbial transformation and a vascular association leave open the cellular steps that would link them in humans. Mechanisms proposed on the basis of other work must retain their own provenance: they do not become observations of this cohort because they make its reading attractive. The philosophy of knowledge comes into play here as a discipline of the link. An explanation can draw on several scales; each of its passages requires justification. The point is not to disqualify a still incomplete construction, but to identify the experiment that would make it more constraining. If two explanations predict the same association, which intervention or which measurement would make it possible to decide between them?

14.3.2.4 Cardiological Reading: a promising target remains a therapeutic question

THREE PROPOSITIONS TO KEEP DISTINCT

For the clinician, three propositions must remain distinct: the assay informs about outcome, an intervention modifies the assay value and this intervention improves outcome. Even if the first two are supported, the third requires its own test. A marker may accompany several processes, and the chosen action may produce other effects, favorable or adverse. The population referred for an invasive test also matters. An association in this setting does not immediately provide the probabilities useful for screening a general population. The value of the assay would depend on the decisions it changes and their consequences. The metabolic experiment therefore provides grounds neither for prescribing antibiotics nor for dietary advice. The work thus links a microbiota-dependent transformation to a prognostic association, without closing the causal gap between the two. A contribution to vascular disease remains plausible; the clinical benefit of lowering the marker requires a defined intervention evaluated on relevant events.

REFERRED PATIENTS AND GENERAL SCREENING

14.3.2.5 Integrative Clinical Reading

ONE MORE ASSAY IN THE CONSULTATION

An additional assay can produce information without resolving the problem presented in the consultation. Before including it in a global workup, the decision it would allow to be made differently would need to be specified, and the consequences of that decision then compared with the care already offered. The renal, dietary and therapeutic context belongs to the interpretation of the person; it is not an automatically demonstrated explanation of the measured value. The integrative task is to situate the result within a history, without turning that history into certain causality.

14.3.3 Koeth and the Capacity to Transform a Nutrient

14.3.3.1 Situating the Result

KOETH, CARNITINE AND TMAO

After ingestion of labeled carnitine, volunteers produce labeled TMAO; this production becomes almost undetectable under antibiotics and reappears after they are stopped. A comparison of five vegetarians or vegans and five omnivores shows different responses to the same challenge. In mice, prolonged exposure to carnitine increases the capacity for transformation and atherosclerosis; antibiotics prevent this additional lesion burden. The work also explores reverse cholesterol transport. Finally, a cohort of 2,595 subjects evaluated in cardiology links concentrations to prevalent disease and to subsequent events. These protocols pose complementary questions; their participants and endpoints must not be merged into a single experiment.[3]

14.3.3.2 Critical Reading: distinguishing exposure from the capacity to transform it

Compared with Wang, carnitine provides another substrate, but the issue goes beyond lengthening a list of foods. An ingested dose does not entirely define biological exposure: its transformation depends on an environment whose his-

DIETARY HABITS NOT RANDOMIZED

tory can modify its functioning. A comparison of doses without measurement of products therefore risks treating as equivalent exposures that are not. The human comparison by dietary habit, however, randomizes no diet. It does not allow every difference to be attributed solely to the absence of meat: associated behaviors, duration of habits and individual characteristics remain intertwined. Repeated measurements after ingestion describe a response but do not multiply the number of independent people. A curve in one volunteer does not become a general law through the abundance of its points. The murine experiments reinforce the idea of a modifiable capacity. They do not, however, provide the timing or magnitude of an adaptation in humans. The same people would need to be followed before and after a controlled dietary change, with the transformation actually measured, to separate the assumed history from the observed history.

14.3.3.3 Immunological Reading: an ecological function is not a bacterial identity

NAMING A GROUP IS NOT IDENTIFYING

Associating a microbial group with a metabolite is not enough to identify the organisms that produce it. A group may accompany a function, facilitate its conditions or benefit from the same substrate. It may also contain organisms with different activities. Moving from composition to function therefore requires an additional experiment; naming a community does not replace this demonstration. The immunological question here concerns the metabolic conditions in which cells work. A change in cholesterol elimination may contribute to a lesional environment without demonstrating universal inflammatory activation of macrophages. Transporting a molecule, accumulating lipids and triggering an immune response are linked but distinct operations. Grouping them under the word "inflammation" would lose precisely what the study makes it possible to differentiate. A flux within the organism must also be distinguished from an isolated cellular property. A global measure integrates several passages and several organs. Its change does not automatically locate the defect in the macrophage. A convincing explanation will need to link the local modification to the total flux, then show what changes when this step is specifically perturbed.

A WHOLE-BODY FLUX, AN ISOLATED CELL

14.3.3.4 Cardiological Reading: a plausible pathway does not prescribe a diet

A SUPPLEMENT IS NOT A MEAL

The proposed chain opens a cardiovascular question; it does not directly measure the benefit of a dietary choice. Supplementation in animals is equivalent neither to a human meal nor to a durable prevention strategy. As for the clinical cohort, it sets concentrations against outcome, not two randomly assigned prescriptions. The attenuation of an association after adjustment does not resolve mediation: adding several variables simultaneously does not allow the whole change to be attributed to one of them. Likewise, an association present in one subgroup is not enough to establish a statistical difference between subgroups. The judgment would be strengthened by a controlled human intervention linking dietary change, microbial activity, metabolic exposure and cardiovascular outcome. Meanwhile, this study gives a reason to measure the transformation rather than to infer risk from the name of the nutrient alone. It provides no grounds here for any recommendation of antibiotics, supplementation or dietary exclusion. Discrepancies in sample sizes and in the definition

THE MISSING HUMAN INTERVENTION

of the clinical endpoint limit the quantitative interpretation of the cohort. They preclude inferring a quantified clinical effect from it here.

14.3.3.5 Integrative Clinical Reading

two imperfect dietary labels

The categories "vegetarian" and "omnivore" imperfectly describe what a person eats, why they eat it and what they could durably change. Their comparison does not yield an overall ranking of ways of living. A general practice reading would ask what concrete change is envisaged, whether it respects the person's needs and which outcomes would allow its value to be appraised. The observed biological difference thus invites a more refined dietary inquiry; it is not enough to replace that inquiry with a label.

14.3.4 Zhu and the Conditions of a Platelet Response

14.3.4.1 Situating the Result

Zhu, platelets and thrombosis

Zhu studies TMAO at several scales. The clinical part reanalyzes the 4,007 people of the Tang cohort for myocardial infarction and stroke: it does not constitute an independent replication. In human platelets, TMAO amplifies the response to submaximal agonists, notably the mobilization of intracellular calcium. In mice, dietary and microbial manipulations modify thrombosis induced by vascular injury. Absence of microbiota blocks the effect of choline; direct supply of TMAO bypasses this step.[4]

14.3.4.2 Critical Reading: the decisive control depends on the question

three experiments, three domains

The value of the design is that it does not ask a single experiment to carry the whole explanation. Adding a substance to a cell preparation tests a capacity for action in that medium. Removing a microbial community probes an upstream dependence. Directly supplying the expected product examines whether this dependence can be bypassed. The answers complement one another, but each

the bypass argument

retains its own domain of validity. The bypass argument deserves a precise reading. If a precursor loses its effect after a perturbation while its product remains active, the hypothesis of an intermediate transformation becomes more constraining. This does not prove that the perturbation has no other effect. Nor does it measure the proportion of the clinical phenomenon that would follow this pathway. A local demonstration can be solid without becoming an

a cohort reanalyzed, not replicated

exhaustive explanation. Finally, analytical novelty must be distinguished from empirical independence. Reexamining a cohort with a more targeted question can produce useful information. Counting this analysis as a second population would, however, give a misleading impression of replication. In a synthesis of the literature, the unit to be counted is not always the article: data can circulate across several publications.

14.3.4.3 Immunological Reading: not forcing a missing step

microbial, inflammatory, platelet-related

An explanation linking the microbiota and thrombosis may interest cardioimmunology without demonstrating that each transition is immune. The words

"microbial," "inflammatory" and "platelet-related" are not synonyms. If a leukocyte relay or a specific cytokine is proposed, this proposal requires its own proof. Thematic proximity is not enough to insert it into the observed mechanism. The notion of sensitivity to a stimulus offers a general lesson here. Amplifying a response does not mean triggering the phenomenon spontaneously. The context of activation is part of the experimental proposition: stimulus intensity, duration of exposure, concentration and medium must accompany the interpretation. A single value within the range observed in humans does not necessarily represent the usual exposure of all patients.

AMPLIFYING IS NOT TRIGGERING

14.3.4.4 Cardiological Reading: induced occlusion and the lived event

An injury model tests the response to a defined insult. It does not encompass the whole history that leads a person to myocardial infarction: emergence of the predisposing background, triggering, competing treatments and consequences after the event. What the model makes comparable must therefore be specified before its result is extrapolated. The therapeutic perspective also requires two distinct evaluations: reducing adverse events and preserving necessary functions. The absence of a harmful effect in an experiment does not demonstrate clinical safety. In particular, promising prevention without a bleeding cost would require a comparison that measures both outcomes, with an appropriate duration and sample size. The experimental modification of platelet reactivity therefore supports a capacity for action under the conditions studied. Its contribution to human vascular risk remains a hypothesis; it does not measure the net benefit of an intervention targeting TMAO. Independent data and a comparison based on clinical events would make it possible to link this biological possibility to a therapeutic question.

EFFICACY AND BLEEDING COST

14.3.4.5 Integrative Clinical Reading

Reducing a mechanism likely to participate in thrombosis is only one goal among those of a prevention pathway. The person may already be receiving several interventions whose combined effects, constraints and adverse consequences would need to be appraised. This article does not provide that comparison. Its integrative contribution is to make a new hypothesis visible while keeping the question of additional benefit: what would actually be gained by acting on this pathway, for whom and at the cost of what other consequences?

ONE PREVENTION GOAL AMONG OTHERS

14.3.5 Patone and the Conditions of a Comparison over Time

14.3.5.1 Situating the Result

This English study uses a self-controlled case series (SCCS) in a source population of 38.6 million vaccinated people aged at least sixteen years, between December 2020 and August 2021. It examines hospitalizations or deaths coded for myocarditis, pericarditis or arrhythmia after vaccination or a positive test for severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2). For myocarditis, the estimated excess over twenty-eight days is 10 events per million after the second dose of mRNA-1273, versus 40 after a positive test. In people

PATONE, A SELF-CONTROLLED SERIES

under forty, the corresponding estimates are 15 and 10 (95% CI: 12 to 16 and 7 to 11). These numbers concern this population and this period; they do not constitute a randomized comparison of two strategies.[5]

14.3.5.2 Critical Reading: what self-control leaves free to vary

WHEN THE EVENT DATES THE EXPOSURE

The SCCS compares, among people who experienced the event, its incidence during exposure windows with its incidence during reference periods. It thus neutralizes constant individual differences. But a hospitalization can delay vaccination or prompt an infection test: the event then helps to determine the date of the recorded exposure. The order of entries in the record does not necessarily reproduce the biological order. The authors isolate a pre-exposure period and day zero, adjust for calendar week and examine several sensitivity analyses. These measures address identified problems; they do not confer on the calendar an independence it does not have. Changing the reference period changes the comparison itself. Stable results across several partitions would be reassuring; variation calls for understanding which process produces this dependence, rather than retaining the most convincing partition. The objection "each person is their own control, so any selection bias disappears" also fails. The source population is vaccinated: it does not automatically represent all infected people. Self-control organizes a comparison within a selected set; it does not cancel the conditions of entry into that set. The absolute excesses require a separate reading: they result from a calculation complementary to the incidence ratios, not from a raw difference provided by the SCCS. Their interval describes the uncertainty attached to this estimate, not all possible errors of diagnosis or dating. A very precise number can therefore remain dependent on debatable assumptions.

OWN CONTROL, BUT SELECTED

14.3.5.3 Immunological Reading: the calendar does not choose the mechanism

A clustering of events after exposure can guide a biological inquiry; it does not name the process that produces them. Two exposures followed by the same diagnosis may act through different pathways. Conversely, distinct timelines are not enough to prove entirely distinct mechanisms. Timing constrains the explanation without completing it. The study uses care and death codes, without establishing for each case a confirmation by imaging or biopsy. It would therefore be excessive to convert each coded event into a demonstration of a precisely characterized immune lesion. Recognizing this limit does not make administrative data useless: they make it possible to search for rare signals on a scale hardly accessible to a clinical series. But the two sources have different functions. A detailed clinical validation could clarify the nature of the signal; it would not, by itself, replace the denominators needed for its estimation.

ADMINISTRATIVE CODES AND CONFIRMATION

14.3.5.4 Cardiological Reading: comparing risks without manufacturing a decision

AGE, PRODUCT, DOSE, WINDOW, ENDPOINT

The difference between the overall estimates and those for people under forty prevents an average from being turned into a proposition valid for everyone. The result must retain its age, its product, its dose number, its time win-

dow and its endpoint. These specifications are the conditions of its meaning, not secondary details. It might then be objected that it suffices to choose, in each subgroup, the smaller number. But the risk after a documented infection is not the risk of a strategy of not being vaccinated: the latter depends notably on the probability of infection and on the consequences to be prevented. Symmetrically, counting myocarditis cases after vaccination does not summarize all the benefits and harms of a vaccination strategy. Finally, myocarditis, arrhythmia and death do not become equivalent because they can be counted. Severity, duration and clinical sequelae remain necessary for the decision. The study provides a historical estimate with its conditions; no current course of action on vaccination can be inferred from these data alone.

risk of infection, risk of a strategy

14.3.5.5 Integrative Clinical Reading

a figure facing a current decision

In a discussion of prevention, a myocarditis figure answers a narrower question than "which strategy best protects this person?" The outcomes the person wishes to avoid, the exposure context and the time horizon considered would also need to be specified. These dimensions must not be filled with estimates invented from the article. The role of the general practitioner here would be to present faithfully the question measured and the additional information needed for a current decision, leaving room for the concerns expressed.

14.3.6 Xie and Outcome after Infection

14.3.6.1 Situating the Result

Xie, the post-acute outcome

Xie and colleagues study 153,760 users of the US Department of Veterans Affairs health care system who were alive thirty days after a positive SARS-CoV-2 test. They compare them with contemporary and historical controls. The weighted analyses find more cardiovascular events during post-acute follow-up. For the composite of all-cause death, myocardial infarction or stroke, the HR is 1.55 (95% CI: 1.50 to 1.60) against contemporary controls; the estimated excess at twelve months is 23.48 per 1,000.[6]

14.3.6.2 Critical Reading: defining the people and periods compared

14.3.6.2.1 Survivors compared with two control groups

starting among survivors

The first decisive choice precedes any calculation: starting the analysis among survivors of the first month. This definition answers a question about post-acute outcome. It does not describe all the consequences since infection, because it leaves early deaths outside its scope. Each reformulation of the result must retain this population. The large sample size does not permit the question to be changed silently. The second choice concerns the control group. A contemporary comparison shares the context of the pandemic but may include unidentified infections. A historical comparison provides a prepandemic reference, while introducing differences of era: movement through the health care system, diagnostic possibilities or medical practices. Their concordance strengthens the interest of the signal; it does not guarantee that every source of bias differs between the two comparisons. Using two control groups therefore does not amount to having two independent experiments.

contemporary and historical controls

14.3.6.2.2 What weighting and controls put to the test

WEIGHTING IS NOT RANDOMIZING

Weighting seeks to make groups comparable on measured characteristics. Its value depends on the quality of the measurements, the overlap of the populations and the variables retained. Adding many covariates does not turn records into randomization. The balance of measured characteristics after weighting applies only to what was actually compared; it does not certify the equivalence of all clinical histories. Before-and-after analyses add a useful test: did the contrast already exist before exposure? Their causal interpretation nevertheless assumes that, without exposure, the trends compared would have been sufficiently comparable. A differential change in health care use could challenge this assumption. Sensitivity analyses and negative controls test certain weaknesses; no single test eliminates all competing explanations. The right question is to specify which weakness each one actually puts to the test.

14.3.6.3 Immunological Reading: a spectrum of diagnoses does not designate a single pathway

A SIGNAL TO EXPLAIN, NOT A PATHWAY

The range of conditions observed may suggest a common process. It may also bring together several paths to disease. This records-based study does not sample tissue or manipulate an immune mediator to decide between these possibilities. The difference between an epidemiological signal to be explained and the explanation that will be proposed for it must therefore be maintained.

14.3.6.4 Cardiological Reading: keeping the absolute scale and the context together

RATIO OF RISKS AND ABSOLUTE EXCESS

The ratio of risks informs about a contrast; the absolute excess helps to gauge its scope in this population. They are not two interchangeable presentations of a personal probability. In particular, an HR cannot be read directly as a proportion of people who will experience the event. The composite also retains different components: not all deaths are attributed to a cardiovascular cause. Finally, extrapolating these estimates requires specifying to whom and to which period. The composition of a population, the history of infections, care and immunity may change. A historical estimate remains instructive without becoming a timeless coefficient applicable to every person. This text invites defining the questions of surveillance and its benefits; it demonstrates the efficacy neither of a screening program nor of any particular preventive treatment.

TO WHOM, TO WHICH PERIOD

14.3.6.5 Integrative Clinical Reading

AN AVERAGE INCREASE, INDIVIDUAL PATHS

An average increase in events after an infection does not, by itself, indicate which tests to offer each person. The consultation must be able to accommodate symptoms and their daily consequences without immediately imposing a single explanation on all difficulties. To evaluate a follow-up pathway, what it actually improves would need to be known: functioning, access to relevant help, outcome and the burden of investigations. The cohort raises a collective question; it does not yet describe the individual pathway that would best answer it.

14.4 Worked Exercises

14.4.1 Exercise on Restoring an Experimental Response

14.4.1.1 Problem

twelve donors, four conditions

Entirely fictional data. An experiment uses paired blood samples from twelve donors. Each sample is divided among four conditions tested in random order, with blinded reading: vehicle, compound X, compound X with added metabolite M and metabolite M alone. X is assumed to reduce M. The platelet response to the same submaximal agonist is measured in arbitrary units. The means are 40, 25, 39 and 54 units, respectively. Under X, the concentration of M decreases by 60%. After addition, it returns to its control value. The preparations are viable. No clinical event is measured; the measures of dispersion are not provided. Question. Can it be claimed that X will prevent myocardial infarction by reducing M? What does the third group demonstrate, and what verification is missing?

14.4.1.2 Reasoned Answer

contrasts without dispersion

The descriptive contrast of X versus vehicle is −15 units. Adding M in the presence of X is accompanied by an increase of 14 units relative to X alone; the response therefore moves closer to control. Without dispersion or analysis of paired differences, neither a statistically established difference nor equivalence between 39 and 40 can be concluded. This restoration is compatible with a role of M in the observed decrease in response. It does not exclude all other explanations: M could offset an action of X through another pathway. The M-alone group, at 54 units, makes visible its capacity to increase the response. It remains to verify that concentrations are truly comparable and that there is no interference with the measurement and, if possible, to modify M by a second method independent of X. A drop in concentration and a change in reactivity are two different results. Even a well-controlled mechanism in this preparation demonstrates neither prevention of myocardial infarction nor absence of bleeding. The result supports an experimental hypothesis; a clinical claim would require its own events and its own comparison.

Synthesis

Linking the heart to the microbiota or to an infection requires following two kinds of passage: between the places where a molecule is transformed and between the periods being compared. Work on TMAO shows how an explanation is built across several scales, from isotopic labeling to the cohort. Each piece retains its own domain: a demonstrated transformation is not a measured pathological contribution, and a prognostic association does not designate the useful intervention. Studies conducted after infection show that a timeline constrains the explanation without completing it. The choice of controls, windows and populations defines the question that each number answers.

Open Questions

Which measurement would make it possible to distinguish, in one person, the intake, production and elimination of the same molecule? Which control would show that an effect attributed to a microbial species is not due to a reorganized community? After an infection, how can a more frequent disease be separated from a disease more often sought? Each of these questions calls for a comparison that remains to be built.

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15

Immune Treatments as Revealers of Disease

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An immune treatment acts on a targeted disease, but it also modifies a balance that the body maintained quietly. When the heart suffers the consequences, the complication becomes both damage to be treated and a source of knowledge. It can shed light on a protection hitherto barely visible, or on a recognition that nothing foreshadowed. This dual nature calls for restraint. A rare alert is not a frequency, a registry is not an exposed population and an animal model is not a patient. Recovery after several treatments does not prove the effect of the last one. And knowledge obtained from a severely ill person never justifies treating that person as an experiment.

15.1 What a Therapeutic Perturbation Brings to Light

15.1.1 Revealing Is Not Merely Making Visible

REVEALING OR CREATING A CONFIGURATION

A treatment can bring to light a previously unknown involvement. The word "revealer" then seems self-evident: on this reading, the intervention made visible what the body already contained. Yet this interpretation combines two different possibilities. The treatment may unmask a preexisting vulnerability; it may also create a new configuration that would not have formed without it. A complication does not prove retrospectively that a silent disease was waiting for its opportunity. A characteristic present before treatment may be a condition of susceptibility without already constituting a disease. Supporting the existence of a prior involvement would require signs independent of that involvement, rather than redefining any susceptibility on the basis of what occurred. The objection is practical: does this distinction change anything? Yes, because searching for an already established disease and searching for a dangerous interaction with an intervention lead to different screening programs. What must be detected depends on the history that is assumed to be preventable.

SUSCEPTIBILITY IS NOT DISEASE

15.1.2 A Therapeutic Perturbation Is Not a Mirror of Ordinary Functioning

A PERTURBATION IS NOT A MIRROR

A complication can shed light on a protective role hitherto barely visible. But concluding that damage following an intervention exactly describes normal functioning would be premature. A perturbation can cross a threshold or make possible relations that are usually absent. The phenomenon observed informs, first of all, about what becomes possible in this transformed situation. It will be objected that perturbation is needed in order to understand. The objection is valid; it does not justify confusing the questions. An experiment can establish that a modification triggers damage in a defined context. Determining whether the same process sustains a spontaneous disease requires another comparison. Toxicity therefore opens an inquiry into the conditions of protection and damage; it does not provide a transparent photograph of the usual state.

15.2 Judging Benefit and Harm Together

15.2.1 A Benefit and a Harm Do Not Prove an Indivisible Mechanism

COMPLICATION AND RESPONSE TO TREATMENT

Consider now a fictional scenario: people with a complication also seem to respond better to treatment. The temptation would be to see the complication as the necessary biological price of efficacy. But the two events could share a condition of occurrence without being inseparable. A longer exposure, for example, could offer more time to observe both. The issue is not only statistical. Calling the damage necessary would prematurely close the search for an intervention that preserves the benefit while reducing the complication. Conversely, assuming that they can be separated does not demonstrate it. It must be determined which common element is indispensable to each effect and which differences might make it possible to act on one without abolishing the other. A temporal resemblance must not set the limits of future treatment.

15.2.2 Knowing without Turning the Person into an Experiment

A serious complication may require action before its explanation is complete. The remaining uncertainty does not disappear after the decision: it determines what still needs to be observed and which reasons would justify a revision.

THE PERSON IS NOT AN INSTRUMENT

Readministration of a treatment could sometimes strengthen a causal attribution. Its informative value does not, by itself, make it justifiable: the person does not become an instrument intended to satisfy a demonstration. It must be possible to build collective knowledge with other comparisons, while acknowledging what they leave uncertain. Judgment thus bears simultaneously on the targeted disease and on the possible harm. Ranking them requires considering their severity, their duration and the available alternatives, with the person concerned. A toxicity can enrich knowledge; its scientific fruitfulness never constitutes its clinical justification.

TARGETED DISEASE AND POSSIBLE HARM

15.3 Texts Put to the Test

15.3.1 Johnson and the Value of a Clinical Alert

15.3.1.1 Situating the Result

JOHNSON, TWO FATAL MYOCARDITIS CASES

Two people treated for melanoma with ipilimumab and nivolumab develop fatal myocarditis associated with myositis. Severe electrical disturbances appear despite preserved or mildly impaired systolic function. Tissue examination and sequencing of lymphocyte receptors link the cardiac, muscle and tumor infiltrates. A manufacturer's safety database provides another level of observation: eight myocarditis cases reported among 2,974 people receiving the combination, versus ten among 17,620 receiving nivolumab alone. The events depend on reporting; no systematic cardiac screening was performed. The report thus combines in-depth observations and a pharmacovigilance signal, without turning them into a comparative trial.[1]

15.3.1.2 Critical Reading: the strength of an alert does not depend on its representativeness

AN ALERT IS NOT A FREQUENCY

A rare event can reveal a biological possibility before its frequency is known. It would be wrong to demand of an initial observation the precision of a cohort before recognizing its value as an alert. The opposite error would be to take the most spectacular forms for the whole of the disease. The two deaths explain why the phenomenon attracts attention; they do not allow its general lethality to be calculated. Timing, lesions and pharmacological plausibility provide different arguments. Their convergence counts for more than the simple sequence "treatment then symptom." It does not, however, eliminate all other explanations. The selection of tissue available after death gives access to remarkable anatomy, while favoring a terminal stage. The order of phenomena cannot be read directly from their coexistence in the final sample. The safety database broadens the view, but a reported proportion also depends on case recognition and reporting. Comparing two denominators guarantees neither an identical search for events nor interchangeable populations. A statistical difference does not, by itself, correct these differences in construction. The signal justifies an organized inquiry; it does not yet measure precisely the causal excess risk of the combination.

A SAFETY DATABASE AND ITS REPORTS

15.3.1.3 Immunological Reading: a shared receptor does not name the antigen

RECEPTOR, TARGET, LESION

The presence of related sequences in several tissues makes a hypothesis of shared recognition interesting. But identifying a receptor sequence, identifying what it recognizes and demonstrating that it produces the lesion are three distinct tasks. Sequencing of the beta chain does not by itself reconstruct the full functional specificity of a lymphocyte. Several scenarios therefore remain open: recognition of a common target, cross-reactivity between different targets or coexistence of cells with differing effector roles. Deciding between these scenarios would require confronting the receptors with defined targets, then measuring a relevant functional response. The anatomical link guides this experiment; it has not already performed it. Virology provides another lesson. The main text reports initial negative searches, then the detection of viral sequences by a broader approach. It therefore cannot be stated that all viral involvement is excluded. Conversely, detecting a sequence establishes neither an active infection nor its causal role. A mechanistic inquiry gains by preserving this precise indeterminacy rather than choosing an exclusive label too early.

VIRAL SEQUENCES AND AN EXCLUSIVE LABEL

15.3.1.4 Cardiological Reading: severity is not reducible to the pump

BEYOND SYSTOLIC FUNCTION

The discordance between electrical involvement and systolic function requires specifying what a reassuring measurement actually makes it possible to rule out. A global index of contraction does not summarize all the vital functions of the heart. This finding does not make the measurement useless: it prevents assigning to it a question it cannot resolve alone. The therapeutic issue then involves two objectives: controlling the toxicity and preserving the oncological benefit. Two fatal trajectories under several treatments do not allow rescue

PROPOSALS FROM 2016, HISTORICAL

strategies to be compared or the one that would have changed the prognosis to be identified. The proposals discussed in 2016 must remain historical proposals, without automatically becoming current prescriptions. The alert is established; a treatment-related immune injury is strongly supported. The responsible antigen, the true frequency and the best intervention remain distinct questions. Standardized prospective observations and experiments on functional recognition could modify these different judgments, each on its own ground.

15.3.1.5 Integrative Clinical Reading

A HANDOVER BETWEEN SPECIALTIES

The combination of cardiac and muscle involvement underlines the value of a handover that does not divide the manifestations among specialties from the outset. In an oncology pathway, relating a new symptom to the treatment timeline can guide the discussion; it does not dispense with examining other explanations. This report makes such coordination intelligible, without having evaluated its effectiveness. The useful message to hand over would combine the alert, its degree of certainty and the person responsible for continuing the evaluation, rather than a diagnosis presented as definitively established.

15.3.2 Mahmood and the Passage from Registry to Prognosis

15.3.2.1 Situating the Result

MAHMOOD, A MYOCARDITIS REGISTRY

The registry gathers 35 cases of myocarditis associated with immune checkpoint inhibitors in eight centers, with 105 treated controls without clinical myocarditis from a single center. The frequency of 1.14% corresponds to 11 cases among 964 people treated at that center, not to the whole registry. Sixteen of the 35 cases experience the major adverse cardiovascular composite studied; six of these events occur with a normal ejection fraction. The so-called final troponin is the value at discharge or the one preceding an event during hospitalization; its prognostic threshold is chosen within these same data.[2]

15.3.2.2 Critical Reading: three populations in one article

15.3.2.2.1 Cases, exposed people and controls

A BORROWED DENOMINATOR

A case registry can describe what happens to the people identified. It provides a frequency among exposed people only if they are identified in the same setting and the same period. A multicenter collection therefore does not permit borrowing the denominator available in a single institution. The largest number is not always the one that answers the question. The comparison with controls introduces another object: which characteristics distinguish the groups constituted? Drawing controls at random from a file is not equivalent to randomizing treatments. If identification routes differ between centers, their differences may contribute to those that would spontaneously be attributed to patients. A useful comparison thus retains its recruitment condition; it does not immediately become an individual risk calculation.

CONTROLS DRAWN AT RANDOM, NOT RANDOMIZED

15.3.2.2.2 The clock and threshold of a prediction

THE CLOCK OF A PREDICTION

The prognostic question then requires a clock. To predict at admission, only information available at admission can enter the rule. A later measurement answers another question. When its timing depends on the care pathway, it must be specified when prediction begins and which people are still at risk of the event. A performance calculated without this clarification risks being understood as an earlier capacity for anticipation than it really is. Finally, choosing a threshold that separates the observed events well, then reporting its performance on the same observations, favors an optimistic estimate. The rule would need to be fixed before being tested in other data. Validation must also cover the assay method, units, timing and the course of action taken after the result. An isolated numerical threshold does not carry all these conditions with it.

A THRESHOLD CHOSEN IN THE SAME DATA

15.3.2.3 Immunological Reading: a complication does not reveal its entire mechanism

TWO EFFECTS OF AN IMMUNE INTERVENTION

The situation requires thinking together about two possible effects of an immune intervention: the result sought against the cancer and an organ injury. Their coexistence invites study of the regulation of the response; it does not demonstrate that the same cells recognize the same targets in each tissue. A clinical series documents manifestations and their therapeutic context, without automatically identifying an antigenic specificity. This caution also applies to the response to treatment of the complication. Improvement under immunosuppression may be informative, but its interpretation depends on the spontaneous course, the timing of the intervention and the other care given. Retrospectively comparing treatment intensities requires understanding why they were chosen. Clinical urgency can justify a decision with incomplete knowledge; it does not turn that decision into an experimental demonstration of its efficacy.

15.3.2.4 Cardiological Reading: preserved function is not a global guarantee

EJECTION FRACTION, ONE DIMENSION

Ejection fraction describes one dimension of cardiac function. Asking it to exclude any dangerous involvement would confuse a measurement with the whole organ. The lesson is not to replace one sovereign number with another, but to specify what each observation makes it possible to rule out, to suspect or to monitor. Likewise, predicting an event among recognized myocarditis cases does not validate screening in all treated people. Screening would also need to measure false signals, additional tests and the oncological consequences of decisions wrongly triggered. The registry documents a potentially serious toxicity and makes it possible to study the contribution of biomarkers to its evolving assessment. It validates neither a universal surveillance threshold nor the comparative efficacy of an immunosuppressive dose. Prospective data collection and independent validation could clarify these propositions.

15.3.2.5 Integrative Clinical Reading

WHAT THE REGISTRY LEAVES IN SHADOW

A registry of severe cases gives particular visibility to recognized complications. It informs less directly about the experience of all exposed people: anxiety raised by an alert, repeated tests or interruption of a treatment plan.

These consequences should belong to the evaluation of screening, alongside the events it would seek to prevent. The point is not to assume them frequent or negligible, but to name them as missing information for judging a complete pathway, beyond the performance of a threshold.

15.3.3 Axelrod and the Recognition of a Cardiac Antigen

15.3.3.1 Situating the Result

Axelrod, CD8 cells and α-myosin

Axelrod and colleagues study spontaneous myocarditis in mice lacking PD-1 and partially deficient in CTLA-4. CD8 depletion improves their survival. Transfer of spleen cells transmits myocarditis to immunodeficient recipients; removing CD8 cells from the mixture prevents this transmission. Reconstructed lymphocyte receptors recognize α-myosin peptides. In three patients with myocarditis under immunotherapy, some clonotypes expanded by these peptides are also found in the injured tissues. Precise recognition is characterized for one human receptor. However, reactive cells can also be expanded from healthy donors. The work thus combines animal perturbation, molecular identification and human correlation; it constitutes neither a therapeutic trial nor a validation of screening.[3]

15.3.3.2 Critical Reading: not adding up experiments that answer different questions

depletion and receptor reconstruction

Depletion asks whether a cell population contributes to the process; receptor reconstruction asks what the receptor recognizes. The concordance is valuable, but it does not automatically turn the second result into a complete explanation of the first. Removing all CD8 cells removes far more than the lymphocytes directed against the identified antigen. A more selective perturbation would be needed to attribute all the observed protection to this particular specificity. The transfer experiment contains an equally important distinction: it is the cell mixture that transmits the disease. That transmission requires CD8 cells does not prove that purified CD8 cells, still less an isolated clonotype, would suffice in any organism. The recipient environment is part of the conditions of the experiment. This precision does not weaken the discovery; it prevents a solitary cause from being substituted for a defined experimental system. The methodological report declares random allocation of the treated mice, but no blinding of the experimenters. The stated impossibility of masking certain manipulations does not remove the value of coded histological reading or of

a fixed age, not a rescue

independently assessed euthanasia criteria. The timing also deserves attention: depletion injections begin at twenty-one days of life. An intervention at a fixed age does not directly answer the question of rescue after clinical diagnosis of established myocarditis. Moving from one to the other requires reexamining the time of action and the state of the tissues, rather than merely carrying over the name of the target.

15.3.3.3 Immunological Reading: recognition is a relation, not a condemnation

a receptor does not recognize a disease

A receptor does not recognize a disease: it interacts with a peptide presented in a molecular context. Identifying this relation provides a remarkable experi-

mental handle. It makes it possible to formulate precise comparisons, instead of indiscriminately invoking a "dysregulated" immunity. But it does not yet say why a capacity for recognition becomes injurious here and remains contained elsewhere. The objection is immediate: if the same reactivity exists without disease, does the identified antigen lose its relevance? No. A condition can participate in damage without being sufficient to produce it. This result invites a search for the conditions under which the shift to attack occurs, rather than rejection of antigenic specificity. Conversely, invoking after the fact an additional condition for each exception would not advance the explanation: these conditions must become observable or manipulable independently of the final diagnosis. Expansion in culture calls for another distinction. It reveals a capacity in an environment built to favor certain cells; it does not directly reproduce their frequency or their activity before manipulation. A test of presence, a test of potential and a test of clinical danger therefore require three different validations. Confusing them would make a mechanistic discovery pass for an individual prediction already available.

PRESENCE, POTENTIAL, CLINICAL DANGER

15.3.3.4 Cardiological Reading: building therapeutic selectivity as an experimental question

TWO PROPERTIES FOR SELECTIVITY

The identification of an antigen opens an ambition: reducing cardiac injury without abolishing the antitumor benefit. This ambition nevertheless requires verifying two distinct propositions. The envisaged intervention must actually interrupt the injurious mechanism; it must also preserve the response that the oncological treatment sought to obtain. The molecular precision of the target guarantees, by itself, neither of these two properties. Finally, the genetic model requires specifying what is being extrapolated. A deficiency present during development and a pharmacological blockade introduced in an adult do not have the same history. Their comparison becomes convincing when human observations support the relevant links; it does not rest on an assumed identity of the situations. The useful question then becomes: which intervention would make it possible to separate recognition, activation and tissue destruction, and by what result would this separation be observed? Only under this condition can a candidate antigen become more than a name added to the disease.

15.3.3.5 Integrative Clinical Reading

A DISCOVERY TO TRANSLATE FOR THE PERSON

An antigenic discovery can help to understand a complication without immediately providing information usable for the patient. Announcing a predisposition or an individual danger to that person would require validation that does not follow from this study. The integrative perspective here concerns the translation of knowledge: explaining what research makes more intelligible, what remains unknown and what would actually change care. Preserving this distinction makes it possible to share an advance without fabricating a personal certainty or a promise of selective treatment.

15.3.4 Giavridis and the Experimental Dissociation of Effects

15.3.4.1 Situating the Result

Giavridis and CAR-T cells

Giavridis and colleagues build a mouse model of cytokine release syndrome after administration of human lymphocytes carrying a chimeric antigen receptor (CAR-T cells). In this system, host macrophages participate in the toxicity. Interventions on interleukin 1 (IL-1), interleukin 6 (IL-6) or a nitric oxide production pathway attenuate certain manifestations. The study thus proposes a target different from the antitumor cell itself. It opens an experimental possibility of dissociating efficacy from toxicity; it does not yet demonstrate this dissociation in patients.[4]

15.3.4.2 Critical Reading

15.3.4.2.1 What the model makes observable

A SETUP THAT MAKES VISIBLE

The first decisive choice precedes measurement. A model with bone marrow localization did not reproduce the syndrome sought. The authors then use immunodeficient mice bearing a heavy peritoneal tumor burden, then administer the CAR-T cells into the same compartment. The setup makes the phenomenon observable. It also determines the interactions that can be explained: a highly concentrated encounter between tumor, transferred cells and host cells does not summarize all clinical situations. The coexistence of human and murine cells becomes a tool for attribution. Distinguishing cytokines by species helps to identify their origin. This elegance has a counterpart: some human cytokines do not act on the corresponding murine receptors. The absence of an effect in this setup therefore does not allow that effect to be ruled out in humans. The model illuminates the circuits it makes functional; it may remain silent about those it interrupts.

15.3.4.2.2 The timing of intervention and the unit counted

PREVENTING IS NOT TREATING

Timing also changes the therapeutic scope. Anakinra is started five hours before administration of the CAR-T cells. A reduction in mortality in this protocol supports an experimental preventive strategy. It does not constitute proof of rescue after shock has set in. A reading that discreetly replaced "prevent" with "treat" would make the article take a step it has not tested. Finally, the units must remain visible. For some expression analyses, cells from three mice are pooled into one biological sample. Nine mice then give three samples, not nine independent measurements. The apparent power of an analysis depends on what is counted as a replicate.

NINE MICE, THREE SAMPLES

15.3.4.3 Immunological Reading

AMPLIFICATION BY THE HOST

The work shifts attention toward amplification by the host. The CAR-T cell recognizes the tumor, but the severity of the syndrome also depends on cells that do not carry the artificial receptor. This mediation makes conceivable an intervention located downstream of tumor recognition. It invites the question of which step produces the damage, without assimilating all mediators to the same "inflammation." The authors experimentally reinforce an interaction with murine cells by having the CAR-T cells express a suitable ligand. The resulting aggravation shows that such an interaction can amplify toxicity. It

does not prove that this pathway is necessary for every syndrome: the model already develops toxicity without this reinforcement. A capacity for aggravation is not a universal condition of occurrence.

15.3.4.4 Cardiological Reading

HYPOTENSION IS NOT MYOCARDITIS

The cardiovascular scope here concerns the possible mechanisms of a systemic and circulatory toxicity. Hypotension during an inflammatory syndrome does not, by itself, identify myocarditis. This text does not provide direct demonstration of myocardial injury or of restoration of ventricular function. The cardiologist must maintain this difference between a pathway likely to affect the circulation and a cardiac disease actually measured. Preservation of antitumor activity requires the same precision. It is examined with CAR-T cells modified to produce an IL-1 receptor antagonist, in another mouse model, with another tumor implantation and other administration conditions. It is therefore not directly an evaluation of anakinra administered separately. These experiments support a hypothesis of functional separation. They do not establish an overall clinical benefit, and an absence of statistical difference is not a demonstration of equivalence. The next step will need to bring efficacy and toxicity together in a comparison capable of evaluating their trade-off.

ANOTHER MODEL FOR THE ANTITUMOR EFFECT

15.3.4.5 Integrative Clinical Reading

A TRADE-OFF WITHIN ONE TRAJECTORY

The trade-off sought between cancer control and limitation of toxicity must ultimately be appraised within the trajectory of a single person. Results obtained in separate setups do not directly inform this shared experience. A clinical evaluation should notably specify recovery, persistent limitations and treatment burdens, together with tumor outcomes. The ambition of dissociating the effects thus finds a concrete recipient, without the model being credited with the human benefits that remain to be established.

15.3.5 Wadden and the Limits of a Rescue Case Report

15.3.5.1 Situating the Result

WADDEN, A RESCUE CASE

A man treated with pembrolizumab for metastatic urothelial cancer develops cardiogenic shock clinically attributed to immunotherapy-related myocarditis. Without endomyocardial biopsy, he receives circulatory support and corticosteroids, then abatacept and ruxolitinib. Functional recovery is reported, followed by a new cardiopericardial and thromboembolic episode requiring reassessment. At eighteen months, the authors describe cardiac and oncological stability. This case, published in 2025, documents a complex trajectory; it does not compare treatment strategies.[5]

15.3.5.2 Critical Reading: giving uncertainty a chronology

AFTER IS NOT BECAUSE OF

The account of a rescue naturally draws attention to the last treatment added before improvement. Yet “after” is not enough to identify “because of.” Attributing a specific share to each intervention would require knowing the

course that would have occurred without it. A single observation does not provide this alternative trajectory. The delay before earlier treatments take effect, circulatory support and other care prevent the narrative sequence from being turned into an estimate of efficacy. This limit does not make the observation useless. It shifts the question: can what was attempted be described precisely enough to build a future comparison? Several clocks must then be kept: that of anticancer exposure, that of the first symptoms, that of circulatory support and that of each immunomodulator. A future comparison should align the start of follow-up, eligibility and the definition of strategies. Patients classified as treated at the time a given treatment is used have necessarily survived until then; comparing them carelessly with controls followed from admission could introduce selection and immortal time bias. The diagnostic qualification must also be kept open. Urgency can justify a decision before all confirmations are obtained; it does not retrospectively turn the presumed diagnosis into histological certainty. Improvement under immunosuppression does not resolve this problem: several conditions could share an apparent response. The reasoning must therefore keep together clinical plausibility and the incompleteness of its verification.

SEVERAL CLOCKS, IMMORTAL TIME

A DIAGNOSIS THAT REMAINED PRESUMED

15.3.5.3 Immunological Reading: a combination does not demonstrate synergy

Combining two agents acting on different pathways can be rational without its superiority being established. The complementarity sought belongs to the hypothesis; the therapeutic interaction belongs to a comparison. Speaking of synergy would require specifying the expected effect of each intervention separately, the scale on which the effects would be combined and the result observed under their combination. No mechanistic adjective replaces these comparators. The expression “reversing immunotherapy” also deserves scrutiny. The immune system is not a single switch in which a second molecule would exactly cancel the first. Restraining an activation pathway can modify several functions simultaneously. Target engagement, modification of the disease process and benefit for the person would need to be distinguished. These three propositions require different observations; cardiac recovery does not demonstrate all of them.

REVERSING IMMUNOTHERAPY?

15.3.5.4 Cardiological Reading: following the person beyond the first recovery

Success cannot be confined to a single measure of ventricular function. Effective circulation, rhythm, pericardium, symptoms and cancer progression define distinct dimensions. Improvement in one does not dispense with examining the others. Likewise, a late imaging abnormality requires a temporal interpretation: persistence of a lesion, inflammatory activity and clinical recurrence are not synonyms. The choice to publish the whole trajectory, including its complications, guards against an overly smooth story of cure. Even so, attributing each subsequent event to the treatment would be a symmetrical error. A case report makes it possible to identify what needs to be monitored; it estimates neither the frequency of a benefit nor that of an adverse effect. The case report thus teaches something precise: how to convey a therapeutic hope with its conditions of verification. A prospective comparison will need to define the diagnosis, the time of inclusion, the associated care and both cardiac

CONVEYING HOPE AND ITS CONDITIONS

and oncological outcomes. The case helps to pose these questions; it does not by itself provide a prescribing protocol.

15.3.5.5 Integrative Clinical Reading

LIFE AFTER THE RESCUE

The word "rescue" describes a decisive moment, but is not enough to recount the life that follows. This case invites the question of how return to activities, continuity of care and discussion of oncological goals are built after a major complication. These outcomes cannot be inferred from the reported cardiac stability alone. In a handover between hospital and general practice, observed facts, still-open hypotheses and follow-up responsibilities would need to be distinguished, without turning a singular trajectory into a general rule.

15.4 Worked Exercises

15.4.1 Exercise on the Denominators of an Alert

15.4.1.1 Problem

SIX HOSPITALS, THIRTY CASES

All data are fictional. Six hospitals report 30 cases of myocarditis after an anticancer treatment. Twelve have a serious complication. Only hospital A knows all its exposed patients during the period: 500 people, including five cases among the thirty. The other institutions do not provide their number of treated people. Ninety controls without clinical myocarditis are drawn at random from the records of A. An analyst states: "Myocarditis affects 30/500, or 6% of patients. Among the 120 people compared, its frequency is 25%. The two estimates prove a large variation in risk." The analyst also proposes a troponin threshold chosen to best separate the twelve serious cases among the thirty records. Questions. Correct the two frequencies. Can this threshold immediately be used to screen all treated patients?

TWO FREQUENCIES ANNOUNCED

15.4.1.2 Reasoned Answer

THE ONLY MATCHING DENOMINATOR

The only proportion of cases among exposed people that can be calculated with a matching denominator is that of A: 5/500, or 1%. It describes the cases identified in that institution and that period, without guaranteeing that all forms were detected. The thirty cases from the six centers must not be divided by the exposed people of A alone. The proportion 30/120 = 25% is imposed by the selection of three controls per case. It does not measure the frequency in the treated population. By contrast, 12/30 = 40% describes the proportion of complications among the cases collected; it is neither the risk of myocarditis nor the risk of complication in all exposed people. The threshold was chosen and evaluated in the same thirty records: its performance requires independent validation. The date of the assay must also be fixed. A value obtained after deterioration is not a predictor available at the start of care. Finally, distinguishing severe forms among recognized cases does not validate a screening test in people without a diagnosis.

A THRESHOLD WITHOUT INDEPENDENT VALIDATION

Synthesis

A therapeutic perturbation brings to light what becomes possible in a transformed situation; it does not directly describe ordinary functioning. The texts commented on show this at different levels. The initial observation is valuable for its alert, the registry for its description of recognized cases. The animal model makes an antigenic recognition or a mediation testable, and the rescue case documents a trajectory. None alone measures the frequency, the mechanism and the efficacy of a course of action. Judgment must finally bear on the targeted disease and the possible harm together, with the person concerned.

Open Questions

Which independent signs would make it possible to establish that an involvement preceded the treatment, rather than having been created by it? At which step, between recognition, activation and tissue destruction, could an intervention limit the damage without abolishing the antitumor benefit? Which prospective comparison would give a chronology to the uncertainty of rescue treatments? Each calls for its own comparison.

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16

Precision Medicine between Measurement and Interpretation

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Precision medicine promises to distinguish individuals more finely, through an image, a tracer, a profile of circulating proteins or a genotype. Each of these instruments constructs a comparison and delineates classes. Their technical refinement can give the impression that disease can at last be read directly. Between a measurement and its interpretation, however, work remains to be done. A diagnostic convention must remain open to examination, a class must be able to meet what resists it and a disagreement between two instruments may carry information. Recognizing a difference does not suffice to explain how it is produced, and a valid category does not yet prove its usefulness in a care pathway.

16.1 Constructing Testable Measurements and Classes

16.1.1 Measurement Constructs a Comparison

WHAT A NUMBER MEASURES

A precise measurement does not necessarily bring one closer to the right answer. It may reproduce with great fidelity a signal that depends on sampling, on the chosen time point or on the processing procedure. The question is therefore not only "how much?" but "of what property is this number a measure, and under what conditions?" Consider two fictional laboratories. They measure the same sample with excellent repeatability but obtain systematically different values. Multiplying repetitions in each laboratory will refine their means without removing their disagreement. Calibration, preparation and the definition of the measured quantity must be examined, and a correspondence then established if possible. Repetition controls one uncertainty; it does not control all errors. It will be objected that requiring this reconstruction for every measurement would make medicine impracticable. The requirement does not concern each consultation: it concerns the procedure that licenses comparison and the information transmitted with the result. Sufficiently documented standardization makes it unnecessary to repeat the entire inquiry; it does not remove the need to recognize when a measurement falls outside its domain.

TWO LABORATORIES IN DISAGREEMENT

16.1.2 A Class Must Be Able to Meet What Resists It

A CLASS TIED TO ITS VARIABLES

Combining many measurements can reveal groups that are invisible to isolated observation. But a class constructed by an algorithm remains tied to the variables selected and to the way similarity is defined. Changing these choices can shift its boundaries. This does not make it arbitrary: it indicates what must be tested. A demanding test does not consist merely in recovering groups in the data that gave rise to them. It requires fixing an assignment rule and then examining, in new individuals, its reproducibility under comparable conditions, the cases left outside it and its failures relative to the stated use. This reproducibility does not mean that a person must keep the same class if their condition changes: a classification of states may serve precisely to recognize a transition. If every exception immediately receives a new subclass, the classification may become infallible by construction and lose its capacity to meet a real difficulty. The opposite objection deserves attention: a stable classification might be too coarse to uncover a rare disease. Stability is therefore not an absolute end. A revision becomes fruitful when it brings verifiable consequences that the former rule could not predict. Three things then need explaining: what the new category gains, what it loses and which observations would decide between the two proposals. Naming better does not consist merely in multiplying names.

A FIXED RULE, NEW INDIVIDUALS

16.2 Each Instrument Must Keep an Identifiable Question

16.2.1 A Disagreement May Carry Information

two discordant instruments

Two instruments may produce discordant results without one of them simply being wrong. They may observe different properties, in distinct compartments or at times that are not interchangeable. Before asking which should prevail, it is necessary to specify what they were supposed to show together. In a fictional example, instrument A describes a structural change and instrument B a momentary activity. A positive result for A and a negative result for B form a contradiction only if the theory requires the change and the activity to be present simultaneously. Otherwise, their discordance may provide information about a history, a timing or a limit of observation. These possibilities are hypotheses to be decided between, not established explanations. It would nevertheless be too convenient to rescue every result by invoking different objects. The expected concordances must be stated, together with the conditions under which their absence would put the model in difficulty. The plurality of instruments does not license reconciling everything. It allows a more instructive confrontation, provided that each instrument keeps an identifiable question.

one question per instrument

16.2.2 A Category May Be Valid without Explaining Everything

16.2.2.1 Recognizing a difference without explaining how it is produced

Sweatt and Khera, two classified objects

Sweatt's profiles and Khera's scores do not classify the same object. The former sort patients into groups according to measurements of circulating proteins; the latter aggregate genetic associations. One recovers an organization of profiles in another cohort, the other evaluates the score in participants distinct from those used to choose it (see 16.3.4 and 19.3.3).[1,2] These approaches produce comparisons. They do not promise, by their precision alone, the same explanation of the disease. Must a complete etiology then be awaited before a category is taken seriously? That requirement would confuse two achievements: regularly recognizing a relevant difference and fully explaining how it is produced. A category can already serve to form groups defined by shared criteria, to organize research or to test a prediction. To grant it this value, its rule must be explicit, the measurements must suit its object and an evaluation adapted to its use must not rest solely on the data that gave rise to it. Its justification concerns a defined task, not a hidden identity that it would at last have revealed.

recognizing without explaining everything

16.2.2.2 Knowing what operation each classification performs

two opacities

The opposite objection is strong: does accepting a classification without a complete mechanism not legitimize a black box? Two opacities must be distinguished. Part of the biology may be unknown while the classification rule, its inputs and its observed errors are known. If this information is missing as well, the difficulty is no longer only an unfinished biological explanation. The user no longer knows what operation is being entrusted to the instrument. Requiring a controllable procedure therefore does not amount to requiring a finished science. This distinction clarifies the confrontation between the two studies. The same rank on a score does not entail a complete resemblance between people. A close molecular profile does not guarantee a shared history. Bringing them together becomes fruitful if one asks what each classification makes it possible to compare that the other leaves aside. It may prepare an

inquiry into the links between susceptibility and measured state; it does not assume that one is the translation of the other.

16.2.2.3 Validity of the classification and usefulness of the pathway

two departments, one category

There remains the clinical decision. Consider a fictional category whose assignment is reproducible and which correctly distinguishes two trajectories. In one department, its result changes no care; in another, it triggers an additional test. The category does not become scientifically false in the first department. Its practical contribution differs because its use differs. In the second, what the test contributes and what it costs must be weighed: information, delay, burden, possible harm. The validity of the classification and the usefulness of the pathway are therefore neither identical nor independent. It will be objected that such an evaluation delays the use of a promising tool. But moving forward does not mean granting it every use from the outset. A category can be adopted now for a research comparison that it makes more rigorous, while its use to trigger an intervention remains to be tested. Progress consists in authorizing the justified use, specifying the still hypothetical use and organizing the test that could bring them together. An incomplete etiology does not condemn knowledge; it requires choosing exactly what is to be done with it.

16.3 Texts Put to the Test

16.3.1 Ferreira and the Conditions for Imaging Evidence

16.3.1.1 Situating the Result

the revised Lake Louise criteria

The expert group's 2018 document proposes a revision of the Lake Louise criteria. In a suggestive clinical context, the combination of a T2-based marker and a T1-based marker supports the existence of acute or active myocardial inflammation. The text examines the techniques, their combination and their limits. It brings together earlier work; it does not report a trial in which the new rule was compared with the former one on patient outcomes.[3]

16.3.1.2 Critical Reading: the diagnostic convention must remain open to examination

16.3.1.2.1 Examining the passage from studies to rule

a literature turned into a rule

A guideline transforms a heterogeneous literature into a usable rule. This operation is necessary for collective practice: without explicit criteria, two readers may use the same name for different observations. But consensus status does not replace examination of the passage from the selected studies to the proposed rule. The populations compared, the diagnostic references and the way performance was summarized must be known. The appendix provides an instructive detail here. To compare certain combinations, the authors use an indicator named AUC*, calculated as the mean of sensitivity and specificity. This summary does not describe the whole curve obtained by varying a continuous threshold. Its similarity in name to the area under that curve may obscure this change of object. Before commenting on how high a result is, its operational definition must therefore be read.

AUC*, an indicator to be read closely

16.3.1.2.2 Populations and a reference to be questioned

THE REFERENCE AND WHAT IT MISSES

Pooling several performance estimates does not make their populations interchangeable. A technique evaluated mainly in typical acute forms and another evaluated in less clear-cut presentations are not necessarily engaged in a fair competition. A direct comparison, in the same patients and with independently organized readings, would better answer the question of their respective contributions. The synthesis reveals this need; it has not already met it. The diagnostic reference is itself an operation of knowledge. A biopsy examines a sample, whereas imaging explores a broader tissue distribution. A disagreement therefore does not automatically identify the faulty test. It may arise from the territory studied, the timing of the examination or the definition adopted for the disease. Stating that a method is compared with a reference requires describing what that reference may miss.

16.3.1.3 Immunological Reading: seeing a consequence does not suffice to identify its driver

THE TISSUE, NOT THE EFFECTOR CELL

Imaging parameters provide information about tissue properties and their transformations. They do not directly name the effector cell, the antigen or the pathway responsible. Several mechanisms may produce similar consequences; conversely, the same mechanism may leave different images depending on its intensity and how long it has been present. Late enhancement illustrates this difficulty: it may persist after the active phase. A lasting trace does not, on its own, attest that the injury is ongoing. Inferring from an abnormal image the existence of a currently accessible immune target would therefore add two inferences: from trace to activity, then from activity to mechanism. Each requires its own support. A combination of markers can make an interpretation more robust without abolishing this distance. The immunological interest of the image then also lies in its capacity to guide sampling and comparisons: where to look, when to observe, which hypotheses to put to the test?

16.3.1.4 Cardiological Reading: a transportable measurement must retain its conditions

A MEASUREMENT AND ITS TECHNICAL CONDITIONS

The text stresses the dependence of measurements on techniques and the need for appropriate reference values. This requirement guards against treating a number as if it carried its own norm. To interpret a change over time, it is also necessary to know whether acquisition and analysis are comparable. A change in equipment can mimic a change in the patient; technical stability helps make the clinical course interpretable. Finally, recognizing an abnormality better and treating better are not equivalent outcomes. A diagnostic strategy must be evaluated through the decisions it entails, with their benefits and harms. The consensus shows how a rule was constructed at a given moment. It does not remove the need to study its subsequent validations or to check current guidelines before clinical use.

RECOGNIZING IS NOT TREATING

16.3.1.5 Integrative Clinical Reading

FROM REPORT TO CONSULTATION

Receiving an imaging report does not mean that the person already has an explanation of their symptoms or a care plan they understand. Moving from the technical result to the consultation requires specifying its scope, the remain-

ing uncertainties and the reason for any further step. The consensus does not measure the quality of this transmission. The question therefore remains how expert language can become information usable by the person and by the caregivers who ensure the continuity of their care pathway.

16.3.2 Tarkin and the Comparison of Tracers

16.3.2.1 Situating the Result

Tarkin, DOTATATE and FDG

In a 2017 study, Tarkin and colleagues evaluate positron emission tomography using gallium-68 DOTATATE, which targets somatostatin receptor subtype 2 (SST2), in 42 people with atherosclerosis. They bring together imaging, cellular data and carotid samples. The signal distinguishes arteries considered responsible for a recent event from arteries that are not. Coronary images are interpretable with this tracer, whereas 27 fluorodeoxyglucose (FDG) scans are hampered by myocardial signal. However, no prolonged myocardial suppression preparation is used for this comparator. The study does not demonstrate improved prediction of future events or improved outcomes of care.[4]

16.3.2.2 Critical Reading: the comparison depends on what is being sought

16.3.2.2.1 An image taken after the event

three closely related questions

Three questions lie close together in this study but are not equivalent: distinguishing an artery linked to an event that has already occurred, recognizing a morphology judged concerning and predicting a future event. Succeeding at the first task does not establish the third. After the acute event, the process observed may include its consequences and those of the interventions performed. An image obtained at that moment does not automatically recount the state that preceded it. Comparing arteries within the same person has merit: it reduces certain general differences between individuals. It nevertheless leaves in place the local differences that are to be interpreted. Here, several coronary lesions had already received a stent before the scan.[4] The critical alternative is therefore precise: what part of the signal reflects the initial lesion, and what part may be related to its recent history? An internal comparison does not remove this chronological question.

16.3.2.2.2 Preparing the comparator, counting the right units

interpretability as a result

Interpretability also deserves to be considered a result, and not a nuisance to be forgotten. Comparing only technically available measurements may select the situations in which a tracer already works. But announcing superiority without recalling how the comparator was prepared would risk the opposite error: attributing entirely to the molecule a difference that also depends on the protocol. The advantage observed under these conditions must therefore

patient, artery, segment

be distinguished from its generalization to all competing procedures. Finally, counting segments is not equivalent to counting independent people. For each analysis, it must be asked whether the unit is the patient, the artery, the segment or the region of interest. Apparent precision becomes difficult to judge when these units shift silently from one to another.

16.3.2.3 Immunological Reading: a molecular target does not constitute a complete inventory

A SIGNAL, NOT AN INVENTORY

Bringing together receptor expression, tracer uptake and the presence of cells in a tissue makes the biological interpretation better argued than a mere resemblance of images. The exact question that each observation answers must nevertheless be kept in view. A signal compatible with certain cells is not an exhaustive measure of all the immune functions present in the lesion. Consider a hypothetical situation: a population becomes less abundant, but each cell expresses more of the recognized target. The overall signal might change little. Conversely, a decrease in signal might result from a change in expression without an equivalent disappearance of cells. These possibilities show why a target and a quantity of cells are not synonyms.

16.3.2.4 Cardiological Reading: a local result still has to be linked to a decision

CAROTID SURGERY SPECIMENS

Tissue validation here concerns carotid arteries that underwent surgery.[4] It provides real support, but extrapolation to the coronary arteries must keep this difference in territory and sampling in view. A technology may be biologically convincing without each anatomical application being validated in the same way. To turn this information into a decision, what changes once the result is known would then need to be defined: surveillance, an intervention, additional exposure to its risks. The useful question becomes comparative: does a strategy using the image improve an important outcome compared with a strategy that does not? Recognizing a lesion provides information; demonstrating the value of the action taken on the basis of that information is a distinct step.

16.3.2.5 Integrative Clinical Reading

A FEASIBLE CARE PATHWAY

Greater visibility of a lesion in a specialized setting does not suffice to establish the place of a test in a health care system. It would be necessary to specify who has access to it, what it replaces, what travel or delays it imposes and which decisions it modifies. These dimensions are not arguments for or against the tracer before they have been studied. They define the problem of appropriate use: an innovation must be compared with a feasible care pathway, not only with another image.

16.3.3 Grayson and the Disagreement between Imaging and Clinical Assessment

16.3.3.1 Situating the Result

GRAYSON, IMAGING AND REMISSION

The prospective cohort of Grayson and colleagues, published in 2018, includes 115 people, 56 of whom have large-vessel vasculitis, and 170 fluorodeoxyglucose scans. The comparator group notably includes diseases that can mimic vasculitis. Among 71 scans performed during clinical remission, 41 are interpreted as consistent with active vasculitis. Clinicians and image readers perform separate assessments. A global arterial uptake score (PETVAS) is constructed and explored for the risk of relapse. These observations reveal

a discordance to be explained; they do not suffice to recommend score-guided treatment intensification.[5]

16.3.3.2 Critical Reading: the reference transforms the meaning of an error

16.3.3.2.1 The reference and the chosen comparators

FALSE POSITIVE AGAINST WHICH REFERENCE

A scan considered positive in a person in remission may be labeled a false positive if clinical assessment is taken as the reference. But this designation describes a disagreement with the reference; it does not prove that the entire signal is biologically false. The opposite hypothesis, that the image would necessarily reveal hidden disease, would be just as premature. To decide between interpretations, information is needed that is not merely a repetition of the initial disagreement. The choice of comparators makes this difficulty concrete. Distinguishing patients with a disease from healthy volunteers does not exactly represent the situation in which several diagnoses are plausible. Including look-alike conditions brings the test closer to a real clinical question. This may change observed performance without any change in the device: what changes is the difficulty of the distinction required.

LOOK-ALIKE COMPARATORS

16.3.3.2.2 A score to validate, scans to count

PETVAS, A SCORE TO VALIDATE

The study constructs a score from arterial territories selected on the basis of the data. A threshold separating clinical activity from remission is then explored for future relapses.[5] This construction requires independent validation. Without it, good separation in the observations that helped define the tool may be overly favorable. Giving a name and a threshold to a set of measurements does not suffice to turn that set into a stabilized rule. Finally, repeated scanning requires keeping denominators visible. The number of images is not the number of individual trajectories. A succession of measurements in the same person may help in understanding a course, but it does not create as many independent confirmations. The statistical method must address this structure of the observations, and the reporting must allow it to be recognized.

16.3.3.3 Immunological Reading: metabolic activity calls for several explanations

A PERSISTENT SIGNAL, SEVERAL CAUSES

Suppose that a signal persists although symptoms have resolved. Possible explanations include residual pathological activity, a tissue change following the initial episode or several simultaneous phenomena. These possibilities are not equivalent for treatment. Naming them demonstrates none of them; it does, however, prevent an abnormality from being turned too quickly into an indication. The timing at which information is collected becomes essential. The meaning of an image depends on what happened before it, notably on the treatments received. An association between a dose and uptake does not, on its own, constitute an experiment comparing that dose with another. The prescribing history remains part of the interpretation.

16.3.3.4 Cardiological Reading: anticipating does not yet mean knowing how to intervene

MONITORING OR TREATING

An association with future relapse could make a signal useful for organizing a clinical inquiry. It does not automatically determine the response to adopt. It would be necessary to distinguish a monitoring strategy from a treatment strategy, to define their expected consequences and also to measure the possible harms of an additional intervention. The cardiovascular scope here concerns large vessels; it must not be tacitly extended to the myocardium or to all coronary inflammation. Mechanism, territory and the decision under consideration remain linked. This text shows that a discordance can be scientifically productive: it reveals the limits of the clinical categories of remission and activity, without licensing the immediate replacement of one by the other.

A PRODUCTIVE DISCORDANCE

16.3.3.5 Integrative Clinical Reading

THE PERSON AND THE IMAGE

A discordance between a person who reports feeling better and an image that remains abnormal must not erase either piece of information. The first concerns, among other things, lived experience; the second, a particular measurement whose meaning remains to be examined. An overall decision would have to specify what is to be improved and what consequences an intensification of care would entail. The article does not settle this trade-off. It helps to explain why listening to the person and interpreting the image are two complementary requirements, with no automatic hierarchy between them.

16.3.4 Sweatt and the Construction of Immune Groups

16.3.4.1 Situating the Result

SWEATT, FOUR IMMUNE GROUPS

Sweatt and colleagues study 48 circulating proteins in 281 patients with pulmonary arterial hypertension, then in 104 patients from another center. An unsupervised classification, reapplied separately in each cohort, recovers four groups with comparable profiles and different prognoses. The endpoint is death or transplantation in the discovery cohort and death in the validation cohort. Measurements rely on fluorescence intensities, not on absolute concentrations. The networks describe partial correlations. This observational work does not compare treatments assigned according to group.[1]

16.3.4.2 Critical Reading: three different meanings of the word "validation"

INTERNAL STABILITY AND NATURAL KINDS

The question is not only whether four groups appear. It must be asked what remains stable when the people, the laboratory or the procedure change. A classification may be robust to resampling of the same dataset while depending heavily on the choice of proteins. Internal stability informs about a property of the computation within a given framework; it does not demonstrate that the disease comprises exactly four natural kinds. The term "unsupervised" means here that clinical characteristics are not used to assign the groups. It does not make the analysis free of choices. The panel, the normalization, the distance between profiles and the algorithm define what can be recognized as similarity.

REPLICATION AND INDIVIDUAL ASSIGNMENT

These decisions are not defects to be abolished: they are the conditions to be disclosed in order to understand the result and allow it to be tested. Recovering an analogous organization elsewhere is a more demanding test than redrawing the same map. But this replication does not yet answer the practical question: where should an individual person be classified tomorrow? Rebuilding the groups in a new population allows their boundaries to shift. Applying a rule fixed in advance requires, by contrast, keeping the initial boundaries. Both exercises are useful; they evaluate different objects.

16.3.4.3 Immunological Reading: centrality is not biological control

STATISTICAL CENTRALITY, BIOLOGICAL TARGET

In a statistical network, a protein may occupy a central position because it varies with many others. Designating it as a target requires a different test: modifying its activity and observing a relevant consequence. A crossroads on the map of associations is not necessarily the point through which the disease process passes.

THE NAME OF A GROUP

The same caution applies to the name given to the group whose cytokine levels remain low, close to those of controls. A name such as "low-inflammatory profile" must remain tied to the compartment, the panel and the time point studied. The absence of a measured signal does not locate all immune phenomena and does not trace their history. Conversely, a combination of elevated signals does not decide between damage, repair and compensatory response. Language must keep this plurality open without emptying the observation of its value. To investigate a group further, successive samples could be compared with tissue observations and experimental perturbations. A persistent identity would then need to be distinguished from a transient state. A change of class might reflect a biological change, but also measurement variability; interpretation requires being able to decide between these possibilities.

16.3.4.4 Cardiological Reading: a category useful for research must still prove its use

PROGNOSIS AND EXPECTED BENEFIT

In cardiovascular practice, additional information must be evaluated against what is already known and against the decision it could change. An association with prognosis does not automatically specify who will benefit more from an intervention. A high-risk group may offer more room for benefit, but also more competing risks or toxicity. The next step could compare explicit strategies: usual care or care guided by a rule fixed beforehand. Relevant outcomes would include events, quality of life and the harms related to additional decisions. A better description of patients justifies this research; it does not prejudge its result.

16.3.4.5 Integrative Clinical Reading

A CATEGORY, NOT A DESTINY

An immune category can organize research without becoming a person's entire medical identity. Two members of the same group may have different priorities, functional capacities and care constraints. To use this classification in consultation, it would be necessary to show what it adds to the judgment already formed from this information. It also remains to be seen how to present this classification without giving the impression of a fixed destiny: a differ-

ence in prognosis between groups does not foretell each individual's future with certainty.

16.3.5 Beck and the Reorganization of Diseases by Genotype

16.3.5.1 Situating the Result

Beck, UBA1 and VEXAS

In 2020, Beck and colleagues describe twenty-five men with somatic UBA1 variants and a late-onset inflammatory syndrome, named VEXAS (vacuoles, E1 enzyme, X-linked, autoinflammatory, somatic). An initial genotype-driven discovery is extended by recruitment guided by clinical resemblance. The study combines the distribution of variants across cell populations, biochemical analyses and a zebrafish model. It links the alteration of the enzyme's cytoplasmic isoform to inflammatory dysregulation. The clinical picture includes venous thromboembolic events, but this is not a study devoted to cardiac mechanisms. This discovery work does not compare randomized treatments.[6]

16.3.5.2 Critical Reading: discovering an entity without erasing recruitment

a discovery that changes its method

A discovery can modify its own research method. Once an initial link has been established between a molecular abnormality and a clinical picture, investigators know better whom to examine. This adaptation is fruitful; it nevertheless makes it incorrect to present the whole as a phenotype-agnostic screen. The final series results from several points of entry. Its homogeneity therefore cannot be attributed entirely to the unifying power of the genotype: clinical selection also contributes to producing it. This does not invalidate the discovery. It changes the questions its numbers can answer. Describing the patients identified is not measuring the frequency of the disease in a general population; observing a frequent manifestation among them does not provide its predictive value in ordinary practice. Turning a discovery into a case-finding rule would require knowing the people not retained, the sensitivity of the detection procedure and the less typical forms. The absence of women in an initial series calls for the same discipline. It describes the observed recruitment; it does not suffice to define a universal biological impossibility. A founding classification must retain the capacity to accommodate the observations that will widen its boundaries. Its strength lies in what it explains, without requiring the first cases to exhaust all possible forms.

describing is not measuring a frequency

an initial series without women

16.3.5.3 Immunological Reading: locating the mutation does not suffice to locate all its effects

where, when, which function

The word "genetic" can obscure three questions: in which cells is the abnormality found, when does it appear and which function does it alter? A variant acquired in a fraction of cells requires specifying the tissue studied and the proportion of cells involved. The coexistence of populations, their selection and their interactions must be considered together. The disease belongs to the person; its molecular cause may be unevenly distributed across that person's tissues. The convergence of several approaches then becomes decisive. An association between a variant and a syndrome gains weight when experiments link that variant to a functional disturbance. Each experiment nevertheless re-

mains a translation of the question. Here, inactivation in the developing fish does not identically reproduce a late-onset human somatic mosaicism. It tests a biological relationship; it does not reconstruct the patient's entire natural history.

16.3.5.4 Cardiological Reading: linking organs without attributing too quickly

A DISEASE THAT CROSSES SPECIALTIES

For the cardiologist, the methodological interest lies in recognizing that a systemic disease may cut across several specialty categories. The hematopoietic abnormality provides a reason to look for relationships between manifestations; it does not remove the need to demonstrate those relationships. Faced with associated cardiac involvement, several levels would remain to be distinguished: coexistence, indirect consequence of the systemic disease, treatment effect or a lesion arising from a mechanism of its own. A venous event does not demonstrate myocarditis, inflammatory coronary disease or direct cardiomyocyte injury. Moving from one to the other would require specific characterization of the events, a timeline and appropriate comparisons. This article therefore offers the cardiologist a school of reasoning, not a management protocol.

A VENOUS EVENT, NOT MYOCARDITIS

16.3.5.5 Integrative Clinical Reading

Bringing together manifestations previously considered separate can make a medical history more intelligible. But recognizing a systemic disease does not mean that every later symptom has the same cause. Follow-up must preserve the possibility of another condition or of a consequence of care. The integrative interest of this discovery thus lies in a twofold movement: uniting what deserves to be united and keeping the inquiry open where the link remains uncertain. A unifying explanation earns its value through its precision, not through its capacity to absorb everything.

16.3.6 Dou and the Conditions of an Immune Difference

16.3.6.1 Situating the Result

DOU, XIST AND GENETIC BACKGROUND

Dou and colleagues study male mice expressing a truncated version of Xist, designed to form ribonucleoprotein complexes without exerting its usual chromosome-inactivation activity. After pristane induction, pathological worsening depends on genetic background: it appears in the susceptible SJL/J strain, whereas the resistant C57BL/6J strain does not develop comparable lesions. Cellular and serological analyses complement the experiment; human sera also recognize XIST-associated proteins. Histology is assessed blinded. The threshold separating low and high disease is, however, constructed from pristane-treated female controls, and some missing organs lead to exclusions. The study addresses a conditional contribution to autoimmunity, without demonstrating specific cardiac involvement.[7]

16.3.6.2 Critical Reading

ISOLATING A FACTOR IN THE MALE

The experimental design spares a global comparison between males and females the task of settling what it could not settle alone. Adding a molecular component to male animals makes it possible to ask a more precise question: can this component modify the disease under the conditions studied? The intervention does not, however, turn the animal into an experimental stand-in for all the characteristics of a female. That would convert a contrast built to isolate one factor into a general biological equivalence. An objection remains: an artificial construct might produce an effect that does not exist with the natural molecule. This objection does not destroy the experiment; it defines its next control. The relevant consequences would need to be reproduced with other manipulations, and those depending on the function of interest distinguished from those specific to the construct. The modification makes the experiment informative precisely because it separates functions; that separation also requires checking what it unintentionally alters. The boundary between low and high disease calls for further vigilance. A threshold chosen within the results may make a comparison legible, but that fact alone does not give it universal biological meaning. The distribution of scores and the robustness of the contrast to other cutoffs must be examined. Blinded assessment limits the influence of expectations on tissue reading; it does not make a threshold defined afterward independent. These two methodological qualities address different problems.

A THRESHOLD CHOSEN WITHIN THE RESULTS

16.3.6.3 Immunological Reading

A CAUSE THAT IS NOT SELF-SUFFICIENT

The resistance of one genetic context provides positive information: it prevents Xist from being treated as a self-sufficient cause. It might be objected that a factor dependent on other factors ultimately explains nothing. That would demand from biology a cause capable of producing the same effect everywhere. An explanation becomes more precise, on the contrary, when it indicates the conditions in which a contribution manifests itself and those in which it fails. What, under these conditions, permits or prevents progression then remains to be determined. Gain of function does not, however, demonstrate that removing the component would abolish all spontaneous disease. This new question requires a different intervention, with its own difficulties: deleting an element with several functions could make the animal noncomparable for reasons unrelated to the mechanism studied. The necessity of a factor cannot be asserted by verbally inverting the result of adding it. Serological recognition must likewise keep its status. A recognized antigen may take part in the response, become visible after injury or accompany its spread. Distinguishing these possibilities would require temporal observations and discriminating perturbations. A human association gives the experimental hypothesis an anchor point; it does not replace testing the steps that would link exposure, response and damage.

A RECOGNIZED ANTIGEN, THREE STATUSES

16.3.6.4 Cardiological Reading

A QUESTION OF SUSCEPTIBILITY

For cardioimmunology, the interest here lies in formulating a question of susceptibility more clearly. It would be insufficient to import an autoimmune signature into a cardiac population and regard its detection as an explanation of the lesions. It would be necessary to specify the targeted involvement, compare relevant patients and determine whether the information adds any-

thing beyond the systemic manifestations already known. Extrapolation to the heart remains a separate question. Understanding an autoimmune susceptibility demonstrates neither its contribution to a cardiac disease nor the usefulness of an intervention in the people concerned. This work prepares an inquiry into risk; it provides neither a validated cardiac screening test nor a justification for a treatment directed against XIST. Precision medicine begins here with precision in the question.

16.3.6.5 Integrative Clinical Reading

susceptibility and a person's sex

A result concerning biological susceptibility must not become an overall prediction based on a person's sex. The study measures neither all individual differences nor the social conditions of access to care. From a general practice perspective, its interest is to refine the questions asked, without deciding in advance who is ill or who should be reassured. Explaining an experimental difference and listening equitably in the clinic belong to distinct registers, which must be articulated rather than substituted for one another.

16.4 Worked Exercises

16.4.1 Exercise on the Reproducibility and Usefulness of a Classification

16.4.1.1 Problem

two laboratories, 200 people

Entirely fictional data, unrelated to Sweatt's sample sizes or results. A team fixes a rule that assigns two immune classes, A and B. Two laboratories analyze aliquots of the same sample from 200 people. They assign the same class in 180 cases. A third team then recreates two groups from its own data, without applying the initial rule; their mean profiles resemble A and B. In a separate trial, 400 people are randomly allocated to usual care or classification-guided care, 200 per group. With complete one-year follow-up, a clinical event occurs in 24 people under usual care and 22 under guided care. The latter leads to 60 additional treatments. The data specify neither adverse effects nor the uncertainty around the difference in events. Questions. What does the agreement between laboratories measure? Does the third team validate individual assignment? Does the trial result allow the classification to be declared clinically useful or useless?

16.4.1.2 Reasoned Answer

90% agreement

Raw agreement is 180/200, or 90%. It informs about the reproducibility of assignment with two laboratories and the same sample. It demonstrates neither biological stability over time nor the existence of two distinct diseases. Without a cross-tabulation of assignments and the class proportions, neither the structure of the disagreement nor its correction for chance-expected agreement can be assessed. The third team recovers a similar organization; it does not directly test the initial rule. That would require applying the rule without relearning it, then examining the assignments and their relevance. The observed risks are 12% and 11%, a difference of −1 percentage point in favor

an interval that does not decide

of guided care. From these numbers, a normal approximation for two independent proportions gives a 95% confidence interval (95% CI) of approximately −7.3 to +5.3 points. This teaching calculation assumes that each person contributes only once and that the groups are independent; it corrects no bias in the trial. The interval remains compatible with a benefit or a harm of non-negligible size. This estimate alone therefore demonstrates neither a benefit nor equivalence. The 60 additional treatments are not, in themselves, an improvement in care. Usefulness depends on the uncertainty, the benefits judged relevant and the harms. The classification may be reproducible without its use having yet shown a clinical advantage.

Synthesis

A measurement is worth only as much as the property it designates and the conditions that make it comparable. The commented texts illustrate this requirement in several forms. An imaging consensus turns a heterogeneous literature into a rule, without removing the need to examine that passage. A tracer, an uptake score or an unsupervised classification must specify their units, their reference and the meaning given to the word "validation." A genotype-driven discovery retains the trace of its recruitment; an experiment on an immune difference delimits its conditions. Validity of a category, explanation of the disease and usefulness of a decision remain three distinct achievements.

Open Questions

What independent test should be required of a diagnostic rule before it serves as the reference for other instruments? When an image remains abnormal in a person who reports feeling better, what information would make it possible to decide between residual activity and a lasting trace? At what point can a classification useful for research be tested as a guide to decision-making? Each of these questions involves both the measurement and its use.

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17

Transplantation and the Knowledge of Rejection

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After transplantation, every result is read in light of a specific fear: rejection. The word seems to designate a simple event, one that biopsy would confirm and a blood marker might announce earlier. Yet it carries a hypothesis about a pathological relationship between the organ and the organism that receives it. Knowing what licenses speaking of rejection involves several operations. A shared reference is needed, aware of its conventions, together with new instruments confronted with that reference without replacing it by declaration. Follow-up is needed that gives each value its place in a trajectory, along with decisions that protect the graft without losing sight of the person.

17.1 What Licenses Speaking of Rejection

17.1.1 A Donated Organ, a History That Has Become Shared

DONOR, RECIPIENT, GRAFT

The vocabulary of transplantation maintains two origins: donor and recipient. This distinction is indispensable for following what comes from each. It does not, however, suffice to describe what becomes of the graft. Once transplanted, the graft must be understood in its relationships with the organism that receives it and with the care that makes this relationship possible. Its origin remains; its conditions of existence change. Three questions must therefore be distinguished: where does this element come from, within what whole does it now function and what process is damaging it? An exact answer to the first does not resolve the other two. Recognizing the origin of a signal does not yet identify the cause of the damage it might indicate. It will be objected that this distinction distracts from the urgency of diagnosis. Above all, it guards against a premature conclusion: attributing every injury to the graft to its different origin. The word "rejection" carries a hypothesis about a pathological relationship; it must not become the automatic name for everything that goes wrong in the transplanted organ. The identity of an organ is not a diagnosis.

IDENTITY IS NOT A DIAGNOSIS

17.1.2 The Histological Reference: An Agreement Answerable to Tissue

CONVENTION, NOT FICTION

Defining a histological grade requires deciding which features to retain, how to combine them and where to place limits. In this sense, a classification involves a convention. But convention does not mean fiction: readers cannot freely choose the structures present on a slide or the consequences their categories will have in patient follow-up. The objection is strong: if the reference results from human agreement, how can it adjudicate between new instruments? The answer is that a reference can be justified without being infallible. Its value lies in the public nature of its rules, in the possibility of examining disagreements and in its verifiable relationships with other observations. It organizes comparison; it does not forbid its own revision. It remains essential to distinguish the tissue observed from the whole organ. The question "what is seen in this sample?" is not identical to "what is happening throughout the graft?" A disagreement between a circulating signal and a biopsy therefore requires examining several explanations. Declaring the former false from the outset because it contradicts the latter would turn the reference tool into an authority exempt from inquiry. Declaring it superior without further checks would commit the symmetrical error.

A JUSTIFIED BUT FALLIBLE REFERENCE

17.2 Monitoring, Deciding and Revising after Transplantation

17.2.1 Follow-Up Changes the Question Asked of Each Result

TIME SINCE TRANSPLANTATION

Time since transplantation is not merely an additional variable in a table. It places an observation within a history of interventions, events and earlier results. Depending on that history, a value may be interpreted as a stable level, a transition or a break. None of these interpretations should be chosen after the fact merely because it seems reassuring. In an entirely fictional example, two people have the same laboratory result. In the first, it has been decreasing steadily since an intervention; in the second, it is rising after several low results. Equal values do not make the trajectories equivalent. Nor does this equality prove that the second person has rejection: the other possible explanations for this course must be tested. It might be thought that more frequent monitoring solves everything. It also produces more opportunities for fluctuations and decisions. Surveillance must specify what constitutes a relevant change and what triggers confirmation. Without a prospective rule, multiplying observations can increase felt certainty faster than knowledge.

SAME VALUE, DIFFERENT TRAJECTORIES

17.2.2 Protecting the Graft without Losing Sight of the Person

TWO POSSIBLE ERRORS

A post-transplant decision involves several possible futures. Error does not consist only in missing a rejection; it may also consist in intervening on a misattributed alert. The consequences of these errors are neither identical nor fixed. Their weight depends on the situation, the possibilities for confirmation and the risk attached to delay. This asymmetry explains why the action threshold does not necessarily coincide with the boundary of a diagnostic category. It does not license calling "certain" what has merely been decided to treat as a precaution. Naming the degree of knowledge and the reason for acting separately makes the decision open to discussion, transmissible and revisable. The objection would be that such caution paralyzes. The answer must take the form of concrete commitments: specifying what justifies immediate action, what is still being sought and which observation would lead to changing course. Protecting the graft is not an end detached from the person's life. It is within that life that the benefits, burdens and harms of the strategies must ultimately be weighed.

ACTION THRESHOLD, DIAGNOSTIC BOUNDARY

THE LIFE OF THE TRANSPLANT RECIPIENT

17.3 Texts Put to the Test

17.3.1 Stewart and the Construction of a Histological Reference

17.3.1.1 Situating the Result

STEWART AND THE ISHLT, 2005

The International Society for Heart and Lung Transplantation (ISHLT) consensus published in 2005 revises a 1990 nomenclature. It merges several former categories of cellular rejection, retains four revised grades from 0 R to 3 R and clarifies the examination of antibody-mediated rejection. The revision results from multidisciplinary work conducted in 2004. It responds to difficulties of interpretation between readers and to accumulated knowledge. It also addresses lesions that can mimic rejection and the technical conditions of sampling. This document organizes a common language; it does not report a trial showing that use of the new nomenclature improves survival.[1]

17.3.1.2 Critical Reading: a common definition creates the possibility of comparison

17.3.1.2.1 Agreeing on a vocabulary, testing its application

To compare treatments between centers, it is necessary to know whether their results concern the same objects. Two teams may report an identical number of rejections while classifying similar biopsies differently. The difference in results may then stem from the treatment, the patients or the reading rule. Standardization is therefore a condition of inquiry, and not merely an administrative convenience. But agreeing on a vocabulary does not guarantee that it will be applied reproducibly. The consensus states what readers should recognize; a reproducibility study examines what they actually recognize. Confusing these steps would make the very question that prompted the revision disappear. The rule would need to be tested with different readers, tissues of variable quality and borderline cases, and the persistent disagreements then explained.

AGREEING, THEN TESTING

17.3.1.2.2 Merging grades, changing versions

MERGING GRADES

Merging categories can increase agreement by removing certain fragile distinctions. This gain has a possible cost: a biologically or clinically informative difference may become less visible. It must therefore be asked whether the distinctions removed changed prognosis or management enough to deserve preservation. A useful nomenclature does not necessarily maximize the number of classes; it organizes the differences that can be recognized and whose significance can be defended. Finally, the change of version raises a historical problem. Comparing older and newer series requires reconstructing their correspondence. The same grade number does not necessarily designate the same lesion. An apparent decrease in the frequency of severe forms may result partly from a change in classification. The date and version of the definition are thus among the indispensable data of a study.

17.3.1.3 Immunological Reading: the infiltrate is not its complete explanation

CALLING A LESION REJECTION

Reading the tissue raises a question of attribution: which observations justify calling the lesion rejection? The presence of inflammatory cells alone does not answer it. Their distribution, their relationship to the injured structures and the other causes compatible with what is seen must be examined. The article notably distinguishes ischemic injury, infection and the Quilty effect; it also acknowledges that cellular rejection and antibody-mediated rejection can coexist. Technique conditions this attribution. Examining more sections of the same fragment can clarify an architecture; it does not replace sampling another territory. Increasing the number of observations therefore does not always mean increasing representativeness. This material distinction becomes essential when a biopsy serves as the reference for evaluating a blood biomarker.

17.3.1.4 Cardiological Reading: a grade informs a decision that goes beyond it

ERRORS WITH REAL CONSEQUENCES

The transplant recipient presents several vulnerabilities simultaneously. An erroneous interpretation may lead to intensifying immunosuppression for a lesion that does not arise from the assumed mechanism; an insufficient in-

A GRADE IS NOT A PRESCRIPTION

terpretation may delay a useful intervention. The quality of the classification matters because its errors do not have abstract consequences. The decision nevertheless requires more than a classification: temporal context, graft function, other results and treatment risks must remain open to discussion. A common rule makes it possible to transmit the observation; it does not turn each grade into a timeless prescription. The 2005 proposals on antibody-mediated rejection, in particular, must be placed back in their history and not used as current guidelines.

17.3.1.5 Integrative Clinical Reading

This contextualized decision must then be able to circulate between the transplant center, other caregivers and the transplant recipient. An isolated grade may seem more decisive than it is. Its name and version should accompany an explanation of what was observed, what the team concludes and what remains to be monitored. Transmission thus makes the collective use of histology intelligible, without turning the classification into a self-standing instruction.

17.3.2 Agbor-Enoh and the Testing of the Reference

17.3.2.1 Situating the Result

AGBOR-ENOH, DONOR-DERIVED CELL-FREE DNA

In this prospective multicenter study, repeated plasma samples are compared with heart graft surveillance. The fraction of cell-free DNA attributed to the donor is associated with biopsy-defined rejection. After the twenty-eighth day, the reported area under the curve for acute rejection reaches 0.92; at a threshold of 0.25%, the negative predictive value approaches 99% in the observations analyzed. The authors envisage a reduction in the use of biopsies. This clinical strategy has not, however, been compared by randomization with usual surveillance. Several discrepancies in sample sizes within the article make it inadvisable to reuse its totals as if they were unambiguous.[2]

17.3.2.2 Critical Reading: the reference is not replaced by declaration

THE REVERSED REFERENCE

The most fruitful issue goes beyond the performance of a test. At the start of the reasoning, biopsy serves to define the events the marker must recognize. Later, a high fraction of the marker itself becomes the definition of graft injury, against which biopsy is compared. This reversal may open a useful hypothesis: the histological reference might miss certain phenomena. It does not constitute independent demonstration that every disagreement vindicates the blood sample. Consider two instruments exploring the same object at different scales. Their discordance may reflect the failure of the first, that of the second or a difference between the phenomena captured. Declaring one accurate by construction makes this question disappear at the very moment it becomes interesting. Evidence must therefore be sought that is not simply a reformulation of the test under study: functional trajectory, tissue course, clinical event or explicit independent adjudication. Time also participates in constructing performance. Several postoperative time points are planned; the twenty-eighth day is selected for subsequent analyses because it provides the best discrimination in these data. Planning the options does not make their

THE DAY SELECTED AFTER THE FACT

selection independent of the results. Subsequent validation should fix this rule before the observations to which it will be applied are known.

17.3.2.3 Immunological Reading: an identified origin, a cause still open

ORIGIN IS NOT MECHANISM

Attributing the DNA to the donor is strong information about its origin. It does not yet attribute its release to a particular alloimmune attack. Between origin and mechanism lie several steps: the type of cell affected, the timing of injury, the modes of release and clearance. A blood measurement condenses these steps without distinguishing them all. A signal preceding a positive biopsy makes earlier detection plausible. However, "earlier" requires a definition: earlier than sampling, than the lesion accessible to that sampling or than its consequences? These propositions are not equivalent. Likewise, an elevation in the face of a negative biopsy remains a discordance to be explained; it does not automatically become subclinical humoral rejection.

17.3.2.4 Cardiological Reading: sparing a procedure without missing a useful opportunity

A PREDICTIVE VALUE CARRIED OVER

A high negative predictive value is naturally of interest for surveillance. Its meaning nevertheless depends on the frequency of rejection, the time after transplantation, the patients included and the threshold selected. Carrying it over to another organization of care requires reexamining these conditions, not merely keeping the same number printed on the report. Calculating potentially avoidable biopsies answers an intermediate question. The clinical question is more demanding: do the procedures actually avoided offset delayed lesions, tests triggered by false alarms and the consequences of therapeutic decisions? Complete strategies would need to be compared, with fallback rules and outcomes that matter to patients. The study establishes a discriminating association within the setting studied. It makes plausible the marker's contribution to combined surveillance, without demonstrating the safety of a generalized abandonment of biopsies or identifying with certainty the mechanism of each lesion. To take this step, the rules would need to be validated in an independent population, and the strategies they guide then compared prospectively.

AVOIDABLE BIOPSIES, COMPLETE STRATEGIES

17.3.2.5 Integrative Clinical Reading

Avoiding a procedure can represent an important benefit for a person, but a simpler sample does not guarantee a simpler care pathway. The added checks, the alerts and the decisions they trigger must be counted. The value of a strategy would also depend on the possibility of reestablishing contact when a result is uncertain. The study does not evaluate all these conditions. It allows a precise integrative question to be asked: does the proposed surveillance reduce the lived burden of care while preserving its clinical objectives?

17.3.3 Khush and the Scope of Predictive Values

17.3.3.1 Situating the Result

Khush and the D-OAR registry

D-OAR follows 740 heart transplant recipients in 26 centers. Of 2,447 samples, 841 are matched to a biopsy; 35 correspond to rejection according to the definition adopted. At a threshold of 0.2% donor-derived cell-free DNA, the study reports a sensitivity of 44%, a specificity of 80%, a positive predictive value of 8.9% and a negative predictive value of 97.1%. Biopsies are interpreted locally. The registry combines surveillance and tests prompted by clinical suspicion. A separate cohort, selected for its risk of humoral rejection, complements the study.[3]

17.3.3.2 Critical Reading: the right denominator changes the conclusion

predictive value and sensitivity

The negative predictive value answers a conditional question: among negative results, what proportion does not correspond to rejection as defined by the reference? Sensitivity asks another: among rejections, what proportion does the test detect? Their coexistence requires reading both. When rejection is rare, a large number of negative results can be correctly reassuring, while a substantial share of rejections escapes the threshold. A high percentage therefore does not suffice to qualify the test as safe for every decision. Three objects must then be distinguished: the patient, the visit and the sample. Resampling a person provides a new observation, but not a new independent person. Closely spaced samples may share the same graft state, the same treatment and the same rejection episode. To assess the precision of performance estimates, it must be asked how this dependence was taken into account. Multiplying tubes does not mechanically replace the diversity of trajectories. During the study, the protocol modifies the circumstances of sampling to favor clinical suspicion. Separating surveillance analyses from suspicion analyses is therefore essential. Performance obtained in one care pathway cannot be carried over simply by changing the laboratory's address: the prior probability, the reasons for testing and the manifestations sought must also be comparable. Finally, not all observations undergo the same verification. A series describing levels in stable people is not identical to the series comparing each assay with a biopsy. Presenting their denominators separately prevents lending the whole set the diagnostic strength of the verified subset. It also leads to examining why some tests are matched and others are not.

patient, visit, sample

surveillance and suspicion

17.3.3.3 Immunological Reading: identifying an origin does not designate a cause

an origin, not a cause

The origin of the DNA fragment is remarkable information: it links the signal to donor cells. But this attribution does not, on its own, name the mechanism of their injury. Between "derived from the graft" and "produced by immune rejection," an explanatory step remains. A rise does not automatically indicate which immune pathway to inhibit. The result is, moreover, a fraction. Its interpretation must take account of both numerator and denominator: a proportion may decrease when the total quantity increases, without the donor-derived

quantity falling. The reasoning must therefore bear on the quantity actually measured and on the conditions that can shift it.

17.3.3.4 Cardiological Reading: monitoring is a strategy, not a succession of isolated results

THE COST OF AN ERROR

The cost of an error depends on the decision that follows. Forgoing a biopsy, bringing a check forward or increasing immunosuppression does not carry the same expected benefits or the same harms. A concordance study does not demonstrate that any of these courses of action improves the outcome for the graft or the person. A strategy would genuinely be put to the test by fixing the schedule, the response to a rise, the course of action in the face of a discordance and the conditions for tissue examination. Its evaluation should count delayed rejections as much as avoided procedures, together with complications related to additional interventions. Sparing an invasive sample is worthwhile only if the care pathway remains sufficiently safe.

A TESTED SURVEILLANCE STRATEGY

17.3.3.5 Integrative Clinical Reading

A NEGATIVE RESULT TRANSMITTED

The meaning of a negative result must be transmissible beyond the laboratory. Saying that rejection is less probable in a defined setting is not equivalent to declaring that every new problem is excluded. In a shared care pathway, it would be necessary to clarify what the result makes it possible to defer and which observations justify reassessment. The registry alone does not validate this organization. The issue thus concerns the continuity of reasoning between two consultations, and not merely the quality of a test at one visit.

17.3.4 Griffith and Knowledge from a First-in-Human Case

17.3.4.1 Situating the Result

GRIFFITH, A PIG HEART

The 2022 report describes the transplantation of a pig heart carrying ten genetic modifications into a man for whom no conventional option was retained. The protocol notably combines immunosuppression based on CD40 blockade with intensive monitoring. After a period of function allowing withdrawal of extracorporeal membrane oxygenation (ECMO), failure occurs on day 49, with wall thickening despite a preserved ejection fraction. Support is finally withdrawn on day 60. Attribution remains open in this report: the day 50 biopsy is not classified as rejection; the day 56 biopsy is classified as grade 1 pathological antibody-mediated rejection (pAMR1), with a caveat about the specificity of certain stains. Viral signals and antibody fluctuations also complicate the analysis. This case documents a time-limited physiological possibility, without a controlled comparison of treatments.[4]

17.3.4.2 Critical Reading

A CASE TEACHES A POSSIBILITY

A case can teach a great deal without estimating the efficacy of a strategy. Here, observing that a graft actually supports the circulation answers a question of concrete possibility. It provides neither the expected frequency of this success nor the survival gain relative to another course of action. An absolute impos-

THE RIGHT UNIT OF CONCLUSION

sibility can be challenged by a single observation; a general promise of benefit requires much more. The classic objection is that everything would be confounded: donor modifications, treatments, preservation and care. That is true for attributing the effect to a single component. It does not make the course unobservable. The right unit of conclusion must then be chosen: a set of means made a clinical sequence possible, some of whose mechanisms remain undetermined. The protocol does not constitute an experiment separating the effect of each of its elements. The richness of the chronology does not replace this separation. When a treatment changes because the patient's condition worsens, simply comparing before and after mixes the effect of the treatment with that of the situation that prompted its prescription. Detailed documentation makes it possible to reconstruct hypotheses and identify missing information; it does not spontaneously turn successive decisions into comparative trials.

17.3.4.3 Immunological Reading

A DEFINITION OUTSIDE ITS HOME GROUND

The most instructive point is the difficulty of carrying a definition of rejection over into an unusual biological situation. Not finding all the expected features is not equivalent to demonstrating that no immune mechanism contributes to the injury. Conversely, an abnormality that appears compatible with that mechanism does not suffice to attribute all the damage to it. Classification and causal inquiry meet without merging. A binary answer might be demanded to make the case intelligible: rejection or no rejection. But this simplification would erase precisely what the observation puts to the test. The pAMR1 classification reported late in the course rules out summarizing the entire evolution as "no rejection." The caveat about the specificity of the staining rules out converting this classification into a definitively established mechanism. The result, the timing of the sample and its debated interpretation must be kept together. The same requirement applies to viral signals. Detecting genetic material, locating its replication and establishing its contribution to an injury are distinct questions. Their sequence must be demonstrated. Nor does an earlier negative result certify that an agent was absent everywhere: its scope depends on the sample and the method. This discipline makes it possible to look for an infectious cause without proclaiming one on the mere presence of a signal.

NEITHER REJECTION NOR ITS ABSENCE

17.3.4.4 Cardiological Reading

WHAT IS CALLED FUNCTION

This case requires specifying what is meant by "function." The ejection fraction provides information about a proportion; it does not summarize all the conditions under which the heart sustains the circulation. The discordance with the graft failure must therefore be studied as a physiological question, and not resolved by choosing the most reassuring figure. Here, the clinical picture gives the interpretation of measurements its scope. Nor should the withdrawal of support become a mere stopwatch of graft durability. It belongs to a decision situated within the patient's entire history. Distinguishing organ function, the possibility of recovery and the choice to continue support respects both scientific precision and the person concerned. Finally, the value of this report lies in what it makes open to reexamination. A later publication on the same trajectory may enrich attribution; it does not provide a second independent patient. Reading the 2022 text means understanding one stage of this inquiry, without freezing its uncertainties or retrospectively crediting it with answers it did not contain.

A TRAJECTORY OPEN TO REEXAMINATION

17.3.4.5 Rereading the same case in light of later investigations

Mohiuddin, three possible contributions

The analysis published by Mohiuddin and colleagues in 2023 revisits the same transplantation. It examines three possible, nonexclusive contributions to the endothelial injury: antibody-mediated rejection, effects of administered immunoglobulins and reactivation of a porcine virus in the graft. Immunoglobulins bind to donor cells in the reported experiments, without complement-dependent cytotoxicity being demonstrated in this in vitro experiment. Binding therefore does not suffice to reconstruct the clinical damage. Likewise, detection of viral DNA in recipient samples does not demonstrate productive infection of the recipient's cells: viral transcription is found in the graft but is not detected in the human tissues examined.[5] This inquiry enriches the explanation without identifying a single cause. Transcriptomic profiles provide clues; thousands of cells analyzed in one person do not become thousands of patients confirming the mechanism. Revision of judgment rests here on new observations of the same trajectory. It constitutes neither an independent replication nor a comparison demonstrating the efficacy of a therapeutic course of action.

17.3.4.6 Integrative Clinical Reading

a first-in-human intervention

A first-in-human intervention involves goals that cannot be reduced to how long an organ functions. Understanding of uncertainty, the person's expectations and the possibility of revising the goals of care belong to the evaluation of this trajectory. This report does not license reconstructing his private preferences or substituting scientific enthusiasm for them. The knowledge value of the case and the singularity of the person who received the graft must be held together, without turning him into the mere vehicle of a proof of feasibility.

17.3.5 SCHEDULE and the Different Effects of a Strategy

17.3.5.1 Situating the Result

SCHEDULE, the kidney and rejections

In SCHEDULE, 115 adults who have newly received a heart transplant are randomized between a strategy involving everolimus with delayed withdrawal of cyclosporine and a strategy maintaining cyclosporine. At twelve months, measured glomerular filtration rate, the primary endpoint, averages 79.8 versus 61.5 mL/min/1.73 m^2. From weeks 7 to 11 through month twelve, biopsy-confirmed rejections are more frequent in the group with cyclosporine withdrawal than in the group maintaining it. The trial therefore compares therapeutic trade-offs whose consequences cannot be reduced to renal improvement.[6]

17.3.5.2 Critical Reading: which effect does randomization make it possible to attribute?

17.3.5.2.1 An assigned strategy with its adaptations

The randomized object is a strategy: several treatment components and their adaptations differ. Withdrawal is delayed and conditional; cyclosporine is resumed in some patients. These events are not merely imperfections surrounding an ideal intervention. They belong to the way this intervention can be applied. An analysis restricted to patients who remained off cyclosporine

would select trajectories arising after allocation and would risk losing initial comparability. To judge a care policy, its capacity to organize a fallback matters as much as its announced regimen.

17.3.5.2.2 A post-randomization variable, partial blinding

ADJUSTING FOR A POST-RANDOMIZATION VARIABLE

One analytical feature nevertheless deserves attention: adjustment of the renal endpoint uses a measurement taken seven to eleven weeks after transplantation, hence after randomization. This measurement may already be influenced by treatment. It therefore does not play the same role as a characteristic recorded before any intervention. Adjusting for a post-randomization variable can change the question being estimated and, depending on the causal relationships, introduce bias. This finding does not suffice to invalidate the result; it invites distinguishing the contrast observed at twelve months from the adjusted effect, and asking what each analysis precisely estimates. The trial is open-label, whereas central reading of intravascular imaging is performed blinded. Calling a trial "open-label" therefore does not remove the need to examine each measurement separately. It must be asked who knows the allocation, which decision that knowledge can influence and whether that decision affects the endpoint studied. Methodological protection is a property of concrete operations, not a single adjective attached to the article.

OPEN-LABEL, BUT BLINDED READING

17.3.5.3 Immunological Reading: pharmacological sparing is not tolerance

SPARING IS NOT TOLERANCE

The same intervention can shift several balances without producing a uniformly better immune state. Reducing a pharmacological exposure means neither reducing all defenses nor restoring selective recognition of the graft. Tolerance would be a much more demanding proposition: it concerns the lasting relationship between the immune system and the transplanted organ, not merely the possibility of withdrawing one component of a combined treatment. Likewise, a renal functional gain does not, on its own, demonstrate an intrinsic reparative action of the other drug. Several explanations may remain compatible with this gain: reduction of a harmful exposure, differences between combinations or adaptations made necessary by follow-up. Attributing all the benefit to a single molecular pathway would require a comparison capable of deciding between these mechanisms. Randomization here strengthens inference about the assigned strategy; it does not automatically isolate each of its mediators.

17.3.5.4 Cardiological Reading: several organs, several horizons

A PERSON, NOT A KIDNEY

The recipient of treatment is a person carrying a graft, not a kidney separated from a heart. A renal improvement and an increase in histological rejections must therefore remain visible simultaneously. Neither merging them into an overall judgment of success nor equating every histological abnormality with immediate clinical failure would do justice to the question. The severity of events, their recurrence, their treatment and their later consequences must be examined. Imaging of the coronary wall provides other information, but a morphological change does not become, through its association with prognosis alone, a demonstrated surrogate for survival. A strategy could improve this measurement while adversely affecting another risk pathway. Conversely, the

absence of a detectable clinical difference over a limited period would not prove that long-term trajectories are equivalent. SCHEDULE makes it possible to examine a treatment-sparing approach; it does not provide a universal verdict on which immunosuppressive regimen to prefer.

17.3.5.5 Integrative Clinical Reading

WHICH TRADE-OFFS FOR WHOM

Comparing renal consequences and rejections leads to a question that averages alone do not resolve: which trade-offs are acceptable for a given person? This appraisal requires knowing the severity, reversibility and impact of events, as well as the burdens of follow-up and the priorities expressed. The trial sheds light on certain components of this discussion. It does not assign each person a universal weighting of benefits and harms. Integrating organs also implies making explicit the judgment that connects their outcomes.

17.4 Worked Exercises

17.4.1 Exercise on Visits and Patients

17.4.1.1 Problem

A THOUSAND VISITS, FOUR HUNDRED PATIENTS

Entirely fictional data, with no correspondence to D-OAR. A registry includes 1,000 visits from 400 transplant recipients, with a matched assay and biopsy at each visit. Fifty visits meet the histological criterion for rejection. The test is positive for 20 of these visits and for 190 visits without rejection. It is negative at the other 790 visits. All biopsies were performed independently of the test. Individual results by patient are not provided. An investigator states: "The negative predictive value exceeds 96%, so biopsies can be dropped after a negative result; 790 patients would be spared." Check the numbers and discuss this conclusion.

17.4.1.2 Reasoned Answer

THE FOUR CELLS OF THE TABLE

The table contains 20 true positives, 30 false negatives, 190 false positives and 760 true negatives. Sensitivity is 20/50, or 40%, and specificity 760/950, or 80%. The positive predictive value is 20/210, or about 9.5%; the negative predictive value, 760/790, is about 96.2%. The stated result is therefore arithmetically correct, but its reading is incomplete. Thirty of the fifty visits with rejection have a negative test: a high negative predictive value coexists with 60% of rejections missed at this threshold. This coexistence stems notably from the low proportion of visits with rejection in the set studied. The 790 negative results correspond to visits, not to 790 people: the registry includes only 400. Without an individual breakdown, it is unknown how many patients have at least one negative result, how many account for the discordances and how many distinct episodes they represent. Nor can an uncertainty accounting for repeated measurements be correctly calculated from these totals alone. Finally, no biopsy was actually avoided. Dropping procedures after a negative test would have left thirty visits with rejection without this immediate verification. This does not make it possible to predict how many clinical complications would have

VISITS AND PEOPLE

NO BIOPSY AVOIDED

occurred, but it rules out proclaiming the strategy safe on the basis of predictive value alone. Defined care pathways and their consequences must be compared.

Synthesis

Knowing rejection requires distinguishing the origin of a signal, the whole within which the graft functions and the process that damages it. Histological nomenclature provides a common language; its reproducibility, versions and groupings remain to be tested. Donor-derived cell-free DNA identifies an origin, not a cause, and its predictive values depend on the denominator, the timing and the care pathway. The first-in-human case teaches a possibility without estimating efficacy; the randomized trial attributes an effect to a strategy, not to each of its mediators. Naming the degree of knowledge and the reason for acting separately makes the decision revisable.

Open Questions

What independent observation would make it possible to decide between a negative biopsy and an elevated blood marker? How can a surveillance strategy be evaluated that spares procedures without delaying clinically important rejections? What criteria would make it possible to recognize rejection in an unusual biological situation, without forcing the definition or abandoning it? These questions involve the organization of care as much as the biology of the graft.

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PART V

Forming and Transmitting Medical Judgment

A result never stays within the study that produced it. It becomes a number that is cited, a reason to act for a particular person, an agreement among experts, a sentence that is taught. At each passage, the question changes while the data do not: what does the result allow one to assert, for whom and under what conditions? A confidence interval does not settle the importance of a difference. A group average does not designate the persons who will avoid an event. A vote does not measure truth, and a simplification can deprive its recipient of the means to judge. These gaps condemn neither calculation, nor extrapolation, nor consensus, nor teaching. They require making the reasons for each passage visible, exposing them to the objections of others and stating what would lead to their revision. Knowing then entails a responsibility: being able to answer for what the data are made to say.

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18

What Numbers Allow One to Conclude

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A published number bears the trace of decisions made before it: what was chosen to be counted, the comparison adopted, the moment at which the analysis was fixed. It also bears the assumptions under which its uncertainty was calculated. Read without these decisions, it appears firmer than it is; read with them, it keeps its strength and finds its place. A p value, a confidence interval or a trend across doses each answers a precise question. None of them states on its own the importance of an effect for patients, or what should be decided. Judgment begins where calculation ceases to answer, without being able to do without it.

18.1 The Question Precedes the Calculation

18.1.1 The Number Depends on What Was Chosen to Be Counted

REDUCING HOSPITALIZATIONS, THREE OBJECTS

A numerical result sometimes seems more determinate than the question that produced it. Saying that a treatment reduces hospitalizations may nevertheless designate several effects: fewer people hospitalized at least once, fewer stays in total or more time spent alive out of hospital. These objects are not interchangeable. Before comparing numbers, it is necessary to determine which ones answer the stated clinical concern. In an entirely fictional example, two groups of one hundred people, all followed for the same period, each include ten hospitalized people. The first accumulates twelve stays, the second thirty. The proportion of people hospitalized is identical, at 10%; the number of stays differs. No calculation should erase this difference between the two questions. If observation times or deaths differ, further specifications become necessary.

DECOMPOSING A COMPOSITE ENDPOINT

The objection is that any definition leaves out part of lived experience. That is true; the answer is not to construct a number claiming to contain everything. A defensible principal object must be chosen, and the complementary results that make its limits visible must be retained. A composite endpoint thus deserves to be decomposed: its movement may be dominated by a frequent event whose importance does not match that of the other components.

18.1.2 Choosing before Seeing Does Not Dispense with Discovering Afterward

ANNOUNCED ANALYSIS, CHOSEN ANALYSIS

An analysis announced before access to the results exposes itself to a disappointing answer. An analysis chosen after examining them may, by contrast, retain the question to which the data respond most favorably. Even when correctly performed, the calculation does not carry the same weight in these two situations. The problem is the selection of the conclusion from among several possible conclusions. It will be objected that exploration has produced major discoveries. It remains indispensable. The requirement is not to borrow for it, retrospectively, the guarantees of a planned test. An unexpected observation may justify a new hypothesis, a transparent reanalysis or a study designed to settle it. It does not become more convincing because its unexpected character is concealed.

TREND AND DOSE-BY-DOSE COMPARISON

In a study with several doses and several endpoints, the structure of the comparisons must therefore be read. A trend test does not exactly answer the question of the effect of each dose against the comparator. Retaining only the most favorable comparison changes the question yet again. The analysis must make this path of choices visible, even when all calculations have been performed without error.

18.2 What Quantified Uncertainty Leaves to Judgment

18.2.1 The Interval Specifies One Uncertainty, Not All Uncertainties

WHAT AN INTERVAL MEASURES

A point estimate proposes a value; its interval indicates the precision allowed by the procedure used, under its assumptions. The interval is neither a closure of all biological possibilities nor an inventory of biases. A large number of measurements can narrow sampling uncertainty without correcting a systematic selection or a poor definition of the outcome. The 95% frequency describes the long-run coverage of the procedure, under the assumptions adopted; it does not directly assign a 95% probability to the statement that the parameter lies within this realized interval. This distinction is not a gratuitous refinement: it prevents a conditional guarantee of method from being converted into certainty attached to the particular result.

18.2.2 A Statistical Error Does Not Yet Have a Clinical Cost

18.2.2.1 Magnitude, precision and clinical importance

STATISTICAL THRESHOLD, CLINICAL IMPORTANCE

Crossing a statistical threshold can organize a rule for drawing conclusions. It does not determine the importance of a difference for patients. A minimal effect may be estimated precisely; a potentially important effect may remain poorly estimated. Magnitude, uncertainty and relevance must therefore remain legible together. Three entirely fictional results make this distinction visible. They concern the risk difference for the same adverse event at one year, calculated as studied strategy minus comparator. For this exercise only, a decrease of at least two percentage points is set, before the results are read, as the threshold for a benefit judged important. This threshold is a teaching convention, not a clinical recommendation.

THREE FICTIONAL RESULTS, A FIXED THRESHOLD

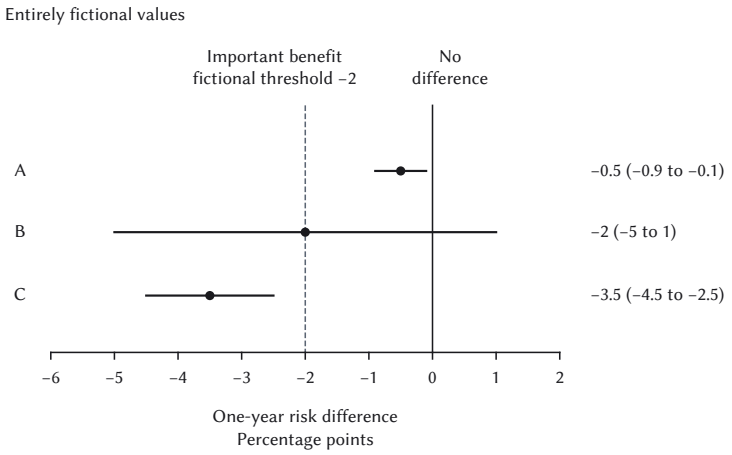


Figure 18.1. Estimate and clinical importance Points represent the estimates and segments their 95% confidence intervals. The solid line marks no difference; the dashed line marks the fictional threshold for important benefit. Original construction from invented values, with no patient data.

In A, the interval excludes zero but remains far from the benefit judged important. In B, it includes an important benefit, zero and a harm: the result remains imprecise. In C, the entire interval lies on the side of important benefit according to the chosen threshold. This reading of magnitude and precision judges neither biases nor the other consequences of the intervention. It sheds light on one endpoint; it does not suffice for a decision.[1]

18.2.2.2 Deciding before a precise estimate

ACTING BEFORE PRECISION

A practical objection remains: it is sometimes necessary to act before a precise estimate is available. Urgency does not abolish uncertainty; it changes the cost of delay. To justify a decision, the consequences of intervening wrongly and of failing to intervene in time must be considered, along with the other available options. These consequences are not contained in a p value. The scientific conclusion and the decision may then differ without contradicting each other. A team may acknowledge that a benefit remains uncertain while judging an action proportionate to a given situation. It must explain this step and plan for its revision. Numbers discipline judgment when they make its reasons more precise; they do not replace it by assigning weight, on their own, to what matters.

18.3 Texts Put to the Test

18.3.1 Everett and the Scope of a Dose Trend

18.3.1.1 Situating the Result

EVERETT, AN EXPLORATORY ANALYSIS OF CANTOS

This prespecified exploratory analysis of CANTOS examines hospitalization for heart failure among the 10,061 participants already randomized. After central adjudication, 592 events are retained in 385 people; the main analysis concerns time to the first event. For 300 mg of canakinumab versus placebo, the hazard ratio (HR) is 0.76, with a 95% confidence interval (95% CI) of 0.57 to 1.01, while the dose trend test gives p = 0.025. No dose, considered separately, significantly reduces this endpoint. The authors report a violation of the proportional hazards assumption and describe their conclusions as hypothesis-generating. Analyses according to on-treatment C-reactive protein (CRP) do not preserve the purely randomized comparison. This is a new perspective on the same trial, not an independent replication of CANTOS.[2]

18.3.1.2 Critical Reading

18.3.1.2.1 A prespecified trend is not a confirmation

TWO STATISTICAL QUESTIONS

The result brings two statistical questions into play. Comparing one dose with placebo evaluates a particular contrast. The trend test examines a pattern of results across treatment levels. Their conclusions can differ without arithmetic inconsistency. But the second cannot be reformulated as though the first had established the efficacy of each dose. A trend justifies an inquiry; it does not automatically select the regimen to prescribe. The prespecified status must also be preserved in both directions. It would be unfair to present this analysis as an inquiry improvised entirely after the results. It would be equally excessive to convert its anticipation into confirmatory demonstration. The intention to ask a question guarantees neither sufficient precision nor a control of multiple comparisons equivalent to that applied to the trial's primary endpoint. What had been planned must be read first, and then what the observed events actually make it possible to estimate.

18.3.1.2.2 What time to first event summarizes

A HAZARD RATIO CALLED INTO QUESTION

The HR requires additional attention here: it summarizes a comparison whose constancy over time is called into question. Looking for a difference between phases can be informative, but choosing where to separate them on the basis of the observed events is not equivalent to testing an independently announced window of efficacy. A temporal explanation becomes convincing when it can meet new data, not merely when it elegantly divides the old. Finally, 592 events do not represent 592 hospitalized people. Analyzing the first event answers an identifiable question; it does not measure the full burden of recurrences. The choice is not an error. The error would be to use its conclusion as though every subsequent hospitalization had contributed equally to the main result.

18.3.1.3 Immunological Reading

RESPONDERS NOT RANDOMIZED

A therapeutic comparison and a comparison of biological responses do not enjoy the same guarantees. After treatment, people whose CRP has fallen substantially may differ from others for several reasons. Their membership in the responder group was not randomized. Even when the analysis adjusts for known characteristics, it does not demonstrate that bringing patients across the biological threshold would suffice to produce the expected clinical improvement. The objection is that rejecting this interpretation would lose valuable mechanistic information. The information must be kept while changing the strength of the statement. The biological response can help build a hypothesis about variability in benefit; it does not yet validate a rule of continuing or stopping the drug according to that result alone. Such a rule would constitute a new strategy to be tested. The distinction also matters for the target. A signal compatible with a contribution of interleukin-1β (IL-1β) does not, on its own, specify what share of the course passes through the myocardium, ischemic events or other processes. The clinical result retains its value without becoming complete demonstration of an intracardiac mechanism.

IL-1β AND THE ROLE OF THE MYOCARDIUM

18.3.1.4 Cardiological Reading

HOSPITALIZATION, A CARE DECISION

Hospitalization combines a biological evolution and a care decision. Its definition and adjudication make it possible to compare sufficiently similar events; they do not describe every form of heart failure. The lack of data distinguishing ejection fractions here limits the precision of the population to which the signal can be attached. To make progress, a study would be needed whose population, endpoint and strategy are built around this cardiac question. The large size of the original trial does not abolish this requirement. It offers fertile ground for discovering a lead, without turning every lead into an indication. This text thus teaches how to keep a promising result without hardening it. The interval, the trend and the biological analyses are not three independent votes in favor of the treatment: they are three partially linked readings of the same observations.

18.3.1.5 Integrative Clinical Reading

LIFE BETWEEN EPISODES

Finally, choosing the first stay as the endpoint leaves open a question for the person: what becomes of their life between episodes? Recurrences, recovery and the burden on relatives could change the importance of the expected ben-

efit. These dimensions would need to be documented, without assuming that canakinumab improves them. They specify what a future comparison devoted to heart failure should learn, beyond a statistical trend across doses.

18.3.2 Rothman and the Constraints of a Negative Result

18.3.2.1 Situating the Result

ROTHMAN, BLOOD PRESSURE ON CANAKINUMAB

This post hoc secondary analysis of CANTOS examines blood pressure and incident hypertension. Of the 10,061 randomized participants, 9,549 have baseline and three-month measurements. No significant reduction in blood pressure is observed with canakinumab at the visits studied. The HR for incident hypertension is 0.96 (95% CI: 0.85 to 1.08). These data concern a population with prior myocardial infarction and residual inflammation; most had known hypertension. They come from the same trial as Everett's analysis: they address a different question, without constituting independent confirmation.[3]

18.3.2.2 Critical Reading

18.3.2.2.1 What a negative result allows, object by object

WHAT A NEGATIVE RESULT CONSTRAINS

A negative result can constrain an explanation without abolishing every possibility. The interval for incident hypertension remains compatible with anything from a 15% decrease to an 8% increase in the hazard rate. The clinical importance of these magnitudes must be discussed; it is not defined merely by whether the statistical threshold is crossed. Claiming equivalence would require a predefined relevant margin and an appropriate design. Failing to cross a threshold does not provide that demonstration. Several objects must also be distinguished: the measured blood pressure level, its change and the occurrence of a diagnosis in people who did not initially have it. Their denominators and uncertainties differ. A large cohort does not guarantee high precision for every subgroup, event or contrast. Measurements repeated at visits offer standardized observation; they do not constitute a continuous record of all blood pressure exposure.

18.3.2.2.2 Subgroups and their limited guarantees

AN INTERACTION NOT ESTABLISHED

Post hoc status does not invalidate the observations. It limits the confirmatory strength attached to questions selected after the trial. Likewise, the absence of a statistically significant interaction between baseline blood pressure categories and treatment effect does not demonstrate that the effects are exactly identical. It indicates that this analysis did not establish a difference between them. Directly comparing the effects matters more than the alternation of significant and nonsignificant results across subgroups. Finally, allocation of the drug does not randomize all the categories used afterward. A high baseline blood pressure remains an observed characteristic; a low on-treatment CRP remains a state subsequent to allocation. Comparing these groups does not automatically transfer to them the guarantees of the randomized comparison. How the groups were formed must be traced for each new question.

NONRANDOMIZED CATEGORIES

18.3.2.3 Immunological Reading

A CAUSAL CHAIN PUT TO THE TEST

The study tests one possible explanation of the cardiovascular benefit: that it would pass through a reduction in blood pressure. If the treatment modifies inflammatory markers and certain events without a detectable fall in blood pressure, this simple chain becomes less convincing. This result is useful precisely because it requires specifying what the proposed mechanism should have made observable. The objection is that an imperfect measurement may miss a real change. It must be taken seriously without making the explanation irrefutable. It would be necessary to state which unobserved change would remain plausible, at what time, with what magnitude and how it could be detected. A mediation analysis would additionally require specific assumptions and data. The dissociation observed does not prove the absence of any causal pathway involving blood pressure. Another generalization would be excessive: concluding that immunity does not contribute to hypertension. Selective blockade of IL-1β is not an intervention on the entire immune system. A post-infarction population, treated and selected according to its CRP, does not represent every phase of the disease or all its forms. A mechanistic hypothesis is specified by the intervention that tests it; it is not tested in all its versions at once.

IL-1β IS NOT THE WHOLE OF IMMUNITY

18.3.2.4 Cardiological Reading

SEPARATING OBJECTIVES

The most instructive signal is the separation of objectives. An intervention may be of value for certain events without becoming a treatment for blood pressure. Conversely, the absence of an effect on blood pressure does not suffice to judge its overall clinical balance. Each conclusion must retain its endpoint, its population and its time horizon. A higher baseline risk may also produce a greater absolute benefit without any demonstrated change in the relative effect. This difference does not, on its own, designate a blood pressure threshold for selecting treatment candidates. A selection rule would need its own evaluation, incorporating the precision of the benefit and the adverse effects. Rothman's analysis complements Everett's: the same dataset can feed several relevant inquiries. Their biological proximity turns them neither into independent repetitions nor into arguments that can be added up without caution. Knowledge advances here by removing apparent precision from one explanation, in order to give greater precision to the next question.

TWO ANALYSES, ONE TRIAL

18.3.2.5 Integrative Clinical Reading

WHAT EACH INTERVENTION HAS SHOWN

This separation of objectives takes on concrete meaning when a person expects a treatment to improve several associated problems. A cardiovascular result does not suffice to promise that person lower blood pressure. Explaining what each intervention aims at and what it has demonstrated helps maintain coherent care: success in one domain must mask neither the other needs nor the arguments that are still fragile.

18.3.3 Yoshikawa and the Comparisons behind Randomization

18.3.3.1 Situating the Result

Yoshikawa, immunoadsorption on trial

This Japanese trial randomizes 44 people with severe dilated cardiomyopathy to immediate immunoadsorption or to an intervention delayed by three months. The primary endpoint pools the before-and-after treatment changes from both sequences: ejection fraction measured by radionuclide ventriculography does not improve significantly ($p = 0.0605$). Several secondary measures change favorably, notably echocardiographic ejection fraction. An autoantibody score also distinguishes subgroups with apparently different responses. These observations justify further inquiry into the intervention; they establish neither a reduction in serious events nor a validated individual predictive rule.[4]

18.3.3.2 Critical Reading: recovering the contrast behind the number

18.3.3.2.1 Two architectures in one trial

initial allocation, later comparisons

Two architectures coexist. At the start, one group receives the intervention while the other waits: their comparison during this period benefits from randomized allocation. Then the delayed group in turn receives treatment. Comparing its state before and after this step answers a different question, exposed to spontaneous evolution, changes in management and period effects. The word "randomized" describes the initial allocation; it does not make all the statistical operations performed afterward equivalent. The primary endpoint combines changes after treatment, whether it began immediately or later. Pooling these observations increases the amount of data on a treated trajectory; it does not create a concurrent control group for each of them. The paired analysis therefore has its own scope, narrower than that of a causal contrast preserving the initial allocation. What is being compared must be stated precisely before the test result is interpreted.

18.3.3.2.2 Discordant measurements and multiple endpoints

switching measures after the fact

The discordance between two measurement methods is a second lesson. A favorable echocardiographic result does not retrospectively replace the primary radionuclide result. This does not make echocardiography uninteresting: it may generate an important hypothesis about the sensitivity of the measurement or about functional trajectory. But switching to a measure because it seems to respond requires independent confirmation. Otherwise, the verification tool gradually becomes the one that grants the desired conclusion. The same restraint applies to the multiplicity of secondary results. Their possible convergence is informative; their number also increases the opportunities to observe a difference. To distinguish these two possibilities, the planned analyses, their dependencies and the confirmation rules would need to be made explicit. An isolated threshold does not do this work. Finally, a nonsignificant primary result does not establish the absence of any effect; above all, it rules out announcing the demonstration that this endpoint was meant to provide.

multiple secondary endpoints

18.3.3.3 Immunological Reading: removing antibodies does not suffice to identify those responsible

A COLUMN DOES NOT ISOLATE AN ANTIBODY

The intervention makes it possible to perturb the humoral compartment. This property brings the study closer to a mechanistic experiment, but without isolating a single antibody. An adsorption column modifies a set of circulating substances; observing their reduction does not demonstrate that each was pathogenic, or that any clinical change stems exclusively from that reduction. It is also important to distinguish recognition from activity. An assay directed against an antigen reports binding under the conditions of the analysis. It does not automatically measure the capacity of serum to modify a cellular function in the heart. Between antibody titer and mechanism of injury lie questions of accessibility, biological effect and tissue context. The composite score may help formulate a selection hypothesis. However, observing a significant improvement in one subgroup and not in the other does not prove that their responses differ significantly. To support an ability to predict a difference in treatment effect, the interaction between treatment and score would need to be examined on a specified scale, within a comparison preserving the randomized contrast. It would then need confirmation with a rule fixed in advance. A median split in a small sample does not, on its own, provide a transportable clinical threshold.

A SCORE SPLIT AT THE MEDIAN

18.3.3.4 Cardiological Reading: a therapeutic possibility remains a question

EVALUATING THE ENTIRE CARE PATHWAY

The management studied involves an extracorporeal procedure, a hospital schedule and medication adjustments. Its evaluation must address this concrete whole. A change in a functional index summarizes neither the tolerability of the pathway nor its effect on the patient's life. Symptoms matter, but their interpretation also requires considering the conditions of observation and the absence of a sham procedure. The small number of participants limits the examination of rare events and differences in prognosis. A larger study would provide more information; it would not guarantee a favorable result. This distinction prevents insufficient precision from being turned into promised benefit. Feasibility and secondary signals therefore justify a question about the intervention, without establishing its prognostic benefit. The hypothesis of an effect differing according to humoral characteristics still requires a selection rule that is defined and independently verified. A trial preserving an explicit comparative contrast and hierarchical endpoints could distinguish this hypothesis from favorable selection in a small sample.

A SIGNAL STILL TO BE RESOLVED

18.3.3.5 Integrative Clinical Reading

THE PRACTICAL BURDEN OF THE PROCEDURE

To this uncertainty about benefit is added the practical burden of the procedure: travel, scheduling and adjustments of care. Their individual weight is not quantified here, and their acceptability cannot be decided for each person from the trial averages. An improvement in an index, especially a secondary one, therefore does not suffice to establish that the pathway as a whole brings the person a net benefit.

18.4 Worked Exercises

18.4.1 Exercise on the Hierarchy of Endpoints

18.4.1.1 Problem

A FICTIONAL TRIAL, A PLANNED HIERARCHY

All the following data are fictional; they reproduce none of the trials discussed. A trial randomizes 120 people after myocardial infarction to treatment X or placebo, 60 per group. The protocol specifies a primary endpoint: the change in indexed left ventricular end-systolic volume at three months. A negative value of the X minus placebo difference is favorable. A hierarchical procedure then specifies testing hospitalizations for heart failure, at the 5% level, only if the primary test is significant at that level. Results: mean difference on the primary endpoint of −1.0 mL/m² (95% CI: −4.0 to +2.0); p = 0.51. Hospitalization is observed in 2 people on X and 9 on placebo. For this secondary endpoint, the software gives a nominal p of 0.03, without taking the hierarchical rule into account. The authors propose: "Despite the absence of an effect on remodeling, X prevents heart failure." Questions. Calculate the absolute difference in the proportions hospitalized. Does the conclusion comply with the protocol? What wording would preserve the information without increasing its scope?

18.4.1.2 Reasoned Answer

2/60 VERSUS 9/60

The proportions are 2/60 = 3.3% and 9/60 = 15.0%. Their difference is −11.7 percentage points, with the rounding shown. This estimate deserves examination, but its point magnitude settles neither its precision nor its status in the analysis plan. Since the first step did not cross the planned threshold, the procedure does not permit a confirmatory conclusion on the secondary endpoint. The nominal p of 0.03 does not reactivate an interrupted step. This result may feed a hypothesis; it must not be presented as though protection against multiplicity had been preserved. The first half of the sentence is also excessive: the interval for the primary endpoint remains compatible with differences in both directions. The trial has not demonstrated an absence of effect on remodeling.

AN INTERRUPTED STEP IS NOT REACTIVATED

A DEFENSIBLE WORDING

A defensible wording would be: "The primary endpoint did not establish the effect sought. Fewer participants were hospitalized on X in this sample; this secondary observation remains exploratory according to the planned hierarchy." Confirming it requires a study designed around this question, with an estimate of its precision and an assessment of tolerability.

Synthesis

A numerical result answers the question it was constructed to ask: an object counted, a comparison chosen, an analysis planned or discovered after the fact. Its interval specifies sampling uncertainty under assumptions; it measures neither biases nor clinical importance, which requires a justified threshold. The three texts studied teach this together. A dose trend, a post hoc analysis and a nonsignificant primary endpoint retain real information, provided their status is kept. Two analyses of the same trial are not two independent confirmations; a comparison based on a state subsequent to allocation does not receive the

guarantees of randomization. The decision may then differ from the scientific conclusion, provided that it sets out its reasons and plans for its revision.

Open Questions

What threshold of clinical importance can be fixed before reading without becoming an arbitrary convention? How can an unexpected discovery be distinguished from a conclusion chosen among several possible ones, when the analysis had not been announced? At what point does a temporal explanation, or a biological response observed on treatment, deserve a study designed to test it? These questions take on a different significance when the result must reach a person.

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19

From the Study Population to the Person Receiving Care

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A trial or a cohort describes groups: means, proportions, risks at a given horizon. The person seeking care is not a mean, and yet that person cannot be treated without this collective knowledge. A shared diagnosis, a biological category or a score brings situations together; it does not yet say which of them resemble one another for the decision at hand. The passage requires explicit reasoning. It must state which differences could change the benefit, the harms or the possibility of following a strategy, and on which scale an effect can be extrapolated. It must also recognize that the person's preferences modify the choice without modifying the estimated efficacy.

19.1 From the Observed Group to the Person Present

19.1.1 Relevant Resemblance Remains to Be Justified

GOING BEYOND THE OBSERVED GROUP

A study never meets all the people to whom its result might be useful. Providing care therefore requires going beyond the boundaries of the observed group. This passage consists neither in finding an exact double of the average participant nor in declaring each person too singular to benefit from collective knowledge. It consists in examining which differences could change the answer to the question asked. Consider a fictional trial that evaluated a strategy in people living independently and followed up every week. A person with the same disease lives far from the center and depends on someone else for transport. The shared diagnosis supports an initial analogy. It is not enough to make the two situations interchangeable: if weekly monitoring governs treatment adjustment, its absence modifies the strategy actually available. The problem is not that this person is a lesser candidate for care; it is a difference in the

RANKING THE DIFFERENCES

conditions under which care can be delivered. It will be objected that listing every difference makes any extrapolation impossible. The differences must indeed be ranked. A difference does not become relevant merely because it exists: it must be shown how it could modify the benefit, the harms or the possibility of following the strategy. This justification may remain uncertain. It is nonetheless more informative than a list of resemblances unrelated to the decision. The validity of the passage thus depends on explicit reasoning about the differences that matter. Descriptive proximity is not yet clinical proximity; a visible difference is not automatically an obstacle.

19.1.2 Belonging to a Group Does Not Dictate a Trajectory

SEVERAL GROUPS AT ONCE

A person belongs simultaneously to several groups: age band, diagnostic category, treatment history, functional situation. Each organizes a possible comparison. None alone delivers that person's future. Choosing a reference population means favoring certain information for a specific question, without claiming that the rest has disappeared. In a second fictional example, two people carry the same diagnosis and share a biological characteristic retained in a study. One can continue regular monitoring; the other has already stopped several treatments because of poorly tolerated effects. Describing them by the same category may be reasonable for certain biological expectations and insufficient for organizing their care. The category is not false: its function is

SINGULARITY AS PROTECTION

limited. The opposite objection is serious. By multiplying particularities, it is possible to construct after the fact a reason to exempt any person from an inconvenient result. Singularity then becomes a protection for the clinician's preference. To avoid this drift, it must be specified before the decision which particularity leads to adjusting the expectation and what would lead to aban-

A MEAN IS NOT A PROPERTY

doning that adjustment. A mean difference between groups must therefore not become a necessary property of each member. But refusing this automatism does not justify erasing documented collective differences. The task is to preserve their information, then to examine its relevance for the person present. Individualizing does not mean replacing knowledge with intuition; it means making the reasons for its application open to discussion.

19.2 What the Person Brings to the Decision

19.2.1 A Preference Changes the Choice, Not the Result Retroactively

LIVED CONSEQUENCES

The decision concerns lived consequences. The same risk may be assessed differently depending on what a person wishes to preserve, what that person fears and the burdens that person can accept. This difference is not a failure of understanding to be corrected until the expected agreement is obtained. Consider two fictional strategies whose expected benefits are close, but one of which requires frequent travel and the other a daily constraint at home. One person may favor the first to keep regular contact with the team; another, the second to continue working. If the information has been understood, these choices may be equally reasonable. The preferences have not changed the estimated efficacy: they have changed the value attributed to the consequences, and therefore the choice between the options. It will be objected that some preferences rest on inaccurate information. Dialogue should then allow them to be corrected, without equating every persistent disagreement with an error. A factual belief open to discussion (for example, what the treatment allows a person to hope for) must be distinguished from a personal priority that cannot

A FALSE PERSONALIZATION

be deduced from a table of results. This distinction also protects the person against a false personalization. An option unavailable for lack of organization does not become the person's choice simply because the person ends up accepting it. Knowing preferences requires making visible the real possibilities, their constraints and the margins for change.

19.2.2 Following a Person Does Not Turn Each Change into Evidence

A TRAJECTORY WITHOUT COMPARISON

After the decision, the person brings a new history. An improvement, a difficulty or a changed priority may justify reexamining the course of care. But observing this trajectory does not automatically provide the comparison that was missing before treatment. A favorable course informs about what happened; it does not by itself indicate what would have happened otherwise. Individual follow-up nonetheless retains a value of its own. It can reveal that a strategy has become impracticable, that a goal no longer fits the person's life or that an assumption about application was ill founded. These findings are sometimes enough to modify a decision, without resolving the whole general

THE MEMORY OF REASONS

question of efficacy. The objection would be to see in this a permanent instability. Revising, however, is not starting from zero at every event. The initial question must be preserved, what has changed must be recorded and why this new element weighs on the choice must be explained. Continuity of care rests on this memory of reasons. An individualized decision thus remains transmissible to another clinician and understandable to the person concerned: it has a history that can be examined, rather than an authority that would have to be believed.

19.3 Texts Put to the Test

19.3.1 Pujades-Rodriguez and the History of Exposures

19.3.1.1 Situating the Result

Pujades-Rodriguez, oral glucocorticoids

This English cohort includes 87,794 adults with immune-mediated inflammatory diseases and no identified prior cardiovascular disease. Oral glucocorticoid prescriptions are reconstructed over the course of follow-up and set against the first occurrence of cardiovascular events. Daily exposure and cumulative exposure are studied separately. The analyses report an association with risk, including in the low-dose category. The authors acknowledge the possibility of confounding by indication. The supplementary methods specify how they reconstruct flares and replace missing data. Discrepancies between the abstract, the text and the tables argue here against retaining a single effect figure as if all presentations agreed.[1]

19.3.1.2 Critical Reading: the reason for treating accompanies the treatment

19.3.1.2.1 Dating exposure does not explain the dose

dose and disease state confounded

A dose rarely increases without reason. It may respond to a worsening that itself modifies cardiovascular risk. Comparing periods of high exposure with periods of low exposure then amounts to comparing treatments and disease states simultaneously. Detailed follow-up improves the description of this history; it does not automatically erase the clinical reason that led to each prescription. The choice of a time-varying exposure is therefore valuable. A person does not remain artificially in the initial category when the prescription changes. But following the change correctly is not yet identifying what produced it. A variable can be well dated and remain insufficiently explained. Chronological quality and causal identification are two complementary requirements.

19.3.1.2.2 Variables reconstructed under assumptions

flares reconstructed from the dose

A particular difficulty arises when a flare is reconstructed, in part, from the dose increase itself. The adjustment then seeks to represent disease activity through a therapeutic decision already linked to the exposure studied. This pragmatic solution is not equivalent to an independent measurement of inflammation. It may reduce certain imbalances without guaranteeing their disappearance. It would be necessary to examine how the conclusions change with activity measured otherwise, and with a better-documented clinical history. Missing data raise another question: what world is being reconstructed when the cells are filled in? Imputation does not manufacture additional observations. It constructs several plausible versions under specified assumptions. Its value depends on the quality of these assumptions, especially when the absence of measurement depends on the care received or on severity. Likewise, coding absent information as a negative response requires a justification different from that of a statistical imputation. A large sample size does not make these choices secondary.

imputation and coded absence

19.3.1.3 Immunological Reading: the intervention and the disease can act together

EFFECTS IN OPPOSITE DIRECTIONS

The biological argument cannot be reduced to "less inflammation means less risk." A treatment may favorably modify one process while acting unfavorably on other cardiovascular determinants. Its overall effect depends on what occurs simultaneously, in which organism and for how long. This coexistence explains why an anti-inflammatory hypothesis is not enough to predict the net balance of a drug exposure. Conversely, an unfavorable association does not show that controlling the disease would be useless. The course without treatment is not directly observed in the same person at the same moment. The consequences of inflammatory activity, those of the exposure and those of the strategy linking them would have to be distinguished. A cohort contributes to this inquiry; it does not automatically measure all its components. Dose equivalence facilitates comparisons between prescriptions. It does not mean that all molecules and all exposure histories are biologically identical. Two trajectories may reach the same cumulative dose while differing in their peaks, their duration and their relation to flares. The cumulative dose is a useful representation of the history; it does not contain that whole history.

THE CUMULATIVE DOSE IS NOT THE HISTORY

19.3.1.4 Cardiological Reading: from the collective signal to a situated decision

A composite cardiovascular event brings together situations whose mechanisms, severity and consequences differ. An overall estimate can signal a vulnerability without specifying, for a given person, which event would occur or which intervention would prevent it. Nor does the first occurrence studied describe the whole burden of subsequent recurrences. The clinician therefore cannot mechanically subtract the risk associated with a dose to calculate the benefit of withdrawing it. Changing the dose can change disease control, other treatments and the circumstances of follow-up. The result useful for the decision would be that of realistic strategies, compared in sufficiently similar people and with their benefits as well as their harms. The population-level association thus justifies attention to risk during the different periods of exposure. It remains compatible with a contribution of both the treatment and the disease, without estimating their causal shares separately. To inform an individual decision, comparable strategies, an independent measure of activity and sensitivity analyses for missing information would be needed.

NO MECHANICAL SUBTRACTION

19.3.1.5 Integrative Clinical Reading

LINKING EPISODES AND INTERLOCUTORS

Reconstructing this history finally requires linking the episodes and decisions of several interlocutors. The day's dose says neither why it changed nor what other measures accompanied it. This continuity of information does not turn the observed association into proof that a reduction would benefit everyone. It makes it possible to formulate the twofold question that the statistical coefficient does not settle: preserving disease control and limiting the harms of the care pathway.

19.3.2 Loiseau and the Reference Population

19.3.2.1 Situating the Result

Loiseau, a Danish Sjögren cohort

This Danish cohort compares 4,697 people with primary Sjögren's syndrome with 46,970 comparison entries matched on age, sex and year. Recorded cardiovascular history is excluded. Several arterial and venous events are more frequent after adjustment; the estimates for heart failure and cardiovascular mortality do not establish an increase in these two endpoints. Median follow-up is 7.6 years, with a maximum of 23 years. This is an association observed in registries, without randomized intervention or demonstration of a preventive strategy.[2]

19.3.2.2 Critical Reading: who begins to be observed, and when?

19.3.2.2.1 The starting point and the population retained

the first hospital diagnosis

The starting point is the first recorded hospital diagnosis. It is not necessarily the onset of the disease. A diagnosis may follow several years of symptoms, or be brought forward by a care episode that multiplies investigations. The administrative chronology provides a reproducible reference point; it does not by itself fix the biological age of the disease. An event occurring shortly after this reference point therefore cannot be automatically attributed to an initial phase of the immune process. The exclusion of cardiovascular history also sharpens the question. What is studied here is the onset of events in people without recorded history, not the whole cardiovascular trajectory of the disease. This restriction facilitates certain comparisons; it also removes the people whose complications preceded the diagnosis. Extrapolating the result to a patient already affected would require another justification.

19.3.2.2.2 Comparators, codes and definitions

matched is not identical

The comparison does not set people made identical against each other. Matching and adjustment address specified characteristics. They guarantee neither the same exposure to all cardiovascular factors nor the same intensity of surveillance; nor do they correct an imperfect measurement of lifestyle. Diagnoses and prescriptions are valuable information, but an absent code does not always mean an absent exposure. The number of records does not replace the precision of their content. Finally, two methodological details prevent an overly hasty reading. Comparators are selected with replacement: the entries should not be spontaneously equated with as many unique people. "Cardiovascular death" is defined as death within 30 days of a studied event; this temporal definition is not an adjudication of the cause of each death. The definition must travel with the number.

comparators sampled with replacement

19.3.2.3 Immunological Reading: a named disease is not a measured mechanism

one label, several diseases

Sjögren's syndrome makes an immune contribution to vascular risk plausible. However, the overall diagnosis does not quantify the biological pathway responsible for each event. The same label may cover different durations, systemic manifestations and treatments. An association persisting after ad-

justment does not allow their contributions to be apportioned exactly. The distinction between arterial and venous manifestations is useful: grouping them under the term "cardiovascular" does not turn them into identical lesions. A mechanistic interpretation must state which phenomenon it explains and which observation could contradict it. Otherwise, every increase in risk ends up confirming the same general idea of inflammation, without really putting it to the test. Likewise, a difference between treated and untreated people would not suffice to designate the treatment as responsible: the severity that leads to its prescription must remain in the reasoning. The registry can guide the inquiry; a molecular attribution requires other observations and appropriate comparisons.

19.3.2.4 Cardiological Reading: keeping the scale and horizon of risk

THE HORIZON OF CUMULATIVE RISK

A risk accumulated over a long period is read neither as an annual risk nor as the risk of each participant over the median duration of follow-up. Not all people are observed for the same length of time. Competing deaths and changes in care also modify the opportunities to observe an event. These dimensions become decisive when moving from a population curve to an individual conversation. The study invites examination of risk in a systemic disease; it does not show that a particular screening or a given intervention improves outcomes. That question would require comparing strategies and their consequences. Nor does it allow the same excess risk to be attributed directly to all inflammatory diseases: moving from Sjögren's syndrome to rheumatoid arthritis would be a new extrapolation, not a reading of the result. The association therefore remains attached to this population and its data collection system. Several immune, therapeutic and contextual contributions remain possible; no single causal fraction and no individual preventive benefit are deduced from it. Finer clinical measurements would clarify the explanation, whereas preventive usefulness would require its own comparison of strategies.

EXCESS RISK WITHOUT A PROVEN STRATEGY

19.3.2.5 Integrative Clinical Reading

AN EXCESS RISK PRESENTED WITH RESTRAINT

Information about an excess risk can usefully extend attention beyond the manifestations that led to the diagnosis of Sjögren's syndrome. It can also become a worrying label if the horizon and the events concerned remain undefined. Loiseau invites a proportionate presentation: the cohort describes associations, not the personal preventive program that would follow from them. An integrative reading keeps the person's other problems and priorities in view, while asking which additional decisions would be truly useful rather than automatically adding a succession of tests.

19.3.3 Khera and the Scope of Genetic Ranking

19.3.3.1 Situating the Result

KHERA, POLYGENIC SCORES

Khera and colleagues construct polygenic scores for five diseases, including coronary artery disease and inflammatory bowel disease. They select the scores in a set of 120,280 UK Biobank participants, then evaluate them in 288,978 other participants. For coronary artery disease, 8% of this set belong

to a category associated with an odds ratio (OR) of at least three, compared with the rest of the population. The analysis concerns prevalent diseases and a population of white British ancestry. It shows a ranking capacity; it does not test a prevention strategy assigned according to the score.[3]

19.3.3.2 Critical Reading

19.3.3.2.1 Selecting, evaluating and attributing performance

SELECT, THEN EVALUATE ELSEWHERE

The separation between model selection and evaluation is an essential strength. If the best-performing score is chosen among several and its performance is then measured in the same people, particularities of that sample risk being rewarded. Reserving other participants for evaluation limits this problem. It does not yet constitute validation in a different health care system, however: the two sets belong to the same biobank and share recruitment conditions.

A MODEL'S AREA UNDER THE CURVE

The instrument to which the announced performance belongs must also be examined. The area under the curve presented in the main table results from a model including the score, age, sex and other covariates. Attributing it to DNA alone would overstate what this measure makes it possible to ascribe to it. To appreciate the additional contribution of the score, it must be compared with an explicit reference model, and it must be examined whether this gain changes useful decisions.

19.3.3.2.2 The language of risk and its transportability

The vocabulary of risk calls for another precaution. An OR relating to a prevalent disease does not directly provide the probability of developing that disease over the next ten years. The percentile ranks a person among the reference participants; it is not a percentage probability of disease. These two confusions would produce an impression of personal precision that the analysis has not constructed. Finally, recruitment selects the people from whom the estimates come. The representation of certain ancestries, age and the characteristics of volunteers matter for application to other populations. The objection is that a biological mechanism may be shared despite these differences. That is possible, but the transportability of a score also depends on variant frequencies, their correlations and the environment in which the disease is recognized. A shared biology does not license presuming identical performance.

RECRUITMENT AND TRANSPORTABILITY

19.3.3.3 Immunological Reading

The score aggregates associations. A high result does not necessarily localize a dominant inflammatory pathway in the person concerned. Several sets of variants can lead to a similar ranking while involving different processes. The numerical precision of a total does not automatically become mechanistic precision. This distinction protects cardioimmunology from a shortcut: interpreting any genetic coronary susceptibility as a measure of present immune activity. An inherited predisposition and an inflammatory state measured at a given moment have neither the same temporality nor the same meaning.

A SHARED MECHANISM YET TO BE SHOWN

Confronting them can be instructive, but how they relate must be studied. Nor does the study of several diseases establish that their scores describe a shared mechanism. To support such a proposition, the shared contributions would have to be identified, their effects examined and the explanation confronted with other types of data. A tool can rank usefully without having resolved all

these questions. Recognizing this limit guards against both devaluing prediction and turning it into a complete explanation.

19.3.3.4 Cardiological Reading

The comparison with a monogenic mutation draws attention to the magnitude of the associations. It does not demonstrate an equivalence of mechanisms, of individual prognosis or of response to an intervention. Two categories may show a similar statistical contrast while calling for different clinical investigations. Moving to care would require an additional question: does using the score improve decisions and their consequences, compared with the information already available? The events prevented, the additional treatments, the adverse effects and the consequences of a misleadingly reassuring classification would have to be examined. The benefit of an intervention in other studies does not show that the strategy of triggering it by this score brings an additional benefit.

does the score improve decisions?

19.3.3.5 Integrative Clinical Reading

a single number to explain

Before using this ranking, it is also necessary to check what the person understands of the result. A single number can produce false reassurance or excessive worry, whereas its meaning depends on the reference population and the intended use. These consequences are not measured here as a benefit of a strategy. The score remains information to be situated, not an anticipated biography or an imposed care priority.

19.3.4 Fairweather and Differences between Groups

19.3.4.1 Situating the Result

Fairweather, sex and gender

This 2023 review addresses sex and gender in myocarditis and dilated cardiomyopathy. It brings together epidemiology, genetics, experimental models, clinical presentation and treatments. It is neither a new trial nor a systematic synthesis with an explicit, reproducible selection procedure. The authors acknowledge the lack of studies specifically designed to examine gender. Table 1 mainly assembles populations in which men are the majority, but it does not show a uniformly worse prognosis in men. The mechanistic figure rests mainly on one particular mouse model. Finally, the therapeutic discussion draws largely on the general heart failure literature. These differences in provenance determine the limits of extrapolation.[4]

19.3.4.2 Critical Reading

composition is not prognosis

The first lesson is to read the tables against the temptation of a general sentence. Counting more men among the cases and concluding that affected men always have a worse prognosis are two distinct operations. The first concerns the composition of the observed populations; the second requires a comparison of trajectories, according to an endpoint and a duration. Table 1 provides precisely the means to resist their fusion. A synthesis becomes more instructive when its discrepancies are preserved instead of being smoothed out. Next,

a review without reproducible selection

what the review makes it possible to verify must be recognized. An abundant bibliography organizes a field; it does not guarantee that all relevant studies were selected according to the same rules. Without a reproducible description of this selection, the arguments can be followed without the absences being fully reconstructible. This does not make the review useless. It gives it a function of connecting and orienting, different from that of an exhaustive estimate of effects. A more specific difficulty appears in the section devoted to angiotensin-converting enzyme inhibitors: the argument that a benefit is significant in men but not in women is not enough, by itself, to establish a difference in their response. The effects must be compared between groups. This reservation concerns the strength of the argument presented, without claiming to reanalyze the cited trials here or to show that their effects would be identical.

19.3.4.3 Immunological Reading

A MOUSE MODEL, NOT A SEX

A mechanism can explain a difference in a model without becoming the immune identity of a sex. The legend of Figure 1 is decisive: it indicates where the scheme mainly comes from. Erasing this indication would turn a situated experimental construction into a general representation of men and women.

THE COMPONENT MANIPULATED OR MEASURED

The objection is that such caution would prevent any biological explanation. It leads instead to a more demanding question: which component of the difference was actually manipulated or measured? A comparison by sex, a hormonal intervention and a chromosomal study do not answer exactly the same question. Grouping these results must preserve the conditions under which they were produced. A broad category can guide the inquiry without replacing

GENDER IS NOT A REMAINDER

the variables that make the mechanism testable. It is also necessary to avoid treating gender as the name for everything that hormones do not explain. Conditions of exposure, social practices or different access to care require their own observations. In the absence of such measurements, their contribution remains an open question; it is neither demonstrated by default nor canceled by an experimental result. Recognizing two dimensions does not amount to claiming to have separated them in the data.

19.3.4.4 Cardiological Reading

THREE QUESTIONS TO KEEP APART

Moving to the person requires keeping three questions distinct: the probability of developing a disease, the course once the disease is present and the expected effect of an intervention. A signal concerning the first does not automatically provide the answer to the other two. Confusing them could lead to less investigation of a disease in a less represented group, and then to taking this lower detection as confirmation of its rarity. The other passage to watch runs from heart failure to a specific myocarditis. A general therapeutic literature can inform management, but its inclusion in this review does not make it a trial specific to each inflammatory cause. A difference in dose or response proposed for a group must be tested in the population and the strategy

THE CASE THAT RESISTS

it claims to guide. The clinical requirement is therefore not to choose between a medicine indifferent to differences and a medicine that would prescribe by category. It consists in asking which additional information truly improves the decision, for which person and under which conditions. Knowledge of groups becomes useful when it opens a better-targeted investigation; it becomes reductive when it serves as a reason not to look at the case that resists it.

19.3.4.5 Integrative Clinical Reading

GROUP MEMBERSHIP AND STEREOTYPE

Knowledge of differences between groups can help to hear a clinical presentation better; it can also produce a new stereotype. Fairweather leads to examining this tension at the level of the encounter: which questions are opened by group membership, and which risk being closed too quickly? The integrative reading leaves room for social context and expressed experience without attributing their effects to this review alone. It seeks additional information relevant to the person, rather than a course of action dictated by a category.

19.4 Worked Exercises

19.4.1 Exercise on Extrapolating the Absolute Benefit

19.4.1.1 Problem

TWO DEPARTMENTS, TWO BASELINE RISKS

All data are fictional. They serve the reasoning, without representing a real therapeutic estimate. A trial in people with an inflammatory disease estimates a five-year risk of cardiovascular events of 10% under usual care and 8% under a strategy X. Assume, for this exercise only, that these numbers represent the risks exactly and that their difference is causal. Two departments wish to extrapolate this result. Department A cares for a population whose five-year risk under usual care would be 4%; that of department B would be 20%. A reader of the trial proposes applying everywhere the two-point absolute reduction observed in the trial. Questions. What other projection is obtained by assuming the risk ratio to be transportable? How many fewer events would it predict per 1,000 people in each department? What must be checked before regarding this projection as clinical knowledge?

19.4.1.2 Reasoned Answer

CONSTANT RATIO, DIFFERENT BENEFITS

The trial's risk ratio is 8/10 = 0.80. If this ratio is transportable, the projected risk under X is 3.2% in A and 16% in B. The absolute reductions would therefore be 0.8 and 4 points: 8 and 40 fewer events per 1,000 people at five years, respectively. Applying two points everywhere would produce 20 fewer events in each department. Both calculations are arithmetically consistent; they rest on different constancy assumptions. Neither the initial randomization nor the ease of calculation alone chooses the transportable assumption.

CHOOSING THE TRANSPORTABLE ASSUMPTION

Populations, inflammatory severity, concomitant treatments, access to care, event definitions and observation horizon must be compared. A prognostic factor can modify the absolute benefit even when the risk ratio remains constant; speaking of effect modification therefore requires specifying the scale. The baseline risks themselves must be estimated in comparable populations, with their uncertainty.

WHO WILL AVOID THE EVENT?

Finally, this projection describes a group average. It does not designate the people who will avoid an event. It incorporates here neither adverse effects nor the uncertainty of the relative effect, deliberately removed from the problem statement. Restoring these dimensions is indispensable before any decision: the conditional calculation makes the assumption visible; it does not validate it.

Synthesis

Extrapolating a result consists in justifying a relevant resemblance, not in noting a descriptive one. Belonging to a group informs an expectation without dictating a trajectory; a particularity justifies a departure only if it is named before the decision. Preferences change the value attributed to consequences, not the estimated effect, and following a person does not provide the missing comparison. The four texts studied clarify these passages. A well-dated exposure is not an explained exposure; a first recorded diagnosis is not the onset of the disease; a genetic ranking is not a prevention strategy; a difference in composition is not a difference in prognosis. An absolute benefit is extrapolated according to a scale assumption that must be made visible.

Open Questions

What difference between a trial population and the person present is enough to adjust the expectation, and how can it be stated before deciding? On which scale should an effect be assumed constant when the baseline risk changes? How can group information, whether diagnostic, genetic or related to sex, open an investigation without closing a question too quickly? These questions also involve the way several clinicians agree on an answer.

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20

Objectivity as Collective Work

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Medical knowledge passes through many hands before it guides care. Teams produce the observations, others analyze them, gather them into syntheses, draw guidelines from them by vote and sometimes correct them years later. At each stage, the robustness of the result depends on what can still reach it: an objection, a verification, a correction. A broad agreement may then reassure wrongly, and a discreet correction may leave standing what it should have modified. The question becomes practical: which verifications genuinely cross-check one another, what does a vote measure and how does a revision make its consequences visible?

20.1 Objecting and Verifying Together

20.1.1 An Objection Must Be Able to Modify the Work

RESISTANCE INDEPENDENT OF ANYONE'S WILL

Objectivity does not require researchers to be devoid of convictions. It requires that their conclusions can meet a resistance that does not depend on their goodwill. A team may sincerely wish to verify its results while organizing its exchanges in such a way that certain objections never reach the decision.

A REMARK LEFT WITHOUT EFFECT

Consider a fictional meeting. A result seems convincing to almost all participants. A junior researcher notes that two series assumed to be distinct might contain the same observations. The team leader thanks her, then continues the presentation without examining the remark. She was allowed to speak; the criticism had no effect. The meeting would become more objective if it provided for checking this point, designating those responsible and revisiting the conclusion according to the result of that check. It will be objected that not every objection deserves to halt the work. That is correct: a criticism must specify what it contests and why this could change the conclusion. But its examination cannot depend solely on the rank of the person who raises it. A defensible organization answers reasons with reasons and distinguishes a refuted objection from one that has merely been dismissed. The diversity of participants has a precise function here. It can make visible difficulties that overly similar skills, experiences or interests leave unnoticed. It does not mechanically deliver the truth. The different perspectives must also have real purchase on the inquiry: access to the relevant elements, time for examination and the possibility of having a response recorded.

OBJECTION REFUTED, OBJECTION DISMISSED

THE FUNCTION OF DIVERSITY

20.1.2 Justified Trust Distributes the Verifications

20.1.2.1 What trust rests on

UNAVOIDABLE TRUST

No clinician can redo all the experiments on which that clinician's practice depends. No team possesses all the skills needed for its own instruments. Medical knowledge therefore rests on a division of labor that makes trust unavoidable. The problem is not to abolish it, but to know what it rests on and how far it extends. Trusting a specialist to interpret an image does not mean delegating the whole of clinical judgment to that specialist without limit. Trust bears on a competence, a task and conditions of practice. It gains robustness when everyone knows which verification was performed, by whom and which question remains open. A collective signature does not replace this allocation of responsibilities.

20.1.2.2 Independent checks that overlap

WHO CHECKS THE CHECKERS?

The objection is that such a system would merely displace the problem: who checks those who check? Demanding an ultimate controller would indeed lead to an infinite regress. A practicable answer is to organize checks that overlap without all depending on the same blind spot. The team that knows the setup can explain how it works; an outside reader can test the coherence of the inference; another team can examine whether the result recurs under new conditions. These operations are not interchangeable. Useful independence is therefore not reduced to the absence of a personal relationship. It also concerns the origin of the observations, the analytic choices and the interests liable to steer judgment. Declaring an interest makes a possible influence visible; it proves neither that the influence had no effect nor that the result is false. How this information changes the evaluation must be organized.

20.2 Agreeing in Order to Act, Revising to Remain Reliable

20.2.1 Agreement Enables Action without Becoming Additional Evidence

20.2.1.1 Consensus and the counting of opinions

THE LEGITIMATE FUNCTION OF CONSENSUS

A group must sometimes establish a common rule before all uncertainties are resolved. This necessity gives consensus a legitimate function: coordinating actions, stabilizing a language and specifying what appears defensible in light of available knowledge. It does not turn the number of signatures into a measure of truth. In a fictional situation, eight experts approve a proposal and two contest it. This count does not say whether the eight examined different reasons or followed the same initial opinion. Nor does it say whether the two opponents bring a decisive element or an objection already resolved. The content of the disagreement must be known. The minority is not right because it is a minority; the majority does not gain an additional observation through its numbers. It will be objected that a conclusion accompanied by all its dissents would be unusable. The report can remain concise: state the proposal adopted, the main reasons, the unresolved difficulty and the conditions for revision. A guideline thus becomes usable without concealing what belongs to well-established knowledge and what belongs to a prudent choice in the face of uncertainty.

EIGHT EXPERTS AGAINST TWO

20.2.1.2 Three disagreements in a fictional committee

TWO TOOLS, A FICTIONAL COMMITTEE

To clarify this passage, consider an entirely fictional situation with no clinical bearing. A committee must choose between two tools for training in reasoning. The first lets fewer serious errors through in the exercises, but interrupts students more often with useless alerts. The second offers smoother work, at the cost of more unflagged errors. Their evaluation remains imperfect. A first disagreement concerns the facts: do the interruptions of the first tool really reduce errors in a later exercise without assistance? The available data can be reanalyzed; an additional comparison could decide between the interpretations. This disagreement calls for an inquiry into learning, not a vote on the importance of safety. Now suppose the estimated effects are accepted by all. Some members give priority to preventing errors; others judge the interruptions penalizing enough to prefer the second tool. A more precise measurement might still shift their choice, but will not, by itself, fix the weight to give each consequence. The students who will bear these consequences must also be heard. Calling

A DISAGREEMENT ABOUT FACTS

A DISAGREEMENT ABOUT PRIORITIES

this disagreement "subjective" in order to set it aside would amount to silently imposing a weighting. It remains to turn these priorities into an applicable rule. The committee may seek the best average balance between errors and interruptions. It may also require that a tool first satisfy a justified limit on serious errors, then compare interruptions among the admissible tools. These rules are not equivalent: in the second, a gain in fluency does not compensate for crossing the limit. The rule therefore expresses a normative commitment; it is not a technical operation added after the values. Under uncertainty, it must also be stated on what evidence the limit will be judged satisfied. It will be objected that this decomposition delays a decision that is nonetheless urgent. It can help to choose the action suited to the disagreement at hand: checking a disputed consequence, discussing a priority or making explicit a condition of admissibility. A common rule then authorizes the choice without turning it into a unanimous scientific verdict. The minutes can preserve what each member accepts of the data, the compromise adopted and the objection that remains. The decision becomes defensible through its public reasons, without every divergence having to disappear.

THE RULE, A NORMATIVE COMMITMENT

20.2.2 Revision Must Have Visible Consequences

A scientific community does not become reliable by claiming never to be wrong. It must be able to recognize that a reason once judged sufficient no longer is, and to modify what depends on it. This capacity requires more than a general declaration of openness to criticism. Correcting an error presupposes delimiting its scope. A mistranscribed value, a flaw affecting one analysis and a problem calling an entire result into question do not call for the same response. Judgment must follow the established elements. It cannot automatically turn every correction into total disqualification, nor minimize a problem on the grounds that the work remains famous. The objection is that a public revision could weaken trust. Yet trust founded on the apparent absence of error is fragile: it depends on silence. Better-justified trust depends on the capacity to explain what has changed and to draw the consequences in the syntheses, teaching and decisions that used the result. A collective requirement remains, often a difficult one: allowing revision without requiring the person who reports the problem to bear its cost alone. Examining an objection takes time, may delay a publication and may contradict an authority. If these costs make criticism impracticable, the proclaimed openness remains formal. Objectivity is then recognized by an effective power: that of having a conclusion reconsidered when its reasons no longer hold.

DELIMITING THE SCOPE OF AN ERROR

THE COST OF CRITICISM

20.2.3 Mill: Discussing without Equalizing the Evidence

20.2.3.1 What Mill defends, and what he does not provide

In Chapter II of *On Liberty*, Mill distinguishes two losses: setting aside an opinion that might be true and holding a true opinion without understanding its reasons. He nonetheless refuses to keep all controversies artificially alive: the stabilization of knowledge is progress, provided that teaching still makes its difficulties intelligible. In Chapter V, regarding the concentration of skills in the governmental apparatus, Mill defends the means of forming, outside that apparatus, capacities of judgment able to criticize it.[1] These are arguments about the conditions of judgment, not a method for evaluating clinical trials.

MILL, *On Liberty*

20.2.3.2 The right to question and the credit granted

A SYMMETRY THE EVIDENCE REFUSES

The medical problem begins precisely at this boundary. Consider, in a fictional situation, a meeting in which a hypothesis supported by several controlled experiments confronts an assertion without verifiable method. Granting each side half of the conclusion in the name of dialogue would manufacture a symmetry that the evidence does not justify. However, refusing to hear any objection because the dominant conclusion is firmly established would confuse the current strength of a result with immunity to revision. The right to question and the credit granted to an answer are not the same thing. One objection remains: whoever selects the admissible questions already holds the power to stifle criticism. The answer is not to abolish selection; no team can examine everything at once. It is to make the reasons for priority explicit and to maintain a channel for contesting them. A precise objection bearing on an identifiable bias may deserve verification even if its author does not belong to the recognized group. An unchanged repetition, already answered by the accessible data, does not create a new scientific uncertainty through insistence alone.

20.2.3.3 Transmitting reasons rather than an authority

TEACHING AN OLD OBJECTION

This distinction also transforms teaching. Presenting an old objection seriously does not require suggesting that its answer remains unknown. It is possible to reconstruct what made it plausible, show the test that weakened it and specify what remains open. Students thus learn why a conclusion deserves trust, instead of learning only which side should prevail. This use of Mill remains critical. His Chapter I excludes peoples he judges insufficiently "civilized" from the liberty he defends. This colonial restriction is rejected here; it forbids using his name as a guarantee of an accomplished universalism. An author provides reasons to be examined here, not an authority that would make testing them unnecessary.

A COLONIAL RESTRICTION REJECTED

20.3 Texts Put to the Test

20.3.1 Nielsen and the Conditions of a Synthesis

20.3.1.1 Situating the Result

NIELSEN, TWO BODIES OF DOCUMENTS

Nielsen and colleagues bring together two distinct bodies of documents. A meta-analysis covers 589 trials and 83,315 unique participants exposed to immune checkpoint inhibitors. A separate search gathers clinical observations devoted to the manifestations and treatments of toxicities. Among the myocarditis cases with reported outcome, myocarditis-associated mortality concerns 83 of 220 patients. This last ratio describes published cases; it does not measure the probability of dying for every person receiving immunotherapy. The main article reports a literature search closed in April 2023. The article thus juxtaposes a frequency estimate in trials and a therapeutic exploration based on case observations, whose demonstrative powers differ.[2]

20.3.1.2 Critical Reading: the denominator is part of the evidence

THE DENOMINATOR OF PUBLISHED CASES

Before comparing two numbers, how their populations were constituted must be specified. In a safety study, the denominator is the people exposed and followed up to detect an event. In a collection of myocarditis cases, it is the patients diagnosed, then described and published. This second selection may depend on the severity, the unusual character or the therapeutic interest of the case. An exact death ratio in this collection therefore remains unsuitable for describing, without reservation, all the cases of myocarditis encountered in practice. The distinction becomes even more demanding for trials without events. Systematically setting aside their results would inflate the estimated frequency; equating them all with a careful search that remained negative could lower it artificially. A zero must be accompanied by its observation procedure. The duration of surveillance, the tests performed and the diagnostic definition are constitutive information of the result. A statistical method can organize rare proportions; it cannot retrieve an injury that no one looked for. A comparison between proportions from different trials does not, through calculation alone, retain the advantages of the randomization present in some of those trials. Cancer populations, concomitant treatments and schedules may differ. The apparent ranking of the molecules therefore calls for examining the contrasts actually available. Likewise, the absence of a detected difference between two strategies does not show their equivalence, either in the frequency or in the severity of adverse events.

ZERO AND ITS PROCEDURE

20.3.1.3 Immunological Reading: not confusing the coherence of a target with the attribution of success

THE LAST MOLECULE ADDED

The possibility of interrupting a harmful immune activation provides a biological reason to study an intervention. It does not allow every improvement observed after its administration to be attributed to it. A patient often receives a succession of treatments whose effects overlap. Recovery may also depend on the delay to recognition, initial severity and the support of failing functions. In this configuration, the most recently added molecule easily benefits from an overly simple causal narrative. The temptation to convert a table of responses into a league table of immunosuppressants must therefore be resisted. A treatment reserved for the most severe situations may show more deaths while bringing a benefit; a drug given to selected patients may appear superior without being so. The mechanistic explanation helps to build the future comparison, but does not correct this retrospective imbalance. A strategy combining several interventions would further require distinguishing its overall effect from the specific contribution of each.

NO LEAGUE TABLE OF IMMUNOSUPPRESSANTS

20.3.1.4 Cardiological Reading: organizing the knowledge needed for a decision

A useful synthesis must keep several questions separate: how many people develop an injury, which forms progress unfavorably and which treatments modify that course? The same dataset rarely answers all three. A low frequency does not make a complication negligible; its severity does not, however, license announcing a therapeutic efficacy that remains uncertain. To progress, a reg-

WHAT A REGISTRY SHOULD SPECIFY

istry should specify eligible patients, diagnostic criteria, time zero, successive interventions and competing outcomes. Prospective follow-up would improve the quality of observation without automatically becoming a causal trial. The comparison must also incorporate oncological consequences: protecting the heart and maintaining tumor control are linked objectives, without being interchangeable. The examination establishes the need to preserve the origin of each estimate. It makes the value of targeted immunological strategies plausible, without settling their ranking. This judgment would change in the face of appropriate prospective comparisons, harmonized diagnoses and explicitly documented surveillance. This requirement does not demand suspending care while awaiting perfect certainty; it prevents passing on as demonstrated what remains a decision under uncertainty.

THE ORIGIN OF EACH ESTIMATE

20.3.1.5 Integrative Clinical Reading

This comparison must finally connect with the goals of the person and of the teams following that person. An isolated toxicity rate does not set their priority between cardiac protection and tumor control. The known risks, those that remain poorly compared and the outcomes that matter at the horizon considered must be specified together. The synthesis informs this coordination under uncertainty; it provides no universal ranking between the two goals.

20.3.2 Birnie and the Construction of a Consensus

20.3.2.1 Situating the Result

BIRNIE AND THE HRS, 2014

The Heart Rhythm Society (HRS) consensus on arrhythmias in cardiac sarcoidosis brings together an international group of experts. All its recommendations carry level of evidence C. Their adoption required more than 75% of the votes. The text borrows the classes of recommendation of the American College of Cardiology and the American Heart Association (ACC/AHA), while stating that it follows the HRS process. A proposal for Holter screening, supported by ten of fourteen members, thus did not become a formal recommendation. The authors explicitly distinguish consensus, unanimity and empirical confirmation. The document also proposes diagnostic criteria and sets out knowledge gaps. It should be read as a collective construction dated 2014, without its therapeutic proposals becoming current instructions.[3]

20.3.2.2 Critical Reading: the threshold decides status, not truth

A VOTE CHANGES, THE DATA REMAIN

The Holter example allows a precise thought experiment. Suppose a member had changed his or her vote without any new observation being available: the editorial status of the proposal could have changed, while the data would have remained identical. The threshold therefore organizes responsibility for a collective statement. It measures neither the probability of benefit nor that of being right. Conversely, remaining below this threshold does not show that a test is useless. It could be objected that an agreement among informed experts is itself information. That is correct: several judgments can detect difficulties that escape an isolated examination. But this value depends on their reasons and their dependencies. Fourteen people who learned the same practice from

FOURTEEN OPINIONS, HOW MANY INDEPENDENT?

the same teams do not necessarily represent fourteen independent confirmations. The vote becomes instructive when the arguments preceding it can be traced. The coexistence of a strong recommendation and a low level of evidence is not, by itself, contradictory. The first expresses a practical orientation; the second indicates its empirical support. A decision may be judged preferable in a serious situation despite a fragile estimate of its consequences. It must then be asked what justifies this preference: anticipated benefit, danger of waiting, feasibility or extrapolation. Confusing these dimensions would give prescriptive strength a scientific precision it does not possess.

STRONG RECOMMENDATION, WEAK EVIDENCE

20.3.2.3 Immunological Reading: a classification must not become its own validation

Among the elements of the proposed clinical diagnosis is a cardiomyopathy or conduction block responsive to corticosteroids or an immunosuppressant. This choice raises a particular methodological difficulty. If patients are selected because they respond to a treatment, and their response is then used to establish that this treatment works in this disease, part of the conclusion has been built into the selection. The objection does not condemn every diagnostic use of a therapeutic response. Such a response may modify a clinical assessment, provided it does not then supply, without correction, independent evidence of efficacy. Studying that efficacy would call in particular for entry criteria established before the response, a relevant comparison and an assessment of outcomes that does not recycle the initial definition. An analysis distinguishing patients classified with or without this criterion would already make the problem visible. This distinction also preserves the meaning of the immunological label. Recognizing an inflammatory component does not automatically settle how much of the damage is still modifiable. The causal argument, the classification of a patient and the promise of a treatment remain three different commitments, even when they appear in the same document.

SELECTED BECAUSE THEY RESPOND

20.3.2.4 Cardiological Reading: putting the limits of a recommendation to work

To assess a strategy against arrhythmias, judgment must specify the outcome sought. Reducing an imaging signal, avoiding a rhythm episode and prolonging life are not three interchangeable formulations of success. A recommendation gains intelligibility when the passage from the available data to the expected benefit can be identified in it, and what remains to be verified can be named. The practical objection is strong: the patient cannot wait until each passage is demonstrated. The answer is to make the decision revisable, not to demand an impossible prior certainty. The question that motivated the action must be preserved, and what would justify its reevaluation must be defined. A favorable course does not retrospectively validate all the assumptions that accompanied it; nor does a complication suffice to disqualify the whole strategy. The lasting interest of this consensus then lies in its critical legibility: its accepted and rejected proposals allow examination of how a common orientation was produced. This traceability allows a more precise discussion than the alternative between obedience to the group and general distrust of expertise.

IMAGING, RHYTHM OR SURVIVAL

A REVISABLE DECISION

20.3.2.5 Integrative Clinical Reading

A COMMON LANGUAGE, A DATE

A consensus can give a common language to professionals who do not see the person at the same moment. Birnie makes it possible to examine this coordinating function without confusing collective agreement with empirical certainty. The integrative reading requires that the proposal, its rationale and its limits can be passed from one interlocutor to another. It also leaves room for preferences and constraints that are not resolved by an expert vote. The history of this consensus is not equivalent to a current course of action applicable without reevaluation.

20.3.3 ADVOCATE: Rereading a Trial after Its Retraction

20.3.3.1 Situating the Result: the historical document

ADVOCATE, 2021, THEN THE 2026 NOTICE

A correction published in 2024 had rectified conversions of corticosteroid doses. The retraction of June 29, 2026 is a distinct change: the article is now examined for its history and its method, without being retained as valid evidence of efficacy.[4] In 2021, ADVOCATE presented a randomized comparison of avacopan and a prednisone taper, with concomitant immunosuppression. The report defined remission at 26 weeks and sustained remission, requiring remission at weeks 26 and 52 without relapse in between, as well as a noninferiority margin of 20 percentage points. The official 2026 notice states that the two academic authors requested the retraction. It refers to the Food and Drug Administration (FDA) investigation then under way. According to this investigation, primary endpoint assessments in nine patients had been readjudicated after database lock and unblinding, without the knowledge of these authors and without disclosure in the article. The editors retracted the publication. This information describes the status of the documents; it does not constitute a new analysis of the individual data.[5,6]

20.3.3.2 Critical Reading: two distinct conditions of validity

THE NONINFERIORITY MARGIN

Noninferiority requires defining the maximum loss of efficacy judged acceptable, then examining whether the data allow it to be excluded. It means neither equality of treatments nor absence of any loss. A margin in percentage points must be discussed on that scale: its clinical importance depends in particular on the endpoint, the baseline risk and the advantages expected in return. The justification of the margin belongs to clinical reasoning as much as to statistical calculation.

TRACEABILITY BEFORE THE FORMULA

Another condition, however, precedes this calculation: the observations analyzed must have been produced and modified according to a verifiable procedure. A confidence interval describes uncertainty under measurement and analysis assumptions; it does not repair a break in traceability. Redoing the calculation on the same values does not verify how they were obtained. The quality of the formula and that of the documentary chain are therefore two distinct objects of verification.

THE NUMBER OF RECORDS IS NOT ENOUGH

The number of records involved is not enough to decide that the anomaly would be negligible. It would be necessary to know the assessments before and after modification, the documented reasons, the people authorized to intervene, the timing of the decisions and their consequences for the planned analyses. Conversely, this number does

not allow a new unfavorable estimate to be manufactured. Withdrawing trust from a published conclusion does not license inventing the opposite conclusion. The notice reports an identified break; it offers neither the data needed for an independent reanalysis nor an exhaustive attribution of responsibilities. The scope of the retraction must remain distinct from these two questions.

20.3.3.3 Immunological Reading: maintaining the separation between hypothesis and evidence

A HYPOTHESIS STRIPPED OF ITS SUPPORT

A biological hypothesis may remain intelligible when the clinical study invoked to support it loses its probative status. The reasoning must then be rebuilt from its independent components: mechanistic experiments, other trials, target measurements and clinical observations. Their existence and quality must be verified separately. Citing them in the same article makes them neither automatically invalid nor automatically sufficient. For cardioimmunology, this case teaches above all how to avoid circular validation. The coherence of a mechanism does not guarantee the reliability of a clinical result, and that result can no longer serve to confirm the mechanism once its documentary support is withdrawn. The map of arguments must make it possible to identify exactly which arrow depended on the retracted text.

20.3.3.4 Cardiological Reading and the revision of knowledge

REMISSION IS NOT PREVENTION

A conclusion of remission in a vasculitis would not, by itself, answer a question of cardiovascular prevention. The retracted status here reinforces an already necessary separation between disease activity, organ damage and clinical events. No supposed cardiac benefit should therefore be preserved by presenting it as an "obvious" indirect consequence of a result now unusable as evidence. Editorial revision consists in finding all the sentences that depended on this reference, withdrawing the claims that have become unsupported and indicating their new justification when one exists. Deleting only the bibliographic entry would let orphaned conclusions survive. Keeping the article in a historical commentary requires, on the contrary, that its status accompany each use. The retraction thus requires reexamining the syntheses that used this publication as support. It does not, by itself, allow the real efficacy of the treatment to be estimated, its current regulatory status to be known or all individual responsibilities to be attributed. The revision of knowledge must follow the propositions actually affected, with the date and the reason for the change.

ORPHANED CONCLUSIONS

20.3.3.5 Integrative Clinical Reading

MAKING REEVALUATION INTELLIGIBLE

A retraction also concerns the trust of people to whom an explanation has already been given. The integrative reading distinguishes acknowledging a change in knowledge from announcing, without evidence, that each past decision was at fault. The aim is to make the reevaluation intelligible: which support has disappeared, which other knowledge remains available and who takes up the clinical question? This requirement of continuity licenses neither a prescription nor a discontinuation inferred from the notice alone; above all, it prevents the documentary correction from leaving the patient without explanation.

20.4 Worked Exercises

20.4.1 Exercise: Correcting a Synthesis without Erasing Its History

20.4.1.1 Problem

THREE TRIALS, ONE SENTENCE

An entirely fictional situation, unrelated to the ADVOCATE data. A teaching review includes three independent trials and one sentence: "The treatment reduces the number of events in all three trials." The first version of its table contains, for each group, 100 participants followed completely for the same duration within each trial. Each number designates people who had at least one event, counted only once: trial A, 12 in the treatment group and 20 in the control group; trial B, 9 and 15; trial C, 8 and 14. An official correction of trial B then states that the group columns had been inverted in the published table. The group sizes remain identical: the corrected result is therefore 15 events in the treatment group and 9 in the control group. The authors maintain the article and explain the presentation error. You have read the correction and verified that it does concern the endpoint used in your review. No new element concerns A or C. Questions. What change should be made to the descriptive calculations? How should the conclusion be revised? Should trial B be removed, or the corrected table kept silently? What documentary elements are missing to carry out a formal quantitative synthesis?

20.4.1.2 Reasoned Answer

A REVERSED DIRECTION, TWELVE POINTS

In B, the risk difference, treatment minus control, goes from −6 to +6 percentage points. The direction of the estimate is reversed; this is not a change of six points, but of twelve points between the two versions. A and C remain at −8 and −6 points, respectively. These calculations do not by themselves indicate the uncertainty of the results. The initial sentence becomes false. An acceptable formulation would be: "The estimates of trials A and C favor the treatment; after correction, that of B favors the control." This divergence calls for analysis,

CORRECTING WITHOUT ERASING

without automatically demonstrating a real difference between trials. The correction does not, by itself, mandate the exclusion of B. Its reference is kept, the data concerned are replaced and the two versions are explicitly linked to the correcting document. The revision log indicates the date, the modified cells and the conclusions affected. A silent correction would make it impossible to understand disagreements between readers holding different versions.

CORRECT, THEN DECIDE ON USE

Before any meta-analysis, the comparability of populations, endpoints, follow-up durations and estimates must still be examined, as well as the risks of bias. Mechanically adding up events does not replace this work. Correcting a data point and deciding how to use it are two distinct operations.

Synthesis

Objectivity rests on an organization: objections capable of modifying the work, distributed verifications that do not share the same blind spot, revisions whose consequences remain visible. Consensus coordinates action without adding evidence; its threshold sets the status of a proposal, not its truth. The texts studied teach this together. A synthesis stands by its denominators and the

origin of each estimate; a consensus, by the reasons for its votes and its rejected proposals; a retracted trial recalls that an exact calculation does not repair a break in traceability. Discussing does not mean equalizing the evidence: the right to question and the credit granted to an answer remain distinct.

Open Questions

When does an objection deserve to suspend a conclusion rather than simply be recorded? How should the independence of verifications that share training, data or interests be judged? How far back should a retraction reach into the syntheses, teaching and decisions that relied on the withdrawn text? These questions concern everyone who transmits knowledge.

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21

The Responsibility to Know and to Transmit

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An established conclusion does not thereby stop circulating. It is taught, summarized, cited in a review, taken up in a consultation, revised when new results arrive. At each retelling, a few words can turn an experimental lead into an available treatment, or a nonsignificant result into proof that a target is of no interest. Transmitting therefore carries a responsibility of its own. It consists in saying what is known and what is not, simplifying without removing the means of judging, dating what was proposed and explaining what a revision changes. A well-transmitted hypothesis remains a hypothesis; a well-conducted revision names what it abandons.

21.1 Saying What Is Known and What Is Not

21.1.1 Answering for the Step a Learner Is Led to Take

21.1.1.1 Changing audience without changing the evidence

LABORATORY, LECTURE, CONSULTATION

Changing the audience can change the use of a result, without changing what the data make it possible to assert. In a laboratory, a hypothesis can serve to choose the next experiment. In a lecture, it can become the explanation that students will remember. In a consultation, its wording can steer a decision. The responsibility of transmitting consists in maintaining the scope of the evidence while explaining what justifies this new use. Consider an entirely fictional situation. An experiment shows that an intervention modifies a cellular activity in a model. A teacher wants to explain its interest and concludes: "We now know how to protect the heart." The difficulty lies in the step taken: an experimental possibility becomes an available therapeutic capacity. The teacher has not merely shortened a method. The teacher has changed what the students may legitimately believe themselves assured of.

A TEACHER CONCLUDES TOO QUICKLY

21.1.1.2 The limit taught and the role assumed

It will be objected that a lecture cannot convey every laboratory precaution. This objection requires ranking the information, not suppressing the conditions that change the conclusion. The short formula could keep both the real gain and what remains open: the experiment provides a mechanistic lead whose relevance to patients remains to be established. The limit is part of the content taught when it prevents a different conclusion. This responsibility is proportionate to the role assumed. Presenting a hypothesis as such, defending a personal interpretation and recommending an action do not involve the same commitment. Whoever takes an additional step must be able to explain the reasons for it. Authority acquired over the observation does not automatically transfer to all its possible consequences.

A RESPONSIBILITY ACCORDING TO ROLE

21.1.2 Qualified Ignorance Guides the Work

21.1.2.1 Naming ignorance and its reason

FOUR WAYS OF NOT KNOWING

Saying "we do not know" can designate a question never studied, results too imprecise, contradictory observations or a document that has not been consulted. These situations call for different responses. An experiment does not remedy a failure to read; an additional reading does not create the follow-up missing from a study. Qualifying ignorance consists in naming the proposition that remains undecidable, the reason for this indecision and what could reasonably reduce it. A personal limit must also be distinguished from a limit of available knowledge. Not having found a study does not license asserting

that it does not exist. Conversely, knowing that a publication exists does not mean possessing the answer attributed to it.

21.1.2.2 A stopping rule that is not an alibi

A JUSTIFIED STOPPING RULE

A practical objection arises: does this requirement not risk prolonging the inquiry indefinitely? An additional search has a cost and does not always bring information liable to change the judgment. Responsibility therefore also includes a justified stopping rule. Before continuing, it must be asked what difference the next verification could make. If no plausible answer can modify the wording adopted, pursuing this point may become less urgent than examining a decisive uncertainty elsewhere. This rule could, however, serve as an alibi: it would suffice to declare that no expected information would change the opinion already formed. In a fictional example, a team decides not to check the blinding of a measurement because the effect seems convincing. Its stopping rule is circular if a plausible measurement bias could produce precisely this effect. It must be stated what an unfavorable answer would change, and the priority of the check must then be justified by this consequence and its cost. Qualified ignorance thus allocates attention without permitting the choice of only reassuring verifications.

21.2 Simplifying, Allocating, Revising without Losing the Reasons

21.2.1 Simplifying without Removing the Means of Judging

WHAT A SIMPLIFICATION LEAVES POSSIBLE

A teaching explanation selects. It may omit a technical step, use an image or begin with a simple case. Its quality depends on what this selection leaves possible. Can the person receiving the explanation then recognize the limits of the model and understand why a different situation requires different reasoning?

THE IMAGE OF THE SWITCH

Return to the fictional example. Comparing an immune pathway to a switch facilitates a first understanding of inhibition. The image becomes misleading if it installs the idea that switching off a single function would be enough to eliminate all adverse effects. The image need not necessarily be abandoned. Its use can be specified: here it represents the action studied on a target, without describing the whole system or predicting the clinical result. It will be objected that this correction makes the image less memorable. But remembering an error that is easy to retain is not a teaching success. The alternative is to build a progression: first the useful relation, then the counterexample that reveals what it omits, finally the more exact reformulation. Students then learn why the model changes, instead of receiving in succession a simplified certainty and then its repudiation. A check of understanding should bear on this capacity for transfer. Asking for a limit to be recited sometimes rewards a purely verbal caution. Asking which conclusion would change in another context obliges the student to use the limit. The criterion of success becomes more demanding: the student can redo a line of reasoning and identify the moment when it no longer applies.

TESTING TRANSFER, NOT RECITATION

21.2.2 Allocating Responsibilities without Dissolving Them

21.2.2.1 Judging each step at the moment of choice

A CHAIN OF RESPONSIBILITIES

An observation, an inference, a decision and an act of teaching form a chain, but no step exempts the others from their own responsibilities. A correctly obtained measurement can receive an excessive interpretation. A reasonable interpretation can lead to an ill-suited action. A defensible decision can later be told as the demonstration of a certainty that did not exist. It would nonetheless be unfair to evaluate each step solely from the final outcome. An unfavorable result does not prove that the initial judgment was at fault; a happy outcome does not turn negligence into method. The evaluation must recover what was accessible at the moment of choice, the alternatives actually conceivable and the reasons put forward at the time. It must also ask whether important information was set aside without examination. It will be objected that such an assessment opens the door to after-the-fact justifications. That is why the reasons must be formulated early enough not to be reconstructed according to the result. The point is not to produce a defensive file for every thought, but to keep the choices that really change the interpretation or the action. The later explanation can then be confronted with an earlier record.

THE OUTCOME DOES NOT JUDGE THE CHOICE

21.2.2.2 What makes a revision progress

21.2.2.2.1 Explaining the difficulty a revision resolves

CORRECTING IS NOT FAILING

Recognizing this responsibility makes it possible to judge progress itself better. Correcting a conclusion is not a failure to be concealed; but changing vocabulary is not by definition an advance either. To appreciate a revision, it must be explained which difficulty it resolves and which earlier results it requires to be kept, restricted or abandoned. Consider a fictional example. A team grouped two observed profiles under the same name. A new method separates them; the researchers announce an additional disease. Yet the distinction might come solely from sample preparation. A first advance would be to show that it persists when this preparation is controlled. This control would strengthen the reality of the measured difference; its interpretation as a distinct disease would still need justifying. Another advance would be to predict, then observe, a difference that the old classification left undetermined. That is the new work the distinction makes possible, and the answer it brings to a defined difficulty.

TWO PROFILES, ONE MORE DISEASE?

21.2.2.2.2 Changing the question, counting the losses

A DISCOVERY THAT CHANGES THE QUESTION

It will be objected that this requirement favors old questions and overlooks discoveries that transform what is being sought. The objection is fair if every research effort is required to perfect the same instrument. A discovery can also make visible a difference previously inaccessible. It then gains its own justification when this difference becomes controllable and fruitful for an inquiry. But changing the question does not prove that the earlier answer was false: it must be specified whether an error is being corrected, a domain extended or another problem opened. This balance sheet may include gains and losses. A new classification might better describe certain cases while making historical series less directly comparable. Recognizing this cost does not amount to refusing the discovery; it designates the work needed to make it transmissible knowledge. Scientific progress is thus reduced neither to the accumulation of publications nor to the story of an entirely mistaken past. It becomes defensible when the new gain, its reasons and the questions it leaves open can be set out. Transmitting this comparison teaches how to evaluate an advance, instead of simply believing its announcement.

21.3 Texts Put to the Test

21.3.1 Mann and the Revision of a Hypothesis

21.3.1.1 Situating the Result

Mann, 2015, disappointing trials

In this 2015 review, Douglas Mann sets the experimental arguments for a role of inflammation in heart failure against the disappointing results of clinical interventions. He considers several explanations: a marker that might not be causal, a contributing pathway insufficiently decisive for events or an intervention whose unfavorable effects might cancel out a benefit. The selection of patients by biomarkers appears as a perspective to be studied. The text discusses in particular the hypothesis of complement-related toxicity to explain certain infliximab results; it does not demonstrate it in the patients concerned. This is a historical narrative review, not a new trial or a validation of a therapeutic algorithm.[1]

21.3.1.2 Critical Reading: revising without indefinitely protecting a hypothesis

rescuing an idea after the fact

An intervention that fails tests a precise proposition: under the conditions studied, this strategy was supposed to improve this outcome. It does not directly test all possible relations between inflammation and disease. However, multiplying after the fact the explanations liable to rescue an idea can make it impossible to contradict. A useful scientific revision must therefore go beyond the formula "it was the wrong treatment." Each new explanation can be asked to produce a testable expectation. If the timing of intervention is invoked, what comparison of schedules would distinguish this explanation from an absence of effect? If a toxicity cancels a benefit, what independent observations would separate the two? If only certain patients are concerned, how can they be recognized before their response is known? These questions turn a retrospective defense into a program liable to meet new objections.

a testable expectation per explanation

21.3.1.3 Immunological Reading: naming operations rather than a single adversary

"anti-inflammatory," too broad a word

Saying that a treatment is "anti-inflammatory" leaves several questions open: which target is reached, in which cells, at what dose and for how long? A molecule is not a simple logical negation of the biological pathway to which it is attached. It has a distribution, associated effects and interactions with the pathological context. General vocabulary can therefore bring together experiments that their actual functioning distinguishes. This remark does not allow the most favorable explanation to be chosen arbitrarily. The failure of a treatment may be compatible with several mechanisms without establishing any of them. The verbs corresponding to each step must be preserved: an observation suggests, an experiment supports within its model, a clinical comparison estimates an effect in its population. Moving to "demonstrates the mechanism in humans" requires an additional argument. The same discipline applies to biomarkers. A worse prognosis does not prove increased biological sensitivity to the targeted treatment. It may nonetheless correspond to a greater absolute

prognostic enrichment, relative effect

benefit when the relative effect remains constant, as the exercise in Chapter 19 shows. Distinguishing this prognostic enrichment from a modification of the relative effect prevents the same evidence from being demanded for two different propositions. The second would require a direct comparison of effects according to the marker, on a stated scale and with appropriate confirmation. A clinical selection rule must, in both cases, justify the expected benefit, the harms and its use in the target population.

21.3.1.4 Cardiological Reading: preserving the date and the audience of knowledge

the 2015 horizon

Reading this text today requires preserving its 2015 horizon. Later trials cannot be presented as arguments the author already had at hand. Conversely, the therapeutic proposals reported at that date do not become contemporary advice through their presence in a recognized review. A teaching presentation must indicate what belongs to a history of research and what actually supports a current decision. The review thus compares arguments of different levels and several explanations of a disappointing therapeutic result. It does not make it possible to determine which one actually explains a particular failure, or which patient would benefit from a new strategy. Each explanation gains scientific content when it announces an observation capable of distinguishing it from the others.

21.3.1.5 Integrative Clinical Reading

the word "failure"

Revising the explanation, however, makes neither the disease nor the person's expectations disappear. The word "failure" must designate the proposition put in difficulty, without encompassing all the goals of care. Between specialists and general practice, this precision makes it possible to transmit the weakened arguments, the knowledge that remains and the new questions. Understanding why a lead did not hold does not mean that all understanding is lacking.

21.3.2 Frangogiannis and the Conditions of Translation

21.3.2.1 Situating the Result

Frangogiannis, a 2014 review

In this 2014 review, Frangogiannis organizes the relations between inflammation, repair and remodeling after infarction. He brings together experimental work and human observations, while returning to the difficulties of therapeutic translation. The search described covers English-language publications from 1970 to January 2014. The text proposes in particular distinguishing excessive inflammatory and fibrogenic responses to guide future targeted strategies. Circulating biomarkers are considered as possible instruments, with an explicit reservation about their myocardial specificity. This is neither a trial of this selection strategy nor a demonstration of its clinical benefit. It is precisely this distance between explanatory model and research program that transmission must preserve.[2]

21.3.2.2 Critical Reading: a synthesis produces an order, not a repetition of the experiments

describing, connecting, proposing

Reading a review does not amount to verifying separately all the studies it draws on. Its own contribution consists in bringing results together, bringing out their tensions and proposing questions. This operation is indispensable to knowledge; it nonetheless adds inferences to the initial observations. It must be possible to recognize the moments when the author describes a result, connects several studies or advances a new explanation. A particularly clear diagram can give the impression that the relations represented all have the same degree of certainty. To transmit it faithfully, each arrow could be accompanied by a question: does it summarize an experimental intervention, a human association or a proposition still to be tested? The figure would remain useful, but it would cease to function as uniform evidence. The elegance of a whole does not erase the inequality of its supports.

a question for each arrow

21.3.2.3 Immunological Reading: keeping the context in the sentence

a "protects" without its context

A piece of biological knowledge loses part of its content when the tissue, the moment and the setup in which it was obtained are removed. Saying that a pathway "protects" or "damages" can be an acceptable teaching shortcut if the conditions remain close. Repeating this shortcut in another context sometimes changes its meaning. The problem is not only imprecise vocabulary: it is the transformation of a conditional relation into a supposedly permanent property. Transmission should therefore bring out the exact subject of the statement. Is a mediator modified throughout the organism, a receptor in certain cells or a response during a delimited period? Is a cellular composition, a tissue structure or a function being measured? These details prevent the consequences of a situated intervention from being attributed to an abstract target. They also help to bring two different results together without declaring too quickly that they contradict each other. Nor does an explanatory typology constitute an available test. Naming two biological profiles helps to design research; it does not prove that each patient can be classified reproducibly, nor that these categories predict a difference in treatment response. This step requires additional observations and comparisons. The relevance of the concept and the usefulness of its measuring instrument must remain separate.

a typology is not a test

21.3.2.4 Cardiological Reading: the proposed future must not become a demonstrated past

a perspective turned strategy

Over time, a proposal may be cited as if its realization went without saying. A "strategy" is then encountered without its being clear whether it is an experimental possibility, an evaluated procedure or a validated clinical use. Preserving the date of the source and the verbal mood of its conclusion prevents this drift. A research perspective deserves to be transmitted for its fruitfulness, without retrospectively attributing to it an efficacy it lacked the means to demonstrate. To teach this difference, commentary on a therapeutic proposal should specify the expected outcome, the endpoint that would measure it and the comparison that would allow it to be attributed to the treatment. To claim therapeutic usefulness, a better classification must improve a

PROPOSED PROGRAM, DEMONSTRATED BENEFIT

decision whose consequences have been examined. This historical framework can guide better-targeted experiments and trials; it does not demonstrate the benefit of an individual treatment algorithm. That would require independent validation of the measurements, then a comparison of the strategies they guide. Transmission must preserve this distance between the proposed program and the demonstrated benefit.

21.3.2.5 Integrative Clinical Reading

THREE HORIZONS

The words "repair" and "remodeling" can evoke global recovery whereas they designate processes that must be situated. Frangogiannis invites transmitting their conditions and limits before relating them to a person's life. The integrative reading thus articulates three horizons: what the model helps to explain, what a clinical strategy would have to demonstrate and what the patient hopes to regain. None of these horizons replaces the others; relating them guards against promising complete restoration on the basis of a research perspective.

21.4 Worked Exercises

21.4.1 Exercise: Transmitting a Hypothesis without Promising a Treatment

21.4.1.1 Problem

A MOUSE MODEL AND 60 PATIENTS

An entirely fictional situation. In a mouse model of heart failure, a treatment X reduces a measure of tissue injury. In 60 patients, a randomized study then observes a greater decrease in an inflammatory marker under X than under placebo. The exploratory endpoint is the distance covered in a standardized walk test; a greater distance corresponds to better performance. The mean difference, X minus placebo, is +8 meters, with a 95% confidence interval (95% CI) from −12 to +28 meters. The study is short; it is not designed to establish an effect on mortality. No threshold of clinical relevance was provided in this problem statement. A text intended for the public concludes: "By eliminating the inflammatory cause of heart failure, X improves exercise capacity and protects patients." A professional version states: "The result on exercise capacity is not significant: the biological target is therefore of no interest." Questions. Identify the slippages in the two formulations. Propose a sentence accessible to the public and a transmission note for a research team. Specify which new observation would cause each message to change.

21.4.1.2 Reasoned Answer

21.4.1.2.1 Two formulations that go beyond the data

A MARKER TURNED GENERAL CAUSE

The first formulation turns an intervention on a marker into the elimination of a general cause. It then attributes a functional benefit and clinical protection that are not established by the information provided. The mouse result justifies a question of extrapolation; it does not resolve it. The decrease in the marker shows an effect on the measured variable, without establishing that it explains any possible benefit. The second formulation also goes beyond the data. The interval is compatible with effects on exercise capacity in both directions. It

proves neither benefit nor the absence of any effect, and its clinical relevance remains to be specified. This result does not, by itself, settle the value of the target in all situations.

21.4.1.2.2 Two messages and a threshold set before the trial

A THRESHOLD SET BEFORE THE TRIAL

Possible public sentence: "X modified a marker of inflammation in a small study; its benefit on exercise capacity and on patients' health remains to be established." Scientific note: "Keep the tissue, biological and functional results separate; specify population, duration, safety and clinically important difference before a confirmatory trial." To make this note operative, extend the case with two entirely fictional sequels. Suppose that, before a new randomized trial, the investigators justified a mean gain of 20 meters as an important difference for this population. This value serves only the exercise; it is not a proposed clinical threshold.

21.4.1.2.3 Two fictional sequels, two revisions of the message

A FIRST, FAVORABLE SEQUEL

First sequel: the trial, conducted in a comparable population with a reliable measurement, estimates a difference of +26 meters (95% CI: +22 to +30 meters). Under the assumptions of the analysis, the interval lies entirely beyond the difference retained. The message could now state that X improves, on average, the measured performance under the conditions studied, with an effect supporting its clinical importance. It should still specify the duration observed and the adverse effects. Neither the elimination of a general cause nor protection against mortality becomes established by this functional result. Second sequel, an alternative to the first: the estimated difference is +1 meter, with an interval from −4 to +6 meters. A mean gain of 20 meters then lies outside the interval. If the conditions of validity are met, the message should no longer be limited to "benefit yet to be established": the study provides an argument against a mean benefit of this magnitude in the situation examined. It does not demonstrate an exactly null effect. The modification of the marker would remain a real biological result, whose expected functional value would here be called into question at the magnitude chosen.

AN ARGUMENT AGAINST 20 METERS

21.4.1.2.4 The conditions of a controllable revision

It will be objected that a narrow interval can surround a biased estimate. That is correct: these two revisions assume that the comparison and the measurement allow the stated interpretation. A selective loss of participants or a compromised measurement would first require reexamining this condition. Precision acquires its force only within valid reasoning. Finally, changing the threshold after the fact to rescue the promise would prevent understanding what has changed. Another threshold may be defensible for a new question, provided its reasons are given and the assessment against the initial criterion is kept. Revision thus becomes controllable: it names a proposition, the conditions of its examination and the consequence of the result, rather than a mere change of tone.

NOT MOVING THE THRESHOLD

Synthesis

Transmitting knowledge requires preserving its scope when the audience changes. Qualified ignorance names the undecidable proposition, its reason and what could reduce it; a stopping rule is justified by what an unfavorable answer would change. A simplification succeeds when it allows its limits to be recognized. A revision makes progress when it states which difficulty it

resolves and what it requires to be kept, restricted or abandoned. The two reviews studied teach this together: a revised hypothesis must produce testable expectations, and a research program must not, over time, become a demonstrated benefit. The date, the context and the verbal mood of a conclusion are part of what must be transmitted.

Open Questions

At what point does an explanation added after a failure stop revising a hypothesis and start protecting it? How can a simplification be evaluated by what it enables a learner to redo, rather than by what it enables a learner to recite? What record of reasons must be kept so that a decision is judged at the moment it was made, not by its outcome? These questions accompany every transmission of medical knowledge.

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General Conclusion

Providing the Means to Continue

I. Arguing the Passages, Situating Each Study

SCALES WITHOUT A COMMON LANGUAGE

Cardioimmunology brings together observations that share neither the same scale nor the same function. A cell, a lesion, an image, a clinical trajectory and a trial do not spontaneously speak the same language. Bringing them together becomes knowledge when the passages are argued, with conditions precise enough to be tested. Method therefore does not merely provide rules for recognizing a good study. It teaches how to ask what this study establishes within a broader inquiry. The same publication can strongly support one proposition and leave another question entirely open. Reading with precision makes it possible to retain this contribution without granting it unlimited authority.

II. Holding a Proposition as Supported, Choosing to Act

DECIDING WITHOUT CONCEALING UNCERTAINTY

This discipline does not require waiting for complete knowledge before acting. It requires distinguishing the reasons for holding a proposition as supported from the reasons for choosing an action. The consequences of an error can justify a different decision threshold; they do not, by themselves, provide additional evidence. Holding these two requirements together makes it possible to decide under uncertainty without concealing it or turning it into a license to conclude anything.

III. Exposing One's Reasons to an Objection

THE TEST OF OBJECTION

Judgment becomes autonomous when it can expose its own reasons to an objection. The test is simple to state: choose a conclusion, write down the decisive reason that supports it, then examine a specific objection. At the end of this examination, judgment must be able to explain why it keeps the conclusion, restricts it or abandons it. The expected result is not an obligatory change of mind, but a response to the difficulty raised. It will be objected that it is easy to construct a weak objection and emerge reassured. The exercise is probative only if the objection genuinely threatens a reason on which the credit granted to the conclusion depends; it therefore benefits from being entrusted to a third party. A conclusion maintained after a relevant examination can constitute progress: why it holds up is better understood. Adding an inconsequential reservation, or changing vocabulary without responding, does not produce the same gain. It is this difference that transmission must make recognizable.

AN OBJECTION THAT TRULY THREATENS

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